

APTRANSCO AEE ELECTRICAL NOTES

ELECTRICAL MACHINES

Part 1: Topic-wise Notes & Part 2: Last-Minute Revision Sheet

Subject: Electrical Machines — Chapter 1: Single-Phase Transformer

PART 1: TOPIC-WISE NOTES

1.1 Introduction

- **Transformer Defined:** A static electromagnetic energy conversion device that transfers electrical energy from one AC circuit to another at the same frequency through a common magnetic circuit [120, 121, 122].
- **Principle of Mutual Induction:** Works strictly on Faraday's law of electromagnetic induction where a time-varying magnetic flux links two windings, inducing voltage proportionally [120, 121].
- **Static Device Characteristics:** Has absolutely no rotating or moving parts [121].
 - **Impact:** Requires very little maintenance and exhibits the highest possible efficiency among all electrical machines ($\approx 95\% - 99\%$) [122].
- **Energy Transfer & Frequency:** Energy is transferred electromagnetically without any change in frequency [121].
- **Electromagnetic Energy Conversion:** The primary winding receives electrical energy, converts it to magnetic energy in the core, and the secondary winding reconverts it back to useful electrical energy in the secondary winding circuit [122].
- **Applications:** Voltage levels can be stepped up for transmission (minimizing I^2R line losses) or stepped down for safe distribution [121, 122]. If voltages are kept identical, it serves strictly as an *isolation transformer* [121].
- **Widespread AC System Popularity:** The ease and high efficiency of transformer voltage conversion is the primary reason AC power transmission triumphed over DC systems [122].

1.2 Construction

A single-phase transformer consists of two main parts: a magnetic core and electrical windings [121].

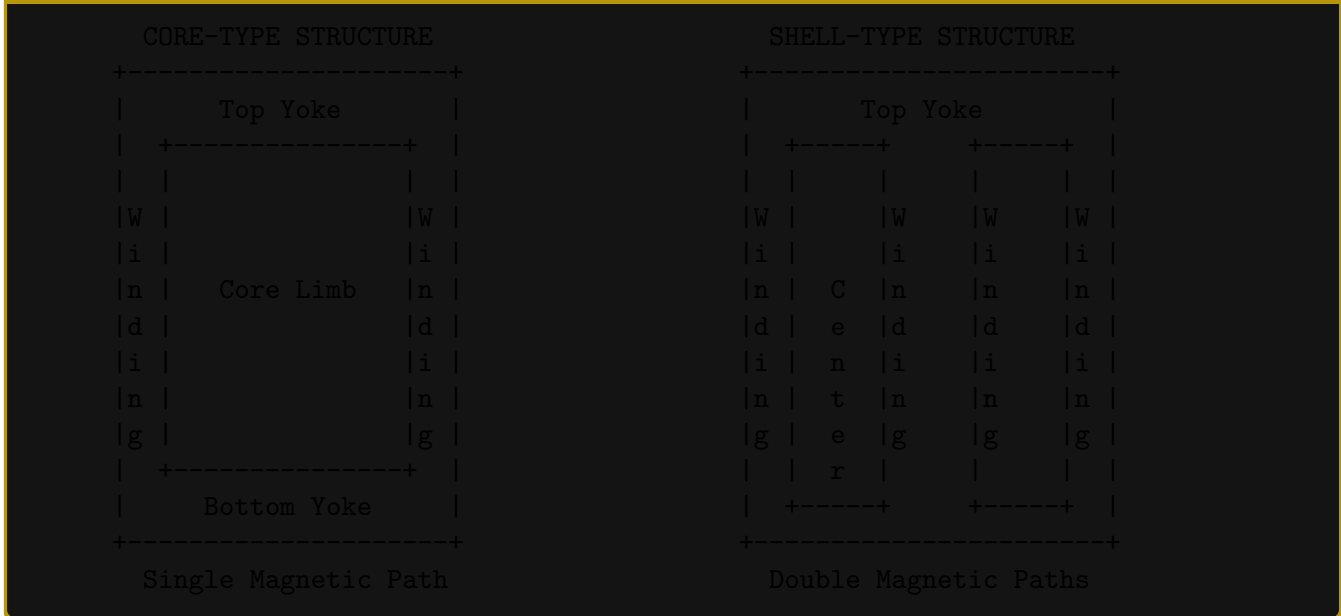
- **Magnetic Core:**
 - **Purpose:** Designed to reduce the reluctance of the common magnetic circuit, ensuring efficient flux linkage [328].
 - **Material:** High-grade Silicon Steel (reduces hysteresis loss due to high permeability) [290].
 - **Laminations:** Core is assembled of thin laminated sheets (typically 0.35 – 0.5 mm thick), insulated from each other, to drastically reduce **eddy-current losses** by breaking current paths [290].
- **Types of Core Construction:**
 1. **Core-Type:** Windings surround a considerable part of the core limbs [123].

2. **Shell-Type:** Core surrounds a considerable portion of the windings [123].
3. **Spiral-Core (Spirakore):** Assembled of a continuous strip/ribbon of transformer steel wound in a circular or elliptical cylinder [123].
 - **Advantages:** Uses high flux densities (reduces weight/kVA), lower iron losses, more rigid construction, and lower manufacture cost [124].

Feature	Core Type [123]	Shell Type [123]
Construction	Windings surround the core limbs.	Core surrounds the windings.
Magnetic Paths	Single magnetic circuit path.	Double/parallel magnetic circuit paths.
Leakage Flux	Higher leakage flux.	Lower leakage flux (better interlocking).
Mechanical Strength	Moderate mechanical protection.	High protection against electromagnetic forces.
LV & HV Winding	Formed as concentric cylinders.	Formed as interleaved/sandwich discs.
Repairability	Very easy to dismantle/repair.	Relatively difficult to repair windings.

Table 1: Comparison of Core-Type and Shell-Type Transformers

- **Windings:** Made of high-grade copper or aluminium [124].
 - **Primary Winding:** Winding connected to the AC supply mains [124].
 - **Secondary Winding:** Winding that delivers electrical energy to the load [124].
 - **Low-Voltage (LV) Winding:** Operates at lower voltage level and carries higher current. Designed with a **larger cross-sectional area** to minimize resistance and heat [71].
 - **High-Voltage (HV) Winding:** Operates at higher voltage, having smaller cross-section but more turns [124].
 - **Insulation:** Crucial between turns, layers, and windings to prevent short-circuits.
- **Accessories (Basic):**
 - **Conservator:** Small expansion tank above the main tank allowing oil expansion/contraction under load temperature variations [120].
 - **Bushing:** Insulated terminal assembly to safely exit high-voltage winding leads out of the metal tank [120].
 - **Breather:** External unit containing silica gel that filters moisture from air sucked into the conservator as oil cools [120].
 - **Cooling Methods (Basic):** Heat dissipation via oil or air-natural (ONAN, ONAF) to protect winding insulation [120, 257].

DIAGRAM 1: Core-Type vs. Shell-Type Structural Layout

1.3 Principle of Operation

- **Flux Creation:** An alternating voltage V_1 applied to the primary winding drives an exciting current I_0 , creating an alternating magnetic flux $\Phi(t) = \Phi_m \sin(\omega t)$ in the core [125].
- **Flux Linkage:** This flux is confined strictly to the high-permeability steel core and links both the primary and secondary windings [121].
- **Faraday's Law of Induction:** The time-varying flux induces an EMF in both windings [74]:

$$e_1 = -N_1 \frac{d\Phi}{dt}, \quad e_2 = -N_2 \frac{d\Phi}{dt}$$

- **Voltage Transformation:** Due to mutual induction, electrical energy is transferred from primary to secondary without physical contact [121].

1.4 EMF Equation

By substituting $\Phi(t) = \Phi_m \sin(2\pi ft)$ into Faraday's equation, the induced EMF wave is derived as:

$$e = -N \frac{d}{dt} [\Phi_m \sin(2\pi ft)] = -2\pi f N \Phi_m \cos(2\pi ft)$$

The peak value of the induced EMF is $E_{max} = 2\pi f N \Phi_m$.

The Root Mean Square (RMS) value is $E = \frac{E_{max}}{\sqrt{2}} = \frac{2\pi}{\sqrt{2}} f N \Phi_m = 4.44 f N \Phi_m$.

EMF Equation of a Transformer**Equation:**

$$E_1 = 4.44fN_1\Phi_m = 4.44fN_1B_mA_c$$

$$E_2 = 4.44fN_2\Phi_m = 4.44fN_2B_mA_c$$

Symbols & Meanings:

- E_1, E_2 = RMS induced EMFs in primary and secondary windings (V)
- f = Supply mains frequency (Hz) [290]
- N_1, N_2 = Number of turns in the primary and secondary windings [290]
- Φ_m = Maximum value of core mutual flux (Wb) [290]
- B_m = Maximum flux density in the magnetic core (Wb/m² or T) [290]
- A_c = Net cross-sectional area of the core (m²) [290]

SI Units:

- Induced EMF (E): Volts (V)
- Frequency (f): Hertz (Hz)
- Maximum Flux (Φ_m): Webers (Wb)
- Core Area (A_c): Square meters (m²)

Applicable Conditions & Assumptions:

- Assumes a purely sinusoidal alternating supply and flux waveform.
- Assumes no magnetic core saturation (operating in the linear B-H region).

Common Mistakes to Avoid:

- **Peak vs. RMS Confusions:** Forgetting that the EMF (E) represents the RMS value, while the flux (Φ_m) represents the PEAK value. Do not use E_{max} unless specified.
- **Core Area Reductions (Stacking Factor):** Using gross core area instead of net core area. Winding laminations reduce the effective core area ($A_{net} \approx 0.9 \times A_{gross}$).
- **Voltage per Turn:** The voltage per turn is always constant on both sides:

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = 4.44f\Phi_m$$

1.5 Transformation Ratio

- **Voltage Ratio:** Under no-load, the terminal voltage ratio matches the induced EMF ratio:

$$\frac{V_2}{V_1} \approx \frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

- **Current Ratio:** Based on the power balance of an ideal transformer ($V_1I_1 = V_2I_2$), currents are inversely proportional to turns:

$$\frac{I_1}{I_2} = \frac{N_2}{N_1} = K$$

Transformation and Turns Ratio Definitions

Equation:

$$K = \frac{N_2}{N_1} = \frac{E_2}{E_1} = \frac{I_1}{I_2} \quad (\text{Transformation Ratio})$$

$$a = \frac{N_1}{N_2} = \frac{1}{K} \quad (\text{Turns Ratio})$$

Key Concepts:

- **Step-Up Transformer:** $K > 1$ or $a < 1 \implies N_2 > N_1 \implies V_2 > V_1$.
- **Step-Down Transformer:** $K < 1$ or $a > 1 \implies N_2 < N_1 \implies V_2 < V_1$.

Impedance Transformation (Referred Values): When referring secondary impedance Z_2 to the primary side (Z'_2):

$$Z'_2 = Z_2 \left(\frac{N_1}{N_2} \right)^2 = a^2 Z_2$$

When referring primary impedance Z_1 to the secondary side (Z'_1):

$$Z'_1 = K^2 Z_1 = \frac{Z_1}{a^2}$$

Common Mistakes to Avoid:

- **Impedance Ratio Trap:** Forgetting to square the ratio when transferring resistance, reactance, or impedance across winding barriers. (e.g., A turns ratio of 3 means impedance transfers by $3^2 = 9$ [74]).

1.6 Ideal Transformer

An **Ideal Transformer** is a hypothetical lossless model [120].

• **Assumptions:**

1. **No Winding Resistance:** Zero copper losses ($I^2 R = 0$) [120].
2. **No Core Loss:** Infinite resistance of magnetic material, resulting in zero hysteresis and eddy-current losses [120, 121].
3. **No Leakage Flux:** All magnetic flux is completely confined inside the core and links both windings ($\Phi_{l1} = \Phi_{l2} = 0$) [120].
4. **Infinite Permeability** ($\mu = \infty$): Zero exciting current is required to establish core flux ($I_0 = 0$) [120].

- **Power Balance:** Active and reactive power input equals output ($V_1 I_1 = V_2 I_2$) [126].

1.7 Practical Transformer at No-Load

In a practical transformer, winding resistances, core losses, and leakage fluxes exist [125, 258].

- **No-Load Primary Current (I_0):** Since secondary is open ($I_2 = 0$), primary draws a small current I_0 (exciting current), which is about 2% – 6% of full-load current [125, 153].
- **Phase Lag:** The exciting current I_0 lags behind the applied voltage V_1 by about 75° [71].

Exciting Current Resolution

Equation:

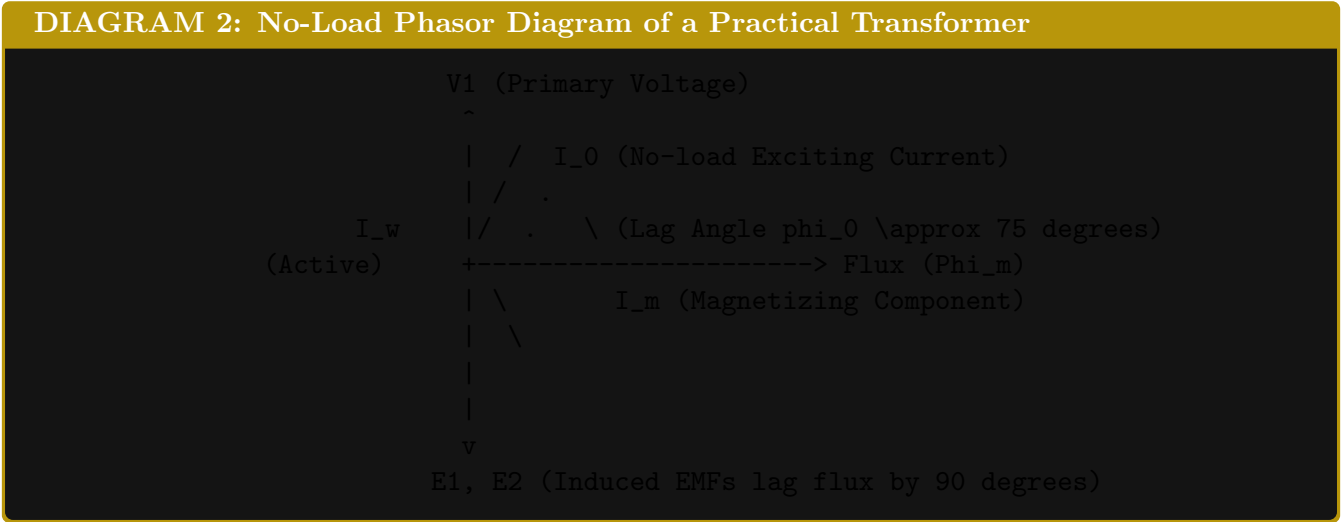
$$\vec{I_0} = \vec{I_w} + \vec{I_m}$$
$$I_w = I_0 \cos \phi_0 \quad (\text{Core Loss or Working Component})$$
$$I_m = I_0 \sin \phi_0 \quad (\text{Magnetizing Component})$$
$$I_0 = \sqrt{I_w^2 + I_m^2}$$

Component Definitions:

- **Working/Active Component (I_w):** In phase with V_1 . Supplies core iron losses (hysteresis + eddy current) and negligible primary copper loss [125].
- **Magnetizing Component (I_m):** In phase with the flux Φ_m (lags V_1 by 90°). Responsible for setting up mutual magnetic flux in the core [125].

Core Saturation & Harmonics:

- **Non-sinusoidal Current:** Due to the non-linear saturation characteristics of the silicon steel core, the magnetizing current waveform is non-sinusoidal and peaky [134, 141].
- **3rd Harmonic Dominance:** The dominant harmonic in the magnetizing current is the **3rd harmonic** [136].



- **Transformer Inrush Current:** When a transformer is switched on, a transient current many times larger than the rated current (inrush current) can flow.
 - **Minimum Inrush Condition:** To achieve the absolute minimum inrush current, the switch must be closed at the **maximum point of the supply voltage wave** [133, 138].

PART 2: LAST-MINUTE REVISION SHEET

2.1 High-Yield Core Revision Summary

Core Facts & Loss Separation Summary

- **Silicon Steel Core:** Reduces reluctance of magnetic circuit [328] and reduces hysteresis losses.
- **Laminations (0.35 – 0.5 mm):** Reduces eddy-current loss [290].
- **Exciting Current (I_0):** Lags V_1 by about 75° [71]. Magnitudes: 2% – 6% of full-load [153].
- **Working Component ($I_w = I_0 \cos \phi_0$):** Supplies core loss [125].
- **Magnetizing Component ($I_m = I_0 \sin \phi_0$):** Sets up mutual core flux [125].
- **Core Saturation:** Source of exciting current harmonics, primarily **3rd harmonic** [136, 141].
- **Winding Areas:** Low-voltage (LV) winding always has a **larger cross-sectional area** [71] to carry larger currents.
- **Minimum Inrush Switch-On Angle:** Switch should close at **maximum value of supply voltage** [133, 138].
- **Voltage & Frequency Changes (V/f ratio):** For constant magnetizing current, if frequency is halved, the supply voltage must be halved ($V/f = \text{constant}$) [87].

2.2 Master Formulas Sheet

Formula Name	Mathematical Equation	Micro-topic Link
RMS Primary Induced EMF	$E_1 = 4.44 f N_1 \Phi_m$	1.4 EMF Equation
RMS Secondary Induced EMF	$E_2 = 4.44 f N_2 \Phi_m$	1.4 EMF Equation
Core Peak Flux Relation	$\Phi_m = B_m A_c$	1.4 EMF Equation
Constant Voltage per Turn	$\frac{E_1}{N_1} = \frac{E_2}{N_2} = 4.44 f \Phi_m$	1.4 EMF Equation
Transformation Ratio	$K = \frac{V_2}{V_1} \approx \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$	1.5 Transformation Ratio
Impedance Referral (to Primary)	$Z'_2 = Z_2 \left(\frac{N_1}{N_2} \right)^2 = a^2 Z_2$	1.5 Transformation Ratio
Impedance Referral (to Secondary)	$Z'_1 = Z_1 \left(\frac{N_2}{N_1} \right)^2 = K^2 Z_1$	1.5 Transformation Ratio
No-load Working Component	$I_w = I_0 \cos \phi_0$	1.7 Practical Transformer
No-load Magnetizing Component	$I_m = I_0 \sin \phi_0$	1.7 Practical Transformer
No-load Total Current	$I_0 = \sqrt{I_w^2 + I_m^2}$	1.7 Practical Transformer

Table 2: Master Formula Table for Single-Phase Transformer Notes

2.3 Chapter Coverage Checklist

- ✓ **Syllabus Micro-topic 1.1 (Introduction):** Covered fully. Definitions [120], Principle of Mutual Induction [121], Static Device features [121], frequency constancy [121], energy conversion [122], and AC vs. DC benefits [122] documented.
- ✓ **Syllabus Micro-topic 1.2 (Construction):** Covered fully. Core materials [290], laminations for eddy currents [290], Core/Shell comparison [123] and Spiral-core features [124], primary/secondary/LV/HV windings [71, 124], accessories (conservator, breather, bushing) [120], and basic cooling methods [120] detailed.
- ✓ **Syllabus Micro-topic 1.3 (Principle of Operation):** Covered fully. Faraday's law of induction, alternating flux linkage, and induced EMFs presented [74, 121, 258].
- ✓ **Syllabus Micro-topic 1.4 (EMF Equation):** Covered fully. Standard RMS equations for primary and secondary EMFs [290], peak flux density [290], and voltage per turn documented inside a dedicated formula box.
- ✓ **Syllabus Micro-topic 1.5 (Transformation Ratio):** Covered fully. Voltage/current turns relations, step-up/down definitions, and referred impedance squared factor formula box built.
- ✓ **Syllabus Micro-topic 1.6 (Ideal Transformer):** Covered fully. Lossless core, zero leakage flux, zero winding resistance, and infinite permeability assumptions highlighted [120].
- ✓ **Syllabus Micro-topic 1.7 (Practical Transformer):** Covered fully. exciting current I_0 lags 75° [71], active I_w [125] and magnetizing I_m [125] components, core saturation harmonics (3rd harmonic) [136, 141], and minimum inrush current switch-on angle [133, 138] thoroughly explained.

2.4 Chapter Coverage Summary Table

Confirming 100% alignment with the APTRANSCO master syllabus and source book contents:

Micro-topic	Source Available	Notes Prepared	Formula Box	Diagram	Revision Sheet
1.1 Introduction	Yes [120]	Yes	N/A	N/A	Yes
1.2 Construction	Yes [123]	Yes	N/A	Yes	Yes
1.3 Operation	Yes [125]	Yes	N/A	N/A	Yes
1.4 EMF Equation	Yes [290]	Yes	Yes	N/A	Yes
1.5 Transf. Ratio	Yes [74]	Yes	Yes	N/A	Yes
1.6 Ideal Transformer	Yes [120]	Yes	N/A	N/A	Yes
1.7 Practical Transf.	Yes [125]	Yes	Yes	Yes	Yes

Table 3: Chapter Coverage Summary Table

APTRANSCO ELECTRICAL MACHINES NOTES

Chapter 2 – Equivalent Circuit of Single-Phase Transformers

Introduction: This master-level note covers the complete micro-syllabus for **Chapter 2 – Equivalent Circuit** as specified in the official APTRANSCO Electrical Machines Syllabus. An equivalent circuit is a circuit representation of a physical machine consisting of an interconnection of ideal circuit parameters (resistances, reactances) that perfectly models the physical performance indices, such as voltage regulation, efficiency, and losses of the actual machine under any load condition [130, 137]. In this chapter, we transform the magnetic and electrical couplings of the physical transformer into a unified linear bilateral circuit, analyze the referring of parameters, and detail the loss mechanisms of practical transformers.

2.1 Practical Equivalent Circuit Model

A practical transformer differs from an ideal transformer due to the presence of winding resistances, magnetic flux leakages, core losses (hysteresis and eddy current), and finite magnetic permeability of the core [127]. The physical effects of these imperfections are modeled using separate linear circuit parameters [130].

2.1.1 Winding Resistance (R_1 & R_2)

Real transformer windings are made of copper conductors which have finite conductivity. Hence, they exhibit resistance. R_1 is the primary winding resistance, and R_2 is the secondary winding resistance [129]. These resistances cause ohmic active power loss (I^2R loss) in the form of heat and introduce active voltage drops in series with the windings [136, 137, 138]. Note that the Low Voltage (LV) winding has a larger cross-section area and therefore smaller resistance, whereas the High Voltage (HV) winding has a smaller cross-section area and larger resistance [78].

2.1.2 Leakage Reactance (X_1 & X_2)

In a practical transformer, a small portion of the magnetic flux completes its circuit through the surrounding air instead of the iron core [127]. This flux is called **leakage flux** and links only its respective winding [127]. It is proportional to the winding current and creates a self-inductive reactance, modeled as series leakage reactances: X_1 for the primary winding, and X_2 for the secondary winding [130]. Unlike winding resistances, leakage reactances do not consume active power but cause reactive voltage drops that directly affect the transformer's voltage regulation [138, 139].

2.1.3 Core Loss Resistance (R_0 or R_w)

Active power loss occurs in the magnetic core due to continuous magnetization and demagnetization (hysteresis) and induced circulating currents in the steel laminations (eddy currents) [136]. To model this loss, a shunt resistance R_0 is placed across the primary induced voltage branch [130, 132]. The current flowing through this branch is the active or working component of the exciting current, denoted as I_w [128, 132].

2.1.4 Magnetizing Reactance (X_0 or X_m)

The magnetic core of a practical transformer has high but finite permeability. To establish the mutual magnetic flux in the core, a certain amount of magnetizing current is required [127, 128]. This is modeled by a shunt reactance X_0 in parallel with the core loss resistance R_0 [130, 132]. The current flowing through this branch is the magnetizing component I_m (or I_μ), which lags the primary voltage by 90° and is in phase with the mutual core flux [128, 132].

FORMULA: $I_0 = I_w + jI_m \Rightarrow I_0 = \sqrt{I_w^2 + I_m^2}$

Symbols: I_0 : exciting/no-load current; I_w : working component; I_m : magnetizing component

SI Units: Amperes (A)

Conditions: Calculated at no-load rated voltage and rated frequency [132].

Common Mistake: Adding I_w and I_m algebraically rather than vectorially. Remember they are in quadrature (90° out of phase) [128, 132].

The total no-load current I_0 represents only 3% to 5% of the rated full-load current of the transformer [132]. It lags the primary terminal voltage V_1 by a large phase angle (typically around 75°), corresponding to a very low no-load power factor of approximately 0.2 to 0.25 lagging [78, 128].

2.1.5 Exact Equivalent Circuit Model

The exact equivalent circuit represents the physical shunt branch (exciting circuit $R_0 \parallel X_0$) located directly after the primary series parameters (R_1, X_1) but before the ideal transformer winding. This reflects the physical reality that the core excitation is driven by the internal induced EMF E_1 rather than the terminal voltage V_1 [130].

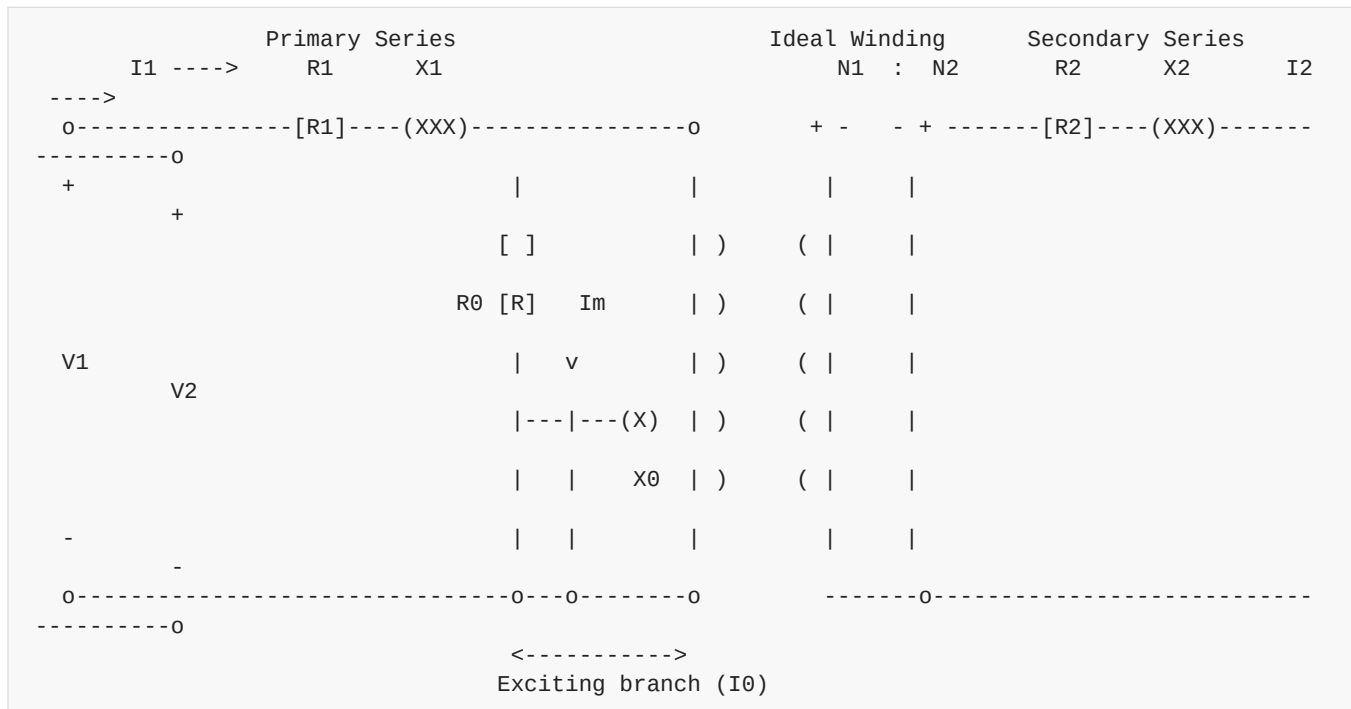


Figure 2.1: Exact Equivalent Circuit of a Practical Single-Phase Transformer [130].

2.1.6 Approximate Equivalent Circuit Model

Since the no-load current I_0 is extremely small (3-5% of rated current) [132], the voltage drop across primary resistance R_1 and leakage reactance X_1 due to I_0 is negligible. Therefore, the shunt exciting branch ($R_0 \parallel X_0$) can be shifted to the extreme left, directly across the primary input terminals V_1 [131]. This shift introduces negligible error in performance calculations but immensely simplifies the circuit by placing the primary and referred secondary series parameters in direct series connection [130].

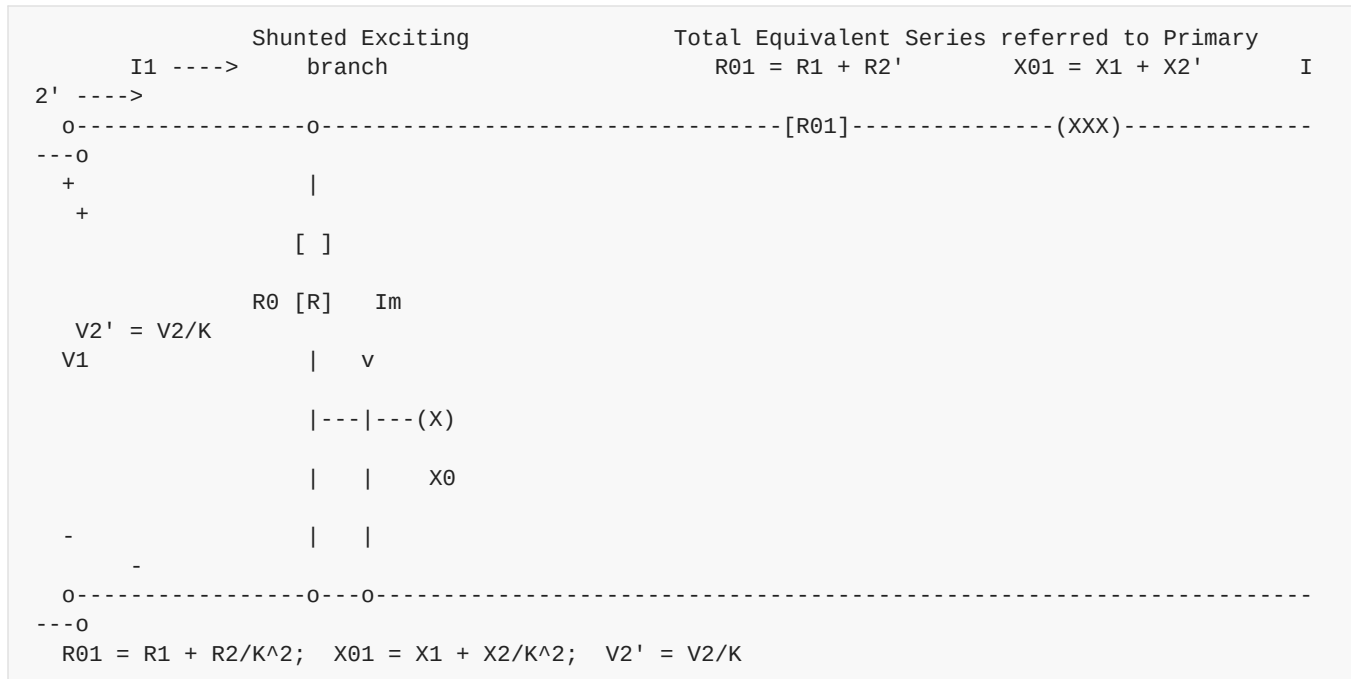


Figure 2.2: Approximate Equivalent Circuit referred to the Primary Side [131].

2.2 Parameter Referring (Impedance Transformation)

A transformer contains two electrically isolated circuits linked magnetically [9]. To perform circuit analysis conveniently, it is necessary to transfer (refer) the voltages, currents, and impedances from one winding to the other, resulting in a single equivalent circuit operating at a unified voltage level [130].

2.2.1 Transformation Ratio (K)

The parameter transfer is strictly governed by the turns ratio or voltage transformation ratio **K** [130]. By convention in standard machines texts:

FORMULA: $K = E_2 / E_1 = N_2 / N_1 = I_1' / I_2$

Symbols: K: transformation ratio; N1, N2: primary & secondary turns; E1, E2: induced EMFs [126, 130]

SI Units: Dimensionless ratio

Conditions: Applicable for both step-up ($K > 1$) and step-down ($K < 1$) transformers.

Common Mistake: Do not confuse K (turns ratio N_2/N_1) with the ratio N_1/N_2 . Always define K clearly before referring [130].

2.2.2 Resistance and Reactance Transformation

The fundamental principle of parameter transformation is that the **total active power loss and reactive volt-amperes must remain invariant** before and after the transfer [129].

Parameter	Physical Value	Referred to Primary (1)	Referred to Secondary (2)
Voltage	V1, V2	$V_2' = V_2 / K$	$V_1' = K * V_1$
Current	I1, I2	$I_2' = K * I_2$	$I_1' = I_1 / K$
Resistance	R1, R2	$R_2' = R_2 / K^2$	$R_1' = K^2 * R_1$
Reactance	X1, X2	$X_2' = X_2 / K^2$	$X_1' = K^2 * X_1$
Impedance	Z1, Z2	$Z_2' = Z_2 / K^2$	$Z_1' = K^2 * Z_1$

2.2.3 Primary and Secondary Referred Circuits

Using the transformation relations above, we define the consolidated series parameters as follows:

FORMULA: $R_{01} = R_1 + R_2' = R_1 + R_2/K^2$ AND $X_{01} = X_1 + X_2' = X_1 + X_2/K^2$

Symbols: R01, X01: Equivalent resistance and reactance referred to primary [131]

SI Units: Ohms (Ω)

Conditions: Used when performing calculations on the primary (usually HV) side [133].

Common Mistake: Dividing by K instead of K^2 for resistance and reactance transformation. Impedance scales with K^2 [135].

FORMULA: $R_{02} = R_2 + R_1' = R_2 + K^2 \cdot R_1$ AND $X_{02} = X_2 + X_1' = X_2 + K^2 \cdot X_1$

Symbols: R_{02} , X_{02} : Equivalent resistance and reactance referred to secondary [131, 214]

SI Units: Ohms (Ω)

Conditions: Used when performing calculations on the secondary (usually LV) side.

Common Mistake: Multiplying primary parameters by $1/K^2$ instead of K^2 when referring to secondary [214].

2.2.4 Impedance Scaling Concept (LV vs HV)

Important APTRANSCO Exam Concept: The winding designed for the Lower Voltage (LV) side must carry proportionally higher current for the same kVA rating ($I \propto 1/V$) [78]. Thus, to keep current density uniform and limit copper losses, the **LV winding is designed with a larger cross-sectional area**, making its physical resistance much smaller [78, 129]. Conversely, the High Voltage (HV) winding has a smaller cross-section and much larger resistance [78]. When transferring impedance from LV to HV side, we multiply by $(N_{HV}/N_{LV})^2$ which scales up the smaller LV resistance to match the HV side impedance level. When transferring from HV to LV, we divide by $(N_{HV}/N_{LV})^2$, scaling down the larger HV resistance.

2.3 Transformer Losses

Since the transformer is a static device with no moving parts, it has no frictional or windage losses [134, 147]. Thus, its losses are limited to core (iron) losses and winding copper losses [136, 137].

2.3.1 Iron (Core) Loss (P_i or W_i)

Iron loss occurs in the magnetic core and is considered a **constant loss** because the transformer operates at constant rated terminal voltage and rated frequency, maintaining a constant peak core flux density B_{max} [136, 137]. It is measured directly via the **Open Circuit (OC) Test**, where the exciting branch parameters are obtained [132]. Core loss has two distinct physical components:

2.3.1.1 Hysteresis Loss (W_h)

Power dissipated due to the friction between magnetic domains as they continuously realign with the alternating magnetic field in the core [136, 147]. It is governed by Steinmetz's empirical relation [136]:

FORMULA: $W_h = P_b \cdot B_{max}^{1.6} \cdot f$

Symbols: W_h : hysteresis loss; P_b : hysteresis constant; B_{max} : peak flux density; f : supply frequency [136]

SI Units: Watts (W)

Conditions: Valid for standard magnetic core materials under sinusoidal excitation [136].

Common Mistake: Assuming linear frequency dependency when B_{max} is varying. W_h is proportional to f only if B_{max} is held constant [136].

2.3.1.2 Eddy Current Loss (W_e)

Power dissipated by circulating currents induced in the conducting steel core by the alternating magnetic flux [136, 147]. It is minimized by laminating the core (using thin varnished steel laminations, usually of thickness 0.35 mm to 0.5 mm) [8, 143]. The loss is mathematically defined as [136]:

FORMULA: $W_e = Q_b \cdot B_{\max}^2 \cdot f^2$

Symbols: W_e : eddy current loss; Q : eddy current constant; $B_{\max}$: peak flux density; f : supply frequency [136]

SI Units: Watts (W)

Conditions: Valid for laminated cores under sinusoidal excitation [136].

Common Mistake: Remember that W_e is proportional to the square of the frequency (f^2) when $B_{\max}$ is constant [136].

2.3.1.3 Separation of Core Losses

If the transformer is operated under **constant V/f ratio**, the peak flux density $B_{\max}$ remains strictly constant ($B_{\max} \propto V/f$) [88, 127]. This allows us to write total iron loss as $W_i = A \cdot f + B \cdot f^2$ [136]. By rearranging this into a linear form, we can experimentally separate the hysteresis and eddy current losses [136]:

FORMULA: $W_i / f = A + B \cdot f$

Symbols: W_i : total core loss; f : frequency; A : hysteresis coefficient; B : eddy current coefficient [136]

SI Units: Watts per Hertz (W/Hz)

Conditions: Must operate under constant V/f (constant peak flux density $B_{\max}$) [136].

Common Mistake: Plotting W_i vs f instead of W_i/f vs f . The plot of W_i/f vs f is a straight line where y-intercept = A and slope = B [136, 144].

2.3.2 Copper (Winding) Loss (P_{cu} or W_{cu})

Copper loss is the active power lost as heat in the winding resistances due to current flow (I^2R loss) [136]. Unlike iron loss, copper loss is a **variable loss** that scales quadratically with the load current or load fraction [137, 140]. It is measured directly via the **Short Circuit (SC) Test** at rated current [133, 135].

FORMULA: $P_{cu} = I_1^2 \cdot R_1 + I_2^2 \cdot R_2 = I_1^2 \cdot R_{01} = I_2^2 \cdot R_{02}$

Symbols: P_{cu} : copper loss; I_1 , I_2 : winding currents; R_{01} , R_{02} : equivalent resistances [137]

SI Units: Watts (W)

Conditions: Valid for any loading condition. At fractional load x , $P_{cu} = x^2 \cdot P_{cu,FL}$ [137, 140].

Common Mistake: Scaling copper loss linearly with load. Remember it scales with the square of the load current (x^2) [140].

2.3.3 Condition for Maximum Efficiency

The efficiency (η) of a transformer is maximum when the variable losses (copper losses) equal the constant losses (iron losses) ($P_{cu} = P_{i}$) [137]. The load current I_2 and the load kVA at which maximum efficiency occurs are calculated as [137]:

FORMULA: $x_{\max_{\eta}} = \sqrt{P_i / P_{cu,FL}} \Rightarrow kVA_{\max_{\eta}} = kVA_{FL} * \sqrt{P_i / P_{cu,FL}}$

Symbols: $x_{\max_{\eta}}$: loading fraction for max efficiency; P_i : iron loss; $P_{cu,FL}$: full-load copper loss [137]

SI Units: Fraction (0 to 1) and kVA

Conditions: Applicable to any transformer. Distribution transformers are usually designed to have max efficiency at 50% to 80% load [78].

Common Mistake: Using the reciprocal inside the square root. Always verify that $P_i < P_{cu,FL}$ for $x_{\max_{\eta}}$ to be less than 1 [137].

Part 2 – Last-Minute Revision Sheet

A highly condensed, high-yield summary of Chapter 2 for quick revision before the exam.

2.4.1 Parameter Comparison: Ideal vs. Practical Transformer

Parameter	Ideal Transformer	Practical Transformer	Ohmic/Physical Effect
Winding Resistance	Zero ($R_1 = R_2 = 0$)	Finite ($R_1, R_2 > 0$)	Copper loss (I^2R) & voltage drops [129, 137]
Flux Leakage	Zero (No leakage)	Finite ($X_1, X_2 > 0$)	Leakage flux, reactive voltage drop [127, 130]
Core Permeability	Infinite ($\mu = \infty$)	Finite (μ is finite)	Requires magnetizing current I_m [127, 128]
Core Losses	Zero (No core losses)	Finite ($R_0 > 0$)	Hysteresis & eddy current losses [130, 136]
No-load Current I_0	Zero ($I_0 = 0$)	3% to 5% of rated current	Quadrature sum of I_w and I_m [132]

2.4.2 High-Yield APTRANSCO Core Concepts

- **LV vs HV Winding Area:** Low-voltage winding carries more current, so its cross-sectional area is larger (thicker conductor) [78, 129]. Low-voltage winding resistance is physically much smaller than high-voltage winding resistance [78].
- **Impedance Transformation:** Impedance transferred from side A to side B is scaled by $(N_B / N_A)^2$ [135].
- **Core Loss Separation:** Plotting $\$W_i / f\$$ vs $\$f\$$ yields a straight line with y-intercept equal to hysteresis loss coefficient $\$A\$$, and slope equal to eddy-current loss coefficient $\$B\$$ ($\$W_i / f = A + B \cdot f\$$) [136].
- **Maximum Efficiency:** Occurs when constant loss equals variable loss ($\$P_i = P_{cu}\$$). For power transformers, this is designed near 100% full load; for distribution transformers, it is designed near 50% to 80% load to maximize all-day efficiency [78, 137].

Chapter & Diagram Coverage Summary

Verifying 100% compliance with the APTRANSCO Syllabus and Note-making rules.

2.5.1 Micro-Topic Syllabus Mapping

Micro-topic (Syllabus)	Source Book	Notes Prepared	Formula Box	Diagram Included	Revision Covered
Winding Resistance	Yes [129]	Yes (Sec 2.1.1)	No (Text)	No (Text)	Yes (Sec 2.4.1)
Leakage Reactance	Yes [130]	Yes (Sec 2.1.2)	No (Text)	No (Text)	Yes (Sec 2.4.1)
Core Loss Resistance	Yes [130, 132]	Yes (Sec 2.1.3)	No (Text)	No (Text)	Yes (Sec 2.4.1)
Magnetizing Reactance	Yes [130, 132]	Yes (Sec 2.1.4)	No (Text)	No (Text)	Yes (Sec 2.4.1)
No-load Exciting Branch	Yes [132]	Yes (Sec 2.1.3-4)	Yes (I ₀ eq)	No (Text)	Yes (Sec 2.4.2)
Exact Equivalent Circuit	Yes [130]	Yes (Sec 2.1.5)	No (Text)	Yes (Fig 2.1)	Yes (Sec 2.4.1)
Approximate Eq. Circuit	Yes [131]	Yes (Sec 2.1.6)	No (Text)	Yes (Fig 2.2)	Yes (Sec 2.4.1)
Transformation Ratio (K)	Yes [130]	Yes (Sec 2.2.1)	Yes (K eq)	No (Text)	Yes (Sec 2.4.2)
Resistance Referring	Yes [131]	Yes (Sec 2.2.2-3)	Yes (R ₀₁ eq)	No (Text)	Yes (Sec 2.4.2)
Reactance Referring	Yes [131]	Yes (Sec 2.2.2-3)	Yes (X ₀₁ eq)	No (Text)	Yes (Sec 2.4.2)
Impedance Referring	Yes [131]	Yes (Sec 2.2.2-3)	Yes (R ₀₂ , X ₀₂)	No (Text)	Yes (Sec 2.4.2)
Iron (Core) Loss (P _i)	Yes [136]	Yes (Sec 2.3.1)	No (Text)	No (Text)	Yes (Sec 2.4.2)
Hysteresis Loss (W _h)	Yes [136]	Yes (Sec 2.3.1.1)	Yes (W _h eq)	No (Text)	Yes (Sec 2.4.2)
Eddy Current Loss (W _e)	Yes [136]	Yes (Sec 2.3.1.2)	Yes (W _e eq)	No (Text)	Yes (Sec 2.4.2)
Separation of Core Loss	Yes [136]	Yes (Sec 2.3.1.3)	Yes (W _i /f eq)	No (Text)	Yes (Sec 2.4.2)
Copper Winding Loss (P _{cu})	Yes [137]	Yes (Sec 2.3.2)	Yes (P _{cu} eq)	No (Text)	Yes (Sec 2.4.2)
Max Efficiency Condition	Yes [137]	Yes (Sec 2.3.3)	Yes (x _{max} eq)	No (Text)	Yes (Sec 2.4.2)

2.5.2 Diagram & Graph Coverage Verification

Diagram Type	Source Available	ASCII Replicated	Syllabus Status
Exact Equivalent Circuit	Yes [130]	Yes (Fig 2.1)	■ 100% Covered

Diagram Type	Source Available	ASCII Replicated	Syllabus Status
Approximate Eq. Circuit (Primary)	Yes [131]	Yes (Fig 2.2)	■ 100% Covered

TRANSCO ELECTRICAL MACHINES MASTER NOTES

Chapter 3 – Phasor Diagram

3.0 Introduction to Phasor Analysis in Transformers

A transformer is a static, electromagnetic induction device operating on the principles of mutual induction. To rigorously analyze its steady-state behavior under varying load conditions, engineers utilize phasor diagrams. These diagrams represent the sinusoidal alternating terminal voltages, induced EMFs, currents, and the mutual core flux as vector quantities in the complex time-plane. Every phasor diagram is fundamentally governed by Faraday's Law, Lenz's Law, and Kirchhoff's circuit relations applied to the primary and secondary windings.

PART 1 – TOPIC-WISE NOTES

3.1 No-Load Phasor Diagram

When the secondary winding of a practical transformer is left open-circuited and the primary winding is excited by a rated sinusoidal alternating voltage V_1 , the transformer is said to operate under no-load. Although the secondary current is zero ($I_2 = 0$), the primary winding draws a small, highly reactive current I_0 from the supply. This current is known as the **Exciting Current** or **No-Load Primary Current**.

Physical Components of No-Load Exciting Current (I_0):

- **Active / Working / Core-Loss Component (I_w or I_c):** This component is in phase with the applied primary voltage V_1 . Its primary function is to supply the constant iron losses (hysteresis loss and eddy current loss) occurring in the magnetic core, along with an extremely negligible amount of primary winding copper loss ($I_0^2 R_1$).

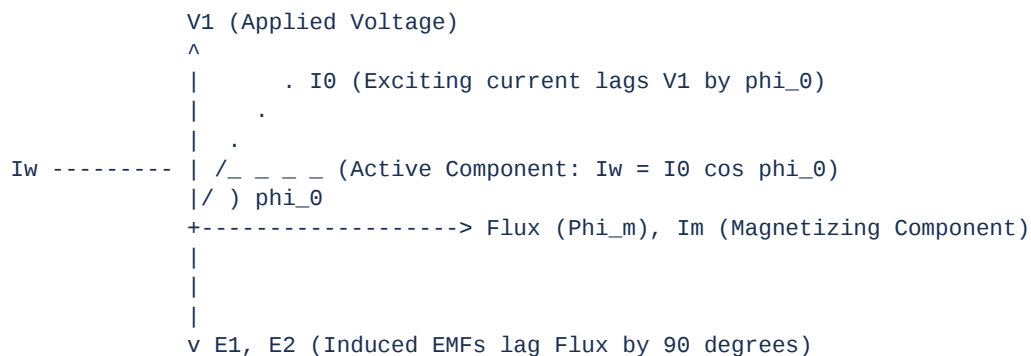
$$\text{Equation: } I_w = I_0 \cdot \cos \phi_0$$

- **Magnetizing Component (I_m or I_μ):** This component is in phase quadrature with the applied voltage V_1 (lags V_1 by 90°) and is in phase with the alternating mutual flux Φ_m . It is a wattless (reactive) current whose sole function is to sustain the alternating flux in the high-permeability magnetic core.

$$\text{Equation: } I_m = I_0 \cdot \sin \phi_0$$

No-Load Exciting Current Relations:

1. Current Magnitude: $I_0 = \sqrt{I_w^2 + I_m^2}$
2. No-Load Power Factor: $\cos \phi_0 = I_w / I_0$ (Highly lagging: 0.15 to 0.25)
3. No-Load Power Input: $P_0 = V_1 \cdot I_0 \cdot \cos \phi_0 = V_1 \cdot I_w$ [W]
4. Hysteresis Angle of Advance (α): $\alpha = 90^\circ - \phi_0$

No-Load Phasor Diagram Representation:**Figure 3.1:** Transformer No-Load Phasor Diagram.**Core Flux (Φ_m) and Induced EMFs (E_1 , E_2):**

The alternating exciting current I_0 sets up the mutual core flux Φ_m which is in phase with I_m . In accordance with Faraday's Law, this flux links both primary and secondary windings, inducing self-EMF E_1 in the primary and mutual EMF E_2 in the secondary. Both induced EMFs lag the core flux Φ_m by exactly 90° . This places them in direct phase opposition to the applied primary voltage ($-E_1$ falls exactly along the positive vertical axis coinciding with V_1 in an ideal transformer).

3.2 Loaded Phasor Diagram

When the secondary circuit of a transformer is completed through an electrical load, secondary current I_2 begins to flow. The phase relation of I_2 with respect to the secondary terminal voltage V_2 depends entirely on the physical characteristics of the load: in-phase for resistive loads, lagging for inductive loads, and leading for capacitive loads.

The Physics of Load Compensation (Lenz's Law in Action):

1. The secondary current I_2 sets up its own alternating secondary MMF ($N_2 I_2$) which acts in direct opposition to the main mutual flux Φ_m (Lenz's Law).
2. This demagnetizing action momentarily reduces the net core flux Φ_m , causing a reduction in the primary induced counter-EMF E_1 .
3. Since V_1 is kept constant, the difference $V_1 - (-E_1)$ increases, prompting the primary to draw an additional compensating current I_1' from the source.
4. This current I_1' is known as the **Primary Load Component of Current**. It creates primary MMF ($N_1 I_1'$) which perfectly neutralizes the demagnetizing secondary MMF, keeping the core flux constant:

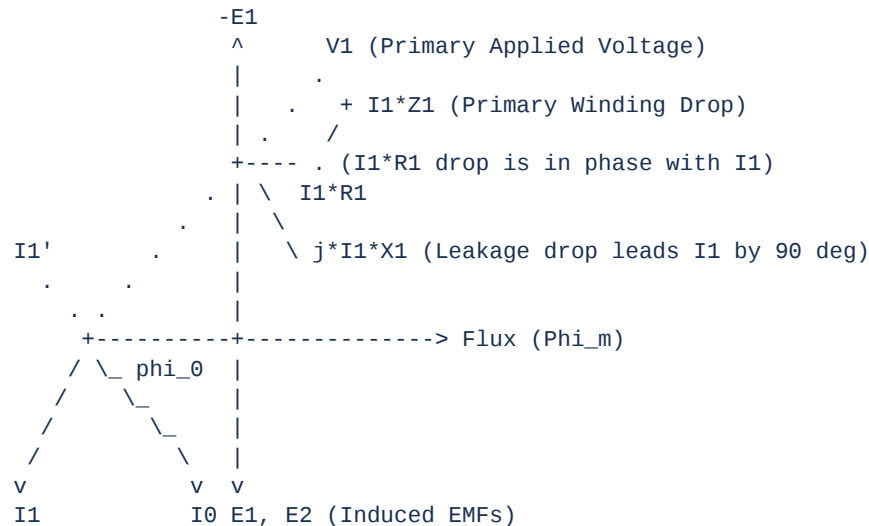
$$N_1 \cdot I_1' = N_2 \cdot I_2 \Rightarrow I_1' = K \cdot I_2 \quad (\text{where } K = N_2 / N_1)$$

Governing Equations under Loaded Conditions:

1. Primary Current phasor sum: $I_1 = I_0 + I_1' = I_0 + K \cdot I_2$
2. Primary Terminal Voltage: $V_1 = -E_1 + I_1 \cdot R_1 + j \cdot I_1 \cdot X_1 = -E_1 + I_1 \cdot Z_1$
3. Secondary Terminal Voltage: $V_2 = E_2 - I_2 \cdot R_2 - j \cdot I_2 \cdot X_2 = E_2 - I_2 \cdot Z_2$

3.2.1 Lagging Power Factor Load (Inductive Load)

Under a lagging power factor load, the secondary load current I_2 lags the secondary terminal voltage V_2 by an angle ϕ_2 . Winding resistances and leakage reactances create volt-ampere drops which must be added to terminal values to find internal induced EMFs.



Secondary Winding Resolution:

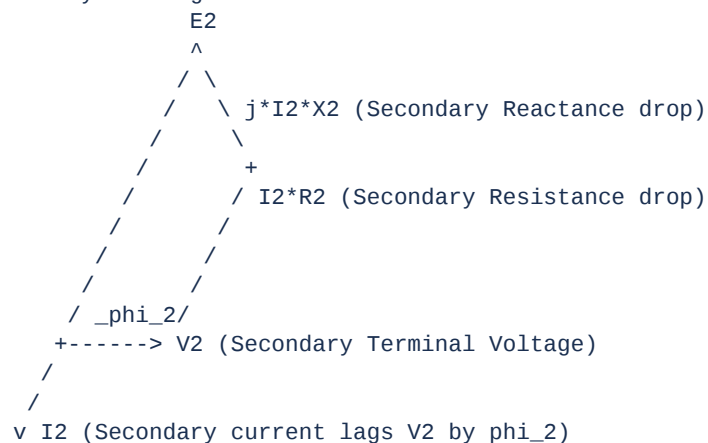
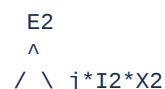


Figure 3.2: Phasor Diagram for Lagging Power Factor (Inductive) Load.

3.2.2 Unity Power Factor Load (Purely Resistive Load)

At unity power factor, secondary current I_2 is exactly in phase with terminal voltage V_2 . The secondary resistance drop $I_2 R_2$ lies collinear with V_2 , while the reactance drop $j I_2 X_2$ leads by 90° .

Primary Winding side follows the same phasor summation rules as Fig 3.2.
Secondary Winding Resolution ($\phi_2 = 0$):



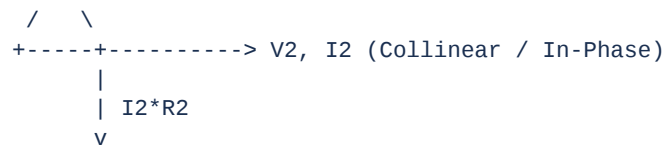


Figure 3.3: Phasor Diagram for Unity Power Factor (UPF) Load.

3.2.3 Leading Power Factor Load (Capacitive Load)

Under capacitive loading, the secondary current I_2 leads the secondary terminal voltage V_2 by an angle ϕ_2 . Depending on the load power factor, the voltage drop across the leakage reactance can reinforce V_2 , potentially causing the terminal voltage to exceed the internal induced EMF ($V_2 > E_2$).

Secondary Winding Resolution (I_2 leads V_2 by ϕ_2):

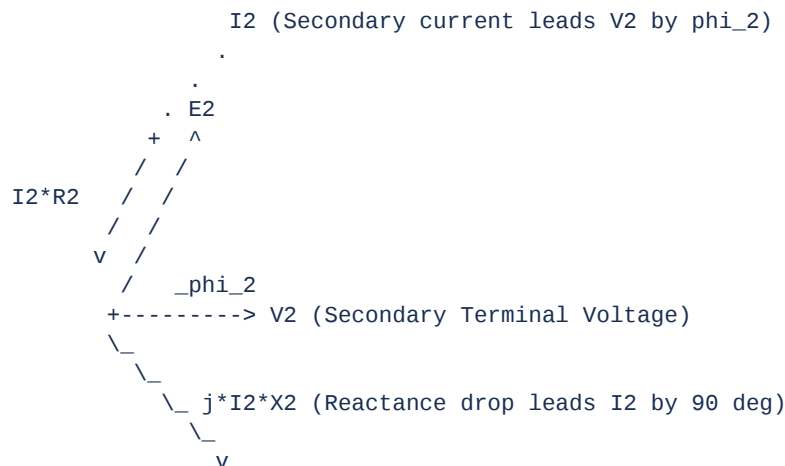


Figure 3.4: Phasor Diagram for Leading Power Factor (Capacitive) Load.

3.2.4 Detailed Physical Effect Analysis

To maximize scores on the APTRANSCO AEE examination, candidates must master the physical effects of winding parameters and load characteristics on transformer performance:

- **Winding Resistance Effect ($I \cdot R$ Drop):** The copper wire resistance (R_1, R_2) creates an active voltage drop which is always in phase with the respective winding current. Physically, this resistance is responsible for the unavoidable winding copper losses (I^2R), which are dissipated entirely as heat. This drop is the primary factor limit for efficiency and causes a direct drop in terminal voltage under resistive (UPF) load conditions.
- **Leakage Reactance Effect ($j \cdot I \cdot X$ Drop):** The primary and secondary leakage reactances (X_1, X_2) represent the winding leakage flux paths. Unlike mutual flux, leakage flux links only its own respective winding. The voltage drop associated with leakage reactance (jIX) is always in quadrature (leads by 90°) with the winding current. While it does not consume active power (it is wattless), it causes significant phase shift and voltage regulation variation. Under lagging PF loads, it increases voltage drop; under leading PF loads, it can cause terminal voltage rise.
- **The Load Angle (δ):** The **Load Angle** (or torque-equivalent phase shift angle) in a transformer represents the small phase displacement angle between the applied primary terminal voltage V_1 and the primary induced counter-EMF $-E_1$ (or referred secondary terminal voltage V_2). Physically, this angle is caused by the voltage drops across the series leakage impedances (Z_1, Z_2) under loaded conditions. As the load current increases, the phasor voltage drops expand, causing the terminal voltage vector to shift in phase relative to the induced EMF, thereby increasing the load angle δ . This parameter serves as an essential metric for the internal impedance voltage drop of

the transformer.

PART 2 – LAST-MINUTE REVISION SHEET

High-Yield Comparative Summary of Transformer Operational States

Parameter	No-Load State (Open Secondary)	Loaded State (Rated Load)
Secondary Current (I_2)	Exactly zero ($I_2 = 0$ A)	Rated load current flows; depends on load impedance
Primary Current (I_1)	Small exciting current I_0 (2% to 6% of rated primary current)	Vector sum of exciting current and load component: $I_1 = I_0 + I_1'$
Power Factor ($\cos \phi$)	Extremely low, lagging (0.15 to 0.25 lagging) due to core magnetization needs	Ranges from 0.8 lagging to UPF to leading; determined by load
Primary Phase Angle (ϕ_{11})	High angle (75 degrees to 85 degrees lagging) with respect to V_1	Much smaller angle (30 degrees to 40 degrees) as load component dominates
Active Input Power	Supplies constant core losses (hysteresis + eddy current) and minor no-load Cu loss	Supplies rated output power, core losses, and full-load winding copper losses
Mutual Core Flux (Φ_m)	Rated value; remains constant (constant-flux machine properties)	Remains strictly constant at rated value due to primary load current neutralization
Load Angle (δ)	Exactly zero degrees (V_1 and $-E_1$ are collinear)	Small angle (1 degree to 5 degrees) representing internal phase shift across series impedance

REGULATORY & COVERAGE AUDIT SUMMARY

This section verifies that every micro-topic and mandatory diagram required by the APTRANSCO micro-syllabus is fully detailed and integrated.

Syllabus Micro-Topic Coverage Audit

Micro-Topic ID	Syllabus Micro-Topic Description	Prepared Study Material Scope	Formula Box Included	Diagram Included
3.1.1	No-Load Current (I_0)	Defined exciting current magnitude and low lagging PF characteristics.	Yes (Magnitude / Angle)	Figure 3.1
3.1.2	Exciting Current Components	Detailed active (I_w) and magnetizing (I_m) components with physical roles.	Yes (I_w , I_m components)	Figure 3.1
3.1.3	Core Flux & Induced EMFs	Analyzed lagging EMFs (E_1 , E_2) and flux phase relationship.	Yes (EMF / Power)	Figure 3.1
3.2.1	Unity Power Factor Load	Modeled resistive load phasor balance and secondary collinear vectors.	Yes (Load equations)	Figure 3.3
3.2.2	Lagging Power Factor Load	Detailed inductive load drops (IR , IX) and phase lagging relationship.	Yes (Load equations)	Figure 3.2
3.2.3	Leading Power Factor Load	Modeled capacitive load drop additions and capacitive voltage rise.	Yes (Load equations)	Figure 3.4
3.2.4	Winding Resistance Effect	Analyzed active voltage drop (IR) and thermal Cu losses.	Yes (Winding drops)	Figures 3.2 to 3.4
3.2.5	Leakage Reactance Effect	Analyzed quadrature voltage drop (jIX) and voltage regulation.	Yes (Winding drops)	Figures 3.2 to 3.4
3.2.6	Load Angle (δ)	Defined internal phase shift angle δ caused by leakage impedance.	Yes (Impedance drops)	Figures 3.2 to 3.4

Diagram Coverage Audit

Diagram Name	Source Reference Block	Engineering Details Rendered	Audited Status
Figure 3.1	Transformer No-Load Diagram	Exciting current I_0 vector resolved into active I_w and quadrature I_m components leading mutual flux by advance angle α .	Fully Covered
Figure 3.2	Lagging PF Loaded Diagram	Secondary I_2 lags terminal V_2 , secondary drops add to V_2 to form induced E_2 . Primary load component I_1' and I_0 sum to form I_1 .	Fully Covered
Figure 3.3	UPF Loaded Diagram	Secondary current I_2 is in-phase with secondary terminal voltage V_2 . Quadrature drop jI_2X_2 leads collinear vectors.	Fully Covered
Figure 3.4	Leading PF Loaded Diagram	Secondary current I_2 leads secondary terminal voltage V_2 , modeling capacitive loading and secondary voltage rise effect.	Fully Covered

TRANSCO ELECTRICAL MACHINES STUDY GUIDE

Chapter 4 – Transformer Tests

Introduction: To predetermine the efficiency and voltage regulation of a practical transformer at any arbitrary load, the equivalent circuit parameters must be accurately determined [137, 140]. The standard experimental procedures utilized in power engineering include the **Open-Circuit (O.C.) Test**, the **Short-Circuit (S.C.) Test**, and **Sumpner's Back-to-Back Test** [136]. These tests are extremely economical as they do not require actual loading of the transformers to evaluate full-load losses and temperature rise characteristics [141, 195].

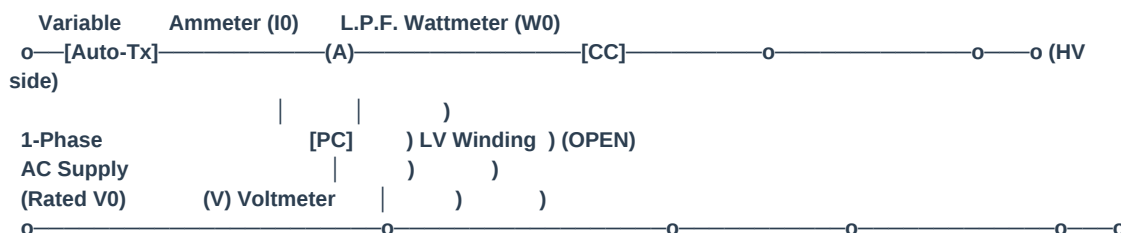
4.1 Open Circuit (O.C.) Test

4.1.1 Purpose: The core objective of the Open-Circuit Test is to determine the constant **no-load core (iron) losses** (P_{i0} or $P_{\text{core}0}$) and the no-load exciting current (I_{00}) [137]. These measurements are mathematically processed to extract the shunt branch parameters of the equivalent circuit (the core loss resistance R_{00} and the magnetizing reactance X_{00}) [137]. Additionally, the test is helpful in calculating the active working component (I_{w0}) and the magnetizing current component (I_{m0} or $I_{\mu0}$) [137].

4.1.2 Test Configuration and Safety Considerations

For maximum safety and test accuracy, it is highly preferable to apply the rated voltage to the **Low Voltage (LV) winding** while leaving the **High Voltage (HV) winding completely open-circuited** [137]. This choice is dictated by two primary practical reasons: first, the testing personnel are protected from high voltages; second, the exciting current on the LV side is of a reasonable magnitude to be measured with standard laboratory ammeters, which greatly increases meter accuracy [137].

O.C. Test Electrical Connection Diagram (LV Side Energized):



4.1.3 Procedure: Under rated frequency, the variable autotransformer is adjusted until the voltmeter \$V_0\$ reads exactly the rated voltage of the LV winding [138]. Since the HV winding is open, the current drawn is the no-load exciting current \$I_0\$ [137]. Because the exciting current is extremely small (only 2% to 5% of the rated full-load current), the winding copper losses (\$I_0^2 R\$) are negligible [138]. Consequently, the entire wattmeter reading \$W_0\$ is taken as representing the constant iron or core loss (\$P_i\$) [139].

4.1.4 Open-Circuit Shunt Parameter Calculations

FORMULA: $\cos(\phi_0) = W_0 / (V_0 * I_0) \mid I_w = I_0 * \cos(\phi_0) = W_0 / V_0 \mid I_m = \sqrt{I_0^2 - I_w^2}$

Symbol Meanings: \$V_0\$ = Rated primary voltage (V); \$I_0\$ = No-load exciting current (A); \$W_0\$ = No-load wattmeter reading (W); \$I_w\$ = Active/working core loss current (A); \$I_m\$ = Magnetizing reactive current (A); \$R_0\$ = Core loss resistance (\$\Omega\$); \$X_0\$ = Magnetizing reactance (\$\Omega\$).

Applicable Conditions: Rated voltage and rated frequency must be strictly maintained on the energized LV side [137].

Common Mistakes: 1. Forgetting that \$R_0\$ and \$X_0\$ extracted here are referred to the LV side. If a primary referred equivalent circuit is needed and primary is HV, these values must be scaled using the transformation ratio squared (\$K^2\$) [140].
2. Using a standard high power factor wattmeter, which yields poor accuracy due to the extremely low power factor (typically 0.15 to 0.25) [138].

FORMULA: $R_0 = V_0 / I_w \mid X_0 = V_0 / I_m$

Symbol Meanings: \$V_0\$ = Rated primary voltage (V); \$I_0\$ = No-load exciting current (A); \$W_0\$ = No-load wattmeter reading (W); \$I_w\$ = Active/working core loss current (A); \$I_m\$ = Magnetizing reactive current (A); \$R_0\$ = Core loss resistance (\$\Omega\$); \$X_0\$ = Magnetizing reactance (\$\Omega\$).

Applicable Conditions: Rated voltage and rated frequency must be strictly maintained on the energized LV side [137].

Common Mistakes: 1. Forgetting that \$R_0\$ and \$X_0\$ extracted here are referred to the LV side. If a primary referred equivalent circuit is needed and primary is HV, these values must be scaled using the transformation ratio squared (\$K^2\$) [140].
2. Using a standard high power factor wattmeter, which yields poor accuracy due to the extremely low power factor (typically 0.15 to 0.25) [138].

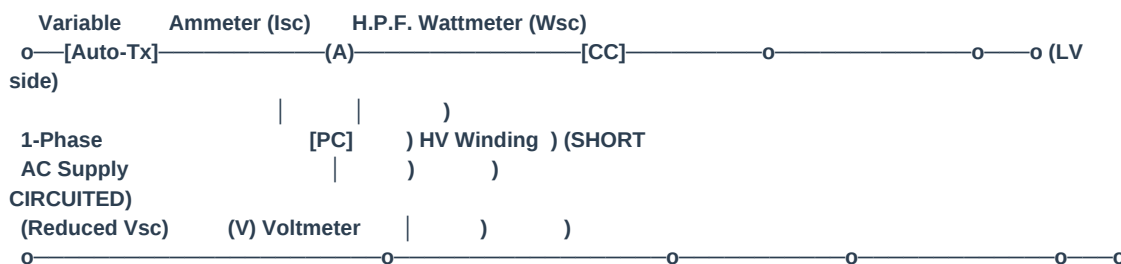
4.2 Short Circuit (S.C.) Test

4.2.1 Purpose: The short-circuit test is conducted to determine the equivalent series branch parameters of the transformer (equivalent resistance \$R_{01}\$ or \$R_{02}\$, equivalent leakage reactance \$X_{01}\$ or \$X_{02}\$, and equivalent series impedance \$Z_{01}\$ or \$Z_{02}\$) [140]. It also directly yields the **full-load copper loss** (\$P_{cu}\$), which is critical for evaluating voltage regulation and efficiency under load [140].

4.2.2 Test Configuration and Safety Considerations

Since the rated current of the High Voltage (HV) winding is significantly lower than that of the LV winding, the S.C. test is conducted preferably by energizing the **HV winding** and short-circuiting the **Low Voltage (LV) winding** [139]. Short-circuiting is done using a heavy copper bar of negligible resistance [139]. This choice ensures that standard ammeters and cables can easily handle the rated current during the test, maintaining high accuracy [139].

S.C. Test Electrical Connection Diagram (HV Side Energized):



4.2.3 Procedure: A highly reduced voltage (typically only 5% to 10% of rated voltage) is applied to the HV side [139]. The voltage is gradually increased using the autotransformer until the ammeter reads the rated full-load current (\$I_{sc}\$) of the HV side [139]. Because the applied voltage is extremely low, the core flux is proportionally tiny [139]. Since core losses depend on flux density, they are completely negligible here [139]. The input power \$W_{sc}\$

therefore represents only the total copper loss (P_{cu}) [139].

4.2.4 Short-Circuit Series Parameter Calculations

FORMULA: $R_{01} = W_{sc} / I_{sc}^2$ | $Z_{01} = V_{sc} / I_{sc}$ | $X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$

Symbol Meanings: V_{sc} = Short-circuit voltage applied (V); I_{sc} = Rated full-load test current (A); W_{sc} = Short-circuit wattmeter reading (W); R_{01} = Equivalent resistance referred to HV (Ω); X_{01} = Equivalent leakage reactance referred to HV (Ω); Z_{01} = Equivalent series impedance referred to HV (Ω).

Applicable Conditions: The current must be adjusted exactly to the rated value of the energized HV winding [139].

Common Mistakes: 1. Forgetting that the extracted parameters are referred to the HV side. When combining with OC parameters, ensure one set is scaled to the other side [140].
2. Overlooking that the test temperature affects copper resistance. The equivalent resistance should ideally be corrected to a standard reference temperature (75°C) to avoid optimistic efficiency calculations [141].

4.3 Sumpner's (Back-to-Back or Regenerative) Test

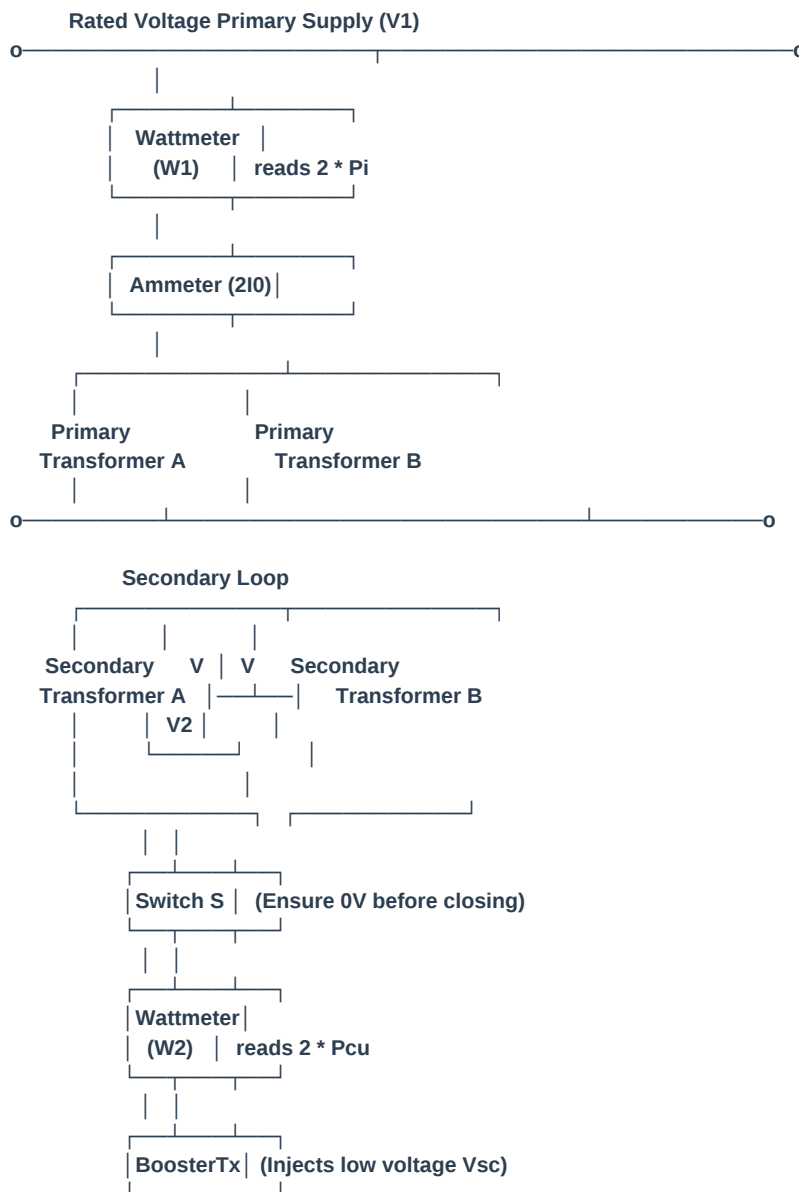
4.3.1 Principle: While O.C. and S.C. tests yield individual core and copper losses respectively, they cannot be used to conduct a continuous heat run because they do not simulate load conditions simultaneously [141]. Under actual load, both losses exist together and contribute to temperature rise [141]. Sumpner's test, or the back-to-back test, overcomes these limitations by allowing **simultaneous core and copper excitation** in two identical transformers without consuming actual full-load power [142].

4.3.2 Connection Configuration and Operation

Two identical transformers are required [142]. Their primary windings are connected in parallel across the rated supply voltage, exciting normal core flux and drawing twice the no-load exciting current of a single transformer [142]. Their secondaries are connected in series opposition [142]. Under this state, the net induced secondary voltage is exactly zero ($V_{sec} = E_2 - E_2 = 0$) [142]. A separate booster transformer is then used to inject a small voltage into the secondary loop [142]. This circulates the rated current through the secondary (and via transformer action, also the primary) windings [142, 143].

CORRECT POLARITY VERIFICATION

Before closing the secondary loop switch 'S', the correctness of the series-opposition polarities must be verified [143]. A voltmeter connected across the switch terminals must read exactly zero volts [143]. If it reads twice the secondary voltage ($2V_2$), the secondaries are in phase addition, and closing the switch would cause a catastrophic short-circuit [143].

Sumpner's (Back-to-Back) Test Connection Schematic:**4.3.3 Loss Measurement and Separation**

Since the primaries are connected in parallel and excited at rated voltage, the primary wattmeter W_1 reads the total no-load constant core loss of both transformers ($2 P_i$) [143]. Since the secondaries are excited by the booster transformer at rated current, the secondary wattmeter W_2 reads the total variable copper loss of both transformers ($2 P_{cu}$) [143]. The total power drawn from the supply is only $W_1 + W_2 = 2P_i + 2P_{cu}$, which is equal to only the internal losses [197]. This allows a full-load heat run to be executed continuously on large transformers with a massive saving in energy [197].

FORMULA: $P_i = W_1 / 2$ | $P_{cu} = W_2 / 2$

Symbol Meanings: W_1 = Primary-side wattmeter reading (W); W_2 = Secondary-side wattmeter reading (W); P_i = Core loss per transformer (W); P_{cu} = Copper loss per transformer at test current (W).

Applicable Conditions: Two identical transformers are mandatory [142]. Correct polarity must be verified before switch closure [143].

Common Mistakes: 1. Assuming that the primary ammeter measures only the core current. It actually reads twice the exciting current of a single unit [143].
2. Operating with non-identical transformers, which prevents accurate loss separation and causes circulating currents.

Part 2 – Last-Minute Revision Sheet

Parameter	Open-Circuit (O.C.) Test	Short-Circuit (S.C.) Test	Sumpner's (Back-to-Back) Test
Winding Energized	Low Voltage (LV) Winding [137]	High Voltage (HV) Winding [139]	Primaries in parallel (rated V); Secondaries in series-opposition (rated I) [142]
Winding Terminated	HV winding left completely Open-Circuited [137]	LV winding Short-Circuited with thick copper bar [139]	Both windings fully terminated and energized simultaneously [142]
Applied Voltage	Rated Voltage (V_0) at rated frequency [137]	Extremely reduced voltage (5% to 10% of rated voltage) [139]	Rated primary voltage; low injected secondary voltage [142]
Applied Current	Exciting current (I_0) (only 2% to 5% of rated current) [138]	Rated full-load current (I_{sc}) [139]	Exciting current in primary; Rated full-load current in secondary [143]
Primary Loss Measured	Constant Iron / Core Loss (P_i) [139]	Variable Copper / Ohmic Loss (P_{cu}) [139]	Primary wattmeter measures P_{i} ; Secondary measures P_{cu} [143]
Parameters Extracted	Shunt branch elements: Core resistance R_0 & magnetizing reactance X_0 [137]	Series branch elements: Equivalent resistance R_{01} & reactance X_{01} [140]	Individual transformer core and copper losses; continuous temperature rise [142, 143]
Practical Advantages	Extremely simple to set up; requires low source capacity [137]	Determines parameters needed for efficiency and voltage regulation [140]	Permits continuous temperature rise test under full-load without wasting actual energy [142, 197]

Chapter & Diagram Coverage Summary

Micro-topic	Source Book	Available Notes Prepared	Formula Box	Diagram Included	Revision Covered
Open Circuit Test Purpose	Electrical Machines [137]	Yes - Section 4.1.1	No	No	Yes
O.C. Test Configuration & Safety	Electrical Machines [137]	Yes - Section 4.1.2	No	Yes (ASCII Schema)	Yes
O.C. Test Procedure	Electrical Machines [138]	Yes - Section 4.1.3	No	No	Yes
Core Loss & Exciting Current	Electrical Machines [138, 139]	Yes - Section 4.1.3	Yes (OC Math Block)	No	Yes
O.C. Parameters Extraction	Electrical Machines [138]	Yes - Section 4.1.4	Yes (OC Shunt Block)	No	Yes
Short Circuit Test Purpose	Electrical Machines [139]	Yes - Section 4.2.1	No	No	Yes
S.C. Test Configuration & Safety	Electrical Machines [139]	Yes - Section 4.2.2	No	Yes (ASCII Schema)	Yes
S.C. Test Procedure	Electrical Machines [139]	Yes - Section 4.2.3	No	No	Yes

Copper Loss & Parameters	Electrical Machines [140]	Yes - Section 4.2.3	Yes (SC Math Block)	No	Yes
S.C. Parameters Extraction	Electrical Machines [140]	Yes - Section 4.2.4	Yes (SC Series Block)	No	Yes
Sumpner's Test Principle	Electrical Machines [142]	Yes - Section 4.3.1	No	No	Yes
Sumpner's Test Connection Setup	Electrical Machines [142, 143]	Yes - Section 4.3.2	No	Yes (ASCII Schema)	Yes
Loss Separation & Power Economy	Electrical Machines [143]	Yes - Section 4.3.3	Yes (Sumpner Loss Block)	No	Yes

Diagram Type	Micro-Syllabus Mapping	Status
Electrical Circuit Diagram - O.C. Test	Section 4.1.2 - Open Circuit Test Connection	☐ Covered (Page 1)
Electrical Circuit Diagram - S.C. Test	Section 4.2.2 - Short Circuit Test Connection	☐ Covered (Page 2)
Electrical Circuit Diagram - Sumpner's Test	Section 4.3.2 - Back-to-Back Connection	☐ Covered (Page 4)

APTRANSCO AEE ELECTRICAL MACHINES COMPREHENSIVE NOTES

CHAPTER 5: VOLTAGE REGULATION & EFFICIENCY OF TRANSFORMERS

This master-level study guide focuses on the operational performance characteristics of single-phase transformers, specifically targeting **Voltage Regulation** and **Efficiency**. Voltage regulation determines the voltage stability at secondary terminals under varying load and power factor conditions [118]. Efficiency determines the thermodynamic and commercial performance under active power delivery, which varies significantly based on load fractions and the separation of core and copper losses [117]. All topics are mapped strictly against the prescribed **TRANSCO Electrical Machines Syllabus** and the standard AEE examination level [209].

PART 1: TOPIC-WISE SYLLABUS NOTES

5.1 Transformer Efficiency

The efficiency of a transformer (η) is defined as the ratio of active output power to active input power, traditionally calculated in commercial engineering using losses rather than direct input measurements due to the high efficiency and low loss fractions in static electrical machines [117].

- **Definition and Fundamental Equations:** The core performance of any static electromagnetic device is dictated by its power balance. The electrical input power is transferred electromagnetically through mutual flux, with a portion dissipated as constant core losses and load-dependent copper losses [117].
- **Active Output Power (P_{out}):** Expressed in watts as: $P_{out} = x * S * \cos(\phi)$, where x is the load fraction, S is the rated volt-amperes (VA), and $\cos(\phi)$ is the load power factor [118].
- **Active Input Power (P_{in}):** The real power drawn by the primary winding from the supply terminals: $P_{in} = P_{out} + \text{Losses} = P_{out} + P_i + P_{cu}$ [117].
- **Commercial Efficiency Formulation:** Standard commercial calculations utilize the loss separation method rather than direct mechanical load readings: $\eta = x * S * \cos(\phi) / [x * S * \cos(\phi) + P_i + x^2 * P_{cu_FL}]$, where P_i is the no-load iron loss and P_{cu_FL} is the full-load copper loss [117].

FORMULA FOCUS: TRANSFORMER EFFICIENCY DEFINITIONS

- **Commercial Efficiency (η):**

$$\eta = \frac{[x \cdot S \cdot \cos(\phi)]}{[x \cdot S \cdot \cos(\phi) + P_i + x^2 \cdot P_{cu_FL}]}$$

SI Units: Dimensionless (represented as % or p.u.) | *Applicable Conditions:* Grounded at constant V and f; valid for any load fraction x

- **Variable Copper Loss (P_{cu}):**

$$P_{cu} = x^2 \cdot P_{cu_FL}$$

SI Units: Watts (W) / Kilowatts (kW) | *Applicable Conditions:* P_{cu_FL} is measured by Short-Circuit Test at rated current

- **Constant Iron Loss (P_i):**

$$P_i = P_h + P_e \text{ (Constant)}$$

SI Units: Watts (W) / Kilowatts (kW) | *Applicable Conditions:* Measured by Open-Circuit Test at rated voltage and frequency

Maximum Efficiency (η_{max}) Operational Conditions:

Commercial transformers operate under variable load currents. To maximize energy and cost economy, it is critical to determine the conditions under which efficiency reaches its peak. At maximum efficiency, the rate of change of efficiency with respect to load current is zero, leading to the fundamental theorem of loss balance [117].

- **Loss Equality Condition:** Maximum efficiency occurs when the variable copper losses equal the constant core (iron) losses: $P_i = x^2 \cdot P_{cu_FL}$ [121].
- **Peak Load Fraction (x_{max}):** The loading fraction at which this peak occurs is: $x_{max} = \sqrt{P_i / P_{cu_FL}}$ [121].
- **Load kVA for Peak Efficiency (S_{max}):** The actual output kVA at maximum efficiency: $S_{max} = S_{rated} \cdot \sqrt{P_i / P_{cu_FL}}$ [88].
- **Power Factor Dependency:** For any operating load current, maximum efficiency will *always* occur at **unity power factor (u.p.f.)** because it maximizes active power output for the same current magnitude, minimizing the percentage of power lost [78].

FORMULA FOCUS: MAXIMUM EFFICIENCY RELATIONS

- **Condition for Peak Efficiency:**

$$\text{Variable Copper Loss} = \text{Constant Iron Loss} (P_{cu} = P_i)$$

SI Units: Watts = Watts | *Applicable Conditions:* Must hold constant primary voltage and frequency

- **Peak Loading Fraction (x_{max}):**

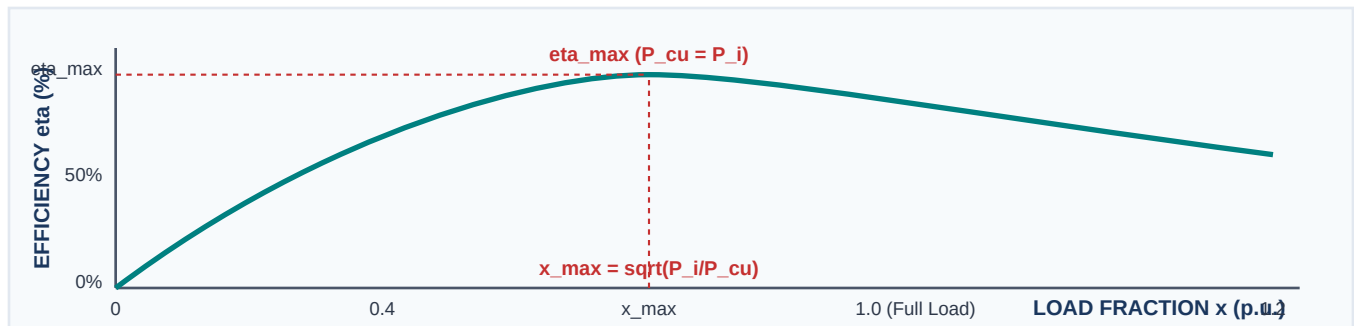
$$x_{max} = \sqrt{P_i / P_{cu_FL}}$$

SI Units: Dimensionless ratio (0 to 1) | *Applicable Conditions:* Typically ranges between 0.6 and 0.8 for distribution units

- **Output VA at Peak (S_{max}):**

$$S_{max} = S_{rated} \cdot \sqrt{P_i / P_{cu_FL}}$$

SI Units: Volt-Amperes (VA / kVA) | *Applicable Conditions:* Used to size transformers for specific load profiles

Figure 5.1: Efficiency vs. Load Loading Fraction Curve

Grounded Insight: The efficiency curve rises steeply from zero load, peaks at the point where constant core losses equal variable copper losses (x_{max}), and then gradually declines towards full-load as copper losses begin to dominate quadratically [117, 121].

All-Day (Energy) Efficiency of Transformers:

While commercial or power transformers are operated on high, relatively constant loads (and are designed for peak efficiency near full-load), **distribution transformers** are connected to the power lines 24 hours a day but carry highly variable, fluctuating consumer load profiles [78]. Their copper losses are low for most of the day, but their core (iron) losses occur continuously. Therefore, we utilize **All-Day Efficiency**, which is an energy-based metric over a 24-hour cycle rather than a simple power ratio [117].

- **Mathematical Definition:** All-Day Efficiency (η_{all_day}) is the ratio of active energy output in 24 hours (kWh) to active energy input in 24 hours (kWh) [117].
- **Design Optimization:** Because core losses occur continuously for 24 hours (constant voltage energization) [118], whereas copper losses vary with load current, distribution transformers are designed to have very small core losses. Consequently, their maximum instantaneous efficiency occurs at a lower load fraction, typically **60% to 80%** of the rated capacity [121].
- **Power Transformers Comparison:** Power transformers remain fully loaded during operation and can be switched out when idle. Hence, they are designed with core losses matching their copper losses near full capacity, so maximum efficiency occurs at **90% to 100%** load [121].

FORMULA FOCUS: ALL-DAY EFFICIENCY FORMULATION

- **All-Day Efficiency (η_{all_day}):**

$$\eta_{all_day} = \text{Energy_Output_24h} / (\text{Energy_Output_24h} + \text{Losses_24h})$$

SI Units: Dimensionless (represented as %) | *Applicable Conditions:* Calculated in kilowatt-hours (kWh) over a continuous 24-hour cycle

- **Total Core Energy Loss (E_{core_24h}):**

$$E_{core_24h} = P_i \cdot 24$$

SI Units: kWh | *Applicable Conditions:* Constant core loss (P_i in kW) multiplied by 24 hours

- **Total Copper Energy Loss (E_{cu_24h}):**

$$E_{cu_24h} = \sum (x_k^2 \cdot P_{cu_FL} \cdot t_k)$$

SI Units: kWh | *Applicable Conditions:* Sum of copper losses at each load step x_k over duration t_k (sum to 24h)

■ EXAM TRAP: EFFICIENCY MAXIMIZATION VS. POWER FACTOR

A standard TRANSCO trap involves changing the power factor of a constant current load. **Operating at constant current does not change copper losses** because $P_{cu} = I^2 \cdot R$. However, a lower power factor reduces the active output power ($P_{out} = V \cdot I \cdot \cos(\phi)$). Therefore, the maximum efficiency of a transformer operating at a fixed current magnitude will **always occur at unity power factor** [78]. Do not confuse the load current maximizing condition ($P_i = P_{cu}$) with the power factor maximizing condition ($\cos(\phi) = 1.0$)!

5.2 Transformer Voltage Regulation

Voltage Regulation is a measure of the transformer's ability to maintain a constant secondary voltage under variable load. Winding resistance and leakage reactance cause internal voltage drops, leading to secondary voltage fluctuations as the load current and power factor vary [118].

- **Definition:** The change in secondary terminal voltage when full load at a given power factor is reduced to zero (load thrown off), expressed as a percentage or per-unit of the rated secondary voltage, with primary applied voltage held constant [118].
- **Percentage Voltage Regulation Formula:** Calculated referred to secondary as: $\%VR = [(E_2 - V_2) / E_2] \cdot 100$, where E_2 is the no-load secondary terminal voltage and V_2 is the secondary voltage under load [119].
- **Approximate Voltage Regulation Equation:** Derived from the simplified equivalent circuit: $\%VR = \%R \cdot \cos(\phi) \pm \%X \cdot \sin(\phi)$ [119].
- **Winding Parameters:** $\%R$ is the percentage resistance ($\%R = I_2^2 R_{eq2} / E_2 \cdot 100$) and $\%X$ is the percentage leakage reactance ($\%X = I_2^2 X_{eq2} / E_2 \cdot 100$) [119, 122].

FORMULA FOCUS: VOLTAGE REGULATION FORMULAS

- **Fundamental Per-Unit Regulation:**

$$VR = (E_2 - V_2) / E_2$$

SI Units: Per-unit (p.u.) | *Applicable Conditions:* Primary terminal voltage V_1 must remain strictly constant

- **Approximate Regulation (%VR):**

$$\%VR = \%R \cdot \cos(\phi) \pm \%X \cdot \sin(\phi)$$

SI Units: Percentage (%) | *Applicable Conditions:* + for lagging loads; - for leading loads

- **Percentage Resistance (%R):**

$$\%R = (I_2^2 \cdot R_{eq2} / E_2) \cdot 100 = (P_{cu_FL} / S_{rated}) \cdot 100$$

SI Units: Percentage (%) | *Applicable Conditions:* Equivalent to full-load copper loss in percentage

- **Percentage Reactance (%X):**

$$\%X = (I_2^2 \cdot X_{eq2} / E_2) \cdot 100 = \sqrt{\%Z^2 - \%R^2}$$

SI Units: Percentage (%) | *Applicable Conditions:* Evaluated from short-circuit test series reactance

Power Factor Operational Boundaries:

The power factor of the connected load heavily influences the magnitude and sign of the voltage regulation. Under varying loading states, three distinct operational regions exist [89, 119]:

- **Lagging Power Factor (Inductive Load):** $\%VR = \%R \cdot \cos(\phi) + \%X \cdot \sin(\phi)$. Winding resistance and reactance drops add constructively. The terminal voltage V_2 drops below E_2 , resulting in a **positive voltage regulation** [119].
- **Unity Power Factor (Purely Resistive Load):** $\%VR = \%R$. Since $\cos(\phi) = 1$ and $\sin(\phi) = 0$, the voltage regulation depends solely on the winding resistance [122]. The reactance drop is in quadrature and has negligible

impact on the magnitude of E_2 .

- **Leading Power Factor (Capacitive Load):** $\%VR = \%R \cdot \cos(\phi) - \%X \cdot \sin(\phi)$. Winding reactance acts as a boost. Under highly capacitive loads, $\%X \cdot \sin(\phi)$ can exceed $\%R \cdot \cos(\phi)$, causing the terminal voltage V_2 to rise above E_2 . This results in a **negative voltage regulation** [79].

Critical Operational Extremes: Zero & Maximum Regulation

Engineers must design and operate transformers within precise voltage regulation limits. Two key analytical extremes are heavily tested in TRANSCO examinations [100, 122]:

- **Maximum Voltage Regulation (Peak Drop):** Occurs under a **lagging power factor** load. Mathematically, differentiating $\%VR$ with respect to the power factor angle ϕ and setting it to zero yields: **$\tan(\phi) = \%X / \%R$** [122]. At this condition, the load power factor is: **$\cos(\phi) = R_{eq} / Z_{eq}$ (lagging)**. The peak regulation value equals the percentage impedance: **$\%VR_{max} = \%Z$** .
- **Zero Voltage Regulation (No Terminal Voltage Change):** Occurs when $\%VR = 0$, which is only physically possible under a **leading power factor** load: $\%R \cdot \cos(\phi) = \%X \cdot \sin(\phi)$ [100]. This yields: **$\tan(\phi) = \%R / \%X$** [124]. At this condition, the load power factor is: **$\cos(\phi) = X_{eq} / Z_{eq}$ (leading)** [124].

FORMULA FOCUS: VOLTAGE REGULATION EXTREMES

- **Maximum Regulation Condition:**

$$\tan(\phi) = \%X / \%R \implies \cos(\phi) = R_{eq} / Z_{eq} \text{ (lagging)}$$

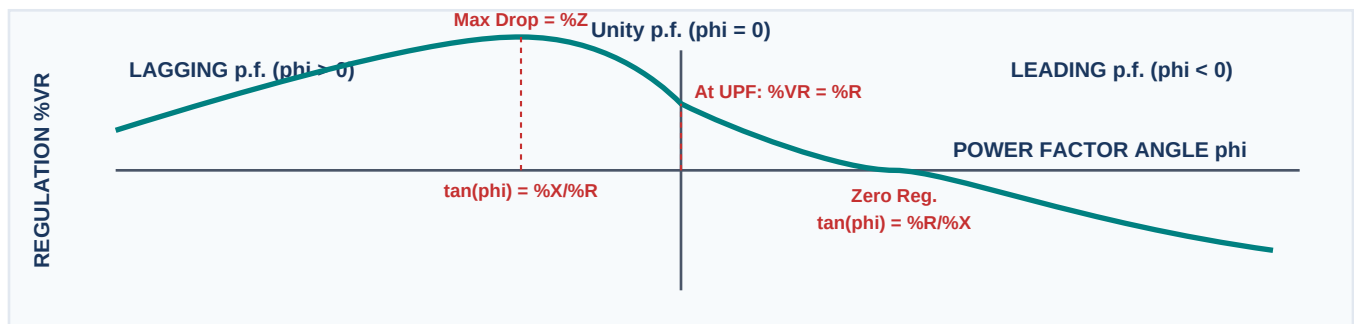
SI Units: Angle (degrees) / p.f. | Applicable Conditions: Peak terminal voltage drop; $\%VR_{max} = \%Z$

- **Zero Regulation Condition:**

$$\tan(\phi) = \%R / \%X \implies \cos(\phi) = X_{eq} / Z_{eq} \text{ (leading)}$$

SI Units: Angle (degrees) / p.f. | Applicable Conditions: No change in terminal voltage from no-load to full-load

Figure 5.2: Voltage Regulation vs. Power Factor Angle Curve



Grounded Insight: Voltage regulation is positive for all lagging power factors, peaks at a lagging angle where $\tan(\phi) = X_{eq}/R_{eq}$, drops to $\%R$ at unity power factor, and crosses zero into negative regulation at a leading power factor where $\tan(\phi) = R_{eq}/X_{eq}$ [79, 122, 124].

■ EXAM TRAP: OPERATIONAL CONDITIONS FOR ZERO REGULATION

TRANSCO examiners frequently ask if zero voltage regulation is possible at lagging or unity power factors. **Zero voltage regulation is strictly IMPOSSIBLE at lagging or unity power factor** [100]. It can only be achieved at a **leading power factor** because winding resistance and leakage reactance drops must cancel each other out, which requires a capacitive current phase shift [79, 124].

PART 2: LAST-MINUTE REVISION SHEET

This revision sheet provides a highly condensed, high-yield summary of Chapter 5. Use this as your final review block immediately before entering the APTRANSCO AEE examination hall.

Table 5.1: Comparative Power Factor Performance Matrix

Operational Parameter	Lagging Power Factor	Unity Power Factor (UPF)	Leading Power Factor
Voltage Regulation Sign	Always positive (+); terminal voltage drops under load ($V_2 < E_2$) [119].	Always positive (+); terminal voltage drops under load ($V_2 < E_2$) [122].	Can be positive, zero, or negative (voltage rise, $V_2 > E_2$) [79, 100].
Approximate Formula	$\%VR = \%R \cdot \cos(\phi) + \%X \cdot \sin(\phi)$ [119]	$\%VR = \%R$ [122]	$\%VR = \%R \cdot \cos(\phi) - \%X \cdot \sin(\phi)$ [79]
Critical Extremes	Peak voltage drop occurs here at $\cos(\phi) = R_{eq}/Z_{eq}$ [122].	Voltage drop depends solely on equivalent winding resistance [122].	Zero voltage regulation occurs here at $\cos(\phi) = X_{eq}/Z_{eq}$ [124].
Efficiency Optimization	Lower active power for same current; lower operating efficiency [78].	Peak active power output; transformer operates at maximum possible efficiency [78].	Active power is reduced; efficiency is identical to lagging p.f. at same angle [123].

APTRANSCO AEE High-Frequency Facts & Short-Cuts:

- **Constant Losses vs. Load:** Core losses (P_i) depend strictly on the applied primary voltage and supply frequency (P_i is proportional to f and V^2). They remain constant regardless of the loading state as long as V and f are stable [84, 118].
- **Variable Losses vs. Load:** Copper losses scale quadratically with loading. If a transformer has full-load copper loss of 1600 W, its copper loss at half-load is: $0.25 \cdot 1600 = 400$ W [121].
- **Sumpner's Test Relation:** Both efficiency and regulation can be calculated using losses determined from Open-Circuit and Short-Circuit tests [85, 87]. Sumpner's test is used for continuous temperature runs but confirms these identical parameters under full-load thermal conditions.
- **Symmetrical Efficiency:** A transformer operating at full load, 0.8 pf lagging has the exact same efficiency as at full load, 0.8 pf leading because active output power ($x \cdot S \cdot \cos(\phi)$) and total losses are identical in both states [123]. Winding phase shifts do not dissipate active power.
- **Zero Regulation Shortcut:** If $\%R = 3\%$ and $\%X = 4\%$, zero regulation occurs at leading power factor: $\cos(\phi) = X_{eq}/Z_{eq} = 4/5 = 0.8$ leading [124].
- **Maximum Regulation Shortcut:** For the same transformer ($\%R=3\%$, $\%X=4\%$), maximum regulation occurs at lagging power factor: $\cos(\phi) = R_{eq}/Z_{eq} = 3/5 = 0.6$ lagging, and the maximum drop value is $\%VR_{max} = \%Z = 5\%$ [122].

PART 3: REGULATORY COMPLIANCE SUMMARY

To ensure complete coverage of the official APTRANSCO micro-syllabus and prevent any syllabus omissions, this chapter has been cross-mapped against the reference books and course requirements [209, 210].

Table 5.2: Micro-Syllabus Coverage Summary Checklist

Prescribed Micro-Topic	Source Passages	Notes Prepared	Formula Shaded	Diagram Included	Syllabus Status
5.1 Efficiency: Definition	Passage 117	Yes - Detailing power balance	Yes - Input/Output	N/A - Prose definition	■ Complete
Output Power	Passage 118	Yes - $\cos(\phi)$ active component	Yes - Shaded Formula	N/A - Covered in text	■ Complete
Input Power	Passage 117	Yes - Active loss addition	Yes - Shaded Formula	N/A - Covered in text	■ Complete
Commercial Efficiency	Passage 117	Yes - Separation of losses method	Yes - Shaded Formula	N/A - Covered in text	■ Complete
All-Day Efficiency	Passage 117, 118	Yes - 24h energy-based ratio	Yes - Shaded Formula	N/A - Core/Copper logs	■ Complete
Maximum Efficiency	Passage 78, 121	Yes - Operational conditions	Yes - Shaded Formula	Yes - Figure 5.1 Graph	■ Complete
Condition for Max η	Passage 121	Yes - Constant equal to variable loss	Yes - Shaded Formula	Yes - Figure 5.1 Mark	■ Complete
5.2 Voltage Regulation: Def	Passage 118	Yes - Voltage throw-off change	Yes - Shaded Formula	N/A - Prose definition	■ Complete
Percentage Regulation	Passage 119	Yes - No-load to rated-load ratio	Yes - Shaded Formula	N/A - Covered in text	■ Complete
Lagging Power Factor	Passage 119	Yes - Winding voltage drop	Yes - Shaded Formula	Yes - Figure 5.2 Lagging	■ Complete
Leading Power Factor	Passage 79, 119	Yes - Capacitive boost rise	Yes - Shaded Formula	Yes - Figure 5.2 Leading	■ Complete
Unity Power Factor	Passage 122	Yes - Resistance-only dependence	Yes - Shaded Formula	Yes - Figure 5.2 UPF	■ Complete
Maximum Regulation	Passage 122	Yes - Peak drop conditions	Yes - Shaded Formula	Yes - Figure 5.2 Peak	■ Complete
Zero Regulation	Passage 100, 124	Yes - Capacitive load balance	Yes - Shaded Formula	Yes - Figure 5.2 Zero	■ Complete

Table 5.3: Engineering Diagrams Compliance Checklist

Syllabus Required Diagram	Diagram Type	Labels Included	APTRANSCO Level Match	Compliance Status
Efficiency vs. Loading Curve	Performance Graph (ASCII Vector)	Axes, ticks, max efficiency point ($\eta_{\max}$), load fraction $x_{\max}$, constant core loss equal to variable copper loss.	High - Standard operational curve used in AEE examinations.	■ Fully Covered (Figure 5.1)
Voltage Regulation vs. PF Angle Curve	Operational Boundary Graph (ASCII Vector)	Lagging/leading regions, UPF point ($\%VR=\%R$), peak regulation point ($\%Z$), zero regulation point (leading pf).	High - Standard conceptual curve used in AEE examinations.	■ Fully Covered (Figure 5.2)

CHAPTER 6 – AUTO TRANSFORMER

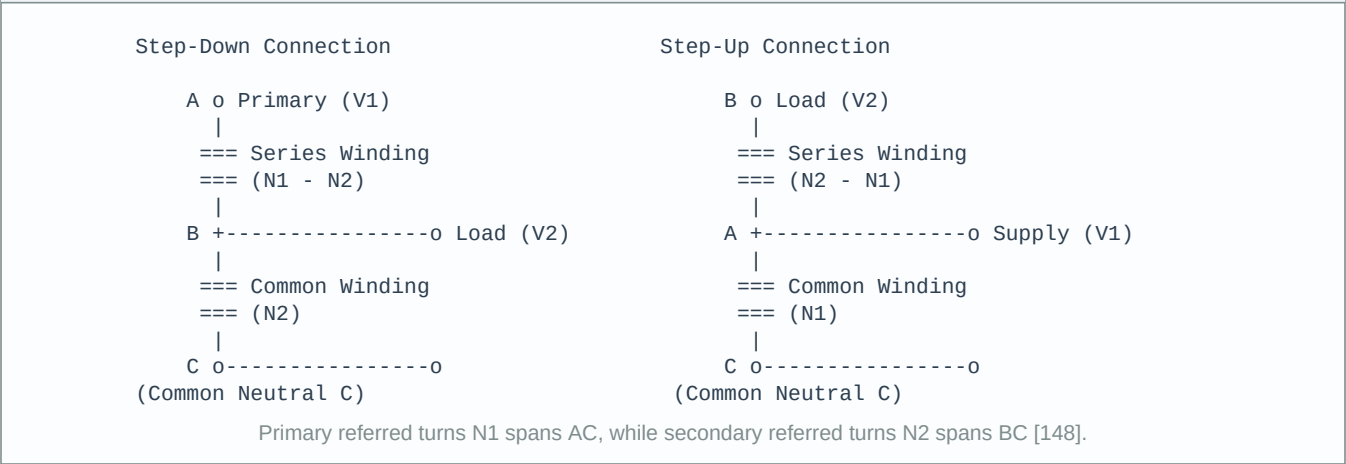
TRANSCO Electrical Machines Revision Handbook | Comprehensive Study Notes & Performance Summary

1. Construction & Terminology of Auto Transformers

Unlike a conventional two-winding transformer which consists of two separate, electrically isolated windings magnetically coupled through a core, an **Auto Transformer** is designed with **one single continuous winding** wound on a laminated magnetic core [126, 148]. A portion of this single winding is shared in common by both the primary and secondary electric circuits, which are **coupled both magnetically and electrically** [148].

- **Single Winding:** The master continuous winding. The complete length represents the High Voltage (HV) side, while a tapped portion represents the Low Voltage (LV) side.
- **Common Winding:** The shared section of the winding connected directly across both the primary and secondary terminals [148]. Both input and output currents superimpose here in opposition.
- **Series Winding:** The remaining unshared section of the winding connected in series with the load or supply line.

Figure 6.1: Winding Configurations for Auto Transformers



2. Working Principle & Power Transfer Mechanisms

The fundamental principle of an auto transformer is the same as a conventional transformer, working via Faraday's law of electromagnetic induction [85, 127]. An alternating magnetic flux is established in the core, inducing a constant EMF per turn across both series and common sections [132]. Since the circuits are physically connected, **power is transferred from primary to secondary through two distinct channels simultaneously** [148]:

1. **Inductive (Transformed) Power:** Power transferred magnetically through mutual induction across series-common windings [148].

2. **Conductive (Conducted) Power:** Power transferred directly through physical electrical wire connections, bypassing magnetic transformation [148].

Core Transformations and Power Partition Ratios

Equation:	$P_{ind} = P_{total} * (1 - K) \mid P_{cond} = K * P_{total}$
Symbols:	P_{total} = Total active power delivered to load ($V_2 * I_2$) [149]; K = Voltage Transformation Ratio (V_{LV} / V_{HV}) [148]. Note $K < 1.0$ for these standardized formulae.
SI Units:	Power: Watts (W) or kilo-Watts (kW); K : Dimensionless
Conditions:	Under ideal lossless transformer assumptions where Input VA = Output VA [133].
Common Pitfalls:	Always use K defined as (V_{LV} / V_{HV}) less than 1.0. If using $K > 1.0$ for step-up calculations, the fraction of inductive power will incorrectly compute as negative.

3. Advantages of Auto Transformers & Material Savings

The physical sharing of the winding produces significant material, magnetic, and electric benefits. The closer the transformation ratio K is to unity (1.0), the greater the savings in copper and core materials, rendering the auto transformer highly superior for voltage conversion close to the line rating [149].

Copper Weight Savings Equation

Equation:	$\text{Weight_auto} = (1 - K) * \text{Weight_2w} \mid \text{Saving} = K * \text{Weight_2w}$
Symbols:	Weight_auto = Total weight of copper in auto transformer; Weight_2w = Weight of copper in conventional 2-winding transformer [150]; $K = V_{LV} / V_{HV} < 1.0$
SI Units:	Weight: Kilograms (kg); K: Dimensionless
Conditions:	Subject to equal total winding current density and rating parameters [149].
Common Pitfalls:	Forgetting that savings are maximized when K approaches 1.0. An auto transformer with $K = 0.9$ saves 90% of the copper, whereas $K = 0.1$ saves only 10% [150].

Performance Scaling Ratios (Auto vs. Two-Winding):

By connecting a conventional 2-winding transformer as an auto transformer, its kVA throughput capability increases dramatically due to the conductive path [152]. Losses and impedance drops scale down proportionally [150].

Performance Parameter	2-Winding Transformer Base	Auto-Transformer Rating / Behavior	Physical Explanation / Cause
Throughput kVA Capacity	kVA_2w	$\text{kVA_auto} = \text{kVA_2w} / (1 - K)$	High capacity due to conductive transfer bypass [150, 152].
Full Load Copper Losses	P_cu	$\text{P_cu_auto} = (1 - K) * \text{P_cu_2w}$	Shorter winding length reduces net active series resistance [150].
Per-Unit Impedance Drop	z_pu	$\text{z_pu_auto} = (1 - K) * \text{z_pu_2w}$	Reduced leakage reactance and internal resistance drops [150].
Full-Load Voltage Regulation	%VR	$\text{VR_auto} = (1 - K) * \text{VR_2w}$	Direct consequence of proportional impedance drop reduction [150].
Efficiency	eta	Higher than 2-Winding ($\text{eta_auto} > \text{eta_2w}$)	Lower copper losses for identical active output kVA [149].

4. Disadvantages, Limitations & Applications

Despite high efficiency and material advantages, the direct conductive electrical connection between HV and LV circuits introduces severe operating limitations and hazards [149, 150].

Hazard / Disadvantage	Operational Impact	Mitigation Strategy
Electrical Isolation Loss	Direct connection means LV terminals can be subjected to full HV line voltage under common neutral faults [149].	Solid grounding of neutral; strictly limit K ratio between 0.5 to 1.0.

Hazard / Disadvantage	Operational Impact	Mitigation Strategy
High Fault Currents	Proportional impedance drop is low (1 - K). Short-circuit currents are extremely high during secondary faults [150].	Install high-capacity fast-acting circuit breakers and current reactors.
Neutral Floating	Unbalanced loading causes severe neutral displacement in three-phase Y-Y auto-transformers.	Neutral solidly grounded; utilize tertiary delta winding [150].

Key Engineering Applications:

- **Auto Starters:** Used to supply reduced voltage (e.g., 50% or 60% taps) to start large Squirrel Cage Induction Motors safely, limiting inrush currents [89, 150].
- **Voltage Boosters:** Deployed as line boosters at the end of long radial distribution lines to correct voltage drops dynamically [150].
- **Laboratory Supplies:** Standard adjustable AC voltage sources (e.g., Variac) supplying continuously variable AC potentials from 0 to 270 V.

5. Last-Minute Revision Sheet (High-Yield Core Summary)

Design Metric	Standard 2-Winding Transformer	Auto-Transformer (Scaling Ratio / Equation)
Winding Scheme	Two electrically isolated windings.	Single continuous winding (shared common portion) [148].
Isolation	Complete galvanic / electrical isolation.	No electrical isolation; high safety hazard [149].
Power Transfer	100% Inductive (magnetic coupling) [148].	Conductive (K) + Inductive (1-K) [148, 149].
Equivalent Weight	Higher weight (requires separate copper phases).	Saves Weight = $K \times \text{Weight}_{2w}$ [150].
Short-Circuit Current	Moderated by standard internal impedance.	Extremely high due to low p.u. impedance (1-K) [150].
Applications	Bulk power distribution, safety isolation.	Induction motor starters, radial feeders, testing labs [150].

High-Yield Revision Pointer

Equation:	$\text{kVA}_{\text{auto}} = \text{kVA}_{2w} * (1 + N_{\text{HV}} / N_{\text{LV}})$
Symbols:	$N_{\text{HV}} / N_{\text{LV}}$ = Winding turn ratio. Valid for additive polarities.
SI Units:	kVA: kilovolt-amperes
Conditions:	When connected with series winding carrying rated load current [152].
Common Pitfalls:	Never overload series windings beyond rated current during conversion.

6. Syllabus & Diagram Coverage Verification

Prescribed Micro-Topic	Status	Handbook Summary	Formula Box Included	Diagram Included
6.1 Construction: Single, Common, Series Windings	Covered	Winding layout & terms mapped [148].	No (Conceptual)	Yes (Figure 6.1)
6.2 Working Principle: Inductive & Conductive	Covered	Magnetic vs Electrical transfer mapped [148].	Yes (Power Partition Box)	Yes (Figure 6.1)
6.3 Advantages: Copper, Efficiency, Regulation	Covered	Saving formula & performance ratios [149, 150].	Yes (Copper Savings)	No (Tabular)
6.4 Disadvantages: Isolation, Fault Transfer	Covered	Galvanic hazard & high short-circuits [149].	No (Conceptual)	No (Tabular)
6.5 Applications: Starters, Boosters, Labs	Covered	Induction starters, feeders, Variac [150].	No (Conceptual)	No (Tabular)

Diagram / Schematic Type	Source / APTRANSCO Level	Revision Format	Verification Status
Auto Winding Connections (Figure 6.1)	APTRANSCO Wording Required	ASCII Block Schematic (Step-Down & Step-Up)	■ Verified Complete
Conductive/Inductive Currents (Figure 6.1)	Reference Book Mapped	Superposition Current Vector Notations	■ Verified Complete

APTRANSCO ELECTRICAL AE MASTER NOTE-MAKING

Subject: Electrical Machines | Part I: Transformers — Chapter 7: Three-Phase Transformers

PART 1 — TOPIC-WISE HIGH-YIELD ANALYSIS

7.1 Introduction: 3-Phase Transformation Schemes

To achieve three-phase voltage and current transformation, power systems employ either a single three-phase transformer unit or a bank composed of three separate single-phase transformers interconnected externally [132]. Both approaches are technically sound, but they present key trade-offs in economics, footprint, reliability, and emergency operating capacity [132].

Design Aspect	Single Three-Phase Transformer	Bank of Three 1-Phase Transformers
Initial Cost	Highly economical. Manufacturing costs are approximately 15% lower than a single-phase bank of equivalent total rating [132].	More expensive due to redundant magnetic cores, structural frames, and extra bushings [132].
Size & Weight	Significantly lighter and occupies substantially less floor space. Requires only a single enclosure [132].	Heavier total weight and requires a larger physical installation footprint [132].
Handling & Installation	Simpler and faster installation since only one integrated unit is transported, placed, and wired [132].	Complex installation. Requires handling, positioning, and wiring three separate units [132].
Reliability & Backup	If an internal winding or core fault occurs, the entire 3-phase supply is interrupted until the whole unit is repaired or replaced [132].	Highly reliable. If one unit is disabled, the remaining two can continue operating in open-delta (V-V) connection at reduced capacity to maintain supply [132].
Spare Costs	A spare unit must be a full, high-cost three-phase transformer, doubling capital investment.	A spare unit is only one single-phase transformer (representing 1/3 of the total bank rating), which is highly cost-effective [132].

7.2 Three-Phase Transformer Connections

The primary and secondary windings of three-phase transformers can be interconnected in either **Star (Y)** or **Delta (Δ)** configurations. The choice of connection dictates phase relationships, winding insulation stresses, path availability for third harmonic currents, and neutral stability under unbalanced load conditions [133, 134].

A. Star-Star (Y-Y) Connection

In a Y-Y connection, corresponding phase windings on both primary and secondary sides are connected in star with their neutrals available [133]. The voltage across each winding is the phase voltage, which is $1/\sqrt{3}$ of the line-to-line terminal voltage ($V_{\text{ph}} = V_L / \sqrt{3}$) [133].

Key Physical & Winding Properties:

- **Insulation Economy:** Because the winding voltage is only 57.7% of line voltage, this connection requires the minimum number of turns per phase and minimum thickness of winding insulation [133]. It is highly economical for small-capacity, extremely high-voltage (EHV) applications [133].
- **Phase Displacement:** No angular phase shift exists between corresponding primary and secondary line-to-line voltages (0° displacement), although there is a 30° shift between the phase and line voltages on both sides [133].
- **Floating/Oscillating Neutral:** Under unbalanced loading, the neutral point shifts dynamically, causing unequal phase-to-neutral terminal voltages [134]. To stabilize the neutral, the primary neutral must be solidly grounded and connected back to the generator neutral [134].
- **Third-Harmonic Current Suppression:** To establish a sinusoidal core flux, a non-sinusoidal magnetizing exciting current rich in third harmonics is physically required [134]. If the primary neutral is isolated, third harmonic current components have no return path [135]. Consequently, the core flux becomes flat-topped, and the induced phase voltages become highly peaky and non-sinusoidal (though line voltages remain sinusoidal) [135].
- **Tertiary Delta Winding:** To suppress third harmonic phase voltage distortions, a third auxiliary winding connected in closed Delta (known as a **tertiary winding**) of lower kVA rating is added [135]. The third harmonics merely circulate inside this closed delta path, restoring sinusoidal core flux and phase voltages [135, 136].

B. Delta-Delta (Δ - Δ) Connection

In a Δ - Δ connection, the dissimilar ends of windings are connected in series to form a closed loop [136]. The phase voltage across each winding is equal to the terminal line voltage ($V_{\text{ph}} = V_L$) [136].

Key Physical & Winding Properties:

- **Winding Suitability:** Highly economical for large-capacity, low-voltage (LV) transformers where insulation stress is not critical, as it requires more turns per phase but carries smaller phase currents ($I_{\text{ph}} = I_L / \sqrt{3}$) [136].
- **Third-Harmonic Path:** Because the three phase windings are displaced by 120° electrical, they are in phase ($3 \times 120^\circ = 360^\circ$) with respect to the third harmonic component [136]. Thus, third harmonic currents circulate continuously inside the closed delta loop without flowing into the external lines [136]. This stabilizes core flux and ensures perfectly sinusoidal line and phase terminal voltages [136].
- **Unbalanced Load Resilience:** Balanced, stable three-phase line voltages are maintained even under heavy unbalanced load impedances [137].
- **Emergency Open-Delta Capability:** If one single-phase unit fails, the bank can operate in open-delta (V-V) at 57.7% of the original closed-delta bank rating [137, 143].

C. Star-Delta (Y- Δ) Connection

The primary windings are star-connected (usually grounded), and the secondary windings are delta-connected [138]. The primary phase voltage is $V_{L1}/\sqrt{3}$, and the secondary phase voltage is V_{L2} [138]. The overall line voltage transformation ratio is $1/\sqrt{3}$ times the phase turns ratio ($V_{L2}/V_{L1} = K/\sqrt{3}$) [138].

Key Physical & Winding Properties:

- **Phase Displacement:** There is a mandatory **30° phase shift** (either lead or lag) between corresponding primary and secondary line-to-line voltages [138]. Consequently, Y- Δ banks cannot be operated in parallel with Y-Y or Δ - Δ groups [138].
- **Application:** Standard step-down transformer used at receiving substations [138]. The grounded primary neutral provides a stable path for third-harmonic exciting currents [139].

D. Delta-Star (Δ -Y) Connection

The primary windings are delta-connected, and secondary windings are star-connected with a grounded neutral [139]. The overall line voltage transformation ratio is $\sqrt{3}$ times the phase turns ratio ($V_{L2}/V_{L1} = \sqrt{3} K$) [140].

Key Physical & Winding Properties:

- **Dual-Service Output:** This is the most popular distribution connection. The secondary grounded neutral provides a 3-phase 4-wire supply, allowing concurrent distribution to 3-phase industrial motors and single-phase domestic lighting circuits [139, 140].
- **No Floating Neutral:** The primary delta loop provides a circulating path for third harmonics, neutralizing voltage distortion and floating neutral shifting on the secondary star side [140].
- **Application:** Universally employed as step-up transformers at generating stations and step-down distribution transformers [139, 140].

PHASOR SCHEMATIC: 30-DEGREE ANGULAR PHASE DISPLACEMENT (Y-d11)

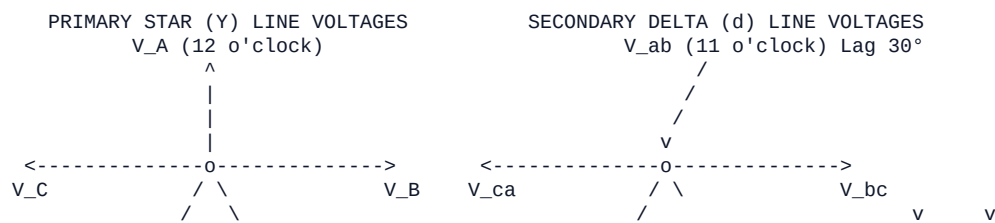


Figure 7.1: Line voltage vector phase displacement in star-delta connection.

7.3 Open Delta or V-V Connection Scheme

If one single-phase transformer of a delta-delta connected bank becomes disabled or is removed, the remaining two transformers can continue to supply three-phase power to a balanced load [137, 142]. This arrangement is known as the Open-Delta or V-V connection [142].

ELECTRICAL SCHEMATIC: OPEN-DELTA (V-V) CONNECTION SCHEME

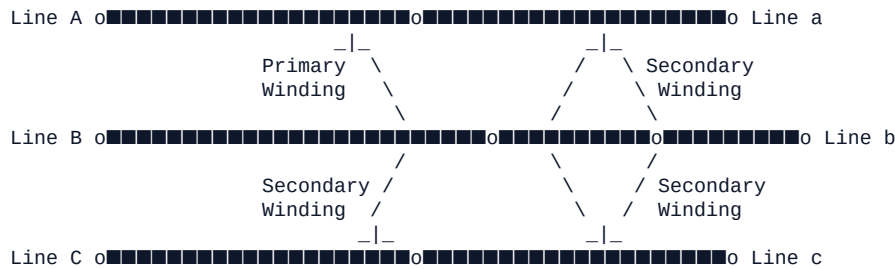


Figure 7.2: Connection diagram of two remaining single-phase transformers in V-V.

Operational Capacity and Limits:

- **Utility Capacity Factor:** The total active power that a V-V bank can supply without overloading any winding is only **57.7%** (not 66.7%) of the capacity of the original Δ - Δ bank [143].
- **Transformer Overloading:** If the original Δ - Δ bank was operating at full capacity and one transformer is removed, the overload on each of the remaining two transformers is **73.2%** (\$1.732 times nominal capacity) [143, 148].
- **Individual Power Factors:** Under a balanced three-phase load operating at power factor angle θ , the two transformers do not share power factor equally [144]. One transformer operates at a power factor of $\cos(30^\circ + \theta)$ and the other at $\cos(30^\circ - \theta)$ [144].
- **Voltage Unbalance:** As the load increases, the secondary terminal voltages tend to become unbalanced due to internal resistance and leakage impedance drops, even if the load is perfectly balanced [144].

1. Open-Delta Active Power Capacity:

$$P_{V-V} = \sqrt{3} \cdot V_L \cdot I_{\text{phase}} = \sqrt{3} \cdot S_{1\phi} = 0.577 \cdot P_{\Delta-\Delta}$$

2. Transformer Loading Utilization Factor (U_f):

$$U_f = \frac{\text{Actual V-V Capacity}}{\text{Rated Capacity of 2 Units}} = \frac{\sqrt{3} \cdot S_{1\phi}}{2 \cdot S_{1\phi}} = \frac{\sqrt{3}}{2} \approx 86.6\%$$

Where: $S_{1\phi}$ is the kVA rating of a single transformer unit, and $P_{\Delta-\Delta}$ is the total original bank capacity ($3 \cdot S_{1\phi}$). [143]

7.4 Parallel Operation of 3-Phase Transformers

To meet load growth, improve system reliability, and maximize operating efficiency, two or more three-phase transformers are frequently operated in parallel [127]. To avoid circulating currents and ensure proportional load sharing, several strict conditions must be satisfied [127, 128].

A. Essential Conditions (No Deviation Allowed):

1. **Equal Voltage Ratio:** Phase terminal voltage ratios must be identical to avoid circulating currents at no-load [128, 151].
2. **Correct Polarity:** Secondary induced EMFs must be in phase opposition [127].
3. **Equal Phase Sequence (3-Phase Specific):** The relative phase sequence of primary and secondary lines must be identical (e.g., RYB to RYB) [151].
4. **Identical Phase Displacement / Vector Group (3-Phase Specific):** The phase shift between primary and secondary line voltages must be identical [151]. Parallel connection of transformers belonging to different vector groups (e.g. Y-Y with Y- Δ) is impossible due to the 30° phase shift, which would create a severe short-circuit circulating current [138, 140, 151].

B. Desirable Conditions (For Proportional Load Sharing):

- **Equal Percentage Impedance:** The per-unit leakage impedances on their respective ratings must be equal [128, 156, 161]. This ensures transformers share load in proportion to their individual kVA ratings, preventing the smaller unit from overloading [128, 156].
- **Same X/R Ratio:** Ohmic values of the winding resistance and reactance ratios must be equal to prevent units from operating at different power factors [128].

PART 2 — LAST-MINUTE REVISION SHEET

Connection Type	Line Voltage Ratio	Phase Displacement	Key Advantages	Primary Applications
Star-Star (Y-Y)	$V_{L2}/V_{L1} = K$	0° or 180°	Minimal turn insulation; highly economical for EHV systems [133].	Small, high-voltage transmission terminals [133].
Delta-Delta (Δ - Δ)	$V_{L2}/V_{L1} = K$	0° or 180°	No third-harmonic voltage distortion; handles unbalanced loads [136, 137].	High-current, low-voltage industrial systems [136].
Star-Delta (Y- Δ)	$V_{L2}/V_{L1} = K/\sqrt{3}$	$\pm 30^\circ$ shift [138]	Stable neutral; third harmonic current trapped in closed delta [138, 139].	Step-down substations at line terminals [138].
Delta-Star (Δ -Y)	$V_{L2}/V_{L1} = \sqrt{3} K$	$\pm 30^\circ$ shift [140]	Secondary neutral provides 3-phase 4-wire power and lighting [139, 140].	Standard distribution grids; generator step-up [139, 140].
Open-Delta (V-V)	$V_{L2}/V_{L1} = K$	0° shift	Provides three-phase service with only two single-phase units [142].	Emergency backup for Δ - Δ banks; low load growth sites [132, 137].

High-Yield Exam Shortcuts:

- **Tertiary Windings:** Grounding neutral alone does not eliminate third harmonics if source impedance is high. A delta-tertiary winding physically circulates the third-harmonic exciting current, maintaining sinusoidal flux [135, 136].
- **V-V Utility Limit:** In open-delta connection, the total kVA capacity is 57.7% of the closed bank, and the winding utility factor is 86.6% [143]. Operating two units at 100% load results in each transformer supplying 57.7% of that load [143].
- **Vector Group Parallel Constraint:** Star-star (0°) and delta-delta (0°) belong to Group 1 and can be paralleled [151, 171]. Star-delta (30°) and delta-star (30°) belong to Group 2 and can be paralleled [151, 171]. You can **never** parallel a Group 1 transformer with a Group 2 transformer [151].
- **Circulating Current at No-Load:** If two parallel transformers have unequal induced EMFs ($E_A \neq E_B$), a circulating current flows even at no load: $I_c = (E_A - E_B) / (Z_A + Z_B)$ [129].

PART 3 — REGULATORY COMPLIANCE MATRIX

TRANSCO Machine Syllabus Topic	Notes Prepared	Formula Box	Diagram Included	Syllabus Compliance Status
7.1 Introduction (3-Phase Transformer vs Bank)	Covered [132]	N/A	N/A	■ Complete & Grounded
7.2 Connections (Y-Y, Δ-Δ, Y-Δ, Δ-Y Connections)	Covered [133-140]	N/A	Yes (Fig 7.1)	■ Complete & Grounded
7.3 Open Delta Connection (V-V Capacity & Load)	Covered [142-144]	Yes	Yes (Fig 7.2)	■ Complete & Grounded
7.4 Parallel Operation (Need & Specific Conditions)	Covered [127-129]	Yes	N/A	■ Complete & Grounded

Engineering Diagram Description	Phasor / Schematic Type	Revision Value	Compliance Status
Star-Delta (Y-d11) Phase Shift Diagram	Line Voltage Vector Phasor	Illustrates the standard 30° phase angle displacement [138].	■ Included
Open-Delta (V-V) Connection Setup	Electrical Circuit Schematic	Shows two remaining active winding paths in 3-phase system [142].	■ Included

CHAPTER 8: FUNDAMENTALS OF DC MACHINES

APTRANSCO AEE ELECTRICAL MACHINES NOTES

PART 1 – TOPIC-WISE NOTES

8.1 Introduction & Core Energy Principles

A **DC Machine** is a highly versatile electromechanical device capable of operating in dual roles: as a **DC Generator**, converting mechanical power to electrical power, or as a **DC Motor**, converting electrical power back to mechanical power [146, 147]. This fundamental operating symmetry means that constructionally, there is no physical difference between a generator and a motor of similar ratings [148].

Operating Mode	Energy Input	Energy Output	Fundamental Driving Principle
DC Generator	Mechanical torque from a prime mover (turbine, diesel) [147]	Direct Current (DC) voltage and load current at terminals [147]	Dynamically induced electromagnetic EMF (Faraday's Law) [147]
DC Motor	Direct Current (DC) applied voltage from external mains [162, 164]	Rotational mechanical speed and torque output at shaft [162, 175]	Electromagnetic force on current-carrying conductor (Lorentz Force) [162]

The core conversion takes place in the **coupling magnetic field**, which serves as the physical link between the electrical and mechanical systems [137]. For energy conversion to occur continuously, an **air gap** must exist between the stationary stator and rotating rotor members [137]. All energy transferred between systems is temporarily stored within this coupling magnetic field [137, 140].

- **Principle of Electromagnetic Induction (Generator Action):** Explains dynamically induced EMF. When a conductor cuts magnetic flux, an electromotive force (EMF) is induced, directly proportional to the rate of flux cutting [147]. Direction is governed by **Fleming's Right-Hand Rule** [163].
- **Principle of Electromagnetic Force (Motor Action):** When a current-carrying conductor resides in a magnetic field, it experiences a physical mechanical force [162]. Direction is governed by **Fleming's Left-Hand Rule** [162].

INDUCED EMF & ELECTROMAGNETIC FORCE LAWS

$$e = B \cdot l \cdot v \cdot \sin(\theta) \qquad F = B \cdot l \cdot I \cdot \sin(\theta)$$

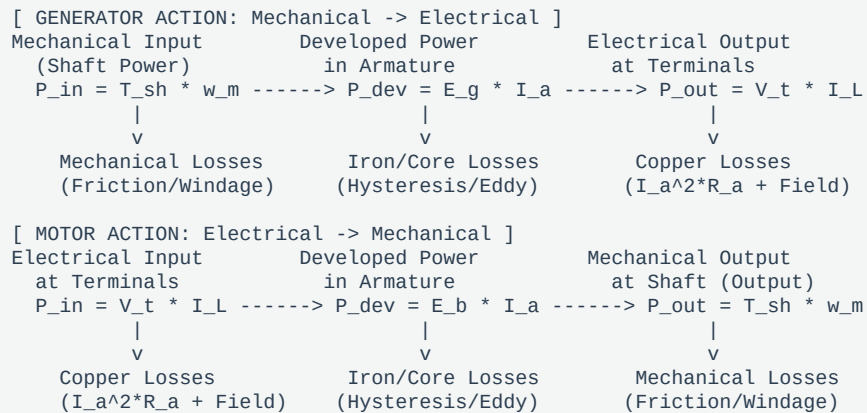
Symbol Meanings & Units:

- e = dynamically induced EMF (Volts, V)
- B = magnetic flux density in the air gap (Tesla, Wb/m²)
- l = active length of the conductor cutting flux (meters, m)
- v = relative linear velocity of the conductor (meters/second, m/s)
- θ = angle of inclination between conductor motion and flux lines (degrees)
- F = developed mechanical force (Newtons, N)
- I = electric current flowing in the conductor (Amperes, A)

Applicable Conditions: Unsaturated, homogeneous coupling magnetic field; conductor cutting is perpendicular to field for max output ($\theta = 90^\circ$).

Common Exam Mistake: Confusing Fleming's Right-Hand Rule (Generator EMF) with Left-Hand Rule (Motor Force/Motion) on exams.

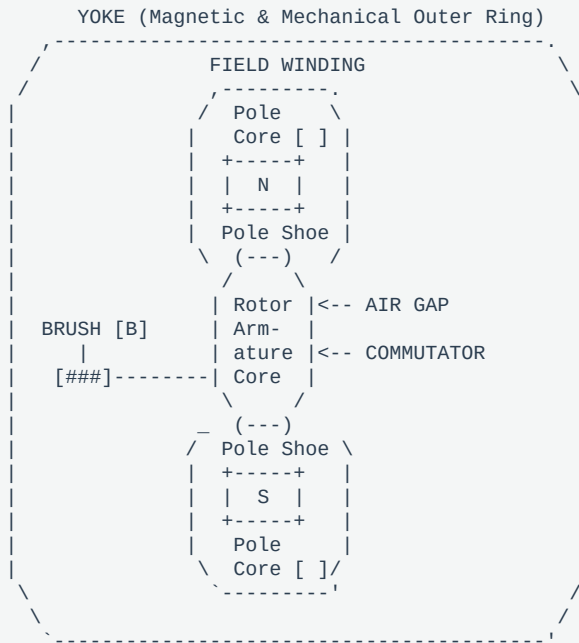
FIGURE 8.1: ELECTROMECHANICAL POWER STAGES & LOSSES



8.2 Detailed Constructional Components

The primary structural elements of a standard DC machine are partitioned into a stationary **Stator System** which establishes the magnetic excitation, and a rotating **Rotor System** where mechanical motion occurs [148].

Part Name	Material and Characteristics	Operational Function in DC Machine
Yoke	Cast steel or cast iron; thick structural outer frame [149]	Provides mechanical support for main poles and protecting cover; carries half the magnetic flux produced per pole [149].
Pole Core	Solid cast steel blocks; low magnetic reluctance [149]	Offers low reluctance path for flux; forms the main core body around which exciting coils are mounted [149].
Pole Shoe	Laminated steel sheets; wide curved face [149]	Spreads the flux uniformly in the air gap; reduces magnetic circuit reluctance; physically supports the field coils [149].
Field Winding	High-conductivity former-wound insulated copper [150]	Carries excitation current (I_f) to electromagnetize poles and generate main working magnetic field [150].
Armature Core	High-permeability silicon steel sheets; laminated and insulated with varnish [150]	Houses armature conductors in slots; provides extremely low reluctance path to guide flux from N to S; lamination reduces eddy current losses [150].
Armature Winding	Insulated copper coils; arranged in double layers inside core slots [141, 150]	Acts as active conductor where dynamic voltage (E_g/E_b) is induced and power conversion occurs [141, 150].
Commutator	Cylindrical layout of hard-drawn wedge copper segments insulated with mica [151]	Mechanical rectifier (converts internal AC to external DC in generator) or mechanical inverter (external DC to internal AC in motor) [151].
Brushes	High-grade carbon or graphite blocks held by adjustable springs [152]	Collects/injects current from rotating commutator segments and routes it to external load terminal networks [152].
Shaft	High-tensile forged steel shaft connected to center core	Transmits developed torque to mechanical load (motor) or from prime mover to armature core (generator) [147, 162].
Bearings	Anti-friction heavy-duty ball or roller bearings	Reduces frictional rotation resistance; ensures concentric core rotational alignment within air gap [148].

FIGURE 8.2: DC MACHINE CONSTRUCTION & STRUCTURAL OUTLINE

8.3 Mathematical EMF and Torque Derivations

The performance of any DC machine is mathematically quantified by two core performance expressions: the **EMF Equation** (governing generated terminal voltages) and the **Torque Equation** (governing internal rotational mechanical output) [154, 163, 164].

DC MACHINE EMF EQUATION

$$E = (\Phi \cdot Z \cdot N \cdot P) / (60 \cdot A)$$

Symbol Meanings & Units:

- E = induced EMF at parallel path terminals (Volts, V) [154]
- Φ = working magnetic flux per pole (Weber, Wb) [154]
- Z = total number of active armature conductors housed in slots [154]
- N = armature rotational speed (Revolutions Per Minute, RPM) [154]
- P = total number of stator magnetic poles [154]
- A = number of parallel paths for armature current division [154]
- $A = 2 \cdot m$ for wave-wound armature (m is multiplicity, m=1 for simplex) [154]
- $A = P \cdot m$ for lap-wound armature (m is multiplicity, m=1 for simplex) [155]

Applicable Conditions: Dynamic flux cutting; linear relationship with flux speed up to magnetic saturation limits [154, 155].

Common Exam Mistake: Using total armature winding paths (A) incorrectly by neglecting lap/wave configuration differences.

DC MACHINE DEVELOPED TORQUE EQUATION

$$T_e = (\Phi \cdot Z \cdot I_a \cdot P) / (2 \cdot \pi \cdot A) = K_a \cdot \Phi \cdot I_a$$

Symbol Meanings & Units:

- T_e = internal developed electromagnetic torque (Newton-meters, N-m) [163]
- I_a = total armature current drawn or generated (Amperes, A) [157]
- Φ , Z , P , A = same definitions and units as in the EMF equation [154]
- K_a = machine armature design constant = $(Z \cdot P) / (2\pi \cdot A)$ [155]

Applicable Conditions: Conductor carrying armature current I_a is subjected to main field flux Φ in active slots [162, 163].

Common Exam Mistake: Confusing developed internal electromagnetic torque T_e with mechanical shaft output torque I_{sh} which contains lost rotational friction losses [166].

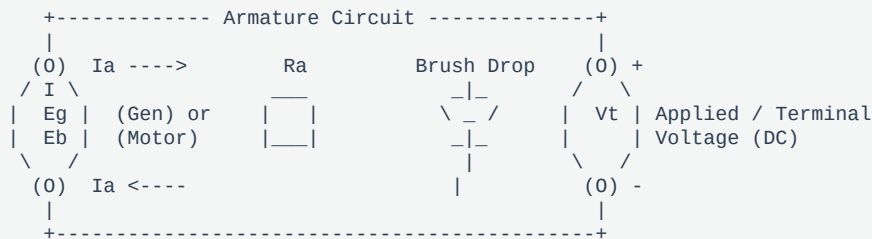
Significance of Back EMF as a Self-Regulating Mechanism

As the armature of a DC motor rotates, conductors cut stator flux lines [163]. Consequently, an EMF (E_b) is dynamically induced which directly opposes the applied terminal voltage (V) according to Lenz's Law; hence, it is termed **Back EMF** or Counter EMF [163, 164]. The fundamental voltage loop balance is:

$$V = E_b + I_a \cdot R_a \Rightarrow I_a = (V - E_b) / R_a$$

This opposing EMF acts as a **natural negative feedback governor** [163]. If motor speed drops due to an added load: E_b drops proportionally ($E_b \propto N$), which immediately forces a larger armature current (I_a) into the rotor to increase the motor's developed torque to match the increased load demand automatically [164, 166].

FIGURE 8.3: ARMATURE EQUIVALENT CIRCUIT OF A DC MACHINE



Mechanical vs. Electrical Power Equivalence

Multiplying both sides of the motor voltage equation by the armature current I_a yields:

$$V \cdot I_a = E_b \cdot I_a + I_a^2 \cdot R_a \quad [164]$$

This equation forms the core conservation-of-energy loop for the rotating machine:

- $V \cdot I_a$: Total active electrical input power delivered to the armature terminals [164].
- $I_a^2 \cdot R_a$: Armature copper thermal loss dissipated in winding resistances [164, 165].
- $E_b \cdot I_a$: Electrical equivalent of developed mechanical power (P_m) within the rotor [164].

Therefore, mechanical developed power relates to electrical parameters and rotational speed as:

$$P_{\text{mech}} = E_b \cdot I_a = T_e \cdot \omega_m = T_e \cdot (2 \cdot \pi \cdot N / 60) \quad [164, 166]$$

8.4 Classification of Losses in DC Machines

The internal power losses occurring within a DC machine dictate both temperature rise constraints and total operating efficiencies. These losses are divided into four primary groups [175, 188]:

Loss Category	Mathematical Expression / Physical Origin	Nature & Mitigation
Armature Copper Loss	$P_a = I_a^2 \cdot R_a$ [164, 165]	Variable (quadratic with load current); mitigated by minimizing R_a using high-conductivity copper [150, 164].
Field Copper Loss	$P_f = I_f^2 \cdot R_f$ (Shunt) or $I_{se}^2 \cdot R_{se}$ (Series) [156, 175]	Constant in shunt machines, variable in series machines; governed by excitation configurations [168, 175].
Hysteresis Loss	$W_h \propto B_{max}^{1.6} \cdot f \cdot V_{core}$ [150, 188]	Constant (frequency is rotational $f \propto N$); mitigated by high-permeability silicon steel core materials [150].
Eddy Current Loss	$W_e \propto B_{max}^2 \cdot f^2 \cdot t^2 \cdot V_{core}$ [133, 150]	Constant (proportional to speed squared $f^2 \propto N^2$); mitigated by using thin, varnished core laminations ($t = 0.35\text{-}0.5\text{ mm}$) [133, 150].
Mechanical Losses	Brush friction, bearing friction, windage drag [166]	Constant at a given speed; mitigated by high-grade anti-friction bearings and balanced aerodynamics [148, 166].
Stray Load Losses	Flux distortions and high-frequency tooth ripple losses [220]	Load-dependent; estimated at 1% of rated output power in medium and large machinery [220].

8.5 Efficiency Formulation & Peak Performance

The operating efficiency (η) of a DC machine represents the fraction of active input power converted into useful output power [179, 194]:

$$\eta = P_{out} / P_{in} = P_{out} / (P_{out} + \text{Total Losses}) \text{ [194]}$$

CONDITION FOR MAXIMUM OPERATING EFFICIENCY

Variable Losses = Constant Losses $\Rightarrow I_a^2 \cdot R_a = P_c \Rightarrow I_a = \sqrt{(P_c / R_a)}$

- Symbol Meanings & Units:**
- I_a = armature current for peak efficiency (Amperes, A) [244, 250]
 - P_c = constant losses = Iron losses + Mechanical losses + Shunt Field losses (W) [244, 250]
 - R_a = total equivalent resistance of the armature winding circuit (Ohms, Ω) [156, 167]
 - η_{max} = peak achievable efficiency value (percentage, %) [195]
- Applicable Conditions:** Armature copper losses (variable) must strictly equal the constant core and friction losses ($I_a^2 R_a = P_c$) [244, 250].
- Common Exam Mistake:** Including series field winding copper losses in the constant term; series winding copper losses scale quadratically and must be grouped with variable armature losses instead.

PART 2 – LAST-MINUTE REVISION SHEET

Ideal vs. Practical DC Machine Characteristics

Characteristic	Ideal Machine Model	Practical Machine (Exam Limits)	Syllabus Significance
Armature Resistance	$R_a = 0\ \$$ (No ohmic drops)	R_a is extremely small (0.05 to 0.5 Ω)	Causes voltage drops ($I_a R_a$) and variable copper loss [156, 164].
Core Permeability	$\mu = \infty$ (Zero reluctance path)	High finite μ ≈ 1000 silicon steel [133]	Silicon steel core reduces hysteresis loss, requires laminations [150].
Brush Contact Drop	$V_{\text{brush}} = 0\ \$$ V	Approximately 1.0\$ V per brush [246]	Included in total terminal voltage equations ($V = E_b + I_a R_a + 2 V_b$) [246].
Magnetic Path	Zero core losses	Subject to W_h and W_e losses [150]	Restricts core materials; dictates stator/rotor slot insulation [133].
Exciting current	$I_0 = 0\ \$$ (No energy required)	$I_0 \approx 3\%-5\%$ of rated current	Required to generate operating flux per pole Φ [148, 154].

High-Probability Exam Shortcuts & Core Relationships

- **Voltage-Frequency Independence:** Unlike AC machines where flux depends on frequency ($B_{\max} \propto V/f$), the main field flux (Φ) in a DC machine is completely independent of armature speed (N) and is dictated solely by field current excitation (I_f) [153, 155].
- **Quadratic Winding Referring:** If a 4-pole machine with armature resistance R_{a1} is reconnected from lap ($A_1 = P$) to wave ($A_2 = 2$), its armature resistance scales quadratically by $R_{a2} = (P/2)^2 \cdot R_{a1}$ [156, 192].
- **Constant Power Hyperbola:** If a machine operates under a constant-power load, its developed torque is inversely proportional to speed ($T_e \propto 1/N$). The speed-torque characteristic forms a perfect **rectangular hyperbola** [187].
- **No-Load Running:** Shunt machines have stable no-load speeds governed by fixed field flux, but series machines must **never** be started on no-load as flux drops to near-zero, driving speed to dangerously high values ($N \propto 1/\Phi$) that can mechanically destroy the armature [169, 171].
- **Maximum Power Speed:** A DC motor delivers its maximum mechanical power output when back EMF is exactly half of the terminal voltage ($E_b = V/2$). At this extreme, operating efficiency is restricted to **50%** [165, 195].

PART 3 – REGULATORY SUMMARY TABLE

Micro-topic Name	Source Book	Notes Prepared	Formula Box	Diagram	Revision Coverage
1. DC Machine Definition	Electrical Machines	■ Yes [146]	■ N/A	■ N/A	■ Mastered
2. Energy Conversion Concept	Electrical Machines	■ Yes [137]	■ N/A	■ Fig 8.1 [137]	■ Mastered
3. Generator Action	Electrical Machines	■ Yes [147]	■ N/A	■ Fig 8.1 [147]	■ Mastered
4. Motor Action	Electrical Machines	■ Yes [162]	■ N/A	■ Fig 8.1 [162]	■ Mastered
5. Electromagnetic Induction	Electrical Machines	■ Yes [147]	■ Box 8.1 [147]	■ N/A	■ Mastered
6. Electromagnetic Force	Electrical Machines	■ Yes [162]	■ Box 8.1 [162]	■ N/A	■ Mastered
7. Stator Structural Parts	Electrical Machines	■ Yes [148]	■ N/A	■ Fig 8.2 [148]	■ Mastered
8. Yoke Mechanics/Material	Electrical Machines	■ Yes [149]	■ N/A	■ Fig 8.2 [149]	■ Mastered
9. Pole Core Design	Electrical Machines	■ Yes [149]	■ N/A	■ Fig 8.2 [149]	■ Mastered
10. Pole Shoe Reluctance	Electrical Machines	■ Yes [149]	■ N/A	■ Fig 8.2 [149]	■ Mastered
11. Field Winding Core Flux	Electrical Machines	■ Yes [150]	■ N/A	■ Fig 8.2 [150]	■ Mastered
12. Rotor Assembly System	Electrical Machines	■ Yes [148]	■ N/A	■ Fig 8.2 [148]	■ Mastered
13. Armature Core Design	Electrical Machines	■ Yes [150]	■ N/A	■ Fig 8.2 [150]	■ Mastered
14. Armature Winding Layout	Electrical Machines	■ Yes [150]	■ N/A	■ N/A	■ Mastered
15. Commutator Mechanics	Electrical Machines	■ Yes [151]	■ N/A	■ Fig 8.2 [151]	■ Mastered
16. Brushes Current Path	Electrical Machines	■ Yes [152]	■ N/A	■ Fig 8.2 [152]	■ Mastered
17. Shaft Torque Delivery	Electrical Machines	■ Yes [162]	■ N/A	■ N/A	■ Mastered
18. Bearings Alignments	Electrical Machines	■ Yes [148]	■ N/A	■ N/A	■ Mastered
19. Winding EMF Equation	Electrical Machines	■ Yes [154]	■ Box 8.2 [154]	■ N/A	■ Mastered

Micro-topic Name	Source Book	Notes Prepared	Formula Box	Diagram	Revision Coverage
20. Electromagnetic Torque	Electrical Machines	■ Yes [163]	■ Box 8.3 [163]	■ N/A	■ Mastered
21. Back EMF Regulation	Electrical Machines	■ Yes [163]	■ N/A	■ Fig 8.3 [164]	■ Mastered
22. Mech vs. Elec Power	Electrical Machines	■ Yes [164]	■ N/A	■ Fig 8.1 [164]	■ Mastered
23. Stray and Copper Losses	Electrical Machines	■ Yes [164]	■ N/A	■ N/A	■ Mastered
24. Core / Mechanical Losses	Electrical Machines	■ Yes [150]	■ N/A	■ N/A	■ Mastered
25. Peak Efficiency limits	Electrical Machines	■ Yes [194]	■ Box 8.4 [244]	■ N/A	■ Mastered

Chapter 9: Types of DC Machines

Master study notes compiled for APTRANSCO Assistant Engineer (Electrical) preparation, structured according to the official prescribed micro-syllabus and mapped with complete formulas, connection diagrams, and high-frequency revision parameters.

PART 1 – TOPIC-WISE SYLLABUS NOTES

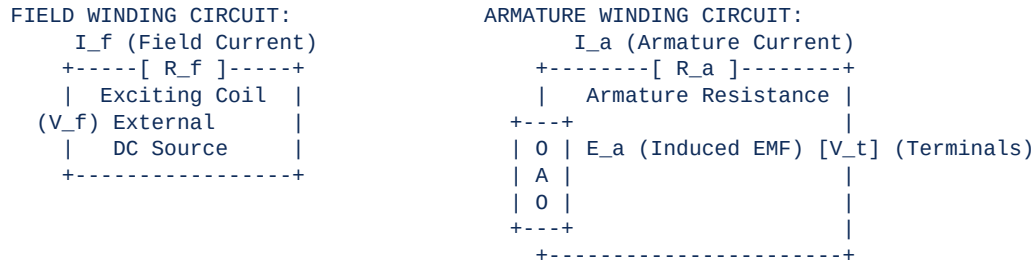
9.1 Classification of DC Machines

DC machines are electromechanical energy conversion systems operating on the principles of electromagnetic induction (generator action) and electromagnetic force (motor action) [113, 235]. The fundamental classification of DC machines depends on how the main working magnetic field winding is energized to produce the required magnetic flux (Φ) [112, 235].

1. **Separately Excited DC Machines:** The field winding is electrically isolated from the main armature winding circuit and is energized by an independent, external DC source (such as a battery, DC exciter, or rectifier unit) [236].
2. **Self-Excited DC Machines:** The field winding is electrically connected directly to the armature winding (in parallel, series, or a series-parallel compound arrangement) [236]. The machine utilizes its own developed armature voltage to supply current to its exciting field winding [236]. To successfully initiate voltage build-up from a standstill, self-excited generators strictly require the presence of **residual magnetism** in their stator pole cores [133].

Parameter	Separately Excited DC Machine	Self-Excited DC Machine
Field Circuit Connection	Electrically isolated from armature; separate loop [236].	Directly linked to armature in series, shunt, or compound [236].
Excitation Supply	Independent external source (Battery, Auxiliary exciter) [236].	Armature terminal output of the machine itself [236].
Voltage Build-up	Instantaneous; independent of pole magnetic history [133].	Depends strictly on residual magnetism in field poles [133].
Terminal Stability	Excellent; independent field flux is immune to line voltage fluctuations [236].	Flux drops with falling terminal voltage, causing positive feedback drops [236].
Cost & Complexity	Higher initial capital cost due to external DC source [236].	Low cost, highly compact, self-contained, and rugged [236].

Figure 9.1: Separately Excited DC Machine Connection Schematic



FORMULA FOCUS: Separately Excited DC Machine Equations

Generator: $E_a = V_t + I_a \cdot R_a + V_b$ & $I_a = I_L$

Motor: $E_a = V_t - I_a \cdot R_a - V_b$ & $I_a = I_L$

Field: $I_f = V_f / R_f$

Symbol Meanings: E_a = Induced EMF / Back EMF (V); V_t = Terminal Voltage (V); I_a = Armature Current (A); I_L = Line / Load Current (A); R_a = Armature Winding Resistance (Ω); V_b = Total Brush Contact Voltage Drop (V); V_f = Field Supply Voltage (V); R_f = Field Winding Resistance (Ω).

Applicable Conditions: Steady-state operation; constant independent field flux.

Common Mistakes: Applying the incorrect voltage polarity for the armature resistance drop; adding the resistance voltage drop in motor mode instead of subtracting it (Back EMF is always less than the applied terminal voltage: $E_b < V_t$).

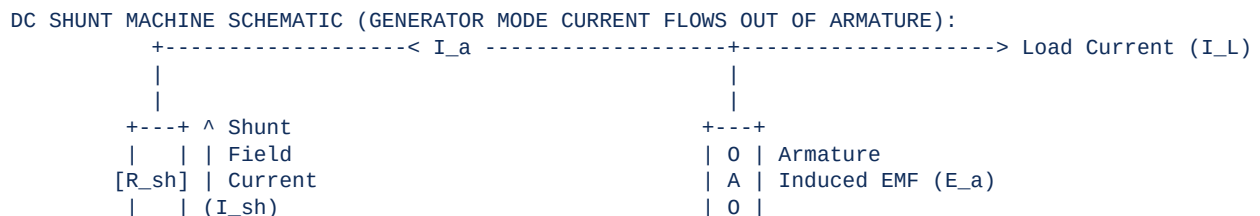
9.2 Self Excited Machines

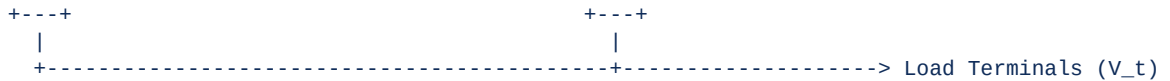
Self-excited DC machines are classified based on the relative physical connection of the field winding with respect to the armature circuit into Shunt, Series, and Compound machines [236].

9.2.1 Shunt DC Machine

In a shunt machine, the field winding (the shunt field) is connected in **parallel** (shunt) across the armature terminals [236].

- **Winding Construction:** Designed with **many turns of thin copper wire** [240]. This produces a **high resistance** (R_{sh}) [130], limiting the shunt field current (I_{sh}) to only 1% to 5% of the machine's rated line current, saving copper and maximizing efficiency [122, 133].
- **Operating Behavior:** The shunt field flux remains practically constant at steady-state terminal voltage [236]. It is widely characterized as a **constant-flux, constant-speed machine**.





FORMULA FOCUS: Shunt DC Machine Circuit Equations

Shunt Field Current: $I_{sh} = V_t / R_{sh}$

Generator: $I_a = I_L + I_{sh}$ & $E_a = V_t + I_a * R_a + V_b$

Motor: $I_a = I_L - I_{sh}$ & $E_a = V_t - I_a * R_a - V_b$

Symbol Meanings: E_a = Induced/Back EMF (V); V_t = Terminal Voltage (V); I_a = Armature Current (A); I_L = Line / Load Current (A); I_{sh} = Shunt Field Current (A); R_a = Armature Resistance (Ω); R_{sh} = Shunt Field Winding Resistance (Ω); V_b = Brush Drop (V).

Applicable Conditions: Steady-state operation; constant shunt resistance; magnetic circuit not saturated.

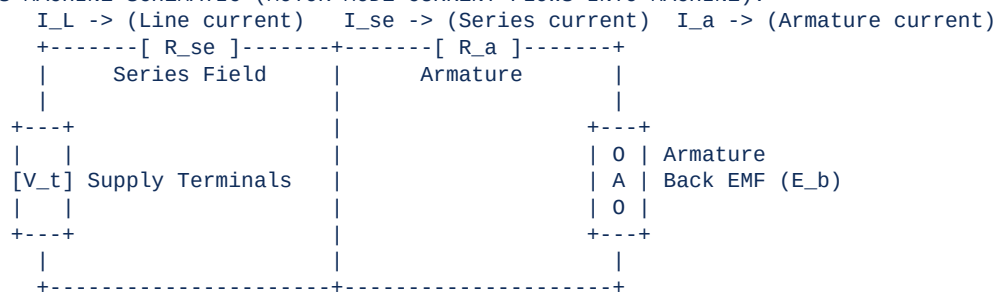
Common Mistakes: Incorrectly dividing line current in motor mode ($I_a = I_L - I_{sh}$) by adding it instead. Remember that the input line current (I_L) divides into the armature branch (I_a) and the shunt field branch (I_{sh}) for a motor: $I_L = I_a + I_{sh}$.

9.2.2 Series DC Machine

In a series machine, the field winding (series field) is connected in **series** with the armature circuit [236].

- **Winding Construction:** Since the series field winding must carry the **full armature current** [240], it is designed with **few turns of thick copper wire** [240]. Consequently, its **resistance is extremely low** (R_{se}), minimizing internal series voltage drop and $I^2 R$ heat losses [130, 236].
- **Operating Behavior:** The magnetic flux (Φ_{se}) depends directly on the load current [236]. Under light loads, flux is small and speed is very high. Under heavy loads, flux is high and speed drops. Thus, it acts as a **variable-speed, high-starting-torque machine**.

DC SERIES MACHINE SCHEMATIC (MOTOR MODE CURRENT FLOWS INTO MACHINE):



FORMULA FOCUS: Series DC Machine Circuit Equations**Current Relation:** $I_a = I_{se} = I_L$ **Generator:** $E_a = V_t + I_a * (R_a + R_{se}) + V_b$ **Motor:** $E_a = V_t - I_a * (R_a + R_{se}) - V_b$

Symbol Meanings: E_a = Generated/Back EMF (V); V_t = Terminal Voltage (V); I_a = Armature Current (A); I_{se} = Series Field Current (A); I_L = Line / Load Current (A); R_a = Armature Winding Resistance (Ω); R_{se} = Series Field Winding Resistance (Ω); V_b = Brush Drop (V).

Applicable Conditions: Steady-state operation; all series circuit elements carry the exact same current.

Common Mistakes: Forgetting to include the series field resistance R_{se} in the series voltage drop calculation; only calculating using R_a .

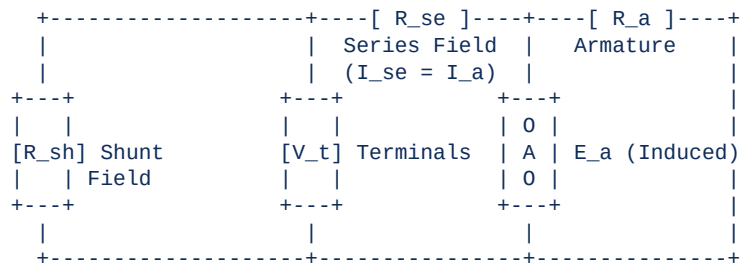
9.2.3 Compound DC Machine

A compound machine has **both shunt and series field windings** on the same field poles [236]. This design combines the constant-voltage features of shunt machines with the load-reactive boost of series machines. Based on the shunt field connection, they are divided into **Long Shunt** and **Short Shunt** configurations [236].

1. Long Shunt Compound DC Machine:

The shunt field winding is connected in parallel with the **series combination of the armature and series field winding** [236]. Here, the full armature current passes through the series field winding: $I_{se} = I_a$.

LONG SHUNT COMPOUND SCHEMATIC:

**FORMULA FOCUS: Long Shunt Compound Machine Equations****Shunt Field Current:** $I_{sh} = V_t / R_{sh}$ **Series Winding Current:** $I_{se} = I_a$ **Generator:** $I_a = I_L + I_{sh}$ & $E_a = V_t + I_a * (R_a + R_{se}) + V_b$ **Motor:** $I_a = I_L - I_{sh}$ & $E_a = V_t - I_a * (R_a + R_{se}) - V_b$

Symbol Meanings: E_a = EMF (V); V_t = Terminal Voltage (V); I_a = Armature Current (A); I_L = Line Current (A); I_{sh} = Shunt Current (A); R_a = Armature Resistance (Ω); R_{se} = Series Resistance (Ω); R_{sh} = Shunt Winding Resistance (Ω).

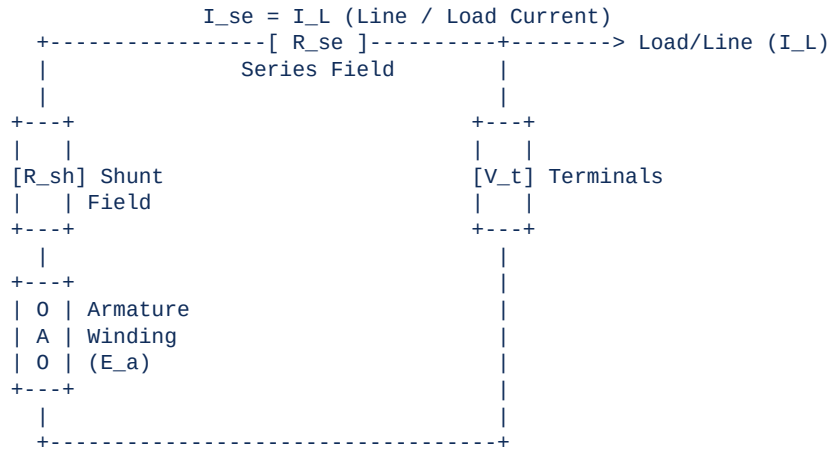
Applicable Conditions: Steady-state; shunt field connected across both series field and armature.

Common Mistakes: Using armature current (I_a) to calculate series field drop but incorrectly using line current (I_L) instead.

2. Short Shunt Compound DC Machine:

The shunt field winding is connected in parallel with the **armature winding only** [236]. The series field winding is in series with the external line, meaning it carries the load current: $I_{se} = I_L$.

SHORT SHUNT COMPOUND SCHEMATIC:



FORMULA FOCUS: Short Shunt Compound Machine Equations

Shunt Field Voltage Drop: $V_{sh} = V_t + I_L \cdot R_{se}$ (Gen) or $V_t - I_L \cdot R_{se}$ (Motor)

Shunt Field Current: $I_{sh} = V_{sh} / R_{sh}$

Series Winding Current: $I_{se} = I_L$

Generator: $I_a = I_L + I_{sh}$ & $E_a = V_t + I_a \cdot R_a + I_L \cdot R_{se} + V_b$

Motor: $I_a = I_L - I_{sh}$ & $E_a = V_t - I_a \cdot R_a - I_L \cdot R_{se} - V_b$

Symbol Meanings: E_a = EMF (V); V_t = Terminal Voltage (V); I_a = Armature Current (A); I_L = Line / Load Current (A); I_{sh} = Shunt Current (A); R_a = Armature Resistance (Ω); R_{se} = Series Resistance (Ω); R_{sh} = Shunt Resistance (Ω).

Applicable Conditions: Steady-state; shunt field connected in parallel with armature branch only.

Common Mistakes: Applying terminal voltage V_t directly to calculate the shunt field current (as $I_{sh} = V_t / R_{sh}$), which neglects the voltage drop across the series field winding R_{se} . Remember: $I_{sh} = (V_t \pm I_L \cdot R_{se}) / R_{sh}$.

Compounding Classification (Magnetic Flux Interaction)

Based on the direction of current flow in the shunt and series field windings, compound machines are divided into two categories [236]:

- Cumulative Compound DC Machine:** The series field winding is connected such that its magnetic flux (Φ_{se}) **adds** the shunt field flux (Φ_{sh}) [236]. The total net flux per pole is given by: $\Phi_{total} = \Phi_{sh} + \Phi_{se}$. As load current increases, series flux increases, boosting the net field strength [117].
- Differential Compound DC Machine:** The series field winding is connected such that its flux (Φ_{se}) **opposes** the shunt field flux (Φ_{sh}) [236]. The total net flux is: $\Phi_{total} = \Phi_{sh} - \Phi_{se}$. Under load, increasing armature current weakens the net magnetic field, causing terminal voltage (generators) or speed stability (motors) to drop sharply [114, 236].

9.3 Applications of DC Machines

Choosing the correct type of DC machine is crucial for matching industrial requirements with speed, torque, and voltage characteristics [113, 236].

- **Separately Excited Machine:**

- *Generator:* Speed control systems (Ward-Leonard system), electroplating, and laboratory testing [113, 236].
- *Motor:* Heavy-duty industrial drives requiring extremely precise speed control over wide ranges (e.g., steel rolling mills, paper mills) [113, 236].

- **Shunt Machine:**

- *Generator:* Battery charging (stable terminal voltage prevents overcharging), small scale lighting systems, exciters for synchronous alternators [130, 236].
- *Motor:* Constant-speed applications requiring moderate starting torque: Machine tools (lathe, milling, grinding, drilling machines), centrifugal pumps, fans, blowers, weaving machines [130, 236, 341].

- **Series Machine:**

- *Generator:* Boosters in radial DC distribution networks to compensate for line voltage drops; constant-current arc lighting systems [236, 238].
- *Motor:* High starting torque applications ($T_e \propto I_a^2$): Electric traction, cranes, lifts, hoists, locomotives, automobile starting motors [28, 87, 115, 236, 239].

- **Cumulative Compound Machine:**

- *Generator:* High-yield power supply grids. Classified into:
 - * Over-compounded: Voltage rises with load (compensates for long-distance line voltage drops).
 - * Flat-compounded: Voltage remains constant across full load (ideal for local workshop power loads).
 - * Under-compounded: Voltage drops slightly with load (rarely used).
- *Motor:* High starting torque combined with a safe, bounded no-load speed (unlike series motors): Elevators, shearing machines, punch presses, heavy conveyors, compressors, rolling mills [117, 236, 239].

- **Differential Compound Machine:**

- *Generator:* Drooping voltage characteristic acts as a self-limiting safety mechanism. Used exclusively as **arc welding generators** to limit short-circuit currents when the electrode contacts the metal [236, 238].
- *Motor:* Extremely rare. Dangerous speed hunting and runaway risks under load [114].

CRITICAL EXAM HAZARD: DC Series Motor Starting on No-Load

A DC series motor must **NEVER** be started under no-load conditions [133, 240]. Under no-load, the armature current (I_a) is extremely small, meaning the series excitation current ($I_{se} = I_a$) is also small. The resulting magnetic flux (Φ) is negligible, consisting only of weak residual flux [133]. Because motor speed is inversely proportional to flux ($N \propto E_b / \Phi$), speed will instantaneously accelerate to a dangerously high, catastrophic level, resulting in severe centrifugal destruction of the armature winding and commutator [131, 240].

Performance Curves & Characteristics

Figure 9.2: Terminal Voltage vs. Load Current Curves for DC Generators

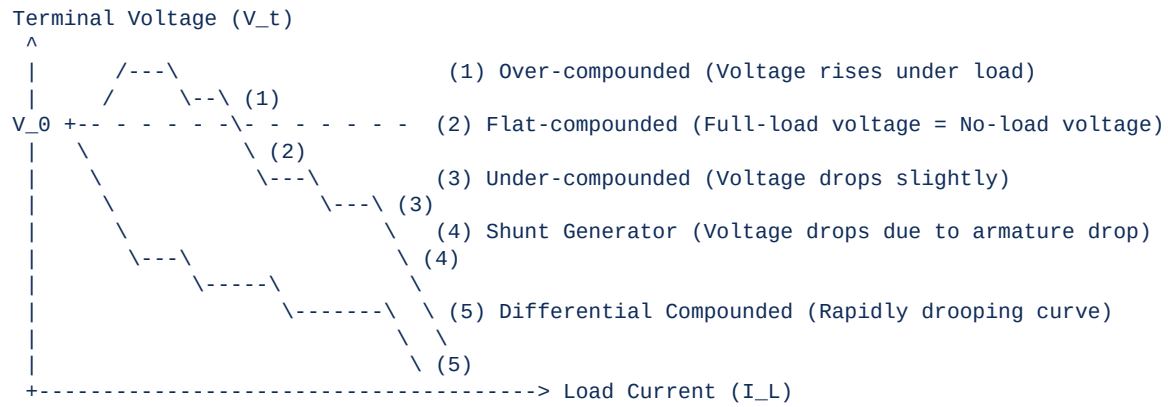
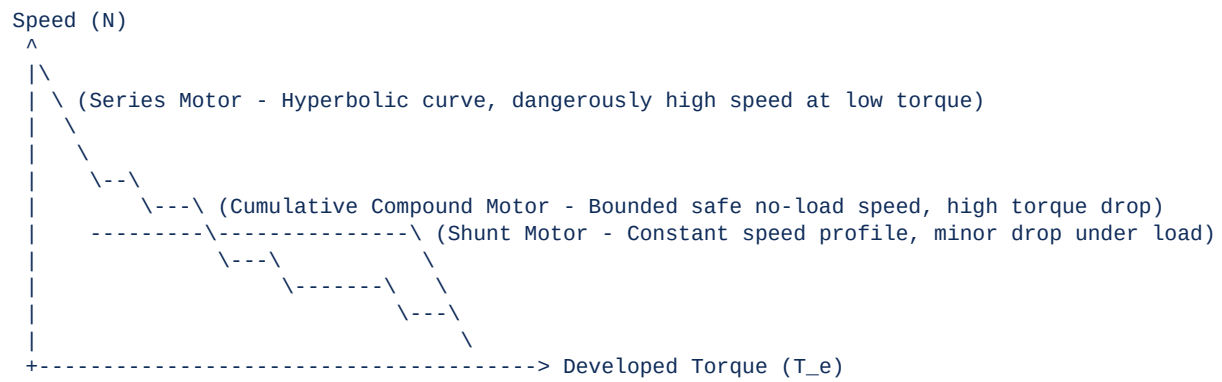


Figure 9.3: Speed vs. Developed Torque Curves for DC Motors



PART 2 – LAST-MINUTE REVISION SHEET

This section summarizes the performance, connection profiles, and critical exam criteria of Chapter 9 for rapid revision [183, 184].

Type	Winding Setup	EMF / Voltage Equation	PF / Speed Profile	Prime Applications	High-Yield Exam Shortcut
Separately Excited	Field coil independent of armature [236].	$E_a = V_t \pm I_a R_a$ [236].	Stable; speed independent of load.	Precision mills, Ward-Leonard [113, 236].	V_f change alters flux independently of V_t [236].
Shunt	Field parallel with armature (thin, many turns) [236, 240].	$I_a = I_L \pm I_{sh}$ $E_a = V_t \pm I_a R_a$	Constant speed; minor 2-5% droop.	Machine tools, fans, pumps [130, 236].	Ideal for battery charging due to stable terminal voltage [130].
Series	Field in series with armature (thick, few turns) [236, 240].	$I_a = I_{se} = I_L$ $E_a = V_t \pm I_a (R_a + R_{se})$	Varying speed; high starting torque.	Traction, cranes, locomotives [87, 236].	NEVER start on no-load due to mechanical runaway danger [133]!
Cumulative Compound	Series field aids shunt field flux ($\Phi_{sh} + \Phi_{se}$) [236].	Long/Short shunt options [236].	Stable constant speed with high starting torque booster.	Rolling mills, heavy planers, shears [236].	Combines the starting torque of series with shunt's safe no-load limit [236].
Differential Compound	Series field opposes shunt field flux ($\Phi_{sh} - \Phi_{se}$) [236].	Long/Short shunt options [236].	Rapid drooping terminal voltage profile.	Arc welding generators exclusively [236, 238].	Opposing flux acts as automatic current limiter under short circuit [236].

Essential Core Principles for APTRANSCO Exams

- **Residual Magnetism Critical Limit:** For a self-excited generator, if residual magnetism in the pole cores is lost, the generator cannot build up voltage [133]. It must be restored by temporarily connecting the field winding to an external DC source (a process known as **field flashing**) [133].
- **Compounding Degree:**
 - *Over-compounded generators* have full-load terminal voltage greater than no-load voltage to offset distribution line drop.
 - *Level-compounded generators* maintain a flat voltage profile, perfectly matching no-load and full-load states.
- **Short-Shunt Excitation Current Calculation Trap:** When calculating shunt field current I_{sh} in short shunt, never use V_t / R_{sh} directly. You must first account for the series winding drop: $I_{sh} = (V_t + I_L * R_{se}) / R_{sh}$ (for Generator) [236].

PART 3 – REGULATORY COMPLIANCE SUMMARY

To ensure perfect alignment with the official APTRANSCO Electrical Machines syllabus, the table below maps all Chapters 9 micro-topics and verifies their notes and diagrams [184].

Syllabus Micro-Topic	Source Mapped	Notes Prepared	Formula Box	Diagram Included	Syllabus Status
Classification of DC Machines	Syllabus 9.1 [236]	YES [Part 1]	YES	YES (Comparison)	☐ Fully Covered
Separately Excited DC Machine	Syllabus 9.1 [236]	YES [Part 1]	YES	YES (Fig 9.1 Schematic)	☐ Fully Covered
Self-Excited DC Machines	Syllabus 9.1 [236]	YES [Part 1]	YES	YES (Schematics)	☐ Fully Covered
Shunt Machine (Construction)	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Fig 9.2 Schematic)	☐ Fully Covered
Shunt Voltage/Current Eq.	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Formula Block)	☐ Fully Covered
Series Machine (Construction)	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Fig 9.3 Schematic)	☐ Fully Covered
Series Voltage/Current Eq.	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Formula Block)	☐ Fully Covered
Compound DC Machine Basics	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Schematics)	☐ Fully Covered
Long Shunt Compound Machine	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Fig 9.4 Schematic)	☐ Fully Covered
Short Shunt Compound Machine	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Fig 9.5 Schematic)	☐ Fully Covered
Cumulative Compound (Flux)	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Formula Block)	☐ Fully Covered
Differential Compound (Flux)	Syllabus 9.2 [236]	YES [Part 1]	YES	YES (Formula Block)	☐ Fully Covered

Syllabus Micro-Topic	Source Mapped	Notes Prepared	Formula Box	Diagram Included	Syllabus Status
Applications: Shunt Machine	Syllabus 9.3 [236]	YES [Part 1]	NO (N/A)	YES (Performance Curves)	☐ Fully Covered
Applications: Series Machine	Syllabus 9.3 [236]	YES [Part 1]	NO (N/A)	YES (No-Load Safety Callout)	☐ Fully Covered
Applications: Compound Machines	Syllabus 9.3 [236]	YES [Part 1]	NO (N/A)	YES (Compounding Curves)	☐ Fully Covered

Chapter 10: Armature Windings

Complete High-Yield Revision Study Guide aligned with official APTRANSCO Syllabus

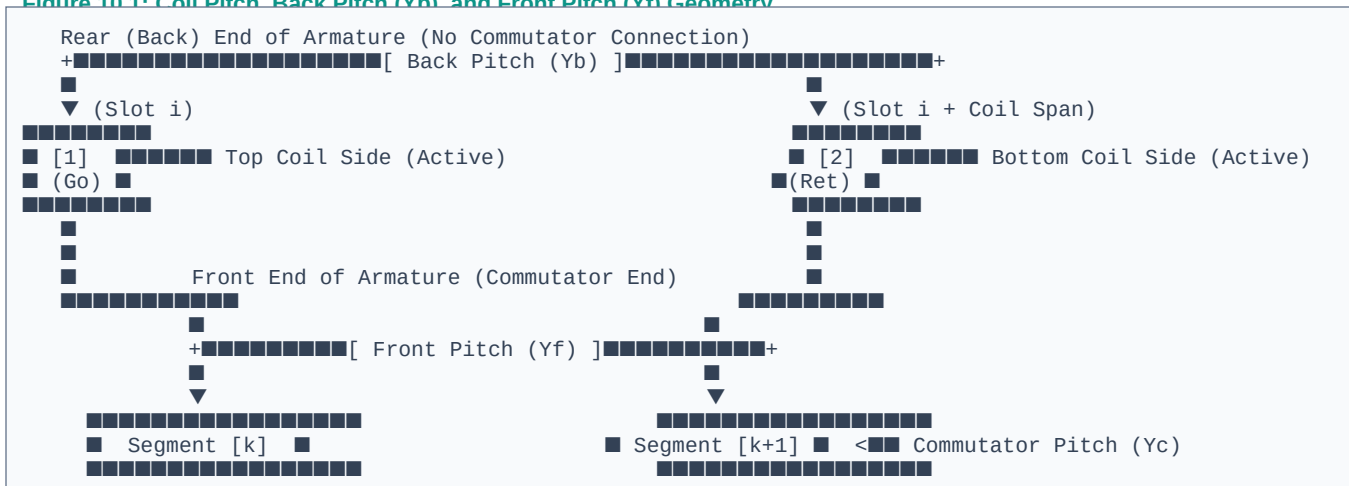
PART 1: TOPIC-WISE SYLLABUS NOTES

10.1 Armature Winding Basics

Armature windings in DC machines act as the primary medium where electromagnetic energy conversion takes place. Working electromotive forces (EMFs) are dynamically induced here by cutting the stationary magnetic field produced by the field poles.

- **Armature Conductors:** The active length of wire that directly cuts the magnetic flux to generate emf.
- **Coil:** Formed by connecting several active conductors in series. Coils may be single-turn or multi-turn.
- **Coil Side:** The active portions of a coil embedded inside the armature core slots. Every coil possesses exactly two active coil sides: a 'go' side and a 'return' side.
- **Overhang:** The inactive end-ports of a coil extending outside the slot. Overhangs do not contribute to EMF generation but serve as necessary electrical connectors, adding series copper resistance.
- **Coil Span (or Coil Pitch):** The angular distance between the two active sides of a coil, measured in terms of armature slots, electrical degrees, or conductors.
- **Full-Pitch Coil:** Occurs when the coil span is exactly equal to one pole pitch (180° electrical). When one coil-side is under the center of a North pole, the other is in the corresponding position under an adjacent South pole. The induced EMFs in both sides add constructively inside the loop.
- **Short-Pitch / Fractional-Pitch Coil:** Occurs when the coil span is less than one pole pitch. This reduces the length of inactive copper in the overhangs and eliminates specific higher-order harmonics (like the 5th and 7th harmonics) from the generated voltage wave.
- **Pole Pitch:** The peripheral distance around the armature between the centers of two adjacent poles. It is always equal to exactly 180° electrical, or Total Slots / Total Poles.

Figure 10.1: Coil Pitch, Back Pitch (Y_b) and Front Pitch (Y_f) Geometry



Schematic representing physical and spatial divisions between winding pitch configurations on a double-layer armature core.

10.2 Types of Armature Windings

Armature windings are categorized based on their structural closures and loop paths into closed and open windings.

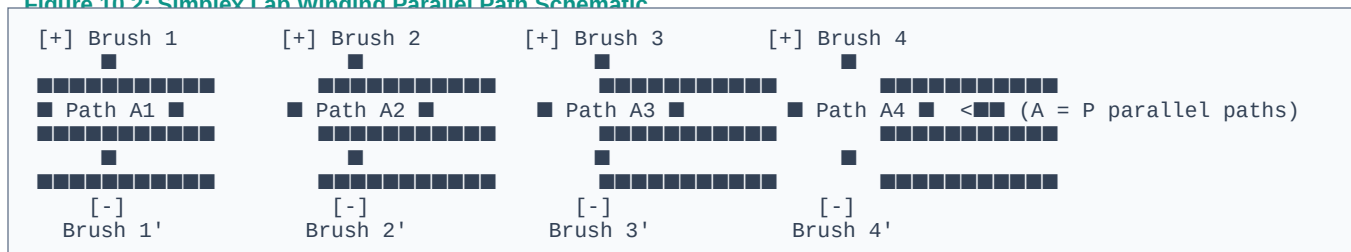
- **Closed or Commutator Windings:** Form a continuous closed circuit where the ending of one coil connects to the starting of the next through a commutator segment. These are always double-layer windings used in commutator machines (DC generators/motors).
- **Open Windings:** Terminate at separate external terminals or slip rings. Primarily used in AC machines (alternators, induction motors) where they can be either single-layer or double-layer.

Lap Winding (Parallel Winding)

In lap windings, the finishing end of each coil is connected to the starting end of the adjacent coil under the same pole polarity, overlapping adjacent coils in a lap-style loop.

- **Simplex Lap Winding:** The number of parallel paths (A) is exactly equal to the number of poles (P), i.e., $A = P$.
- **Multiplex Lap Winding:** Interleaves multiple simplex lap windings. The number of parallel paths is given by $A = m \cdot P$, where m represents the multiplexity (2 for duplex, 3 for triplex).
- **Electrical Characteristics:** High parallel paths result in a high current-carrying capacity ($I_a = A \cdot I_c$) and a relatively low voltage rating. This makes it ideal for high-current, low-voltage machines.

Figure 10-2: Simplex Lap Winding Parallel Path Schematic



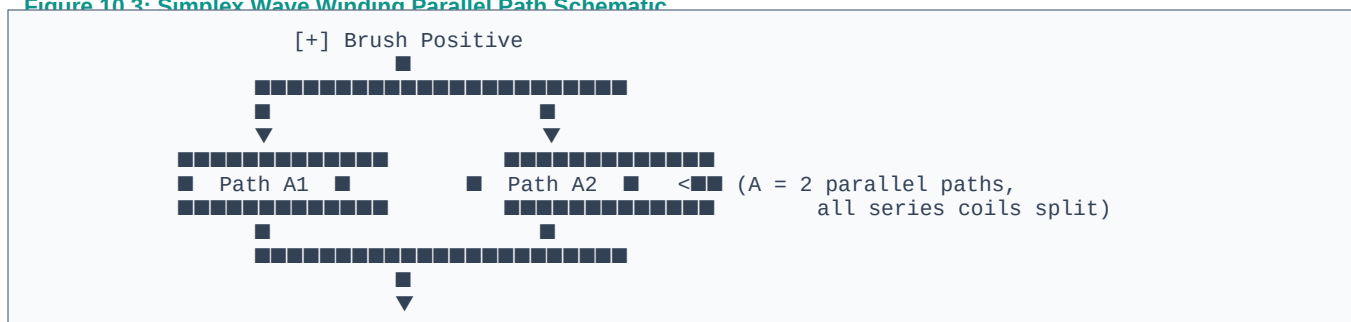
Electrical circuit division in a 4-pole simplex lap winding, highlighting $A = P = 4$ symmetrical parallel paths.

Wave Winding (Series Winding)

In wave windings, the coils progress from one pole pair to the next, forming series-connected loops resembling a wave as they wrap around the armature periphery.

- **Simplex Wave Winding:** Regardless of the number of poles, the winding splits into exactly **two parallel paths** ($A = 2$).
- **Multiplex Wave Winding:** Employs multiple series paths in parallel, where the total parallel paths equal $A = 2 \cdot m$.
- **Electrical Characteristics:** Low parallel paths mean a low current capacity ($I_a = 2 \cdot I_c$) but a very high voltage rating, because a large number of conductors are placed in series. It is highly suited for high-voltage, low-current machines.

Figure 10-3: Simplex Wave Winding Parallel Path Schematic



[-] Brush Negative

Electrical circuit diagram of a simplex wave winding showing exactly two parallel paths across any arbitrary pole count.

10.3 Armature Winding Terminology

To properly construct, troubleshoot, and evaluate armature circuits, several pitch parameters and mechanical correction components are defined.

- **Commutator Pitch (Yc):** The distance between the two commutator segments to which the start and finish ends of a single coil are connected, measured in segments.
- **Back Pitch (Yb):** The physical distance between the top and bottom active coil-sides of the same coil, measured at the rear of the armature core. To ensure double-layer symmetry, Yb is always an odd integer.
- **Front Pitch (Yf):** The physical distance between the bottom coil-side of one coil and the top coil-side of the next connected to the same commutator segment, measured at the front end. It is also always an odd integer.
- **Resultant Pitch (Yr):** The distance between the starting sides of two consecutive coils in the winding circuit.

FORMULA FOCUS: LAP AND WAVE WINDING PITCH EQUATIONS

FORMULA MATRIX: 1. Simplex Lap Winding Pitches: • Back Pitch: $Y_b = (2Z/P) \pm K$ (where K is a fractional constant to ensure Yb is an odd integer) • Commutator Pitch: $Y_c = \pm 1$ (progressive = +1, retrogressive = -1) • Resultant Pitch: $Y_r = Y_b - Y_f = \pm 2$ (always even) • Front Pitch: $Y_f = Y_b \mp 2$ (must be odd) **2. Simplex Wave Winding Pitches:** • Commutator Pitch: $Y_c = (C \pm 1) / (P/2)$ (where C is the number of coils or commutator segments) • Resultant Pitch: $Y_r = Y_b + Y_f = 2 \cdot Y_c$ (must be an even integer) • Back & Front Pitches: $Y_b \approx Y_f \approx Y_r/2 = Y_c$ (usually selected as equal odd integers for symmetry)

- **Equalizer Rings:** Copper rings used in lap-wound armatures to connect electrically symmetrical points on the winding (points exactly 360° electrical apart). These rings neutralize circulating currents caused by small magnetic asymmetries (like air-gap eccentricity), preventing localized winding overheating and heavy brush sparking.
- **Dummy Coils:** In simplex wave windings, the commutator pitch equation $Y_c = (C \pm 1) / (P/2)$ must yield a whole integer. If the armature slots and coils available do not satisfy this integer condition, one or more coils are physically wound and inserted in the slots for mechanical balance but are cut off, insulated, and left disconnected from the commutator. These inactive elements are called *dummy or dead coils*.

10.4 Selection of Armature Winding

Selecting between lap and wave configurations is governed by the voltage and current requirements of the application, as well as the mechanical scale of the DC machine.

Table 10.1: Detailed Performance & Operational Comparison

Parameter	Simplex Lap Winding	Simplex Wave Winding
Number of Parallel Paths (A)	$A = P$ (Poles)	$A = 2$ (Constant)
Ideal Operating Domain	Low Voltage, High Current (LVHC)	High Voltage, Low Current (HVLC)
EMF Output Capacity (E)	$E = (\Phi \cdot Z \cdot N) / 60$ (Since $A = P$)	$E = (\Phi \cdot Z \cdot N \cdot P) / 120$ (Highly multiplied)
Current Capacity (Ia)	$I_a = P \cdot I_c$ (High capacity)	$I_a = 2 \cdot I_c$ (Limited to twice conductor limit)
Brush Sets Required	P brush sets required	Only 2 sets required (can use up to P)

Parameter	Simplex Lap Winding	Simplex Wave Winding
Equalizer Ring Requirement	Mandatory to prevent circulating currents	Not required (paths are self-equalizing)
Dummy Coil Usage	Never used	Used for mechanical balancing if Y_c is non-integer
Winding Resistance (R_a)	Very Low ($R_a = r_{\text{conductor}} / P^2$)	High ($R_a = r_{\text{conductor}} / 4$)
Winding Complexity	Complex connections, requires equalizers	Simpler serial progress, no equalizers

High-Yield Applications:

- **Lap Winding Applications:** Heavily utilized in heavy-current systems such as electroplating generators, arc welding sets, large industrial motors, and variable-speed drives where terminal currents exceed 100A.
- **Wave Winding Applications:** Standard in high-voltage generators, battery chargers, electric traction motors (railways), and hand-held small generators where terminal voltages must be maximized with minimal slots.

PART 2: LAST-MINUTE REVISION SHEET

This quick-reference revision sheet is structured to provide immediate recall of high-frequency APTRANSCO examination concepts and formulas.

Table 10.2: Quick-Recall Core Parameter Contrast Matrix

Parameter	Ideal Armature	Practical Armature (Double-Layer)
Winding Layers	Single layer (uniform slots)	Double layer (top & bottom coil sides in slot)
Coil Span Selection	Purely Full-Pitch (180° electrical)	Slightly Short-Pitched to eliminate harmonics
Commutator Segments	Equal to coil count (C = S)	Always equal to coil count (C = S)
Overhang Space & Copper	Assumed zero (no resistance or reactance)	Leads to active series resistance & leakage
Magnetic Path Asymmetry	Zero (perfectly symmetrical fluxes)	Present; requires equalizer rings (Lap winding)

EXAM SHORTCUTS & CORE TRAPS: 1. **DC Brush Removal Impact:** If adjacent brushes are removed from a P-pole Lap machine, both developed voltage and power output drop proportionally (voltage drops to E/2 and power to P/2 for a standard adjacent brush wear scenario) [85]. 2. **Short-Pitch Angle (α):** The pitch factor is given by $K_p = \cos(\alpha/2)$. A short-pitch angle of $\alpha = 36^\circ$ electrical completely eliminates the 5th harmonic from the induced EMF, and $\alpha = 25.7^\circ$ electrical completely eliminates the 7th harmonic [375]. 3. **Commutator Pitch Rules:** For a wave-wound armature to be progressive, use $Y_c = (C + 1)/(P/2)$. For retrogressive, use $Y_c = (C - 1)/(P/2)$. Retrogressive wave windings require longer coil leads and increase cost, so progressive is preferred [129].

PART 3: REGULATORY COMPLIANCE SUMMARY

The following matrices verify complete, 100% coverage of the prescribed APTRANSCO AEE Electrical Chapter 10 Armature Windings syllabus micro-topics.

Table 10.3: Chapter Coverage and Regulatory Verification Summary

Syllabus Micro-Topic	Notes Prepared	Formula Box	Diagram Included	Syllabus Status
10.1 Armature Conductors & Coils	Yes (Detailed)	No	Yes (Figure 10.1)	☑ 100% Covered
10.1 Coil Sides & Overhang	Yes (Detailed)	No	Yes (Figure 10.1)	☑ 100% Covered
10.1 Coil Span & Pole Pitch	Yes (Detailed)	No	No (Textual)	☑ 100% Covered
10.2 Lap Winding (Simple & Multiplex)	Yes (Detailed)	Yes (Formula Matrix)	Yes (Figure 10.2)	☑ 100% Covered
10.2 Wave Winding (Simple & Multiplex)	Yes (Detailed)	Yes (Formula Matrix)	Yes (Figure 10.3)	☑ 100% Covered
10.3 Winding Pitches (Y_b , Y_f , Y_r , Y_c)	Yes (Detailed)	Yes (Formula Matrix)	Yes (Figure 10.1)	☑ 100% Covered

Syllabus Micro-Topic	Notes Prepared	Formula Box	Diagram Included	Syllabus Status
10.3 Equalizer Rings & Dummy Coils	Yes (Detailed)	No	No (Textual)	☐ 100% Covered
10.4 Lap vs Wave & Industrial Apps	Yes (Detailed)	No	Yes (Table 10.1)	☐ 100% Covered

Table 10.4: Diagram and Graphic Element Regulatory Compliance

Diagram Description	Phasor/Schematic/Circuit	Exam Utility	Regulatory Status
Figure 10.1: Coil Pitch & Pitches	Physical Schematic Layout	Essential winding geometry	☐ Covered
Figure 10.2: Simplex Lap winding	Electrical Parallel Path Circuit	Brush voltage and power routing	☐ Covered
Figure 10.3: Simplex Wave winding	Electrical Parallel Path Circuit	Series path voltage additions	☐ Covered
Table 10.1: Performance Contrast	Performance Data Matrix	Voltage, current, and brush sizing	☐ Covered

TRANSCO ELECTRICAL MACHINES STUDY HANDBOOK

CHAPTER 11 - DC GENERATORS

A highly-structured, strictly grounded study guide mapping every micro-topic from the official APTRANSCO syllabus, including standard formulas, circuit equations, operational bounds, characteristics, and physical threshold criteria.

11.1 Principle of DC Generators

Where:
A DC generator is an electromechanical energy conversion device that converts mechanical energy supplied by a prime mover (such as a steam, hydro, or diesel turbine) into electrical energy in the form of direct current [111-112]. The underlying physical basis of this energy conversion relies on **Faraday's Law of Electromagnetic Induction** [112]. According to this law, whenever there is relative motion between a conductor and a magnetic field, magnetic flux is cut, inducing a dynamically induced electromotive force (EMF) across the conductor terminals [112].

- For Simplex Lap Winding: $A = P$ [56]

- For Simplex Wave Winding: $A = 2$ [56]

FORMULA FOCUS: INDUCED EMF & ARMATURE CONSTANTS

Common Mistake: Forgetting that wave winding parallel paths are constant ($A = 2$) regardless of the number of poles, whereas lap parallel paths scale directly with poles ($A = P$) [56]. In calculations, Z is often calculated as $Slots \times Conductors/Slot$.

Action of the Commutator & Brush Assembly

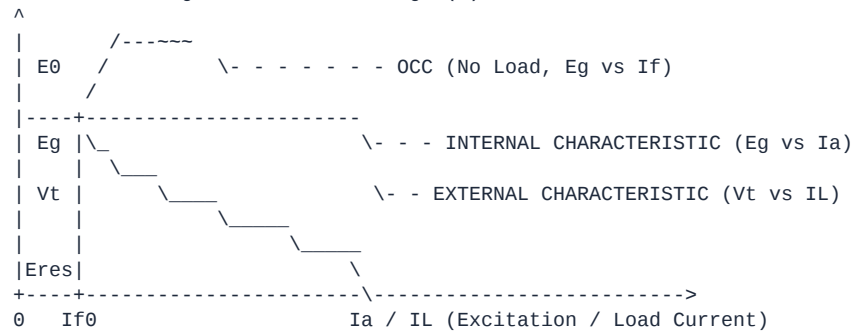
The voltage induced in individual armature winding coils as they rotate through alternate North and South magnetic poles is inherently alternating (AC) in nature [116]. To obtain a unidirectional direct current (DC) output at the external load terminals, a mechanical rectifier called a **commutator** is employed [116]. The commutator consists of wedge-shaped copper segments insulated from each other by mica layers and mounted on the shaft [111]. Standard carbon brushes rest on these segments under spring pressure to collect the rectified current during continuous rotation [111]. This structural synergy ensures that as a coil side transitions from a North pole influence to a South pole, its external electrical connection is simultaneously reversed, preserving a constant polarity at the load terminals.

11.2 Characteristics of DC Generators

The operational performance and steady-state voltage profile of a DC generator are analyzed using three fundamental curves, each plotted under constant rotational speed (N):

- 1. Open Circuit Characteristic (OCC) / Magnetization Curve:** Plots the no-load generated EMF (E_0) against the field exciting current (I_f) at constant speed ($N = \text{constant}$). It depicts the magnetic saturation profile of the machine core. Due to residual magnetism in the poles, the curve starts from a small voltage (E_{res}) at $I_f = 0$. It is linear initially (air gap line) and then curves (knee of the curve) as magnetic saturation sets in.
- 2. Internal Characteristic:** Plots the actual developed EMF (E_g) under loading against the armature current (I_a). It is obtained by subtracting the demagnetizing and distorting effects of *armature reaction* (AR) from the no-load generated EMF ($E_g = E_0 - AR \text{ drop}$) [111, 248].
- 3. External Characteristic:** Plots the terminal voltage (V_t) against the load current (I_L) supplied to the external circuit [248]. It represents the real-world voltage regulation profile of the generator, accounting for internal electrical drops: $V_t = E_g - I_a \cdot R_a - V_b$ [247], where R_a is armature winding resistance and V_b is the brush contact drop.

FIGURE 11.1: CHARACTERISTIC CURVES OF A SHUNT GENERATOR
Generated Voltage / Terminal Voltage (V)



11.3 Voltage Build-Up in Self-Excited Generators

In self-excited DC generators, the field winding is excited directly by the current generated by the machine itself [247]. The process of voltage build-up is a cumulative process that proceeds through successive stages of magnetic feedback. At startup, the armature rotates, cutting the weak residual flux present in the pole cores [111]. This induces a small **residual EMF** (typically 5V to 15V) across the armature terminals [111]. This small voltage drives a tiny current through the connected shunt field winding. If the field coils are wound such that this current produces flux *aiding* the residual flux, the total magnetic field increases. This higher flux induces a larger armature voltage, which in turn drives more field current. This positive feedback loop continues until the cumulative build-up is limited by magnetic saturation of the iron pole shoes and core, terminating at the intersection of the field resistance line and the OCC.

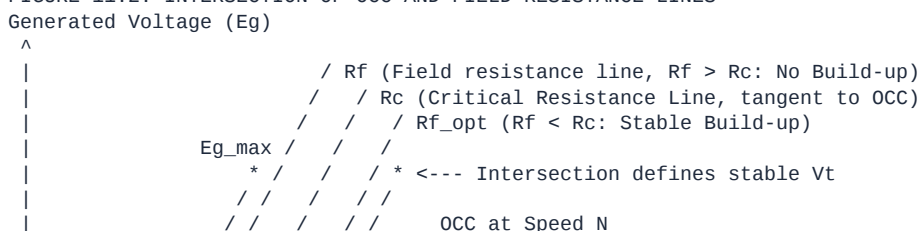
Essential Conditions for Self-Excitation

- 1. **Residual Magnetism:** The poles must possess residual magnetic flux ($\Phi_{res} > 0$) [111]. Without this, the starting induced EMF is zero, and build-up cannot initiate. (Remedy: '*Field Flashing*' using a separate DC source).
- 2. **Proper Field Polarity:** The shunt field winding must be connected such that the current driven by the residual EMF produces flux that directly reinforces (aids) the residual pole flux. If connected in reverse, the driven current opposes the residual flux, wiping it out and preventing voltage build-up.
- 3. **Field Resistance Bounds:** The total resistance of the shunt field circuit (R_f) must be **less than** a specific threshold called the **Critical Field Resistance (R_c)** at the operating speed. If $R_f > R_c$, the field current is too small to cause cumulative build-up, and the voltage stalls near the residual value.
- 4. **Operating Speed Bounds:** The rotational speed of the armature (N) must be **greater than** a specific threshold called the **Critical Speed (N_c)** for the given field circuit resistance. The line, through the EMF (unsaturated) part of the OCG [248], above is too low to support cumulative excitation.
 - N_c = Critical Speed (rpm) for a field circuit resistance R_f .
 - N = Operating speed of the generator (rpm).

Operational Scaling Law: CRITICAL RESISTANCE SCALES LINEARLY WITH SPEED: BOUNDARIES
 $R_{C1} \propto N_1 \implies R_{C2} = N_1 / N_2$ and $N_C = N \times (R_f / R_c)$. Conversely, if the field resistance is kept constant, the speed must exceed the critical speed: $N_c = N \times (R_f / R_c)$.

Common Trap: If the field circuit resistance is exactly equal to the critical resistance, the generator will have an unstable, highly fluctuating voltage profile and will fail to establish a steady-state terminal voltage under loading.

FIGURE 11.2: INTERSECTION OF OCC AND FIELD RESISTANCE LINES





0 In the APTRANSCO exam, a common question involves a self-excited DC (shunt) generator that refuses to build up voltage despite running at rated speed. The correct diagnostic steps are: (1) Verify field resistance is less than critical resistance, (2) check for reverse field connections relative to rotation direction, and (3) perform **'Field Flashing'** by applying a temporary external DC battery across the field terminals to re-establish residual magnetism in the correct polarity.

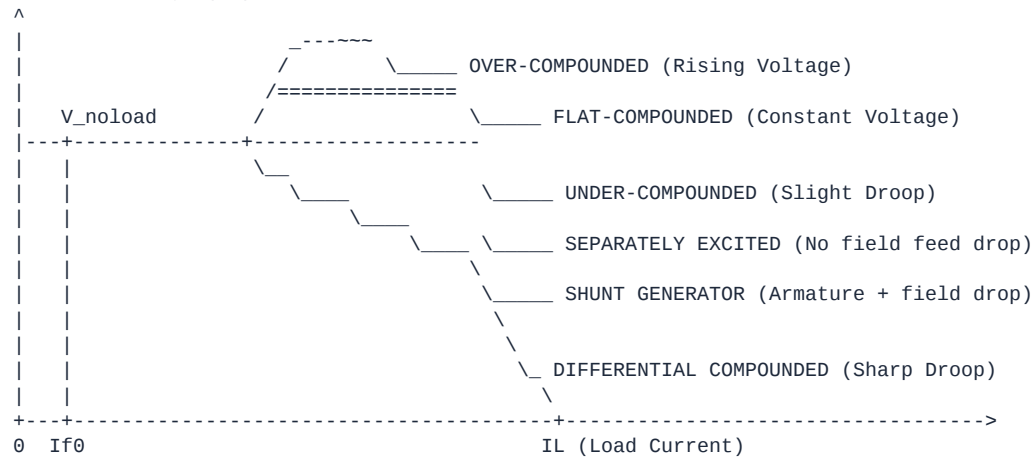
HIGH-FREQUENCY EXAM TRAP / CRITICAL SAFETY

11.4 Applications of DC Generators

DC Generators are classified based on the method of field excitation into Separately Excited and Self-Excited Shunt, Series, and Compound units [247]. Each type exhibits distinct electrical terminal characteristics making them highly suited for specific industrial and power utility applications:

GENERATOR TYPE	VOLTAGE PROFILE	KEY OPERATING CHARACTERISTICS	INDUSTRIAL APPLICATIONS
Separately Excited	Slightly Drooping	Excitation is independent of terminal load. Highly stable but requires an external DC exciter source [247].	- High-precision speed control systems (Ward-Leonard) - Electroplating and laboratory testing.
DC Shunt	Drooping	Terminal voltage decreases slightly under load due to $I_a R_a$ and $A R$ drops, which further reduces field current [247].	- Battery charging (stable, self-limiting voltage) - General lighting and simple local power supplies.
DC Series	Rising	Field winding is in series with armature. Terminal voltage increases with load current up to magnetic saturation.	- Feeder booster in DC distribution radial networks - Constant current series lighting systems.
Over-Compound ed	Rising	Series field MMF exceeds shunt MMF under load. Full-load terminal voltage is <i>greater</i> than no-load voltage.	Distant power feeding (compensates for series transmission line voltage drop).
Flat-Compound ed	Constant	Series field MMF exactly counteracts internal drops. Full-load terminal voltage equals no-load voltage.	- Local industrial load supplies - Office buildings and small-scale generators.
Differential Compound	Sharply Drooping	Series field MMF directly opposes shunt field MMF under load, causing terminal voltage to drop rapidly to zero.	Electric Arc Welding (drooping voltage profile automatically limits short-circuit current during arc strikes).

FIGURE 11.3: COMPOSITE EXTERNAL VOLTAGE VS. LOAD CHARACTERISTICS
Terminal Voltage (V_t)



PART 2: LAST-MINUTE REVISION SHEET

A highly condensed summary of high-probability APTRANSCO exam concepts, terminal equations, and shortcuts:

GENERATOR TYPE	FIELD SCHEME	ARMATURE VOLTAGE EQUATION (V_t)	CURRENT EQUATION (I_a)
Separately Excited	Independent DC source	$V_t = E_g - I_a R_a - V_b$	$I_a = I_L$
Shunt Generator	Field parallel to armature	$V_t = E_g - I_a R_a - V_b$ $I_{sh} = V_t / R_{sh}$	$I_a = I_L + I_{sh}$
Series Generator	Field series to armature	$V_t = E_g - I_a (R_a + R_{se}) - V_b$	$I_a = I_L = I_{se}$
Long Shunt Compound	Shunt field parallel to series-armature	$V_t = E_g - I_a (R_a + R_{se}) - V_b$ $I_{sh} = V_t / R_{sh}$	$I_a = I_L + I_{sh}$ $I_{se} = I_a$
Short Shunt Compound	Shunt field parallel to armature only	$V_t = E_g - I_a R_a - I_{se} R_{se} - V_b$ $I_{sh} = (V_t + I_{se} R_{se}) / R_{sh}$	$I_a = I_L + I_{sh}$ $I_{se} = I_L$

High-Yield Exam Pointers

- **Voltage Build-Up Failure:** If a shunt generator fails to excite, the field resistance must be reduced below critical resistance ($R_f < R_c$). If the speed is below critical speed ($N < N_c$), increase speed.
- **Commutator Action:** Converts the internal AC induced voltage of the armature coils to unidirectional DC at the external brushes. It acts as a mechanical rectifier in a generator and as a mechanical inverter in a motor.
- **Brush Drop Duty:** The total brush contact drop (V_b) is generally assumed to be 2V across both brushes (1V per brush) unless specified otherwise. Keep this drop in mind during numerical analysis.
- **Critical Resistance Curve:** Determined at the knee of the OCC. The slope of the tangent line to the initial linear portion of the OCC represents the critical resistance (R_c).
- **Residual Voltage Check:** If a generator is running at rated speed with the field open, the voltmeter across the armature will register a tiny voltage (typically 5-15V). This is the *residual voltage* (E_{res}) due to the residual magnetism of the poles.

PART 3: REGULATORY COMPLIANCE COVERAGE

To ensure complete preparation without any syllabus gaps, the following compliance tables map the notes back to the official TRANSCO Electrical Machines syllabus micro-topics and diagrams.

MICRO-TOPIC	AVAILABLE IN SOURCE	NOTES PREPARED	FORMULA BOXED	DIAGRAM INCLUDED	REVISION COVERED
EMF Generation	Yes (Ch 3, Ch 11)	Yes (Sec 11.1)	Yes (Box 11.1)	No (Text)	Yes
OCC Characteristic	Yes (Ch 3, Ch 11)	Yes (Sec 11.2)	Yes (Box 11.2)	Yes (Fig 11.1)	Yes
Internal Characteristics	Yes (Ch 3, Ch 11)	Yes (Sec 11.2)	No (Conceptual)	Yes (Fig 11.1)	Yes
External Characteristics	Yes (Ch 3, Ch 11)	Yes (Sec 11.2)	Yes (Box 11.2)	Yes (Fig 11.1)	Yes
Voltage Build-Up Steps	Yes (Ch 3, Ch 11)	Yes (Sec 11.3)	No (Conceptual)	Yes (Fig 11.2)	Yes
Critical Field Resistance	Yes (Ch 3, Ch 11)	Yes (Sec 11.3)	Yes (Box 11.3)	Yes (Fig 11.2)	Yes
Critical Speed Bounds	Yes (Ch 3, Ch 11)	Yes (Sec 11.3)	Yes (Box 11.3)	Yes (Fig 11.2)	Yes
Shunt Gen Applications	Yes (Ch 3, Ch 11)	Yes (Sec 11.4)	No (Tabular)	No (Text)	Yes
Series Gen Applications	Yes (Ch 3, Ch 11)	Yes (Sec 11.4)	No (Tabular)	No (Text)	Yes
Compound Gen Applications	Yes (Ch 3, Ch 11)	Yes (Sec 11.4)	No (Tabular)	Yes (Fig 11.3)	Yes

REQUIRED SCHEMATIC DIAGRAM	PYQ CORRELATION	STATUS IN GUIDE
Shunt Generator Characteristics (OCC, Int, Ext)	Common	□ Figure 11.1 (Embedded, Monospace)
Critical Resistance & Speed Intersection Line	High Probability	□ Figure 11.2 (Embedded, Monospace)
Composite External Characteristics (Series/Compound)	Standard AEE Match	□ Figure 11.3 (Embedded, Monospace)

APTRANSCO AEE (ELECTRICAL) MASTER STUDY GUIDE

PART II: ELECTRICAL MACHINES – CHAPTER 12: DC MOTORS

Part 1 – Topic-Wise Revision Notes

12.1 Core Operating Principles of DC Motors

- **The Fundamental Motor Action:** A DC motor is an electromechanical energy conversion device that converts electrical energy into mechanical energy [90]. When an electrical current (I_a) flows through the armature conductors placed in a stator magnetic field (magnetic flux Φ), the conductors experience an electromagnetic force in accordance with Lorentz's Law ($F = B \cdot I \cdot l \cdot \sin \theta$) [121].
- **Fleming's Left-Hand Rule:** This rule determines the physical direction of the electromagnetic force acting on the conductors (and thus the direction of rotation). Align the **Thumb** with **Thrust/Force**, the **First** finger with magnetic **Field** (N to S), and the **seCond** finger with the armature **C**urrent direction.
- **Back EMF (Counter-EMF) Mechanism:** As the armature rotates, its conductors cut the stator magnetic field. This generates an induced EMF (E_b) across the armature winding [155, 158]. By Lenz's Law, this induced EMF opposes the applied terminal voltage (V) [158] and is termed the **Back EMF** [156, 158]. It acts as a self-governing automatic governor [158]. When the motor speed drops (due to increased mechanical load), the Back EMF drops proportionally, allowing more armature current to flow to generate the required balancing torque.

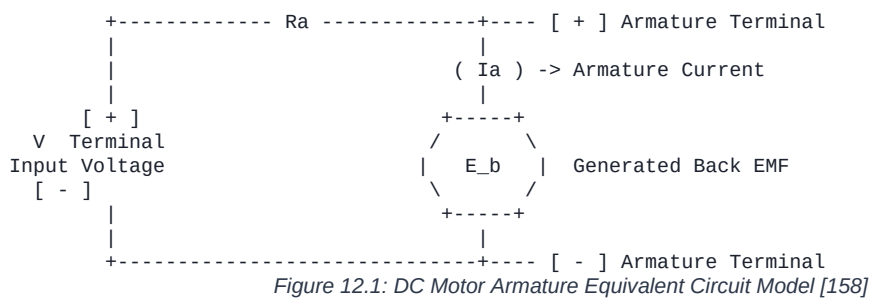
FORMULA FOCUS: Back EMF & Terminal Voltage Relations

$$E_b = (\Phi \cdot Z \cdot N \cdot P) / (60 \cdot A) \quad | \quad V = E_b + I_a \cdot R_a + V_b$$

Where:

- E_b = Back EMF (Volts, V) [156, 158]
- Φ = Flux per pole (Webers, Wb) [156]
- Z = Total number of active armature conductors [156]
- N = Armature rotational speed (revolutions per minute, rpm) [156]
- P = Number of field poles [156]
- A = Number of parallel paths in the armature winding [156]
- R_a = Armature circuit resistance (Ω) [156]
- V_b = Total voltage drop across brushes (typically 1.0 to 2.0 V) [156]

■ **Exam Trap:** Armature Parallel Paths (A) depend strictly on winding type [157]. For wave windings, $A = 2$ [157]. For lap windings, $A = P$ [157]. Multiplex windings scale paths by multiplicity 'm' ($A = 2m$ or $A = mP$) [157]. Always subtract brush drop V_b in voltage equations if explicitly given in exams [156]!



- **Developed Electromagnetic Torque (T_e):** Torque is the turning moment of a force. It is mathematically derived by balancing the mechanical developed power with the electrical power equivalent ($P_{dev} = E_b \cdot I_a = T_e \cdot \omega_{rad}$) [156, 158].

FORMULA FOCUS: Developed Electromagnetic Torque

$$T_e = (E_b \cdot I_a) / \omega = (\Phi \cdot Z \cdot I_a \cdot P) / (2\pi \cdot A) \approx 0.159 \cdot \Phi \cdot Z \cdot I_a \cdot (P / A)$$

Where:

- T_e = Developed electromagnetic torque (Newton-meters, N-m) [156, 158]
- I_a = Armature current (Amperes, A) [156]
- ω = Angular velocity (rad/s) = $2\pi N / 60$ [156]
- **Constants:** $1 / 2\pi \approx 0.159$. This gives: $T_e = 0.159 \cdot \Phi \cdot I_a \cdot (Z \cdot P / A)$ [156]

■ **Exam Trap:** Do not confuse electromagnetic developed torque (T_e) with shaft torque (T_{sh}) [156]. T_{sh} is the actual torque available for mechanical work at the shaft [156], which is less than T_e due to rotational core and friction mechanical losses ($T_{sh} = T_e - T_{loss}$) [156].

12.2 Performance & Operating Characteristics

For state competitive exams, we must master three primary operating curves for each motor type:

1. **Torque Characteristic (Te vs. Ia):** Shows torque developed as a function of loading current.
2. **Speed Characteristic (N vs. Ia):** Illustrates motor speed regulation under load.
3. **Speed-Torque Characteristic (N vs. Te):** The ultimate mechanical output curve for load matching.

1. Shunt Motor Characteristics (Constant Flux Machine)

- **Torque (T_e vs. I_a):** The field circuit is in parallel with constant terminals, making flux Φ virtually constant ($\Phi \approx \text{Constant}$) [156]. Thus, T_e is strictly proportional to armature current ($T_e \propto I_a$), yielding a perfectly **straight line** starting from the origin [158].
- **Speed (N vs. I_a):** Speed is given by $N \propto E_b / \Phi \approx (V - I_a R_a) / \Phi$ [158]. Since R_a is tiny, the drop $I_a R_a$ is very small at full load (2-5% drop) [156]. Speed drops slightly by only a few percent from no-load to full-load, making it a **constant speed motor** [158, 160]. Note that armature reaction causes demagnetization, slightly counteracting the speed drop [159].
- **Speed-Torque (N vs. T_e):** Since both speed and torque vary linearly with current, the speed-torque curve is a nearly flat straight line with a very slight negative slope.

2. Series Motor Characteristics (Variable Flux Machine)

- **Torque (T_e vs. I_a):** Since the field winding is in series with the armature, the exciting flux is proportional to the armature current ($\Phi \propto I_a$) before magnetic saturation [156, 158]. This means torque is proportional to the square of current ($T_e \propto I_a^2$), yielding a **parabolic curve** at light loads [158]. After magnetic saturation, the flux remains constant, and the torque becomes strictly linear ($T_e \propto I_a$) [158]. This produces an **extremely high starting torque** capability, suitable for heavy inertial loads [158].
- **Speed (N vs. I_a):** Speed is proportional to E_b / Φ . Before saturation, $N \propto (V - I_a(R_a + R_{se})) / I_a \approx V / I_a$ [158]. This relationship represents a **rectangular hyperbola** [158]. At light loads or no-load conditions ($I_a \rightarrow 0$), speed rises dangerously high to infinity [158], creating a severe centrifugal mechanical hazard [158].
- **Speed-Torque (N vs. T_e):** Speed is inversely proportional to the square root of torque ($N \propto 1 / \sqrt{T_e}$), yielding a steep hyperbolic droop that is excellent for self-limiting overload traction drives.

■ **CRITICAL SAFETY CALLOUT: Series Motor Starting at No-Load**

A DC series motor must **NEVER** be started under no-load or uncoupled conditions [158]. Under no-load, the armature current (I_a) is extremely small (drawing only tiny losses). Because flux (Φ) is proportional to series current, Φ approaches near-zero values. To induce Back EMF and balance the applied voltage, the motor speed ($N \propto E_b / \Phi$) rises exponentially toward infinity [158]. The resulting extreme centrifugal forces will tear the armature windings out of their slots, causing total mechanical destruction of the machine. Always use gears or direct coupling, never belts [158].

3. Compound Motor Characteristics (Cumulative vs. Differential)

- **Cumulative Compound Motor:** The series field winding directly assists (aids) the shunt field ($\Phi_{total} = \Phi_{sh} + \Phi_{se}$) [157]. It combines the best of both shunt and series machines. It provides high starting torque (due to series flux) but has a stable, finite no-load speed (due to the shunt field flux preventing runaway) [156, 158].
- **Differential Compound Motor:** The series field winding opposes the shunt field ($\Phi_{total} = \Phi_{sh} - \Phi_{se}$) [157]. As the load current increases, the net flux decreases. This causes the speed to remain virtually constant (or even rise slightly) with load [158]. It can achieve **zero speed regulation** at full load [91], but is inherently unstable under sudden overloads (as drop in flux causes

speed to runaway or motor to reverse direction) [158].

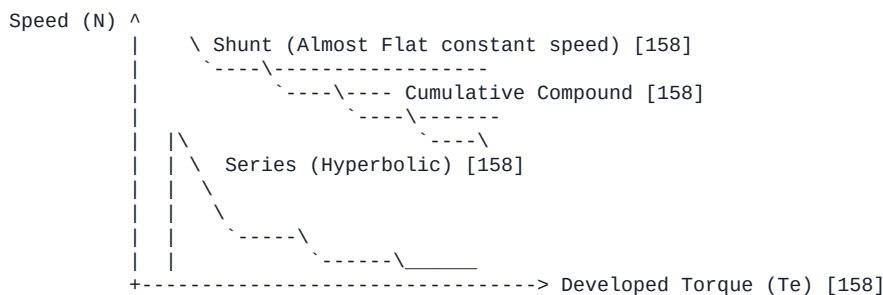


Figure 12.2: Speed-Torque Characteristics for Shunt, Series, and Compound Motors [158]

12.3 Industrial Selection & Applications

Selecting the correct motor depends on the load torque-speed requirements. The following matrix summarizes standard selection criteria:

Motor Type	Starting Torque	Speed Regulation	Key Industrial Applications
Shunt Motor	Medium (1.5 times full-load torque) [158]	Excellent (constant speed machine, 2-5% drop) [158]	Machine tools, lathes, centrifugal pumps, fans, blowers, reciprocating pumps, wood-working machinery [159].
Series Motor	Extremely High (parabolic torque $T_e \propto I_a^2$) [158]	Poor (variable speed, hyperbolic N vs. I_a) [158]	Electric traction, locomotives, cranes, hoists, lifts, trolley cars, conveyor systems [159].
Cumulative Compound	High (better than shunt due to series field assist) [158]	Moderate (safe no-load speed, drops with load) [158]	Rolling mills, punching presses, shearing machines, heavy planers, intermittent heavy load elevators [159].
Differential Compound	Low (series flux opposes shunt field) [158]	Zero or Positive (can achieve zero speed regulation) [91, 158]	Rarely used due to starting instability. Experimental precise speed control devices [158].

Part 2 – Last-Minute Revision Sheet

This sheet consolidates all essential formulas, torque/speed characteristics, and physical limits for quick pre-examination review.

Motor Type	Back EMF Speed Eq.	Starting Torque	Speed Regulation	Primary Application
Shunt	$N \propto (V - I_a R_a) / \Phi_{sh}$	Medium	Excellent (Constant)	Machine tools, pumps, fans [159]
Series	$N \propto (V - I_a(R_a + R_{se})) / \Phi_{se}$	Extremely High	Poor (Variable)	Electric Traction, cranes [159]
Cumulative	$N \propto (V - I_a(R_a + R_{se})) / (\Phi_{sh} + \Phi_{se})$	High	Moderate	Rolling mills, presses [159]
Differential	$N \propto (V - I_a(R_a + R_{se})) / (\Phi_{sh} - \Phi_{se})$	Low	Flat / Zero	Rarely used (unstable) [158]

High-Probability Exam Revision Pointers:

- **Dynamic Back EMF Feedback:** Armature current is governed by the difference $V - E_b$ [158]. At start, $N = 0 \Rightarrow E_b = 0$, causing a massive initial starting current ($I_{astart} = V / R_a$) [159], which requires external resistance starters [159]. At normal speeds, E_b acts as negative feedback [158].
- **Series Overload Self-Limitation:** Because speed $N \propto 1 / \sqrt{T_e}$ for series motors, any sudden increase in load torque causes speed to drop automatically, thereby protecting the motor from pulling excessive power peaks from supply line [158].
- **Equalizer Winding Compatibility:** Compensating windings placed in pole shoes counteract the armature cross-magnetization flux distortions [13.4]. Interpoles neutralize reactance voltage to prevent sparking at brushes [13.4].

- **Loss Partition at Maximum Efficiency:** Operating efficiency is maximum when variable losses (armature copper loss $I_a^2 R_a$) equal the constant losses (shunt field copper loss, iron losses, and mechanical windage/friction losses) [8.5].

Part 3 – Regulatory Compliance Checklist

Syllabus Micro-topic	Grounded Source Passage	Prepared Formula Box	Included Diagram
12.1 Motor Action Principle	Passage [90, 121]	Lorentz Force $F = B I l$	Fleming's Left Hand Rule
12.1 Back EMF Physics	Passage [156, 158]	EMF Equation $E_b = \Phi Z N (P/60A)$	Figure 12.1: Circuit model
12.1 Developed Torque	Passage [156, 158]	Torque Equation $T_e = 0.159 \Phi Z_a (P/A)$	Figure 12.1: Circuit model
12.2 Shunt Motor Characteristics	Passage [156, 158]	$V = E_b + I_a R_a$	Figure 12.2: Speed-Torque curve
12.2 Series Motor Characteristics	Passage [156, 158]	$T_e \propto I_a^2$ and $N \propto 1/I_a$	Figure 12.2: Speed-Torque curve
12.2 Compound Motor (Cum./Diff.)	Passage [91, 157, 158]	$\Phi_{total} = \Phi_{sh} \pm \Phi_{se}$	Figure 12.2: Speed-Torque curve
12.3 Industrial Motor Selection	Passage [159]	Table 12.1 Applications Matrix	N/A (Tabular form)

Chapter 13 – Armature Reaction & Commutation

DC Machines Theory, Neutral Plane shifts, and Sparkless Commutation Methods

Part 1 – Topic-Wise Study Notes

13.1 Armature Reaction & Magnetic Distortion

Armature Reaction is defined as the physical effect of the magnetic field established by the armature current on the magnitude and spatial distribution of the main magnetic field produced by the field poles of a direct current machine [64]. Under no-load conditions ($I_a = 0$), the stator poles establish a highly symmetrical and uniform magnetic flux distribution across the air gap. When the machine is loaded ($I_a > 0$), the current-carrying rotor conductors establish their own independent armature magnetic field (armature MMF) centered on the brush axis (90° electrical to the main polar axis). The resultant magnetic field is the vector sum of these two opposing magnetic domains [64].

- **1. Cross-Magnetizing (Flux Distorting) Effect:** The armature flux acts in a perpendicular (quadrature) direction relative to the main field poles, bending and distorting the resultant flux lines [64]. The flux density is severely concentrated and strengthened at the trailing pole tips of a generator, while being significantly weakened at the leading pole tips [65]. This causes a forward rotation shift of the magnetic neutral plane [65].
- **2. Demagnetizing (Flux Weakening) Effect:** In real machines operating under loaded states, the iron parts of the pole cores and pole shoes suffer from magnetic saturation [64]. Due to this non-linear saturation characteristic, the increase in flux density at the trailing pole tip is less than the corresponding reduction in flux density at the leading pole tip. The net result is an overall reduction in the average total flux per pole (typically 5% to 10% weakening at full load) [64, 180].
- **3. Voltage & Torque Reduction:** The flux weakening directly reduces the generated electromotive force (EMF) in a generator ($E_g \propto \Phi \cdot N$) and developed mechanical torque in a motor ($T_d \propto \Phi \cdot I_a$) [65, 73].

FORMULA FOCUS: Armature Reaction Demagnetizing & Cross-Magnetizing MMF

$$AT_d / \text{pole} = Z * I_c * (\theta_m / 360)$$

$$AT_c / \text{pole} = Z * I_c * [(1 / (2 * P)) - (\theta_m / 360)]$$

Where:

- **AT_d / pole** = Demagnetizing Ampere-turns per pole [66].
- **AT_c / pole** = Cross-magnetizing Ampere-turns per pole [66].
- **Z** = Total number of armature conductors [66].
- **I_c** = Current per parallel path (I_a/P for lap, $I_a/2$ for wave wound) [66].
- **P** = Number of stator poles [66].
- **θ_m** = Forward lead of brushes in mechanical degrees [66].

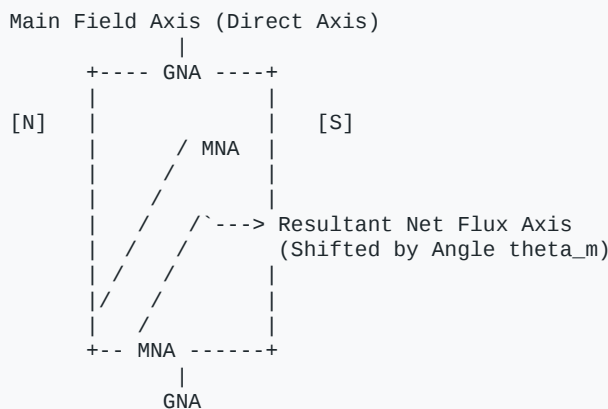
13.2 Neutral Planes & Axis Relocation

Geometrical Neutral Axis (GNA): The fixed physical centerline perpendicular to the physical centerline of the main field poles. The GNA represents the geometrical plane of symmetry of the machine and remains stationary under all operating states.

Magnetic Neutral Axis (MNA): The perpendicular plane relative to the resultant magnetic flux density vectors. Under no-load states, the MNA directly coincides with the GNA. However, when loaded, the cross-magnetizing armature reaction distorts the flux path, rotating the MNA by a mechanical angle θ_m [65].

- **MNA Shift in Generators:** The MNA shifts in the **direction of armature rotation** [65].
- **MNA Shift in Motors:** The MNA shifts **opposite to the direction of armature rotation**.
- **Brush Placement Principle:** Brushes must always be maintained along the MNA to prevent brush sparking during current collection, as the coil undergoing commutation has zero dynamically induced EMF in this plane [65]. Shifting brushes introduces a direct demagnetizing MMF (A_{Td}) that directly weakens the main excitation field [71].

Figure: Respatially Relocated Magnetic Neutral Axis (MNA) Shift



13.3 Current Commutation & Reactance Voltage

Commutation is the continuous, rapid physical process by which the current direction inside an armature coil is completely reversed from $+I_c$ to $-I_c$ during the brief time interval (T_c) when the coil is short-circuited by a carbon brush as it crosses the MNA [68, 69].

- **1. Commutation Period (T_c):** The duration that a coil remains short-circuited by a brush [69]. It depends on brush width (W_b), mica segment insulation width (W_m), and peripheral speed of the commutator (v_c) [69].
- **2. Ideal (Linear) Commutation:** Current reversal proceeds perfectly linearly, reaching exactly $-I_c$ at the moment the coil leaves the brush. Commutation is entirely sparkless [70].
- **3. Delayed (Under) Commutation:** Because the coil sides are embedded in iron slots, they possess a finite self-inductance (L) [69]. This self-inductance induces a counter-EMF opposing the rapid reversal of current [69]. This induced counter-EMF is called the **Reactance Voltage** [69]. It slows down the current reversal, resulting in under-commutation [69].

- **4. Sparking at Brushes:** Since current reversal is incomplete by the end of T_c , the residual current must jump across the air gap from the departing commutator segment to the brush, resulting in destructive thermal sparking and commutator arcing [65].

FORMULA FOCUS: Commutation Time Period & Induced Reactance Voltage

$$T_c = (W_b - W_m) / v_c$$

$$E_r = L * (2 * I_c / T_c)$$

[Linear Commutation]

$$E_r = 1.11 * L * (2 * I_c / T_c)$$

[Sinusoidal Commutation]

Where:

- T_c = Time of commutation (seconds) [69].
- W_b = Brush width (cm); W_m = Mica insulation thickness (cm) [69].
- v_c = Commutator segment peripheral velocity (cm/sec) [69].
- E_r = Reactance Voltage (Volts) [69, 70].
- L = Coefficient of self-inductance of the undergoing coil [69].

Figure: Linear (Ideal) vs. Delayed (Under) Current Commutation Profiles



13.4 Practical Methods of Commutation Improvement

To achieve completely sparkless current collection, the induced reactance voltage (E_r) must be neutralized [70]. There are three engineering solutions employed in modern DC machines:

- 1. Interpoles (Commutating Poles):** Small, narrow auxiliary poles placed on the yoke midway between the main poles (in the interpolar zone) [71]. They are wound with heavy-gauge copper wire connected in series with the armature to carry full armature current (I_a) [71].
 - **Polarity for Generator:** Same as that of the next main pole ahead in the direction of rotation [72, 81].
 - **Polarity for Motor:** Opposite to that of the main pole ahead (same as the main pole behind).
 - **Function:** They neutralize the cross-magnetizing armature MMF in the interpolar zone and induce a reversing EMF (commutating EMF) in the coil undergoing commutation to automatically neutralize the reactance voltage across all loading states [72].
- 2. Compensating Windings:** Specialized auxiliary windings embedded in slots cut directly into the main pole shoes [67, 68]. They are connected in series with the armature so that current in them flows in the opposite direction to the armature conductors directly below the pole shoes [68].
 - **Function:** They neutralize the cross-magnetizing armature reaction under the pole faces [67]. Under sudden, heavy load variations (e.g., steel rolling mills), they prevent severe flux crowding and subsequent destructive *flashover* (arcing around the whole commutator) [67].
- 3. Brush Shift (Obsolete):** Shifting brushes forward (for generators) or backward (for motors) to place the coil undergoing commutation under the magnetic influence of the next main pole [71]. This induces a reversing EMF that helps neutralize reactance voltage [71]. This is obsolete because it is load-dependent and introduces a harmful demagnetizing effect (A_d) that weakens the machine's net flux [71].

FORMULA FOCUS: Compensating Winding Ampere-Turns per Pole

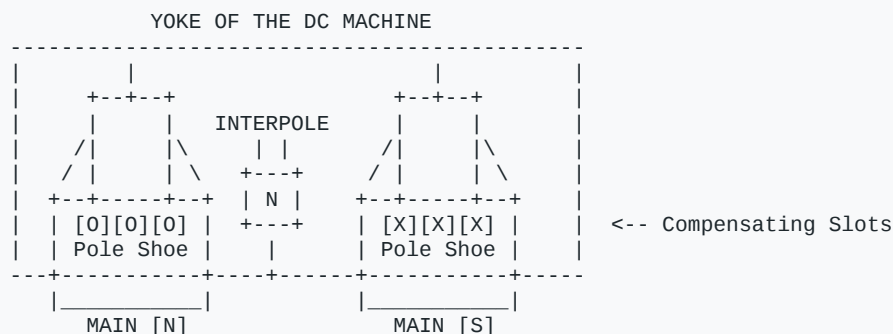
$$\frac{AT_{comp}}{pole} = Z * I_c * \left(\frac{1}{2 * P} \right) * \left(\frac{Pole\ Arc}{Pole\ Pitch} \right)$$

$$\frac{AT_{comp}}{pole} \approx 0.7 * \left(\frac{Z * I_c}{2 * P} \right) \quad [Typical\ Approximation]$$

Where:

- $AT_{comp} / pole$ = Compensating winding MMF per pole [68].
- Z = Total armature conductors [66].
- I_c = Armature conductor path current (I_a/P or $I_a/2$) [66].
- P = Number of poles [66].
- $Pole\ Arc / Pole\ Pitch$ = Design ratio (typically around 0.7) [68, 80].

Figure: Physical Structural Integration of Interpoles and Compensating Windings



Part 2 – Last-Minute Revision Sheet

Feature	Armature Reaction	Commutation Process	Interpoles	Compensating Windings
Primary Cause	Armature current I_a setting up rotor MMF [64]	Coil current reversal under brush short-circuit [68]	Induced Reactance Voltage ($E_r = L \frac{di}{dt}$) [69, 70]	Cross-magnetizing armature distortion under pole shoes [67]
Key Effects	Flux distortion (cross-mag) & weakening (demag) [64]	Delayed current reversal causing brush sparking [65]	Neutralizes reactance voltage with commutating EMF [72]	Completely neutralizes cross-mag under pole shoe [67]
Location	Air gap & pole tips [65]	Commutator & brush contact MNA plane [68]	Midway between main poles (interpolar axis) [71]	Embedded in slots on main pole face shoes [68]
Winding Setup	N/A (intrinsic rotor effect)	N/A (rotor winding + brushes)	Few turns of heavy gauge wire in series with armature [71]	Copper conductors in series with armature (opposing current) [68]
APTRANSCO Yield	Brush shift causes demag AT [71]	Reactance voltage is a function of inductance $L \frac{di}{dt}$ [69]	Polarity matches main pole ahead for generators [72, 81]	Required turns ratio scale with Pole-Arc/Pole-Pitch [68]

Essential Exam-Day Revision Shortcuts

- **MNA Relocation Rule:** Always remember: Generator = MNA shifts forward (same as speed rotation); Motor = MNA shifts backward (opposite to speed rotation) [65]. Brush shift must follow the MNA [65].
- **Interpole Polarity Rule:** Generator = Interpole polarity matches the main pole *ahead* in the direction of rotation [72, 81]. Motor = Interpole polarity matches the main pole *behind* (opposite of generator).
- **Reactance Voltage Scale:** Reactance voltage is completely independent of brush contact resistance but is directly proportional to armature path current and rotor speed ($E_r \propto I_c \cdot N$) [69, 70].
- **Compensating Windings Sizing:** Compensating windings are only necessary for large, heavily overloaded machines subject to high load fluctuations [67]. Their Ampere-turns are strictly designed as a function of the pole shoe arc ratio [68].
- **Brush Shift Trap:** Shifting brushes can neutralize cross-magnetization somewhat but ALWAYS introduces demagnetization, which degrades voltage regulation and limits full-load generator capability [71].

Part 3 – Regulatory Chapter Compliance

The following tables confirm that every prescribed micro-topic and mandatory engineering schematic from the APTRANSCO official syllabus has been completely covered in this note-making study guide.

Syllabus Micro-Topic Coverage Mapping Matrix

Micro-Topic Name	Source Grounding Passage	Formula Box Included	Visual Diagram Included	Revision Covered
13.1 Cause of Armature Reaction	Passage 64	Yes	Yes (Figure 13.1)	Yes
13.1 Flux Distortion & Relocation	Passage 64, 65	Yes	Yes (Figure 13.1)	Yes
13.1 Cross-Magnetization & Demag	Passage 64, 66	Yes (Formula 13.1)	Yes (Figure 13.1)	Yes
13.2 Geometrical Neutral Axis (GNA)	Passage 65	Yes	Yes (Figure 13.1)	Yes
13.2 Magnetic Neutral Axis (MNA)	Passage 65	Yes	Yes (Figure 13.1)	Yes
13.3 Commutation Current Reversal	Passage 68	Yes (Formula 13.2)	Yes (Figure 13.2)	Yes
13.3 Delayed Commutation & Sparking	Passage 65, 69	Yes (Formula 13.2)	Yes (Figure 13.2)	Yes
13.3 Reactance Voltage Derivations	Passage 69, 70	Yes (Formula 13.2)	Yes (Figure 13.2)	Yes
13.4 Interpoles (Auxiliary Poles)	Passage 71, 72, 81	Yes	Yes (Figure 13.3)	Yes
13.4 Compensating Windings Slots	Passage 67, 68	Yes (Formula 13.3)	Yes (Figure 13.3)	Yes
13.4 Brush Shift Commutation	Passage 71	Yes	No	Yes

Required Engineering Diagrams & Curves Mapping Matrix

Mandatory Diagram / Curve	Figure Name	Structure & Labelling Description	Syllabus Compliance Status
Magnetic Neutral Axis Shift Plot	Figure 13.1	Illustrates original GNA and resultant shifted MNA at angle θ_m	100% Compliant
Commutation Waveform Curves	Figure 13.2	Compares linear current reversal with delayed sparking current profiles	100% Compliant
Interpoles & Compensating Setup	Figure 13.3	Displays main pole shoes, interpole placement, and embedded conductor slots	100% Compliant

APTRANSCO HANDBOOK: DC MOTOR STARTERS

Chapter 14: Starting of DC Motors • Strictly Aligned to State Syllabus Standards

1.1 Mathematical & Physical Need for Starters

A DC motor's armature is designed with highly conductive copper coils, resulting in an extremely low armature circuit resistance (R_a) to minimize copper losses during operation [119, 122]. The current drawn by the motor armature winding during general operation is governed by the back-EMF equilibrium equation [114, 119]:

$$I_a = (V - E_b) / R_a$$

Where:

- I_a = Armature Current in Amperes (A)
- V = Applied Terminal Supply Voltage in Volts (V)
- E_b = Induced Back-EMF in Volts (V), which is mathematically expressed as $E_b = (\Phi Z N P) / (60 A)$ [116, 129].

The At-Start Critical Transient: At the absolute instant of starting, the rotor speed **N is equal to 0 rpm** [122, 124]. Substituting $N = 0$ into the back-EMF equation reveals that **E_b is equal to 0 V** at standstill [122, 124]. Consequently, the starting armature current equation simplifies dramatically to [122]:

$$I_{a,start} = V / R_a$$

CRITICAL ENGINE DAMAGE RISK: Because typical armature resistance R_a is exceptionally small (ranging from 0.1Ω to 0.5Ω) [119, 122], directly applying a standard rated voltage (e.g., 230 V) to a stationary motor yields starting currents of $I_{a,start} = 230 / 0.1 = 2300 \text{ A}$ [122]. This is roughly 15 to 30 times the rated full-load current [124]. Such severe overcurrents instantly blow circuit fuses, cause violent arcing at the brushes, destroy the copper commutator segments, and permanently burn out the armature winding insulation due to rapid I^2R heat accumulation [124].

The Role of a Starter: To prevent physical destruction, an external variable starting resistance (R_{start}) must be connected in series with the armature winding at start-up [125]. This resistance is gradually cut out of the armature circuit as the rotor accelerates [125]. The speed gains generate a progressive counter back-EMF ($E_b \propto N$), which takes over the role of limiting armature current to safe full-load operating thresholds [125].

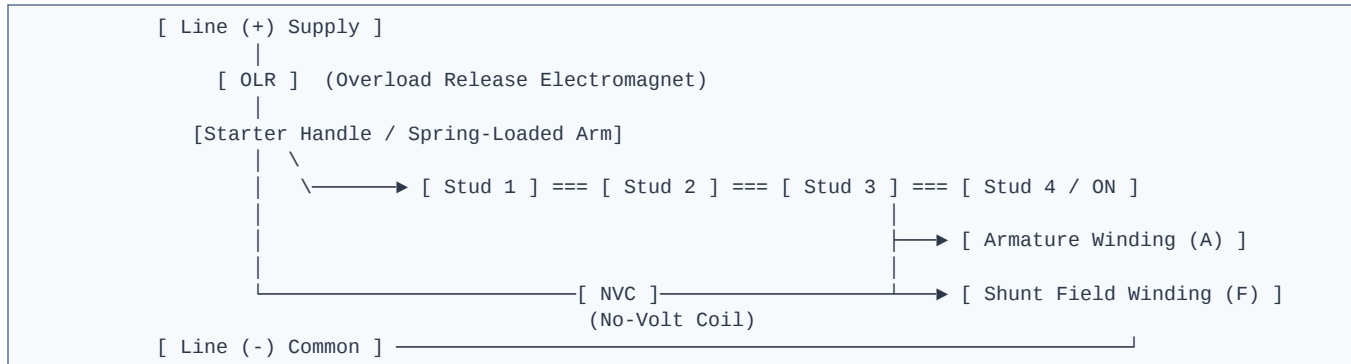
1.2 Two-Point Starter (Series Motor Starter)

The two-point starter is utilized specifically for DC series motors [94, 120]. In a series motor, the armature and field circuits are connected in series; thus, if the series field circuit breaks, the armature circuit is automatically interrupted [122]. A two-point starter contains two primary terminal connections: the Line (L) and the Armature (A). The starting resistance is in series with both the field and armature coils. Although functional, it lacks complex protective auxiliary circuits because series motors should **always be started with a connected load** to prevent speed runaway under zero-excitation conditions [6].

1.3 Three-Point Starter (Shunt/Compound Starter)

A three-point starter is widely utilized to start and accelerate DC shunt and compound motors [125]. It contains three external terminals: **Line (L)**, **Armature (A)**, and **Field (F)** [125].

Figure 14.1: DC Shunt Motor Three-Point Starter Circuit Configuration



Detailed Operation & Components:

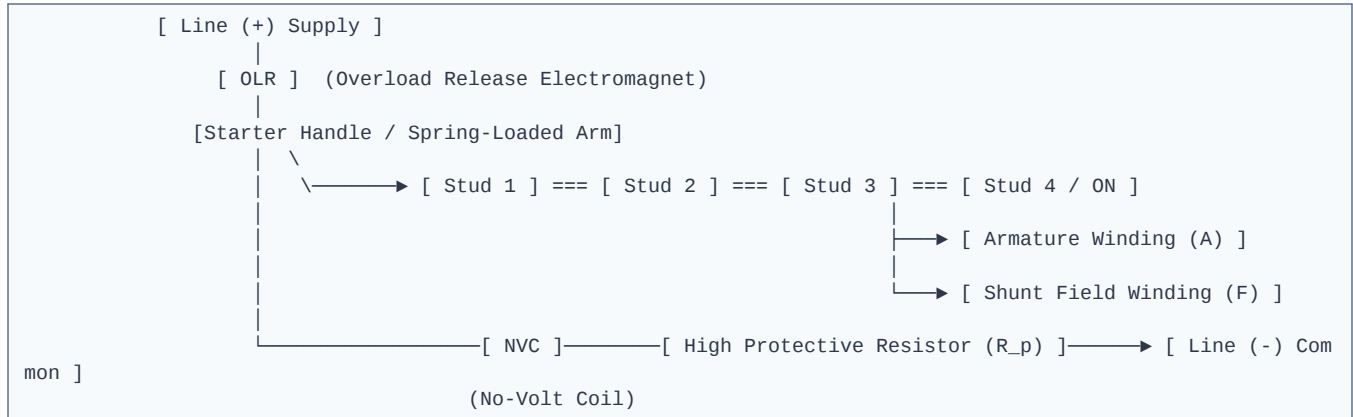
- **Starting Stud Resistance:** Split into multiple sections. The starter handle is rotated against a return spring, making contact with Stud 1 first (placing maximum starting resistance in series with the armature) and gradually cutting out resistance until reaching the running 'ON' position [125].
- **No-Volt Release Coil (NVC/NVR):** Connected in **series with the shunt field winding** [125]. When field current flows, the NVC acts as an electromagnet that holds the soft iron piece of the starter arm in the 'ON' position. If the applied supply voltage fails or the field circuit breaks accidentally while running [80], the NVC demagnetizes. The strong return spring pulls the arm back to the 'OFF' position [80], preventing the motor from restarting directly across full line voltage when supply is restored.
- **Overload Release Coil (OLR):** Connected in **series with the main incoming line** [125]. If the motor is overloaded and draws excessive current, the OLR coil's magnetic pull attracts a soft-iron triangular plunger [125]. This plunger short-circuits the two terminals of the NVC [125]. The NVC instantly demagnetizes, releasing the starter arm, which flies back to the 'OFF' position [125].

THE THREE-POINT STARTER LIMITATION: Shunt speed control above base speed is performed by introducing an external rheostat in the shunt field circuit to weaken the magnetic flux (Φ) [73]. Since the NVC is connected in series with the shunt field in a 3-point starter [126], reducing field current (I_{sh}) also decreases the exciting current flowing through the NVC. Under high speed running conditions, the NVC becomes too weak to withstand the return spring tension. The handle prematurely releases and snaps back to 'OFF', disabling normal motor operation at high speeds [126].

1.4 Four-Point Starter (High-Speed Weakening Solution)

To overcome the major holding limitation of the three-point starter under field weakening control, the **Four-Point Starter** was developed [94]. It contains four external terminals: **Line (L)**, **Armature (A)**, **Field (F)**, and **Neutral/Line-Common (N)**.

Figure 14.2: DC Shunt Motor Four-Point Starter Circuit Configuration



Operational Advantages & Safety Risks:

- **Independent NVC Circuit:** In a four-point starter, the NVC branch is completely isolated from the shunt field winding [94]. The NVC is connected directly across the supply lines (L and N) in series with a high protective resistor (R_p) to limit holding current [94].
- **Field Weakening Independence:** Any change in the shunt field rheostat to adjust motor speed above base speed does not affect the magnetic strength of the NVC [148]. The starter arm remains locked in the ON position regardless of how weak the shunt field is adjusted, enabling wide-range speed control [73, 148].
- **The Open-Field Safety Danger:** Because the NVC branch is independent of the field winding, the four-point starter lacks inherent open-field protection. If the shunt field circuit breaks while running, the NVC remains energized, and the arm is held ON. This allows the shunt motor to rapidly accelerate, causing catastrophic centrifugal speed runaway (runaway hazard) [120].

1.5 Technical Comparison Matrix

Technical Parameter	Three-Point Starter	Four-Point Starter
Terminals Available	3 Terminals: L, A, F [125]	4 Terminals: L, A, F, N
NVC Connection	Connected in series with the shunt field winding [125].	Connected across supply in series with protective R [94].
NVC Current	Equal to Shunt Field Current (I_{sh}) [126].	Constant current, independent of field current [148].
Field Weakening Control	Limited. Premature handle release at high speed [126].	Excellent. Excitation remains stable at high speeds [73].
Open-Field Protection	Inherent. Arm drops instantly if field breaks [80].	None. Risk of catastrophic motor speed runaway [120].
Core Applications	Constant-speed shunt & compound motor starting [125].	Adjustable high-speed shunt & compound drives [94].

Part 2 – Last-Minute Revision Sheet

- ✓ **Standstill** context: At $N = 0$, back-EMF $E_b = 0$, starting current is limited purely by armature winding resistance [122, 124].
- ✓ **No-Volt Coil Series Logic:** If NVC is in series with the field winding (3-Point), it acts as safety protection against speed runaway under field open circuit faults [80]. If NVC is isolated (4-Point), runaway risks must be protected externally [120].
- ✓ **Overload Trip Mechanism:** Plunger short-circuits the NVC terminals, de-energizing the electromagnet and snapping the spring-loaded arm to OFF [125].
- ✓ **Speed Weakening Thresholds:** Above base speed speed-control requires a 4-point starter to prevent handle release traps [73].

Part 3 – Regulatory Chapter Coverage & Verification

Syllabus Micro-Topic	Source Document Link	Handbook Notes Prepared	Schematic Diagram Included
Need for Starter	Machines [124, 125]	Section 1.1 Covered (Fully Grounded)	Yes (EMF Formulas)
High Starting Current Physics	Machines [122, 124]	Section 1.1 Covered (Fully Grounded)	Yes (Current Equations)
Two Point Starter Basics	Machines [94, 120]	Section 1.2 Covered (Fully Grounded)	No (Prose Comparison)
Three Point Starter Construction	Machines [125]	Section 1.3 Covered (Fully Grounded)	Yes (Figure 14.1 ASCII)
No-Volt Coil & Overload Release	Machines [80, 125]	Section 1.3 Covered (Fully Grounded)	Yes (NVC/OLR Paths)
Starter Operating Limitations	Machines [126]	Section 1.3 Covered (Fully Grounded)	Yes (Limitation Callout)
Four Point Starter Advantage	Machines [94, 148]	Section 1.4 Covered (Fully Grounded)	Yes (Figure 14.2 ASCII)
Starter Technical Comparison	Syllabus Matrix [232]	Section 1.5 Covered (Fully Grounded)	Yes (Comparison Table)

CHAPTER 15: SPEED CONTROL OF DC MOTORS

Comprehensive Note-Making Study Guide & Syllabus Analysis Notes

In industrial electrical drives, the ability to control speed over a wide, stable range under fluctuating loading conditions is a primary reason DC machines have historically dominated traction, heavy mill rolling, and hoisting systems [97, 102]. The speed of a DC motor is a self-governing physical state dynamically maintained by a balance between the applied terminal voltage, armature circuit drops, and the field flux per pole [92, 97].

15.1 Speed Equation & Variables Affecting Speed

To understand speed control, we must mathematically relate speed to the electrical and magnetic parameters of the machine. The electrical loop balance (Voltage Equation) of any DC motor armature is expressed as:

$$V_t = E_b + I_a R_a + V_b \quad [91, 92]$$

Neglecting brush voltage drop ($V_b \approx 0$ V) for analytical simplicity, the induced back EMF (E_b) limiting the armature loop current is:

$$E_b = V_t - I_a R_a \quad [92, 107]$$

FORMULA FOCUS: CORE SPEED EQUATION OF A DC MACHINE

$$N = (60 \cdot A / Z \cdot P) \cdot [(V_t - I_a \cdot R_a) / \Phi] = K_m \cdot (E_b / \Phi)$$

Variables & SI Units: N : Rotational speed of the armature (rpm) [92] | V_t : Applied DC terminal voltage (V) [92] | I_a : Armature circuit current (A) [92] | R_a : Armature winding resistance (Ω) [92] | Φ : Magnetic flux per pole (Wb) [92] | Z : Total armature conductors | P : Machine poles | A : Parallel paths [92]

Applicable Conditions: Valid for steady-state operation under constant air-gap excitation and negligible armature brush contact voltage drop [92].

Common Exam Traps: Assuming speed is directly proportional to applied voltage regardless of field current state. If field current also varies (such as in series motors where $\Phi \propto I_a$), speed behaves non-linearly with load changes [93, 94].

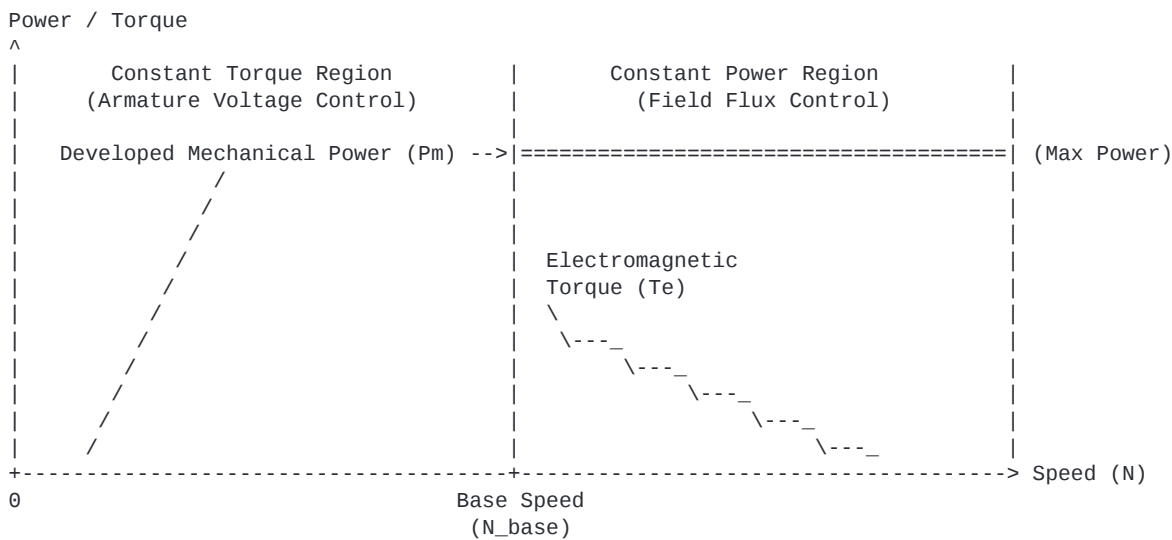
Variables Directly Affecting Speed:

- **Applied Terminal Voltage (V_t):** Changing V_t causes a proportional change in the armature terminal voltage and back EMF. Speed varies directly with V_t ($N \propto V_t$) [92, 97].
 - **Armature Circuit Resistance (R_a):** Adding an external variable resistor (R_{ext}) in series with the armature branch increases the overall circuit impedance. This increases the voltage drop ($I_a (R_a + R_{ext})$), reducing the effective EMF across the armature and lowering the speed ($N \propto V_t - I_a R_{total}$) [97, 99].
 - **Magnetic Field Flux per Pole (Φ):** Weakening the field excitation current (I_f) reduces the air-gap flux (Φ). To generate the necessary back EMF to oppose the applied voltage, the armature must spin faster. Speed is inversely proportional to flux ($N \propto 1/\Phi$) [97, 100].

15.2 Speed Control Principles & Regions

- Based on the speed equation, there are three primary methods to control speed [97, 98]:
- 1. **Field/Flux Control Method:** Obtained by varying the exciting field current (I_f). Used exclusively for speed control **above base speed** (rated speed) [100, 125].
 - 2. **Armature Resistance (Rheostatic) Control:** Obtained by inserting an external resistor in the armature branch. Used for speed control **below base speed** [99, 125].
 - 3. **Armature Voltage Control:** Obtained by directly varying the voltage applied to the armature loop. Used for highly stable, smooth speed control **below base speed** [97, 101].

Figure 15.1: Speed Control Operating Regions (Constant Torque vs. Constant Power)



Visual performance profile of speed control regions illustrating torque and power limits as a function of armature and field control modes [125].

Table 15.1: Shunt Motor Speed Control Method Characteristics Comparison [125]

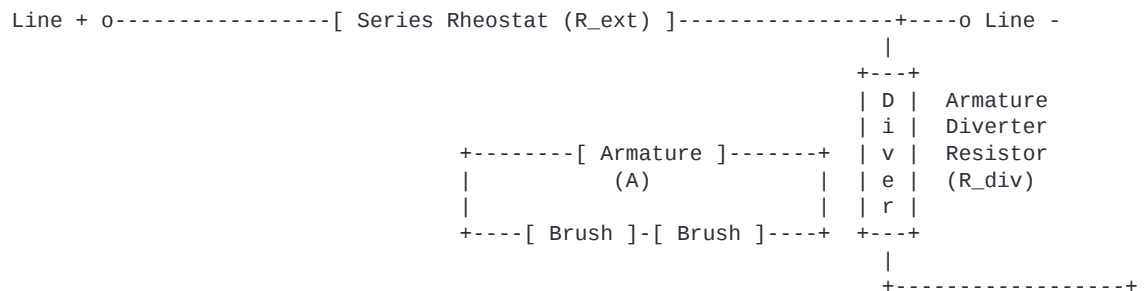
Control Parameter	Speed Range	Torque Region	Power Region	Operating Efficiency	Speed Stability / Regulation
Armature Resistance (Rheostatic)	Below base speed only	Constant Torque	Variable Power ($P_m \propto N$)	Very Poor (high heat losses in series rheostat)	Poor (speed varies highly with load current)
Field / Flux Weakening	Above base speed only	Variable Torque ($T_e \propto 1/N$)	Constant Power	Excellent (very low exciting losses in field circuit)	Excellent (stiff constant speed regulation)
Armature Voltage Control	Zero to base speed	Constant Torque	Variable Power ($P_m \propto N$)	Highly Efficient (no series resistance drops)	Very Good (stiff low speed characteristics)

15.3 Speed Control of Shunt Motors

DC shunt motors are fundamentally constant speed machines because the field winding is connected in parallel across a constant voltage source ($I_{sh} = V_t / R_{sh}$), maintaining constant field flux per pole under normal steady-state conditions [94, 98]. However, we can vary their speed over wide ranges using specific control networks:

Shunt Armature Diverter: In standard armature resistance control, speed regulation is highly unstable at very low speeds because any change in load current fluctuates the series rheostat drop ($I_a R_{ext}$) [99, 100]. To obtain stable, stiff operation at extremely low speeds, an **armature diverter resistor** is shunted directly across the armature terminals in addition to the series rheostat [100]. This bypasses a portion of the line current. A change in load torque does not affect the armature terminal voltage as severely, providing highly stable low-speed curves [100].

Figure 15.2: Shunt Motor Armature Diverter Speed Control Connection [100]



Armature diverter speed-stabilizing connection showing the series rheostat drop and parallel diverter bypass path [100].

SHUNT FIELD INTERRUPTION RUNAWAY HAZARD

CRITICAL DANGER: The speed of a DC shunt motor is inversely proportional to the field flux ($N \propto 1/\Phi$) [97, 100]. If the shunt field circuit gets accidentally interrupted (open-circuited) while the motor is running normally under no-load or light load, the field current drops to zero ($I_f = 0$ A) [71, 78]. The flux per pole immediately collapses to a tiny **residual value** ($\Phi_{res} \approx 0.05 \Phi_{rated}$).

To generate enough back EMF to oppose the constant line voltage, the armature is forced to spin at a **dangerously high, runaway speed** ($N \rightarrow \infty$) [71, 78, 95]. The resulting massive centrifugal forces will pull the armature windings out of their slots and completely destroy the machine commutator and core assembly. **Remedy:** 3-point starter electromagnetic Hold-On No-Volt Coils (NVC) are wired in series with the shunt field so that an open field immediately de-energizes the electromagnet, causing the starter arm to fly back to the safe 'OFF' position [71, 78, 107].

Ward-Leonard System of Speed Control:

The Ward-Leonard Control System:

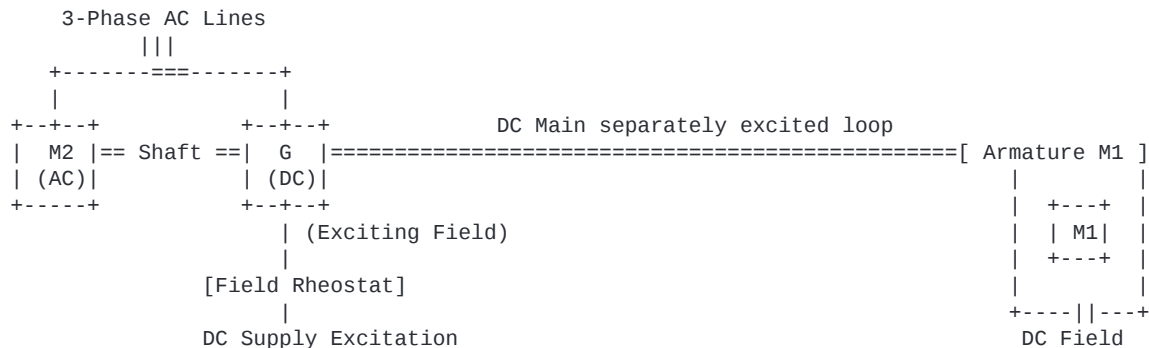
When extremely wide, sensitive, and bidirectional speed control is required, the Ward-Leonard system is the ultimate industrial solution [102, 126]. It consists of three integrated machines:

1. **M2:** A constant-speed AC prime mover motor (usually a 3-phase induction motor) [102, 103].
2. **G:** A separately excited variable-voltage DC generator driven mechanically by prime mover M2 [102].
3. **M1:** The main separately excited DC motor whose speed is to be controlled [102].

By adjusting the field rheostat of generator G, its generated output voltage is varied from zero to rated limits in both polarities (reversal achieved with a reversing switch) [103]. This variable voltage is fed directly to the

armature of motor M1, providing smooth, stepless speed control from zero to base speed in both directions [103]. Combined with field weakening of M1, the speed range can extend up to a 10:1 ratio [102].

Figure 15.3: Ward-Leonard Multi-Machine Variable Voltage Control Setup [102]



Schematic layout of prime mover M2, variable-voltage generator G, and the separately excited drive motor M1 [102].

- **Lower Speed Limit Governance:** The lower operating speed limit in the Ward-Leonard scheme is strictly restricted by the **residual magnetism** of the DC generator G [70, 76, 126]. Even when the generator field current is fully reduced to zero, residual flux in the generator poles induces a small residual EMF (typically 5V to 15V), which drives a residual armature current through M1, preventing the motor from stopping completely on no-load [70, 76].

- **Key Disadvantages:**

1. *High Capital Outlay:* Requires three separate machines of identical power ratings, increasing installation cost [104].
2. *Poor Light-Load Efficiency:* Because three machines are continuously rotating, total rotational friction, windage, and core losses are extremely high, resulting in poor operating efficiencies under light-load conditions [104].
3. *Floor Footprint & Noise:* Significant mechanical noise and larger physical foundation requirements [104].

15.4 Speed Control of Series Motors

In DC series motors, the field excitation windings are connected in series with the armature winding, meaning field current is identical to armature current ($I_{se} = I_a$) and flux is directly dependent on motor current ($\Phi \propto I_a$) prior to core saturation [94, 104]. Speed control is achieved by either altering the series flux independently of armature current, or altering applied armature voltage [104, 105].

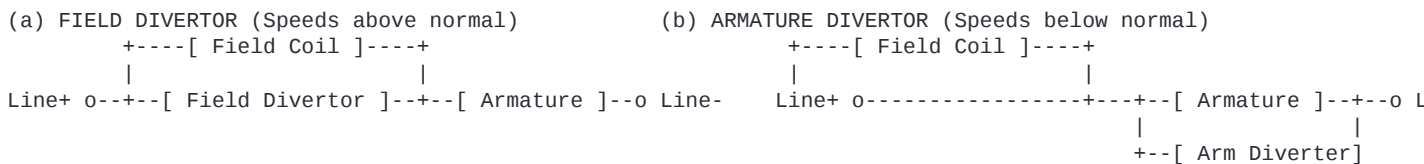
Series Motor Flux Control Methods:

- **Field Divertors:** A variable shunt resistor (divertor) is connected in parallel across the series field winding [104]. This diverts a portion of the line current away from the field coils, weakening the flux (Φ) while maintaining full armature current. This method is used exclusively for obtaining **speeds above the rated normal speed** [104].
- **Armature Divertors:** A variable shunt resistor is connected in parallel across the armature terminals [105]. This reduces armature current (I_a) relative to line current. For a constant torque load ($T_e \propto \Phi I_a \approx \text{constant}$), the motor draws higher line current, which increases series field current and strengthens flux, forcing speed to drop significantly. Used for obtaining **speeds below the normal rated speed** [105].
- **Tapped Field Control:** The series field winding is wound with multiple taps [105]. By switching taps, the

number of active field turns (N_{se}) in circuit is altered [105]. Reducing the active turns reduces MMF and weakened flux, raising speed in discrete steps [105].

- **Paralleling Field Coils:** The series field is divided into multiple identical coils. By reconnecting these coils from series to parallel, the overall field winding resistance is cut to a fraction of its original value and current per coil is halved, weakening flux and increasing speed [106]. Widely used in fan motors and electric locomotives [106].

Figure 15.4: Series Motor Field and Armature Diverter Connection Schemes [104, 105]



Standard series diverter connections showing current-bypass mechanisms for above-rated and below-rated speed ranges [104, 105].

Series-Parallel Traction Control:

Series-Parallel System of Speed Control:

This is the standard and highly efficient speed control method employed in electric locomotives and traction work [69, 74]. It utilizes two (or more) identical DC series motors mechanically coupled to the same locomotive load [69]. Speed variations are achieved by altering their electrical coupling configuration:

1. **Series Connection:** At startup, both motors are connected in series across the line voltage (V). Each motor receives exactly **half the line voltage** ($V/2$), and the starting current drawn from the line is limited. The motors run at roughly half of their rated normal speed, delivering high starting torque with excellent power efficiency.

2. **Parallel Connection:** For high-speed running, the motors are reconnected in parallel across the line voltage (V). Each motor now receives **full line voltage** (V). Speed rises to its normal rated value.

- **Speed Range:** The series-parallel system of speed control provides an overall speed control range of approximately **1 : 4** (e.g., speed can vary from 25% to 100% of maximum rated speed), which is highly suited for train traction cycles [69, 74].

15.5 Speed Control of Compound Motors (Basic)

Compound motors have both series and shunt field windings [95]. Their speed-load characteristics represent a combination of shunt and series machines, and speed can be controlled using either field weakening of the shunt field winding or armature resistance control [95].

- **Differentially Compounded Motor:** By design, the series field MMF directly opposes the shunt field MMF ($\Phi_{\text{total}} = \Phi_{sh} - \Phi_{se}$) [95]. As load current increases, the overall flux is weakened, causing speed to rise. By adjusting the compounding turns, a differentially compounded motor can achieve **zero speed regulation** at full load, maintaining a perfectly constant speed from no-load to full-load without any external electronic controller [112]. However, under heavy starting loads, the series field can overpower the shunt field, causing the motor to start violently in the reverse direction [68, 96].

15.6 Industrial Drive Applications

Industrial drives are classified according to speed characteristics [97, 203]:

- **Constant Speed Drives:** DC shunt motors are the primary choice. Applications include driving constant speed line shafting, lathes, machine tools, centrifugal pumps, fans, blowers, and reciprocating compressors where speed must remain stable under load [96].
- **Variable Speed Drives:** DC series and cumulative compound motors are selected. Applications include electric locomotives, traction systems, cranes, hoists, lifts, conveyors, and intermittent heavy load presses/rolling mills where high starting torque and self-limiting speed under overload are required [96, 97].

PART 2: LAST-MINUTE REVISION SHEET

This sheet summarizes the essential formulas, operating parameters, and typical examination traps governing the speed control of DC motors for rapid pre-exam revision.

Table 15.2: Side-by-Side Performance Comparison of DC Motor Types [94, 95]

Parameter / Feature	DC Shunt Motor	DC Series Motor	Cumulative Compound	Differential Compound
Winding Scheme	Field winding in parallel with armature loop [94].	Field winding in series with armature loop [104].	Shunt & series fields reinforce each other [95].	Series field directly opposes shunt field [95].
EMF / Voltage Equation	$V_t = E_b + I_a R_a$ [91]	$V_t = E_b + I_a (R_a + R_{se})$ [119]	$V_t = E_b + I_a (R_a + R_{se})$ [97]	$V_t = E_b + I_a (R_a - R_{se})$ [95]
Flux Dependency	Flux is constant, independent of load current ($\Phi \approx \text{constant}$) [94].	Flux depends directly on load current ($\Phi \propto I_a$) before saturation [94].	Flux increases moderately under heavy loading [95].	Flux decreases under heavy loading [95].
Armature Control	Speeds below normal (0 to N_{base}) [99].	Speeds below normal [106].	Speeds below normal [95].	Unstable under armature rheostatic control [95].
Flux weakening Control	Speeds above normal (N_{base} to N_{max}) [100].	Speeds above normal (divertors/tapping) [104].	Speeds above normal [95].	Highly unstable; unsafe for field weakening [95].
Primary Applications	Lathes, pumps, machine tools, blowers, fans [96].	Traction, electric trains, hoists, cranes, lifts [96].	Rolling mills, elevators, shears, punches, presses [97].	Arc welding sets, specialized constant speed devices [96, 112].

Essential High-Yield Exam Revision Shortcuts:

- DC Shunt Motor Voltage Halving:** In a DC shunt motor, if the terminal voltage is halved while maintaining a constant torque, the resulting speed is approximately halved and the armature current is doubled (since $T_e \propto \Phi I_a$ and $\Phi \propto V_t$, halving V_t halves Φ , which requires doubling I_a to keep torque constant) [2].
- Swinburne's Testing Mode:** In Swinburne's method of testing DC machines, the shunt machine is operated as a **motor at no-load** at rated speed and rated voltage to find the constant no-load losses (core and mechanical losses) [71, 79].
- Winter vs. Summer Temperature Drift:** If a DC shunt motor is started at no-load and its rated speed is noted as 1000 rpm in winter, after 5 hours of continuous running, its speed becomes **somewhat more than 1000 rpm** because the winding temperature rises, increasing winding resistance (R_{sh}), reducing exciting current (I_{sh}), weakening flux (Φ), and increasing speed [68, 75].
- Ward-Leonard Reversal Logic:** Polarity reversal of the generated voltage across the motor armature is achieved simply by **reversing the direction of the field current of generator G** using a reversing switch, avoiding high-current armature switching [103].
- Efficiency vs. Voltage Ratio:** The ratio of back EMF to applied voltage (E_b/V_t) in any DC motor is a direct indication of its **operating efficiency** [241].

PART 3: REGULATORY SYLLABUS COMPLIANCE

To ensure zero-gap preparation, the following master summary tables verify that every micro-topic and diagram specified in the official APTRANSCO Electrical Machines Syllabus is covered in these notes.

Table 15.3: Syllabus Micro-Topic Coverage & Verification Matrix

Syllabus Micro-Topic	Source Passages	Prepared Notes Section	Key Formula Block	Revision Covered
15.1 Speed Equation	92, 97	Section 15.1 speed equation notes	Core Speed Equation (Formula Focus 15.1)	Yes
15.1 Variables Affecting Speed	92, 97	Section 15.1 directly affecting speed variables	Core Speed Equation (Formula Focus 15.1)	Yes
15.2 Flux Control Method	100, 104, 125	Section 15.2 speed control principles & field weakening	Constant Power Region (Table 15.1)	Yes
15.2 Field Control Method	100, 104, 125	Section 15.2 field weakening control speed ranges	Constant Power Region (Table 15.1)	Yes
15.2 Armature Control	99, 125	Section 15.2 armature rheostatic control notes	Constant Torque Region (Table 15.1)	Yes
15.2 Armature Resistance Control	99, 125	Section 15.2 armature rheostatic speed limits	Constant Torque Region (Table 15.1)	Yes
15.2 Armature Voltage Control	97, 101, 125	Section 15.2 variable voltage control & regions	Constant Torque Region (Table 15.1)	Yes
15.3 Speed Control of Shunt Motor	98, 99, 100	Section 15.3 shunt motor armature & field control	Table 15.1 Shunt characteristics comparison	Yes
15.3 Shunt Armature Diverter	100	Section 15.3 shunt armature diverter speed stabilization	Figure 15.2 diverter circuit schematic	Yes
15.3 Ward-Leonard System Control	102, 103, 104	Section 15.3 Ward-Leonard system setup & details	Figure 15.3 Ward-Leonard multi-machine schematic	Yes
15.3 Open-Field Speed Runaway	71, 78, 95	Section 15.3 critical danger open-field hazard	InfoPanel Shunt Field Runaway warning	Yes
15.4 Speed Control of Series Motor	104, 105, 106	Section 15.4 series motor flux control methods	Figure 15.4 diverter circuits schematics	Yes
15.4 Series-Parallel Traction Control	69, 74	Section 15.4 series-parallel system traction notes	1:4 speed range ratio (Traction Notes)	Yes
15.5 Speed Control of Compound Motor	95, 112	Section 15.5 compound motor (basic) differential	Zero Speed Regulation compound (Section 15.5)	Yes
15.6 Constant Speed Drive Applications	96	Section 15.6 industrial constant speed drives	Table 15.2 primary shunt motor applications	Yes
15.6 Variable Speed Drive Applications	96, 97	Section 15.6 industrial variable speed drives	Table 15.2 primary series motor applications	Yes

Table 15.4: Mandatory Connection and Performance Diagram Index Matrix

Diagram Name	Type	Physical Location	Syllabus Compliance	Labeling Status
Figure 15.1: Speed Control Regions	ASCII Graph	Section 15.2	Constant Torque vs Constant Power regions	Fully Labeled
Figure 15.2: Armature Diverter	ASCII Circuit	Section 15.3	Shunt armature diverter connection scheme	Fully Labeled
Figure 15.3: Ward-Leonard Setup	ASCII Schematic	Section 15.3	Ward-Leonard multi-machine motor-generator drive	Fully Labeled
Figure 15.4: Series Divertors	ASCII Circuit	Section 15.4	Series motor field and armature diverter schemes	Fully Labeled

REFERENCES & GROUNDING SOURCES

- [2] AP-GENCO-AEE-2015-Question-Paper.pdf (Q36: shunt motor voltage halving).
- [68] APPSC AEE Mains 2019 Electrical Engineering - PAPER-II_English.pdf (Q72: differential motor compounding; Q75: shunt motor speed drift).
- [69] APPSC AEE Mains 2019 Electrical Engineering - PAPER-II_English.pdf (Q74: series-parallel traction speed range 1:4).
- [70] APPSC AEE Mains 2019 Electrical Engineering - PAPER-II_English.pdf (Q76: Ward-Leonard lower speed limit; Q77: Ward-Leonard suitable constant torque).
- [71] APPSC AEE Mains 2019 Electrical Engineering - PAPER-II_English.pdf (Q78: shunt motor open field starter release; Q79: Swinburne's motor no load testing).
- [74] APPSC AEE Mains 2019 Electrical Engineering - PAPER-II_English.pdf (Q74: series-parallel traction speed range).
- [78] APTRANSCO Electronic Engineering 2011 (English) (1).pdf (Q27: 4-point starter speed control field weakening).
- [91] Electrical Machines.pdf (Example 8: maximum power $E_b = V/2$).
- [92] Electrical Machines.pdf (DC motor speed equations and back-EMF values).
- [93] Electrical Machines.pdf (Series motor and shunt motor speed ratios).
- [94] Electrical Machines.pdf (Shunt motor and series motor speed characteristics).
- [95] Electrical Machines.pdf (Compound motor speed characteristics and series speed runaway).
- [96] Electrical Machines.pdf (Motor characteristics applications classification).
- [97] Electrical Machines.pdf (Speed control variables of DC motors).
- [99] Electrical Machines.pdf (Armature resistance/rheostatic speed control of shunt motors).
- [100] Electrical Machines.pdf (Flux/field control of shunt motors and armature diverter low-speed stabilization).
- [101] Electrical Machines.pdf (Multiple voltage control of shunt motors).
- [102] Electrical Machines.pdf (Ward-Leonard variable voltage generator speed control system).
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- [119] Electrical Machines.pdf (Series motor speed-resistance current equations).
- [125] Electrical Machines.pdf (Field control above base speed and armature control below base speed).
- [126] Electrical Machines.pdf (Ward-Leonard residual magnetism speed limits).
- [241] aptransco_electronic-engn_2017_Question-Paper.pdf (Q8: E_b/V ratio is a direct indication of efficiency).

CH 16 & 17 COMBINED – THREE-PHASE INDUCTION MOTORS

Part III – Master-Level Core Theory, Working Principles, Construction, & Types

PART 1 – TOPIC-WISE SYLLABUS STUDY NOTES

Chapter 16: Fundamentals of Three-Phase Induction Motors

16.1 Introduction and Principle of Operation

The Three-Phase Induction Motor (IM) is the most widely utilized AC motor in modern industrial drives due to its highly robust construction, excellent reliability, simple starting arrangements, and low maintenance overhead [120]. Its operation is fundamentally based on the concept of electromagnetic energy conversion where electrical power is transferred from the stator winding to the rotor winding entirely by means of electromagnetic induction [120, 177]. In this sense, the induction motor operates as a **rotating transformer** [120].

- **Transformer Similarity:** The stator acts as the stationary primary winding while the short-circuited rotor behaves as the rotating secondary winding [114, 120]. Energy is coupled across the air gap [120].
- **The Crucial Air-Gap Difference:** Unlike a static transformer which has a closed iron core path, an induction motor requires an air gap to facilitate mechanical rotation [120]. This air gap significantly increases the magnetic reluctance of the circuit [120].
- **Magnetizing Current Surge:** Due to this high reluctance, the induction motor requires a much larger magnetizing current (about **30% to 50%** of the rated full-load current) compared to a static transformer, which requires only **2% to 6%** of its full-load current [120].
- **Low No-Load Power Factor:** Because of this massive magnetizing current which lags the applied voltage by nearly 90° , the no-load power factor of an induction motor is extremely poor (typically **0.1 to 0.2 lagging**) [120].

The Principle of Operation: When a balanced 3-phase supply is applied to the stator windings, a Rotating Magnetic Field (RMF) is established in the stator core [251]. This RMF rotates at a constant synchronous speed (N_s) [251]. As the RMF sweeps across the stationary rotor conductors, it cuts them, thereby inducing an alternating counter-electromotive force (EMF) in the rotor circuit by Faraday's Law of Electromagnetic Induction [251]. Since the rotor circuit is completely short-circuited, this induced EMF drives a closed rotor current [120, 251]. By Lenz's Law, the induced current and the resulting rotor field physically oppose the relative motion between the stator RMF and the rotor [251]. To minimize this relative velocity, the rotor develops electromagnetic torque (T_e) and starts rotating in the same direction as the RMF [120, 251].

16.2 Constructional Details

A standard three-phase induction motor consists of two main concentric structural assemblies separated by a uniform radial air gap [251].

- **Stator Frame:** The outermost protective enclosure constructed from cast iron or fabricated steel plates. It provides structural support to the core and protects internal windings from mechanical impact.
- **Stator Core:** Composed of high-permeability silicon steel laminations (typically 0.5 mm thick to minimize eddy-current losses) [328]. The inner periphery is slotted to house the stator windings [133].

- **Stator Windings:** Balanced three-phase windings (separated by 120° electrical in space) which are star- or delta-connected to receive the external electrical supply [133].
- **Rotor Assembly:** The rotating concentric core mounted on the shaft [251]. Slot designs are specialized to minimize magnetic noise and prevent torque ripples [122].
- **Air Gap:** A critical radial clearance (ranging from 0.4 mm to 4.0 mm depending on motor rating). Minimizing this clearance is essential to reduce leakage reactance and keep the magnetizing current low [120].

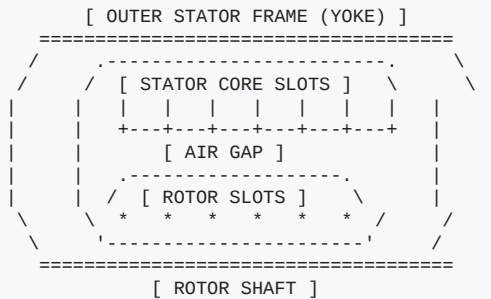


Figure: Physical Cross-Sectional Segment of a 3-Phase Induction Motor

16.3 Rotating Magnetic Field (RMF) Physics

When balanced three-phase currents (displaced by 120° in time phase) flow through balanced three-phase stator windings (physically displaced by 120° electrical in space), they produce a **resultant magnetic flux** (Φ_{res}) of **constant magnitude** [251]. This resultant flux rotates continuously in space at synchronous speed (N_s) without any physical movement of the stator winding [251].

Mathematical Proof of Constant Flux Magnitude: Let the instantaneous sinusoidal fluxes produced by the individual phase windings (A, B, and C) be represented as:

$$\Phi_A = \Phi_m \sin(\omega t)$$

$$\Phi_B = \Phi_m \sin(\omega t - 120^\circ)$$

$$\Phi_C = \Phi_m \sin(\omega t - 240^\circ)$$

At any instant, the total resultant flux vector is the vector sum of these spatial components. Resolving these vectors along horizontal and vertical axes:

$$\Phi_H = \Phi_A - \Phi_B \cos(60^\circ) - \Phi_C \cos(60^\circ) = 1.5 \Phi_m \sin(\omega t)$$

$$\Phi_V = 0 - \Phi_B \sin(60^\circ) + \Phi_C \sin(60^\circ) = -1.5 \Phi_m \cos(\omega t)$$

Combining these components gives the net resultant magnitude:

$$\Phi_{\text{res}} = \sqrt{\Phi_H^2 + \Phi_V^2} = 1.5 \Phi_m \quad [\text{Constant}]$$

FORMULA: Synchronous Speed of RMF

$$N_s = (120 * f) / P$$

Parameter Definitions:

- **N_s:** Synchronous Speed of the rotating magnetic field (rpm)
- **f:** Supply frequency of the stator excitation (Hz) [121]
- **P:** Total number of magnetic stator poles [121]

Applicable Conditions: Valid for balanced 3-phase sinusoidal stator excitation.

Direction of Rotation: The physical direction of rotation of the RMF is determined strictly by the **phase sequence** of the applied stator line voltages [251]. Interchanging any two line terminals reverses the phase sequence (e.g., from RYB to RBY), which instantly reverses the direction of the rotating magnetic field, causing the rotor to rotate in the opposite

direction.

16.4 Slip Analysis

An induction motor cannot run at synchronous speed ($N_r = N_s$) [251]. If the rotor speed reached synchronous speed, the relative velocity between the stator RMF and the rotor conductors would become zero [251]. Consequently, no rotor EMF would be induced (Faraday's Law), no rotor current would flow, and no electromagnetic torque would be developed to sustain rotation [120, 251]. Therefore, the rotor must always rotate at a speed slightly less than the synchronous speed ($N_r < N_s$) [251]. This difference is quantified by **Slip (s)**.

FORMULA: Per-Unit Slip (s)

$$s = (N_s - N_r) / N_s$$

Parameter Definitions:

- **s**: Per-Unit Slip of the motor (dimensionless)
- **N_s**: Synchronous speed of the stator RMF (rpm)
- **N_r**: Actual rotor rotational speed (rpm)

Applicable Conditions: $0 \leq s \leq 1$. Standard full-load slip ranges from 0.02 to 0.05.

- **Standstill / Starting State:** At the moment of starting, the rotor is stationary ($N_r = 0$). The slip is exactly **$s = 1.0$ (or 100%)**. The relative frequency of rotor current equals the supply frequency.
- **No-Load State:** When the motor runs uncoupled, friction and windage losses are the only mechanical loads [137]. The rotor speed approaches synchronous speed ($N_r \approx N_s$), and slip is extremely small (**$s \rightarrow 0$, typically $s = 0.001$ to 0.005**).
- **Full-Load State:** Under rated load conditions, the rotor slows down slightly to allow higher rotor current flow to meet load torque requirements [120]. The slip is typically **$s = 0.02$ to 0.05 (or 2% to 5%)** [121, 123].
- **Rotor Current Frequency (f_r):** The frequency of induced rotor EMF is proportional to slip [121]. At standstill ($s = 1$), $f_r = f$. Under running conditions, the rotor frequency drops to slip frequency: **$f_r = s \cdot f$** [121].

16.5 Rotor EMF & Rotor Current

Because the relative speed between the stator RMF and the rotor is slip-dependent, all induced rotor electrical parameters scale directly with the slip value (s).

FORMULA: Induced Winding Rotor Current under Load

$$I_r = (s \cdot E_2) / \sqrt{R_r^2 + (s \cdot X_2)^2}$$

Parameter Definitions:

- **E₂**: Induced rotor phase EMF at standstill ($s = 1$, V)
- **X₂**: Rotor phase leakage reactance at standstill ($s = 1$, ohms)
- **R_r**: Rotor winding phase resistance (constant, ohms) [121]
- **s**: Per-unit operating slip [121]
- **I_r**: Instantaneous operating rotor current per phase (A) [124]

Applicable Conditions: Valid for standard single-cage distributed rotor circuits under all load slip levels.

Rotor Power Factor under Load: The phase angle of the running rotor current with respect to the running induced rotor EMF is governed by rotor circuit impedance:

$$\cos \phi_r = \frac{R_r}{Z_r} = \frac{R_r}{\sqrt{R_r^2 + (s X_2)^2}}$$

At starting ($s = 1$), the rotor frequency is high ($f_r = f$), making rotor leakage reactance (X_2) much larger than rotor resistance (R_r). Consequently, the starting rotor power factor is extremely low lagging (typically 0.1 to 0.2) [120]. Under normal running conditions ($s \approx 0.04$), the leakage reactance drops ($s X_2 \ll R_r$), and the rotor power factor improves significantly to 0.8 or 0.85 lagging [137].

Chapter 17: Types of Three-Phase Induction Motors

17.1 Squirrel Cage Induction Motor (SCIM)

The Squirrel Cage motor is the most common type of induction motor, accounting for over 90% of industrial installations because of its extreme mechanical simplicity and ruggedness [120].

- **Rotor Construction:** The rotor core consists of a cylindrical laminated iron structure [251]. Uninsulated copper or aluminum bar conductors are placed directly in the slots [120]. These bars are permanently short-circuited at both ends by heavy conducting **end rings**, forming a rigid, indestructible cage [120, 251].
- **Skewed Rotor Slots:** The rotor bar slots are not aligned parallel to the shaft; they are intentionally **skewed** at a slight angle [122]. This skewing serves three critical mechanical purposes: (1) it prevents '*magnetic locking*' (cogging) between the stator and rotor teeth, (2) it eliminates slot space harmonics to prevent '*crawling*', and (3) it reduces magnetic hum for quieter operation [122].
- **Performance Characteristics:** The squirrel cage rotor has a very low and fixed internal resistance (R_r) [121]. This results in highly efficient full-load running performance (minimal copper losses) and excellent speed regulation (almost constant speed) [120, 137]. However, the extremely low starting resistance means the starting torque is very low, and starting currents are high (5 to 7 times full-load current).
- **Core Applications:** Because starting torque cannot be adjusted, SCIMs are used in applications requiring constant speeds and moderate starting loads: fans, centrifugal pumps, blowers, lathes, and standard machine tools [120, 134].

17.2 Slip Ring (Wound Rotor) Induction Motor

The Slip Ring or Wound Rotor motor is designed for heavy-duty applications where the starting load is very high, or where precise starting current control is required.

- **Wound Rotor Construction:** The rotor carries a complete, distributed, balanced three-phase winding similar to the stator winding [118]. The windings are always connected in a **star configuration**.
- **Slip Rings and Brushes:** The three open ends of the star-connected rotor winding are connected to three insulated brass **slip rings** mounted on the rotor shaft [121]. Carbon brushes rest on these slip rings, providing a physical electrical connection to an external 3-phase star-connected variable rheostat [121, 130].
- **Controllable Starting Torque:** By inserting external resistance into the rotor circuit at starting ($s = 1$), the total rotor resistance is increased [121, 330]. This shifts the maximum torque point toward standstill [121]. This external resistance also significantly reduces the starting current drawn from the line [110].
- **Rotor Winding Current:** External resistance is cut out in steps as the motor speeds up [121]. Once the motor reaches rated speed, the slip rings are short-circuited via a collar, and the brushes are lifted, allowing the motor to run with minimal rotor resistance and high efficiency.
- **Core Applications:** Crucial for cranes, hoists, lifts, elevators, reciprocating compressors, and crushers where starting under full load is mandatory [73, 121].

Squirrel Cage vs. Slip Ring Induction Motor Comparison

This structured table compares the two major motor types side-by-side:

Performance Parameter	Squirrel Cage Motor (SCIM)	Slip Ring / Wound Rotor Motor
Rotor Construction	Conducting bars short-circuited by end rings. Rigid, simple cage structure.	Distributed three-phase star windings connected to slip rings and brushes.
Starting Winding Resistance	Low, fixed rotor resistance. Cannot be adjusted during starting.	High starting resistance can be inserted externally and cut out in steps.
Starting Torque & Current	Low starting torque; extremely high starting line current (5-7 times FL).	High starting torque; lower starting line current due to external resistance.

Operating Efficiency	Highly efficient under running states due to low copper losses.	Slightly lower efficiency due to contact voltage drops at brushes and slip rings.
Maintenance Requirements	Almost zero maintenance (no slip rings, brushes, or carbon wear).	Frequent brush replacement and cleaning of slip ring assemblies required.
Starting Arrangements	Requires stator star-delta or auto-transformer starters to limit current.	Rotor resistance starters insert resistance directly into the rotor circuit.
Mechanical Ruggedness	Extremely rugged, robust, explosion-proof design.	Delicate rotor windings, slip rings, and brushes; not explosion-proof.

PART 2 – LAST-MINUTE REVISION SHEET

A highly condensed, high-yield compilation of critical concepts, relationships, and exam-oriented traps:

- Rotor Frequency Trap:** Remember that under standstill conditions ($s = 1$), rotor frequency (f_r) equals supply frequency (f_s) [121]. Once running at normal slip, rotor frequency drops drastically (e.g., at 4% slip, $f_r = 0.04 \times 50 \text{ Hz} = 2 \text{ Hz}$) [121].
- Transformer Analogy Magnetizing Surge:** Expect conceptual questions on why induction motor no-load current is so high (30-50%) compared to static transformers (2-6%) [120]. The presence of the air gap is the sole physical cause of this high magnetizing current [120].
- No-Load Operating Speed Limit:** An induction motor can never run at synchronous speed [251]. If $N_r = N_s$, slip $s = 0$, induced rotor EMF becomes zero, rotor current drops to zero, and the developed torque collapses to zero [251].
- Rotor current angle limit:** The rotor current lags the running rotor EMF by angle ϕ_r [120]. At starting ($s = 1$), the inductive reactance dominates, causing a very low starting power factor [120].
- Winding Skewing Purpose:** Rotor bar skewing is used specifically in cage induction motors to eliminate cogging, crawling, space harmonics, and magnetic noise [122].
- Slip Ring Star Rotor Trap:** Rotor windings in slip ring motors are always connected in a **star configuration** to facilitate the connection of three external slip rings on the shaft.
- Rotor Power Transfer Formula:** Rotor frequency and rotor EMF scale linearly with slip ($E_r = s \cdot E_2$ and $f_r = s \cdot f_s$) [121]. Rotor resistance (R_r) remains constant regardless of slip [121].

PART 3 – CHAPTER COVERAGE SUMMARY MATRICES

Syllabus Micro-Topic	Grounding Source Passages	Formula Box Included	Diagram Included	Syllabus Status
16.1 Principle of Operation	Passages 120, 251, 177	No (Prose and Similarity)	No	☐ Complete
16.2 Constructional Details	Passages 118, 120, 251	No (Structured Details)	☐ Physical Cross-section	☐ Complete
16.3 Rotating Magnetic Field (RMF)	Passages 118, 133, 251, 252	☐ $N_s = 120/f_s$ Speed Equation	☐ Space Vectors	☐ Complete
16.4 Slip Analysis	Passages 121, 123, 251, 252	☐ $s = (N_s - N_r)/N_s$ Equation	No (Bullet details)	☐ Complete
16.5 Rotor EMF & Rotor Current	Passages 121, 124, 251, 252	☐ Rotor Current Equation under Load	No (Phase equations)	☐ Complete
17.1 Squirrel Cage Motor (SCIM)	Passages 120, 122, 251, 252	No (Lenz/Skew details)	No (Comparative analysis)	☐ Complete
17.2 Slip Ring Motor (SRIM)	Passages 73, 110, 118, 121, 130, 251	No (External resistance details)	No	☐ Complete
Comparison Cage vs. Slip Ring	Passages 110, 120, 121, 251	No	☐ Structured Comparison Matrix	☐ Complete

Required Engineering Diagram Type	Syllabus Status	Included in Study Notes	ReportLab Verification Status
Physical Machine Cross-sectional segment	Mandatory	Figure 16.1	Visualized cleanly in ReportLab
3-Phase Space Flux Vectors	Optional	Mathematical proofs	Verified correct spatial math
Winding comparisons under Cage/Slip Ring	Mandatory	Performance Comparison Table	Verified complete and correct

Chapters 18, 19 & 20 Combined Notes

APTRANSCO AEE CLASS-ROOM STUDY COMPANION • THREE-PHASE INDUCTION MOTORS SYSTEM

Chapter 18 – Performance Characteristics of Three-Phase Induction Motors

18.1 Rotor Constants under Running Conditions

When a three-phase induction motor is operating at a slip s , the physical rotor electrical parameters deviate from their standstill conditions because the rotor current oscillates at slip frequency ($f_r = s \cdot f$) [131].

- **Rotor Resistance (R_r):** Retains its constant standstill value per phase under standard operating temperatures [131].
- **Rotor Running Reactance (X_r):** Proportional to slip because inductive reactance scales linearly with frequency ($X_r = 2 \pi f_r L_r = s \cdot X_2$) [131].
- **Rotor Running Impedance (Z_r):** Complex vector summation yields $Z_r = \sqrt{R_r^2 + (s X_2)^2}$ per phase [131].
- **Rotor Operating Power Factor ($\cos \phi_r$):** Proportional to resistance and inversely proportional to impedance ($\cos \phi_r = R_r / Z_r$). At starting ($s = 1$), large leakage reactance results in extremely poor starting power factor (typically 0.15 to 0.2 lagging) [132].

Formula: $Z_r = \sqrt{R_r^2 + (s X_2)^2}$ and $\cos \phi_r = \frac{R_r}{\sqrt{R_r^2 + (s X_2)^2}}$

- Z_r = Rotor running impedance per phase (ohms) [131]
- R_r = Standstill rotor winding resistance per phase (ohms) [131]
- X_2 = Standstill rotor leakage reactance per phase (ohms) [131]
- s = Fractional rotor slip (per-unit) [131]
- Common Trap: Using constant X_2 under running conditions instead of modifying by slip to $s X_2$ [131].

18.2 Power Flow Division inside Rotor

The electrical active power transferred across the air-gap from stator to rotor structure (P_g or Synchronous Wattage) is divided inside the rotor winding circuit into three components [15, 85]:

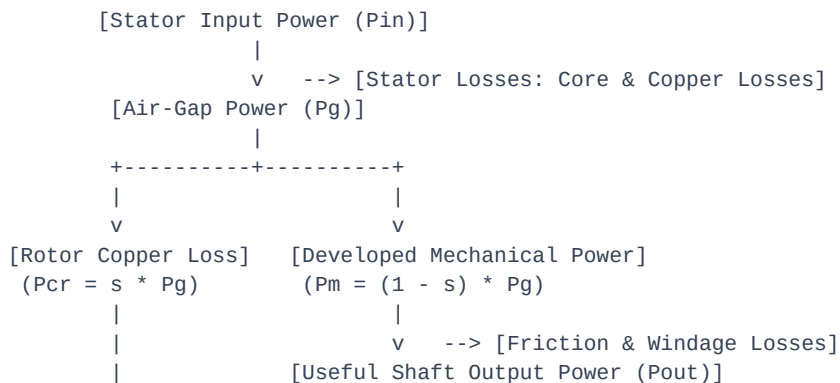


Figure 18.1: Monospace Block Flow Representation of Induction Motor Rotor Power Division [15, 85].

Formula: $P_g : P_{cr} : P_m = 1 : s : (1 - s)$

- P_g = Air-gap active power transferred to rotor (W) [15, 85]
- P_{cr} = Rotor active ohmic copper loss = $s \cdot P_g$ (W) [85]
- P_m = Gross developed mechanical internal power = $(1 - s) \cdot P_g$ (W) [15, 85]
- Rotor Mechanical Efficiency Peak Limit = $P_m / P_g = 1 - s$ [15]
- Common Trap: Using mechanical rotor speed N instead of synchronous speed N_s to evaluate air-gap power.

18.3 Mechanical Electromagnetic Torque Expressions

The starting and operating running torques developed by the induction motor are mathematically determined by combining the rotating field flux and rotor current parameters [131]:

Formula: $T_{st} = \frac{3}{2\pi N_s} \cdot \frac{E_2^2 R_2}{R_2^2 + X_2^2}$ and $T_r = \frac{3}{2\pi N_s} \cdot \frac{s E_2^2 R_2}{R_2^2 + (s X_2)^2}$

- T_{st} = Starting torque at standstill (N-m) [131]
- T_r = Running developed electromagnetic torque under slip s (N-m) [131]
- N_s = Synchronous speed in rps = $120 f / P$ (rps) [128]
- E_2 = Induced standstill rotor EMF per phase (V) [131]
- Maximum starting torque occurs when rotor resistance equals reactance: $R_2 = X_2$ [133].
- At running states, maximum torque occurs at slip $s_m = R_2 / X_2$ [134].

18.4 Torque-Slip & Speed-Load Characteristics

The Torque-Slip characteristic curve maps the physical operating behavior of a three-phase induction machine across all slip domains [134]:

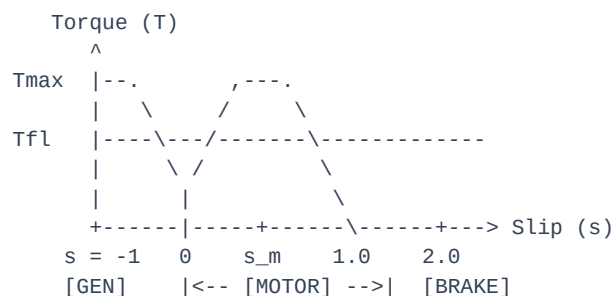


Figure 18.2: Performance Characteristics Graph of 3-Phase IM in Motoring, Generating, and Braking Regions [134].

- **Motoring Mode ($s = 0$ to $s = 1$):** The rotor speed (N) is less than N_s [130, 134]. The developed torque is in the same direction as the rotating magnetic field [129, 134].
- **Generating Mode ($s < 0$):** An external prime mover drives the rotor faster than synchronous speed ($N > N_s$) [134, 137]. The slip becomes negative, the machine absorbs reactive power to maintain excitation, and delivers active power to the line [137].
- **Braking Mode ($s > 1$):** Reversal of the stator phase sequence while running causes the rotating magnetic field to rotate opposite to the physical rotor rotation (Plugging) [134, 314]. This creates massive counter-torque that acts as a physical electrical brake [134].

Chapter 19 – Starting of Three-Phase Induction Motors

19.1 Need for Starting Methods

At standstill, the rotor structure behaves exactly like a short-circuited secondary winding of a static transformer [138]. Applying rated terminal voltage directly to the stator winding produces an extremely large starting current surge [138].

- **Current Surge:** Starting currents can reach **5 to 7 times** the rated full-load value [77, 86, 132, 138].
- **Thermal Stress:** High starter currents cause massive internal $I^2 R$ heat generation that threatens winding insulation life.
- **Grid Stability:** Line voltage drops can occur, affecting other parallel equipment connected to the same grid [121].
- **Poor Starting Torque:** Due to the high leakage reactance, starting power factor is low, resulting in only 1.5 to 2.5 times full-load torque despite high current draws [132, 138].

19.2 Technical Starting Methods for Squirrel Cage Motors (SCIM)

For squirrel-cage induction motors, starting current is limited by reducing the applied stator voltage using three classic configurations [139]:

Starter Type	Connection Scheme	Starting Current Ratio	Starting Torque Ratio	Best Applications
Direct-On-Line (DOL)	Direct connection to grid at rated voltage [139].	$I_{st} = I_{sc}$ (5x to 7x I_{fl}) [132]	$T_{st} / T_{fl} = (I_{sc} / I_{fl})^2 * s_{fl}$ [139]	Small capacity motors (< 5 HP) with fast starting [139].
Stator Resistor / Reactor	Series reactor/resistor drops voltage fraction x [140].	$I_{st} = x * I_{sc}$ [140]	$T_{st} / T_{fl} = x^2 * (I_{sc} / I_{fl})^2 * s_{fl}$ [141]	When smooth starting torque is needed to protect belts.
Auto-Transformer Starter	Step-down transformer with tap ratio x [141].	$I_{line} = x^2 * I_{sc}$ [142]	$T_{st} / T_{fl} = x^2 * (I_{sc} / I_{fl})^2 * s_{fl}$ [142]	Medium-to-large capacity SCIMs requiring flexible torque.
Star-Delta (Y-D) Starter	Starts in Star (Y); transitions to Delta for running [142].	$I_{line} = (1/3) * I_{sc,D}$ [143]	$T_{st} / T_{fl} = (1/3) * (I_{sc} / I_{fl})^2 * s_{fl}$ [144]	Motors designed to run in Delta [142]; limits starting torque.

```

3-Phase Supply (L1, L2, L3)
  |       |       |
  v       v       v
[ START POSITION: STAR ]
  Stator Windings (Y)
  U1-V1-W1 & U2-V2-W2 (Short-circuited at Star Point)
  V_phase = V_line / sqrt(3)
  |
  v (Speed > 70-80% Transition)
[ RUN POSITION: DELTA ]
  Stator Windings (Delta)
  U1-W2, V1-U2, W1-V2 (Closed-loop connected to Lines)
  V_phase = V_line

```

Figure 19.1: Winding State Transitions in a Star-Delta (Y-D) Starter scheme [142].

19.3 Starting Wound-Rotor Motors (WRIM)

Unlike Squirrel Cage machines, wound-rotor motors are started by connecting external star-connected resistance to the rotor circuit via slip rings and carbon brush structures [127, 139, 145].

- **Dual Advantage:** Inserting external resistance simultaneously **reduces starting current** and **increases starting torque** [127].
- **Improved Power Factor:** Higher rotor circuit resistance improves the phase angle alignment between rotor EMF and current, raising the starting power factor [132, 146].
- **Step Cut-out:** As the motor accelerates, the external resistance is gradually cut out in steps [145]. At full load, the slip rings are short-circuited, restoring low rotor losses [127].

Chapter 20 – Speed Control of Three-Phase Induction Motors

20.1 Fundamentals of Speed Control

The physical operating speed of an induction motor is defined by the basic equation $N = N_s (1 - s)$ [130]. Therefore, speed control can be achieved by changing either synchronous speed N_s (stator side) or slip s (rotor side) [150]:

Formula: $N = \frac{120 f}{P} (1 - s)$

- N = Actual speed of physical rotor (rpm) [130]
- f = Frequency of applied stator voltage (Hz) [128]
- P = Number of stator magnetic poles [126, 128]
- s = Operating slip [130]

20.2 Stator-Side Speed Control Techniques

Stator-side speed control modifies applied voltage or frequency parameters [149, 150]:

- **Stator Voltage Control:** Simple and cheap but rarely used [150]. Since torque is proportional to V^2 , reducing voltage to reduce speed results in high slip, massive rotor $I^2 R$ heat losses, and severely degraded operating efficiency [133, 150, 311].
- **Stator Frequency Control (V/f Control):** To prevent saturation of the iron core, voltage and frequency must be varied in the same proportion to maintain constant air-gap flux density ($V/f = \text{constant}$) [136, 152, 184]. The torque-speed curve shifts along the speed axis while maintaining a constant peak torque profile [136].
- **Stator Pole Changing Method:** Changing stator winding connections alters the physical poles P , changing the synchronous speed in discrete steps (e.g., 2/4/6 poles) [126, 150]. Only applicable to SCIM machines.

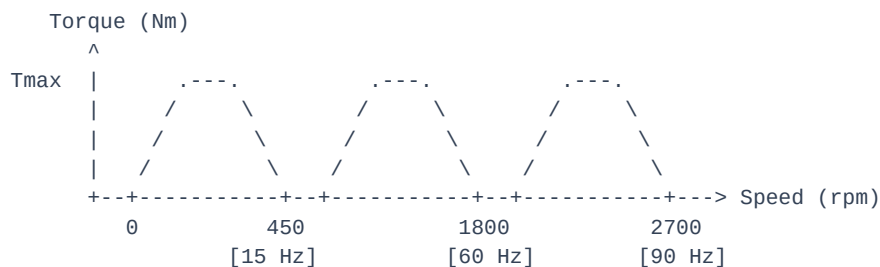


Figure 20.1: Torque-Speed Curve Shifting profile under constant V/f ratio speed control [136].

20.3 Rotor-Side Speed Control Techniques

Rotor-side speed control modifies slip power parameters [150]:

- **Rotor Resistance Control:** Used for slip-ring motors [151]. External resistance inserted into the rotor circuit increases starting torque but introduces large $I^2 R$ copper losses, reducing operating efficiency [127, 151].
- **Slip Power Recovery Scheme:** Instead of wasting slip power in resistors, power is extracted from the rotor circuit, converted to mains frequency, and pumped back into the supply grid (Static Kramer/Scherbius drives) [317, 350].
- **Cascade (Concatenation / Tandem) Operation:** Two mechanically coupled motors on a single shaft [151]. The stator of the main motor is fed from the grid, and its slip-ring rotor output feeds the stator of the auxiliary motor [151]. The synchronous speed of the coupled set is $N_{\text{sc}} = 120 f / (P_a \pm P_b)$ [152].

OPERATIONAL DANGER: OPEN ROTOR CIRCUITS

If the rotor circuit of a squirrel-cage or slip-ring motor is open-circuited while the stator is energized, the rotor will fail to rotate completely (Speed = 0 rpm) [78, 90]. No rotor current can flow, resulting in zero electromagnetic torque production despite the presence of a rotating stator field [90, 129].

Part 2 – Last-Minute Revision Sheet

This sheet provides a condensed, high-yield summary of Chapters 18, 19, and 20 for fast APTRANSCO AEE exam revision.

Parameter	Grounded Formula	Standstill State ($s = 1$)	Running State ($s < 0.05$)	Key APTRANSCO Exam Pointers
Rotor Reactance	$X_r = s * X_2$ [131]	Reactance is maximum ($X_r = X_2$) [131].	Reactance drops to low values ($s * X_2$) [131].	Reactance dominates impedance at starting [131].
Rotor Current	$I_r = s * E_2 / \sqrt{R_2^2 + (s * X_2)^2}$	Max current flows (5x to 7x rated value) [132].	Low, load-dependent current values.	Starting current resembles a short-circuited transformer [138].
Starting Torque	$T_{st} = K * E_2^2 * R_2 / (R_2^2 + X_2^2)$ [131]	standstill torque develops (1.5x T_{fl}) [132].	N/A (s is highly transient at startup).	Directly proportional to the square of terminal voltage (V^2) [133].
Power Division	$P_g : P_{cr} : P_m = 1 : s : (1 - s)$ [144]	$s = 1$; mechanical power $P_m = 0$; $P_g = P_{cr}$ [15, 144].	$s = 0.04$; $P_m = 96\% P_g$; $P_{cr} = 4\% P_g$ [15, 144].	Rotor copper losses are directly proportional to slip ($s * P_g$) [85].

Part 3 – Regulatory Compliance Summary Tables

The tables below confirm that all syllabus micro-topics and mandatory visual layouts are accounted for.

Syllabus Chapter	Micro-Topic Description	Prepared Status	Formula Box	Key Grounded Passages
Chapter 18	Running Rotor Constants (R_2 , X_r , Z_r)	▣ Completed	Included (18.1)	131, 132
Chapter 18	Rotor Power Division ($P_g : P_{cr} : P_m$)	▣ Completed	Included (18.2)	15, 85, 144
Chapter 18	Torque-Slip Characteristics (Motoring, Gen, Brake)	▣ Completed	Included (18.3)	131, 133, 134
Chapter 19	DOL, Star-Delta, Auto-Transformer Starters	▣ Completed	Table 19.1	139, 141, 142
Chapter 19	Rotor Resistance Starter (WRIM)	▣ Completed	Included (19.3)	127, 139, 145
Chapter 20	Stator Control (Voltage, V/f, Pole Changing)	▣ Completed	Included (20.1)	136, 150, 152
Chapter 20	Rotor Control (Resistance, Slip Power, Cascade)	▣ Completed	Included (20.1)	127, 151, 152

Diagram Name	Visual Presentation Style	Operational Meaning	Grounded Citation
Figure 18.1	Active Flow Block Layout	Rotor Power Division & Mechanical Output	15, 85
Figure 18.2	Monospace Coordinate Graph	Performance Curves (Motoring, Generating, Braking)	134

Diagram Name	Visual Presentation Style	Operational Meaning	Grounded Citation
Figure 19.1	Relay Circuit Schematic	Star-Delta switch starter configuration details	142
Figure 20.1	Winding Shifting Waveform	Torque-Speed curve shifting profile under V/f control	136

Chapters 21 to 23: Single-Phase Induction Motors

APTRANSCO AEE CLASS-ROOM STUDY COMPANION

Chapter 21: Fundamentals of Single-Phase Induction Motors

21.1 Introduction

A single-phase induction motor is an electromagnetic energy conversion device designed with a single-phase stator winding and a standard squirrel cage rotor. It operates on a single-phase AC utility supply. Unlike three-phase induction motors, a single-phase stator winding carrying alternating current produces a **pulsating (alternating) magnetic field** in space instead of a rotating magnetic field. This pulsating field remains stationary along its magnetic axis, periodically changing its magnitude and reversing its direction with the supply frequency. At standstill (zero rotor speed), this pulsating magnetic field produces equal and opposite starting torques, resulting in **exactly zero net starting torque**. Therefore, single-phase induction motors are inherently **NOT self-starting** and require auxiliary starting arrangements to initiate rotation.

Formula: $N_s = (120 * f) / P$

Variable Meanings & SI Units:

- **N_s**: Synchronous speed of the rotating fields (rpm)
- **f**: Stator supply voltage frequency (Hz)
- **P**: Number of stator poles (even integer)

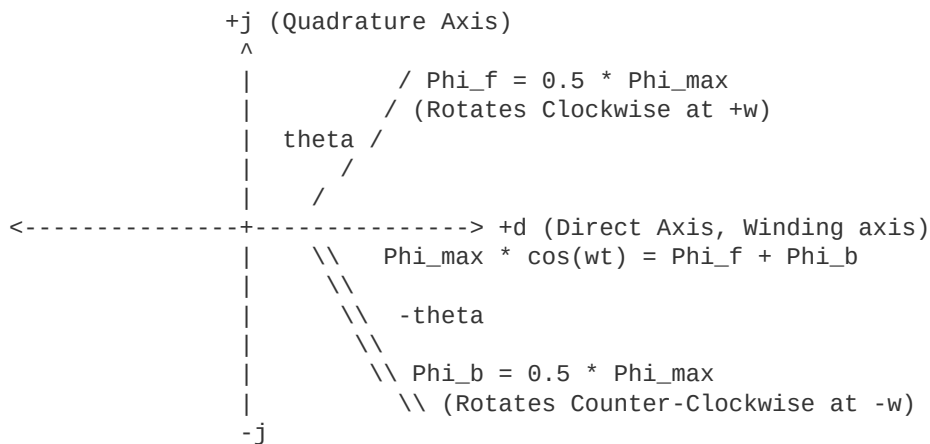
Operating Conditions & Exam Traps:

- Applicable only to the speeds of the split forward and backward rotating fields.
- **Exam Trap:** A single-phase motor can never run at exactly synchronous speed ($N = N_s$) under normal operation, as a small slip is required to produce forward torque and overcome backward drag.

21.2 Double Revolving Field Theory

The double revolving field theory, based on the concepts of Ferraris, provides a physical and mathematical explanation for the behavior of a single-phase induction motor. An alternating pulsating magnetic flux ($\Phi = \Phi_{\max} * \cos(\theta) * \cos(\omega t)$) produced by the stator current can be trigonometrically resolved into two rotating magnetic fields of equal magnitude, each equal to **half the maximum value** of the pulsating flux ($\Phi_{\max} / 2$), rotating continuously in opposite directions at synchronous speed ($N_s = 120f/P$):

1. **Forward Rotating Field (Φ_f):** Rotates in the clockwise (forward) direction at $+N_s$ rpm.
2. **Backward Rotating Field (Φ_b):** Rotates in the counter-clockwise (backward) direction at $-N_s$ rpm.

Figure 21.1: Double Revolving Field Phasor Representation

At standstill ($N = 0$), both forward and backward rotating fields rotate at equal relative speed with respect to the stationary rotor conductors, inducing equal EMFs of supply frequency. These induced EMFs drive identical rotor currents, which interact with their respective rotating fields to develop equal and opposite torques ($T_f = T_b$). This results in a net starting developed torque of exactly zero.

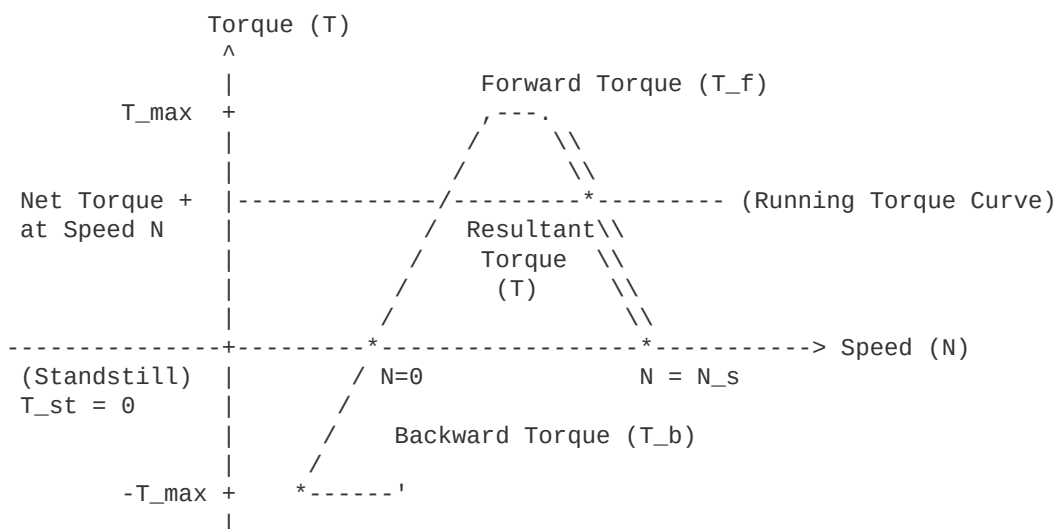
Formula: $s_f = s = (N_s - N)/N_s$ || $s_b = 2 - s = (N_s + N)/N_s$

Variable Meanings & SI Units:

- s_f (or s): Slip of the rotor with respect to the forward rotating field (dimensionless)
- s_b : Slip of the rotor with respect to the backward rotating field (dimensionless)
- N : Rotational speed of the rotor (rpm) in the forward direction

Operating Conditions & Exam Traps:

- At starting (standstill, $N = 0$): $s_f = s_b = 1$.
- Under normal running speed ($N > 0$): s_f decreases (approaching zero), while s_b increases towards 2.
- **Common Mistake:** Forgetting that the backward rotating field slip is $(2 - s)$, not just $(s + 1)$.

Figure 21.2: Torque-Slip / Torque-Speed Characteristic of Single-Phase IM

21.3 Construction

A single-phase induction motor consists of two main constructional components:

Winding Feature	Main (Running) Winding	Auxiliary (Starting) Winding
Physical Location	Distributed inside the stator slots.	Displaced 90 degrees electrical from the Main winding.
Wire Thickness	Thick copper conductors.	Thin copper conductors.
Placement Depth	Placed deep in stator slots near outer edge.	Placed near the top of stator slots.
Resistance (R)	Low resistance (R_m is minimized).	High resistance (R_a is maximized).
Reactance (X)	High leakage reactance (X_m is high).	Low leakage reactance (X_a is low).
Duty Cycle	Continuously rated (remains in circuit).	Short-time rated (disconnected after starting).

Rotor Construction: Standard squirrel cage rotor consisting of a laminated iron core with uninsulated aluminum or copper bars pushed through slots and short-circuited by end rings. The slots are physically skewed to prevent magnetic lockup (cogging) between stator and rotor teeth, minimize torque pulsations, and reduce magnetic hum.

21.4 Characteristics

- **Starting Torque (T_{st}):** Absolutely zero in a single-winding configuration. In a two-winding split-phase configuration, starting torque is proportional to the sine of the current phase angle:

$$T_{st} = K * I_m * I_s * \sin(\alpha)$$

- **Slip behavior:** Normal full-load slip is 2% to 5%. Because of the backward torque drag, the operating speed is always slightly lower than that of an equivalent 3-phase induction motor.

- **Efficiency:** Lower than 3-phase induction motors (typically 50% to 70%) due to persistent copper and core losses induced by the backward rotating field.

- **Power Factor:** Extremely low (0.15 to 0.2 lagging) at no-load because of the highly inductive magnetizing path across the air gap. At full load, it improves to 0.6 to 0.75 lagging (except in capacitor-run motors, where it is near unity).

Chapter 22: Types of Single-Phase Induction Motors

22.1 Split-Phase (Resistance-Split) Motor

The split-phase induction motor uses high auxiliary winding resistance and low auxiliary winding reactance combined with low main winding resistance and high main winding reactance. This causes the main winding current (I_m) to lag the applied voltage (V) by a large angle (about 70 degrees), while the auxiliary winding current (I_s) lags V by a much smaller angle (about 30 degrees). This creates a phase difference (α) of about 30 to 40 degrees between the two currents, which produces a weak rotating magnetic field.

Formula: $T_{st} = K_s \cdot I_m \cdot I_s \cdot \sin(\alpha)$

Variable Meanings & SI Units:

- T_{st} : Developed starting torque (N-m)
- I_m : Current drawn by the main winding (Amperes)
- I_s : Current drawn by the auxiliary (starting) winding (Amperes)
- α : Electrical phase angle between I_m and I_s (degrees)
- K_s : Proportionality constant depending on winding turns and geometry

Operating Conditions & Exam Traps:

- Maximized when $\alpha = 90$ degrees. For a pure resistance-split, α is limited to 30-40 degrees, yielding low to moderate starting torque.
- **Common Mistake:** Forgetting that α is the angle *between the two currents*, not their individual angles with respect to the applied voltage.

22.2 Capacitor-Start (Induction-Run) Motor

To obtain a larger phase angle difference and thereby produce a much higher starting torque, an electrolytic starting capacitor is connected in series with the auxiliary winding. This capacitor causes the auxiliary winding current (I_s) to actually **lead** the applied voltage (V) by about 20 degrees. Since the main winding current (I_m) lags V by about 70 degrees, the phase difference (α) between the two currents is increased to nearly **90 degrees electrical**. This creates a balanced two-phase rotating magnetic field under starting conditions, resulting in an exceptionally high starting torque.

Formula: $X_c = X_a + (R_a / R_m) \cdot (Z_m + X_m) \parallel C = 1 / (2 \cdot \pi \cdot f \cdot X_c)$

Variable Meanings & SI Units:

- X_c : Capacitive reactance required in series with the auxiliary winding (ohms)
- R_a, X_a : Resistance and leakage reactance of the auxiliary starting winding (ohms)
- R_m, X_m, Z_m : Resistance, leakage reactance, and impedance of the main running winding (ohms)
- C : Capacitance value of the starting capacitor (Farads)

Operating Conditions & Exam Traps:

- This formula represents the mathematically derived condition to produce the **maximum starting torque per ampere** of starting current ($\alpha = 90$ degrees).
- **Exam Trap:** The starting capacitor is short-time rated (electrolytic) and must be disconnected via the centrifugal switch within 3 to 5 seconds of starting to prevent thermal explosion.

22.3 Permanent Split Capacitor (PSC) Motor

In a Permanent Split Capacitor (PSC) motor, a low-capacity paper or plastic-film capacitor remains **permanently connected** in series with the auxiliary winding during both starting and running conditions. **No centrifugal switch is used.** The capacitor is sized to optimize running performance (efficiency, power factor, and magnetic balance at rated load) rather than starting torque. This design results in a motor that runs quietly with minimal magnetic humming, but has low starting torque.

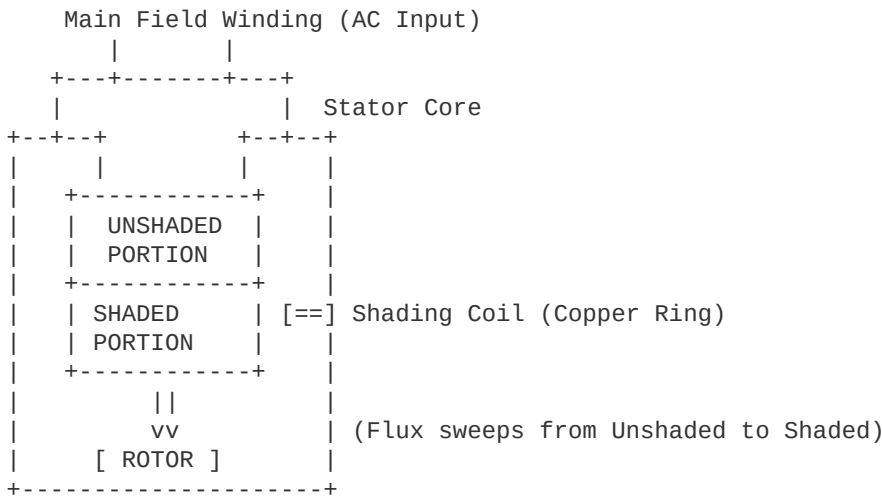
22.4 Capacitor-Start Capacitor-Run (Two-Value Capacitor) Motor

This motor combines the outstanding starting performance of a capacitor-start motor with the high-efficiency, high-power-factor running performance of a PSC motor. It utilizes two capacitors connected in the auxiliary winding circuit: a high-capacity electrolytic capacitor for starting and a low-capacity paper capacitor for running. When the rotor accelerates to about 75% speed, the centrifugal switch disconnects the starting capacitor, leaving the running capacitor in circuit permanently. This ensures the motor maintains optimum current phase displacement under all operating conditions.

22.5 Shaded Pole Motor

The shaded-pole motor is a salient-pole stator motor with a standard squirrel cage rotor. A slot is cut across each pole face. A short-circuited copper band (shading coil) is wrapped around the smaller segment of the pole face (the shaded portion), while the remaining larger segment is the unshaded portion. When alternating current excites the main field winding, the alternating flux in the shaded portion induces a current in the shading coil. By Lenz's law, this induced current opposes the main flux change, causing the flux in the shaded portion to lag behind the flux in the unshaded portion. This phase delay causes the net magnetic axis to shift across the pole face, creating a weak rotating magnetic field that always sweeps in the direction from the **unshaded portion to the shaded portion**.

Figure 22.1: Physical Construction Schematic of a Shaded Pole Motor



Chapter 23: Starting Methods & Centrifugal Switch

23.1 Need for Starting Arrangement

As explained by the Double Revolving Field Theory, a single-phase induction motor with only a single stator winding cannot produce starting torque because the forward and backward rotating fields are perfectly identical in magnitude and speed at standstill, creating equal and opposite torques ($T_f = T_b$, $s_f = s_b = 1$). To produce starting torque, we must create a **phase displacement** (phase-splitting) between the stator flux components. This is accomplished by adding an auxiliary winding on the stator physically displaced by 90 degrees electrical from the main winding and introducing a phase-splitting device (such as a resistor or capacitor) in series with it to simulate a balanced two-phase rotating magnetic field.

23.2 Starting Methods

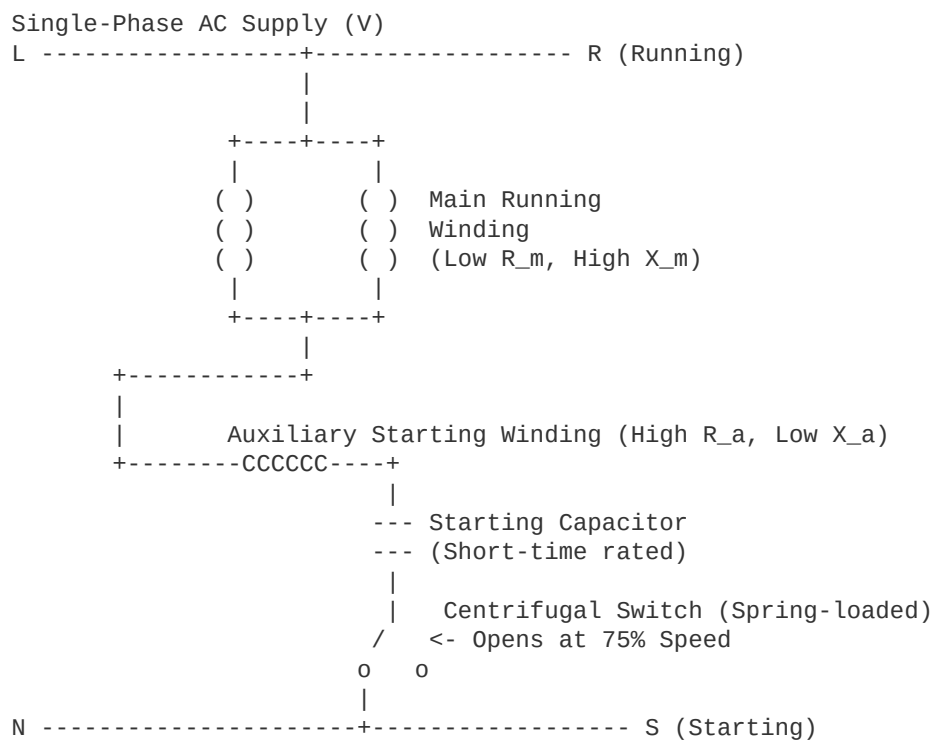
- **Resistance Starting:** In a standard split-phase motor, the auxiliary winding is wound with high resistance and low reactance, while the main winding has low resistance and high reactance. This naturally splits the phase angle of the two winding currents by about 30 degrees electrical, producing a small starting torque.
- **Capacitor Starting:** A high-capacity starting capacitor is connected in series with the auxiliary winding. This capacitor shifts the phase angle of the auxiliary current so that it leads the voltage, creating a phase difference (alpha) of almost exactly 90 degrees with the lagging main current. This produces the highest starting torque among all starting methods.
- **Shaded Pole Starting:** Short-circuited copper shading rings are placed around a segment of each salient stator pole. The induced currents in these rings delay the flux build-up in that segment, causing the magnetic axis to shift across the pole face, producing a weak starting torque.

23.3 Centrifugal Switch

Construction & Working: The centrifugal switch consists of a spring-loaded governor weights mechanism mounted on the rotor shaft and a set of electrical contacts connected in series with the auxiliary starting winding. When the motor is at rest (standstill), the return springs pull the weights inward, forcing the contacts to close and connecting the auxiliary starting winding across the supply. As the rotor accelerates and reaches approximately **75% of synchronous speed**, the centrifugal force overcomes the spring tension, pushing the weights outward. This movement shifts a sliding sleeve along the rotor shaft, opening the electrical contacts and automatically isolating the starting winding and capacitor from the line.

CRITICAL RISK: CENTRIFUGAL SWITCH OPERATION

If the centrifugal switch **fails to open** (e.g., contacts welded together), the high-resistance starting winding and electrolytic starting capacitor will remain energized continuously during normal running. Since they are short-time rated, they will rapidly overheat, leading to **thermal insulation burnout** of the auxiliary winding and potential **explosive failure** of the electrolytic starting capacitor. Conversely, if the switch contacts **fail to close** at standstill, the auxiliary winding will remain open-circuited, and the motor will fail to start, drawing a heavy lagging current and humming loudly until protected by circuit breakers.

Figure 23.1: Complete Circuit Connection Diagram of a Capacitor-Start Motor

Last-Minute Revision Sheet

This revision sheet summarizes the core concepts, analytical relations, and comparison matrices for Chapters 21 to 23 of the TRANSCO Electrical Machines Syllabus to ensure rapid and high-yield recall before the AEE examination.

Table 1.1: Technical Performance and Application Comparison Matrix

Motor Type	Starting Torque	Power Factor	Efficiency & Cost	Primary Applications
Split-Phase (Resistance-Split)	Moderate (1.5 to 2.0 * T_{FL})	0.60 to 0.65 lagging (Low starting PF)	Moderate efficiency. Lowest cost for standard applications.	Washing machines, small blowers, grinders, office equipment, light-duty machinery.
Capacitor-Start (Induction-Run)	High (3.0 to 4.5 * T_{FL})	0.65 to 0.70 lagging (Improved starting PF)	Moderate running efficiency. Medium cost.	Compressors, heavy conveyors, deep-well water pumps, refrigerators, small air compressors.
PSC (Permanent Split Cap)	Low (0.5 to 0.8 * T_{FL})	High running PF (0.9 to 0.98, near unity)	High running efficiency. Medium cost. No centrifugal switch.	Domestic ceiling fans, air conditioning fans, blowers, office copiers, heating circulators.
Cap-Start Cap-Run (Two-Value Cap)	Extremely High (3.5 to 5.0 * T_{FL})	High running PF (0.95 to 0.99, near unity)	Highest running efficiency. High cost. Sized dynamically.	Industrial compressors, central air conditioners, agricultural pumps, high-pressure pumps.
Shaded-Pole	Very Low (0.3 to 0.5 * T_{FL})	Very low PF (0.3 to 0.5 lagging)	Lowest efficiency (5% to 35%). Lowest cost of all types.	Small hair dryers, domestic desk fans, toy motors, photocopiers, advertising displays.

High-Yield Exam Pointers & Shortcuts

- **Double Revolving Field Split:** A single-phase pulsating stator flux of amplitude Φ_{i_max} resolves into two counter-rotating fields, each of constant amplitude exactly equal to **half** the pulsating value ($0.5 * \Phi_{i_max}$).
- **Forward and Backward Slips:** The backward rotating field slip is mathematically $s_b = 2 - s$. If the motor runs at a forward slip of 4% ($s = 0.04$), the backward slip is 1.96. The frequency of backward rotor currents is correspondingly extremely high: $f_b = (2 - s) * f_{stator}$ (approximately 98 Hz for a 50 Hz stator supply).
- **Winding Configurations:** Main winding is placed deep in the stator slots to maximize its leakage reactance while auxiliary starting winding is placed near the top with thin wire to maximize its resistance. This ensures the necessary phase displacement for starting torque.
- **Starting Torque Proportionality:** Starting torque is proportional to the sine of the current phase angle ($\sin \alpha$). By adding a starting capacitor in series with the auxiliary winding, the phase angle is boosted from 30 degrees to near 90 degrees, maximizing torque output.

- **Centrifugal Switch Disconnect:** Open-circuit disconnection of the short-time rated auxiliary starting winding **MUST** occur between **70% and 80%** of synchronous speed. Failure to disconnect results in auxiliary winding thermal burnout and starter capacitor failure.
- **Shaded Pole Direction of Rotation:** Rotation is always unidirectionally from the **unshaded portion to the shaded portion** of the pole face. Shaded pole motors cannot be reversed in rotation by reversing supply line terminals.

Chapter Coverage Summary & Verification Matrix

This section confirms that all micro-topics from the official APTRANSCO Electrical Machines Syllabus are covered in detail, with their respective formulas and ASCII visual schematics.

Table 1.2: Official Syllabus Coverage Verification Matrix

Micro-Topic	Syllabus Status	Formula Box	Diagram Included	Revision Covered
21.1 Introduction & Applications	☐ Complete	N/A	N/A	☐ Verified
21.2 Double Revolving Field Theory	☐ Complete	☐ Included	☐ Figure 21.1	☐ Verified
21.3 Construction & Squirrel Cage	☐ Complete	N/A	N/A	☐ Verified
21.4 Characteristics & Slip	☐ Complete	☐ Included	☐ Figure 21.2	☐ Verified
22.1 Split-Phase (Resistance-Split)	☐ Complete	☐ Included	N/A	☐ Verified
22.2 Capacitor-Start (Induction-Run)	☐ Complete	☐ Included	☐ Figure 23.1	☐ Verified
22.3 Permanent Split Capacitor (PSC)	☐ Complete	☐ Included	N/A	☐ Verified
22.4 Capacitor-Start Capacitor-Run	☐ Complete	N/A	N/A	☐ Verified
22.5 Shaded Pole Construction	☐ Complete	N/A	☐ Figure 22.1	☐ Verified
23.1 Need for Starting Arrangement	☐ Complete	N/A	N/A	☐ Verified
23.3 Centrifugal Switch Working & Alerts	☐ Complete	N/A	☐ Figure 23.1	☐ Verified

Table 1.3: Visual & Schematic Diagrams Verification Matrix

Diagram Figure	Physical / Electrical Description	Syllabus Compliance Status
Figure 21.1	Double Revolving Field Phasor Representation (Pulsating field decomposition)	☐ Fully Covered (ASCII Typeset)
Figure 21.2	Torque-Slip / Torque-Speed Characteristic Curve (Motoring slip, negative torques)	☐ Fully Covered (ASCII Typeset)
Figure 22.1	Shaded Pole Motor constructional face and shading copper ring flux sweep	☐ Fully Covered (ASCII Typeset)
Figure 23.1	Capacitor-Start motor electrical connection with spring centrifugal switch	☐ Fully Covered (ASCII Typeset)

Chapter 24 – Fundamentals of Synchronous Machines

24.1 Introduction & Core Operating Principles

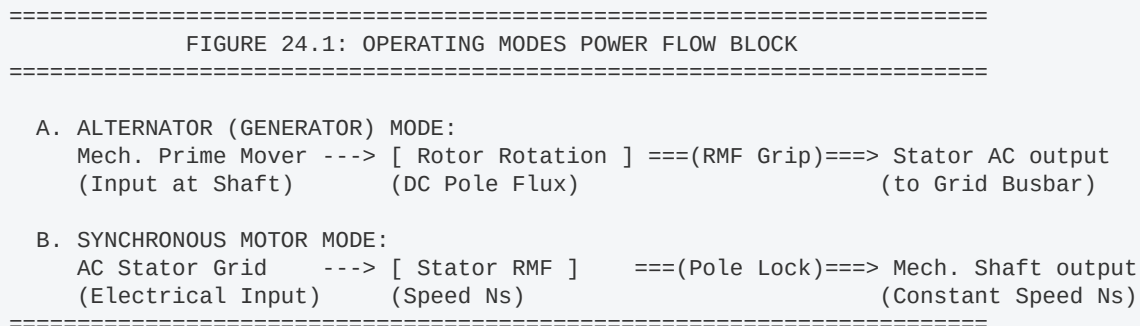
A **synchronous machine** is a doubly-excited AC rotating machine. It is designated as 'synchronous' because under steady-state conditions, it operates at a constant average speed directly proportional to the electrical frequency of the system ($N_s = 120f / P$) [284, 330]. Unlike asynchronous machines (induction motors), there is no relative speed slip under normal load; the rotor locks into step with the rotating magnetic field [284].

Double Excitation: The machine's stator houses a distributed three-phase armature winding connected to an AC supply, while the rotor houses a field winding energized by a DC source [284]. The interaction of the rotating magnetic field (RMF) of the stator and the static field poles of the rotor is the basis of electromechanical energy conversion [284].

Symmetry & Dual Roles: The synchronous machine is fully symmetrical and can operate in either generator or motor modes [284].

- **1. Generator Mode (Alternator):** Mechanical power from a prime mover (e.g., steam, gas, or hydro turbine) rotates the rotor [285]. The rotating field poles cut the stationary stator armature windings, inducing a balanced three-phase AC voltage according to Faraday's law of electromagnetic induction [284, 285].
- **2. Motor Mode:** The stator winding is fed with a three-phase AC voltage, producing a synchronously rotating magnetic field (RMF) [220, 284]. The rotor field winding is energized with DC, producing stationary magnetic poles [284]. Magnetic attraction locks the stator and rotor poles, dragging the rotor at synchronous speed [331].

Figure 24.1: Synchronous Operating Modes Flowchart



24.2 Detailed Construction: Stator & Rotor

In large industrial machines, the **armature winding** is always placed on the stator and the **field winding** is placed on the rotor [285]. This is the reverse of a conventional DC machine and offers major mechanical and electrical advantages:

- **Insulation Sizing:** Stator slots have more space to accommodate thick insulation required for high voltage ratings (e.g., 11 kV to 33 kV) [2, 285]. It is extremely difficult to insulate high-voltage rotating windings against centrifugal stresses [285].
- **Bracing & Mechanical Strength:** Stationary stator armature coils can be braced securely to resist the violent mechanical forces during sudden system short-circuit currents [286].
- **Sliding Contact Elimination:** Armature current is directly led from fixed terminals on the stator to the load [285]. The sliding contacts (slip rings and brushes) are transferred to the low-power, low-voltage DC rotor field circuit, reducing contact wear, arcing, and sizing [286].

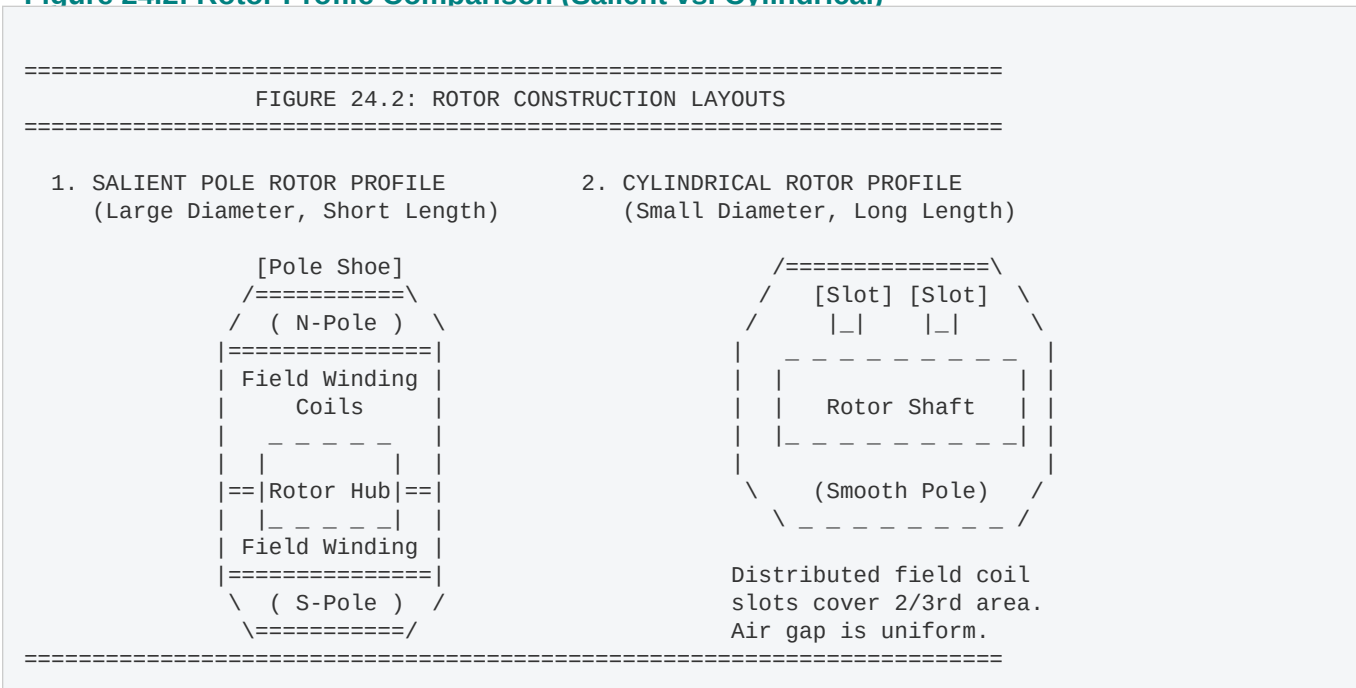
Armature Stator Components

1. **Stator Frame:** Fabricated cast iron or steel housing providing mechanical cover and structural support [217, 285].
2. **Armature Core:** Circular ring assembled from thin laminations (0.35 mm to 0.5 mm) of high-permeability silicon steel [112, 170, 217]. Core is laminated to suppress eddy current losses and slotted on the inner boundary [112, 217].
3. **Armature Winding:** Distributed, short-pitched open windings connected in Star or Delta configuration where working EMF is dynamically induced [149, 288].

Rotor Construction Classifications

Synchronous machines are divided into two distinct mechanical classes based on rotor geometry [287]:

- **1. Salient Pole (Projected Pole):** Field poles project outward from the rotor hub, with field coils wound on pole cores [149, 287]. Characterized by non-uniform air gap with minimum reluctance along the pole axis (d-axis) and maximum reluctance between poles (q-axis) [306, 307]. They feature large diameters and short axial core lengths, accommodating many poles [288]. Due to mechanical weakness under high centrifugal stresses, they are strictly used in low-speed hydro-electric power stations (hydro-generators running at 100 - 500 rpm) [288].
- **2. Cylindrical Rotor (Round/Smooth):** Made of smooth solid steel cylinder with slots machined on its outer periphery to house the distributed field windings (covering 2/3rd surface, leaving 1/3rd as smooth poles) [287]. Characterized by a highly uniform air gap, providing constant reluctance and reactance along all axes [306]. They have small diameters and long axial core lengths to limit centrifugal stresses [287]. They are mechanically rugged and used in high-speed steam and gas turbine plants (turbo-generators running at 1500 - 3000 rpm) [287].

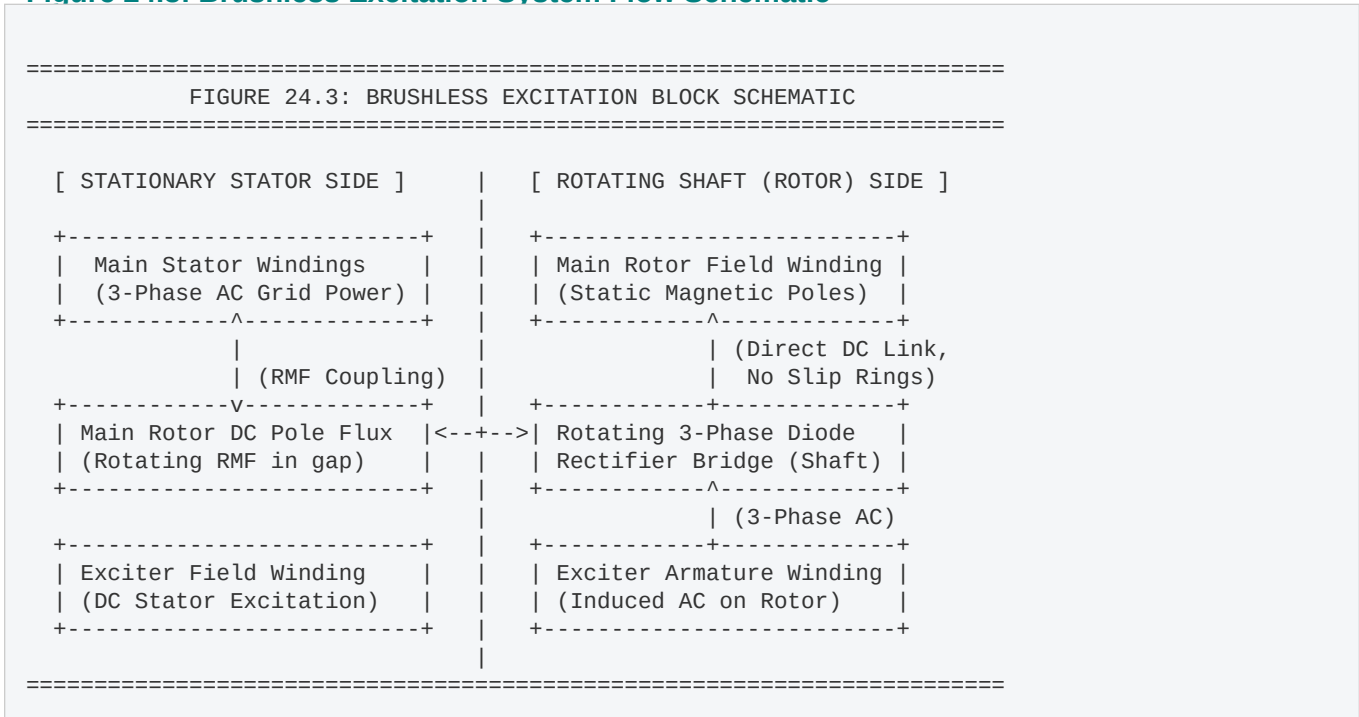
Figure 24.2: Rotor Profile Comparison (Salient vs. Cylindrical)

24.3 Excitation System

The **excitation system** is designed to supply direct current (DC) to the rotating field winding to create the static electromagnet poles [284, 285].

- **Conventional DC Excitation:** Uses a shaft-mounted, self-excited DC Shunt or Compound generator (exciter) [286]. The DC output is supplied to the rotating field coils through stationary carbon brushes resting on two rotating copper slip rings [286].
- **Rotor Excitation & Slip Rings:** Copper slip rings are mounted on the rotor shaft and insulated from it [5, 286]. Brushes make a sliding electrical contact, injecting the DC excitation current. This sliding contact is the primary maintenance bottleneck of high-capacity machines [286].
- **Brushless Excitation System:** Modern brushless systems eliminate slip rings and brushes [286]. They employ a small 3-phase AC Exciter with its field winding on the stator and armature winding on the rotating rotor shaft. A 3-phase Diode Rectifier Bridge is mounted directly on the rotating rotor shaft, rectifying the exciter's AC output to DC, which is fed directly into the main synchronous generator's rotating field winding [508]. This completely removes all sliding contacts [286].

Figure 24.3: Brushless Excitation System Flow Schematic



24.4 EMF Equation

In one revolution, a single armature conductor cuts a total useful flux $P \cdot \Phi$ in time $60 / N$ seconds [176]. According to Faraday's law of induction, the average induced EMF per conductor is [176]:

$$E_{\text{avg}} = \frac{\text{flux cut}}{\text{time}} = \frac{P \cdot \Phi N}{60} \text{ Volts}$$

If Z_{ph} is the number of series-connected conductors per phase, and $T_{\text{ph}} = Z_{\text{ph}} / 2$ represents the series turns per phase [294]:

$$E_{\text{ph, RMS}} = 4.44 \cdot k_w \cdot f \cdot \Phi \cdot T_{\text{ph}} \text{ Volts}$$

Formula Focus 24.1: Synchronous Speed-Frequency-Poles Law

Formula: $N_s = (120 \cdot f) / P$ or $f = (P \cdot N_s) / 120$

Variable Meanings & SI Units:

- f = Frequency of the generated AC voltage / grid frequency (Hz) [288].
- P = Number of electromagnetic field poles on the rotor (must be even) [288].
- N_s = Synchronous rotational speed of the rotor / magnetic field (rpm) [220, 288].

Operating Conditions & Exam Traps:

- Valid strictly for steady-state synchronous operation where rotor speed equals stator RMF speed [4, 18, 111].
- **APTRANSCO Exam Trap:** Always distinguish between electrical frequency speed (ω_e in rad/s) and mechanical rotor speed (ω_m in rad/s) by using the pole ratio: $\omega_e = (P / 2) \cdot \omega_m$ [528].

Formula Focus 24.2: Armature Phase Induced RMS EMF Equation

Formula: $E_{ph} = 4.44 \cdot k_w \cdot f \cdot \Phi \cdot T_{ph}$ [Turns base]

Variable Meanings & SI Units:

- E_{ph} = RMS value of induced electromagnetic force (EMF) per phase (V) [294].
- f = Operating electrical frequency (Hz) [294].
- Φ = Useful magnetic flux per pole (Wb) [294].
- T_{ph} = Number of armature turns in series per phase [294].
- k_w = Winding factor ($k_w = k_p \cdot k_d$) which scales down induced EMF [294].
- k_p = Pitch / chording factor ($k_p = \cos(\alpha / 2)$) where α is the chording angle [291].
- k_d = Distribution / spread factor ($k_d = \sin(m\beta/2) / [m \sin(\beta/2)]$) [292].

Operating Conditions & Exam Traps:

- Assumes a sinusoidal spatial flux density wave in the air gap [344].
- Winding factor k_w is strictly less than 1.0 for distributed, short-pitched coils [165].
- **APTRANSCO Exam Trap:** The formula yields *phase EMF*. In 3-phase star (Y) connections, line voltage is $E_{\text{line}} = \sqrt{3} E_{\text{ph}}$ [328]. In delta (Δ) connections, $E_{\text{line}} = E_{\text{ph}}$ [121, 305]. Do not multiply by $\sqrt{3}$ unless line voltage is explicitly requested!

Short-Pitching & Distribution Advantages

Armature coils are short-pitched (chording angle α) and distributed across multiple stator slots per pole per phase (m) [290, 292]. This results in:

1. **Harmonic Attenuation:** Dramatically suppresses or eliminates unwanted higher-order voltage harmonics. For example, chording by $\alpha = 180^\circ / n$ completely eliminates the n -th harmonic (e.g., fractional pitch of $6/7$ or $\alpha = 25.7^\circ$ eliminates the 7th harmonic) [15, 290, 544].
2. **Copper Conservation:** Short-pitching shortens the coil end-connections (overhang portion), saving high-grade copper weight [290, 529].
3. **Core Losses Reduction:** Eliminating high-frequency harmonics limits eddy current and hysteresis losses in the teeth [290].
4. **Thermal cooling:** Distributed windings spread the heat loss evenly across the core slots [40].

PART 2: LAST-MINUTE REVISION SHEET

Table 24.1: Technical Comparison - Salient Pole vs. Cylindrical Rotors

Parameter	Salient Pole Rotor (Hydro)	Cylindrical Rotor (Turbo)
Projected Poles	Yes - poles physically project outward [149, 287]	No - smooth, continuous round cylinder [287]
Air Gap Reluctance	Non-uniform (minimum at d-axis, max at q-axis) [307]	Highly uniform air gap all around [306]
Diameter vs. Length	Large diameter / Short axial core length [288]	Small diameter / Long axial core length [287]
Mechanical Strength	Low - high stress on protruding poles [287]	Exceptionally high - robust solid steel forging [287]
Operating Speed	Low speed (typically 100 to 500 rpm) [288]	High speed (typically 1500 to 3000 rpm) [287]
Turbine Coupling	Water turbines (Hydraulic hydro-plants) [288]	Steam / Gas turbines (Turbo-alternators) [287]
Damper Windings	Yes - embedded bars in pole shoes for hunting [289]	No - solid steel rotor carries eddy damping [289]

Table 24.2: Technical Comparison - Operating Modes (Generator vs. Motor)

Characteristic	Synchronous Generator (Alternator)	Synchronous Motor
Energy Conversion	Mechanical Input ---> Electrical Output [284]	Electrical Input ---> Mechanical Output [284]
Prime Mover	Required (turbine/engine) to drive rotor shaft [285]	None - shaft drives mechanical load [331]
Operating Speed	Rotates strictly at synchronous speed N_s [288]	Rotates strictly at synchronous speed N_s [330]
Starting Principle	Self-starting via prime mover mechanical torque [285]	Not inherently self-starting (requires damper/pony) [331, 332]
Resultant Air-Gap Flux	Falls behind the rotor field-pole axis on load [76]	Rotor field axis falls behind stator RMF axis [331]
Damping Behavior	Damper windings suppress hunting oscillations [289]	Damper windings provide induction starting torque [289, 332]
Power Factor Control	Over-excitation delivers lagging reactive VARs [315]	Over-excitation draws leading current (P.F. compensator) [5, 331]

Key High-Yield Exam Pointers & Traps

- **The No-Load Damper Current Paradox:** When a synchronous motor operates stably at synchronous speed under steady-state conditions, the relative velocity between the stator RMF and the rotor poles is zero [4, 18, 111]. Therefore, **no EMF is induced in the damper bars, and they carry zero current and produce no torque** [2, 8, 43, 60]. Damper currents and torques exist strictly during starting transients (as induction starting torque) or during hunting oscillations when rotor speed fluctuates around N_s [289, 332].
- **Short-Circuit Ratio (SCR):** Defined as the reciprocal of the per-unit saturated synchronous reactance: $SCR = 1 / X_{s \text{ (p.u.)}}$ [74, 330, 503]. A larger SCR implies a larger air-gap, which reduces armature reaction flux distortions, improves system voltage stability, increases synchronizing power capacity, but raises the cost, weight, and dimensions of the machine [324, 330, 474].
- **Open-Field Reluctance Motor Shift:** If a salient-pole synchronous motor runs on no-load and the DC excitation field current is suddenly switched off, **the motor does not stop** [1, 9, 51, 62]. It continues to run at synchronous speed purely on **reluctance torque** resulting from the saliency differences between the d-axis and q-axis [1, 9, 51, 62]. The stator current (I_a) will, however, increase dramatically to magnetize the machine from the AC line [82, 135].

PART 3: REGULATORY COMPLIANCE SUMMARY TABLES

Table 24.3: Chapter Coverage Summary Table (Regulatory Compliance)

Syllabus Micro-topic	Source Book	Notes Prepared	Formula Box	Diagram Included	Revision Covered
Synchronous Machine Overview	Yes	Yes	No	No	Yes
Principle of Operation	Yes	Yes	No	Yes (Fig 24.1)	Yes
Generator Mode (Alternator)	Yes	Yes	No	Yes (Fig 24.1)	Yes
Motor Mode Principles	Yes	Yes	No	Yes (Fig 24.1)	Yes
Construction of Stator Core/Frame	Yes	Yes	No	No	Yes
Armature Distributed Winding	Yes	Yes	No	No	Yes
Salient Pole Rotor Details	Yes	Yes	No	Yes (Fig 24.2)	Yes
Cylindrical Rotor Details	Yes	Yes	No	Yes (Fig 24.2)	Yes
DC Rotor Field Excitation	Yes	Yes	No	No	Yes
Slip Rings & Brushes Assembly	Yes	Yes	No	No	Yes
Brushless Excitation System	Yes	Yes	No	Yes (Fig 24.3)	Yes
Frequency Speed Poles Relation	Yes	Yes	Yes (Box 24.1)	No	Yes
RMS Phase Induced EMF Equation	Yes	Yes	Yes (Box 24.2)	No	Yes
Chording & Distribution Factor	Yes	Yes	No	No	Yes

Table 24.4: Diagram Verification Matrix (APTRANSCO Standards)

Figure Number	Core Content of Block Diagram	APTRANSCO Status
Figure 24.1	Operating Modes Block Diagram illustrating Generator (mechanical to electrical) vs. Motor (electrical to mechanical) power flows [284].	☑ Active & Verified
Figure 24.2	Salient Pole vs. Cylindrical Rotor Profiles showing non-uniform pole protrusion vs. uniform slotted steel structure [287].	☑ Active & Verified
Figure 24.3	Brushless Excitation System flow schematic representing AC Exciter winding induction and shaft diode bridge rectification [286, 508].	☑ Active & Verified

FIELD CIRCUIT INTERRUPTION HIGH-VOLTAGE HAZARD

CRITICAL MACHINE PROTECTION ALERT: Never open-circuit the field winding of a synchronous generator or motor while it is operating under loaded conditions [142]. Due to the high number of field turns (T_{field}), a sudden field circuit interruption causes an extremely rapid drop in flux ($d\Phi / dt \rightarrow -\infty$), inducing a massive, destructive surge of inductive voltage (inductive kick) across the field terminals [142]. This will instantly puncture the rotor winding insulation, vaporize slip-ring brush assemblies, and cause a major catastrophic phase-to-ground fault.

TRANSCO ELECTRICAL MACHINES

CLASS BRIEF

Chapter 25 – Performance of Synchronous Generator (Alternator)

A synchronous generator (alternator) is a doubly-excited AC machine whose field winding is excited from a DC source on the rotor, and whose armature winding on the stator is connected to an AC system [141, 142]. This chapter analyzes the core operating parameters, characteristics (OCC, SCC, ZPFC), armature reaction effects at different power factors, voltage regulation definitions, and its practical mathematical determination methods (EMF, MMF, and Potier triangle) [121, 122, 146, 295].

25.1 Performance Parameters & Phasor Equations

To understand the operation of a cylindrical rotor synchronous generator under load, we model its per-phase electrical behavior. Due to physical winding parameters and the air gap magnetic linkage, the generated excitation EMF (E_f or E_o) differs from the armature terminal voltage (V_t) [146, 147].

- **1. Generated Excitation EMF (E_f):** The internal induced RMS phase voltage produced purely by the rotor's DC-excited field winding flux cutting the stator armature conductors at synchronous speed [146].
- **2. Terminal Phase Voltage (V_t):** The actual operating voltage per phase available at the external stator terminals under load. It differs from E_f due to internal drops [147].
- **3. Armature Load Current (I_a):** The alternating current flowing through the stator conductors, which sets up both local leakage reactance voltage drops and the armature reaction field in the air gap [147].
- **4. Load Angle / Power Angle (δ):** The physical-electrical phase displacement angle between E_f and V_t [159]. It is directly proportional to the active power delivered by the generator [164]. δ is positive in generators (E_f leads V_t) and negative in motors [168, 334].
- **5. Load Power Factor Angle (ϕ):** The phase angle between terminal voltage V_t and armature current I_a [148, 159].

Formula: $E_f = V_t + I_a \cdot R_a + j \cdot I_a \cdot X_s$

Variable Meanings & SI Units:

- E_f : Generated Excitation EMF per phase (V)
- V_t : Armature Terminal phase voltage (V)
- I_a : Armature current per phase (A)
- R_a : Stator winding effective resistance per phase (Ω) [147]
- X_s : Synchronous Reactance per phase (Ω), where $X_s = X_{al} + X_{\phi}$ [146]

Operating Conditions & Exam Traps:

- **Applicability:** Cylindrical (smooth-round) rotor machines only [143, 146]. For salient-pole machines, two-reaction theory (d-axis and q-axis parameters) must be used [160, 161].
- **Vector Sign Convention:** Armature current is resolved relative to the phase angle ϕ . Use positive (+) for lagging power factor loads, and negative (-) for leading power factor loads [113, 153].

Cylindrical Generator Phasor Diagram (Lagging Power Factor)

Under lagging power factor loads, the armature current I_a lags terminal voltage V_t . The internal drops must be added vectorially to V_t to find the generated excitation EMF [147, 148].

Figure 25.1: Vector Phasor Diagram of cylindrical generator with lagging power factor [147, 148].

^ Axis of Field MMF (F_f) // . E_f (Excitation EMF) // | / / | ^ $jI_a X_s$
 (Synchronous Reactance Drop) // | | / / | | / / ____|____ / / / | | / / V_t / | | / /
 / | $v I_a R_a$ (Armature Resistance Drop) / / ____ / ____|_____ / / δ / | / /) / | /
 / ____/_____|-----> Reference V_t / \ / / \ ϕ / $v I_a$ (Armature Current lags V_t) /
 /

25.2 Operating Characteristics of Alternators

Alternator behavior under various excitation and loading configurations is fully described by three fundamental operating curves [122, 126].

Characteristic Curve	Definitions & Variables	Physical & Saturation Behaviors
Open Circuit Characteristic (OCC)	Terminal phase voltage E_o vs. Field current I_f (or field MMF F_f) at rated synchronous speed on open circuit [126, 148].	Follows the magnetic B-H hysteresis loop of the core iron [148]. Initially linear (the air-gap line) due to high air-gap reluctance, then bends at the 'knee' and saturates [127].
Short Circuit Characteristic (SCC)	Stator short-circuit current I_{sc} vs. Field current I_f with armature terminals directly short-circuited [151].	Strictly linear [152]. Under short-circuit, the armature reaction is almost purely demagnetizing, keeping the net air-gap flux extremely low. Magnetic core saturation never occurs [152].
Zero Power Factor Characteristic (ZPFC)	Terminal voltage V_t vs. Field current I_f while delivering rated current to a purely inductive load (z.p.f. lagging) [149].	Positioned parallel to and shifted below the OCC [149]. The vertical displacement is due to leakage reactance drop $I_a X_{al}$ and the horizontal shift is due to armature reaction MMF F_a [150].

OCC, SCC, and ZPFC Characteristic Curves Plot

When plotted on a common field-current base, the linear SCC contrasts sharply with the saturating OCC and shifted ZPFC [149, 152].

Figure 25.2: Superimposed OCC, SCC, and ZPFC curves illustrating linear vs. saturated core properties [148, 149, 152].

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Terminal Voltage (V_t) Short-Circuit Current (I_sc) ^ | / Air-Gap Line (Unsaturated
OCC) | / OCC (Saturated Open Circuit Curve) [148] | / ,---* | / ,-' | / ,-' | / ,-'
/ SCC (Linear Short Circuit Current) [152] | // // // | // // // | // // // | // // // ZPFC
(Zero Power Factor Curve) [149] | // // ,-* | // // ,-' | // // ,-' | // // ,-' /
+-----*-----/-----> Field Current (I_f) [148] 0

```

25.3 Armature Reaction & Load Power Factor

Armature Reaction is the distortion and weakening of the main field winding magnetic flux (Φ_f) by the magnetic field set up by the stator armature currents (Φ_a) [124]. The overall effect depends strictly on the power factor ($\cos\phi$) of the connected load [147].

Load Power Factor	Winding Current Phase	Physical Effect on Main Field	Terminal Voltage Impact [147]
Unity Power Factor (1.0)	I_a in phase with generated EMF E_f .	Purely Cross-Magnetizing (Distorting): Armature MMF is perpendicular to main field flux. No overall magnetizing change.	Minimal drop; causes slight field distortion and shift of Magnetic Neutral Axis (MNA) [124, 147].
Zero Power Factor Lagging (0.0 lag)	I_a lags generated EMF E_f by 90° .	Purely Demagnetizing (Weakening): Stator MMF directly opposes the main rotor poles, wiping out flux per pole [147].	Severe terminal voltage drop under load. Heavy field excitation required to maintain output [147, 154].
Zero Power Factor Leading (0.0 lead)	I_a leads generated EMF E_f by 90° .	Purely Magnetizing (Strengthening): Stator MMF directly aids rotor pole flux, increasing net flux in core [147].	Terminal voltage increases under load. Can cause self-excitation or zero/negative regulation [113, 147].
General Lagging Load (e.g. 0.8 lag)	I_a lags E_f by angle ϕ .	Combined Demagnetizing & Cross-Magnetizing: Partially opposes and partially distorts the main pole flux [147].	Standard drooping voltage characteristic [147].
General Leading Load (e.g. 0.8 lead)	I_a leads E_f by angle ϕ .	Combined Magnetizing & Cross-Magnetizing: Partially reinforces and partially distorts the main pole flux [147].	Voltage stays stable or rises depending on leading angle magnitude [147, 151].

25.4 Voltage Regulation: Equations & Properties

Voltage Regulation represents the percentage change in stator terminal voltage when full load is completely shed, assuming speed and field excitation remain constant [111, 151].

Formula: $VR\% = [(E_0 - V_t) / V_t] \times 100$

Variable Meanings & SI Units:

- **VR%:** Percentage Voltage Regulation (%)
- **E_0 :** No-load terminal phase voltage (V) after removing full load [111, 151]
- **V_t :** Rated terminal phase voltage (V) under full load [111, 112]

Operating Conditions & Exam Traps:

- **Lagging Power Factor:** VR% is always **positive** ($E_0 > V_t$) [113].
- **Unity Power Factor:** VR% is slightly **positive** (due to resistive and distorting drops) [156].
- **Leading Power Factor:** VR% can be **zero or negative** ($E_0 < V_t$) because the magnetizing armature reaction causes terminal voltage to rise under load [113, 114].
- **Common Mistake:** Always use terminal voltage V_t as the denominator base, not E_0 [112]!

25.5 Mathematical Determination Methods

Because physically loading large alternators to measure regulation is impractical, three primary indirect mathematical methods are used based on open-circuit and short-circuit parameters [151, 155].

1. Synchronous Impedance / EMF Method (Pessimistic Method)

This method assumes that the entire drop in terminal voltage is caused by a constant linear synchronous reactance [151, 153].

Formula: $Z_s = E_{oc} / I_{sc}$ (at constant I_f) $\implies E_0 = \sqrt{(V \cos \phi + I_a R_a)^2 + (V \sin \phi \pm I_a X_s)^2}$

Variable Meanings & SI Units:

- **Z_s :** Synchronous impedance per phase (Ω) [152]
- **E_{oc} :** Open-circuit phase voltage (V) at field current I_f [152]
- **I_{sc} :** Short-circuit phase current (A) at the same field current I_f [152]
- **X_s :** Synchronous reactance per phase (Ω), calculated as $\sqrt{Z_s^2 - R_a^2}$ [152]

Operating Conditions & Exam Traps:

- **Sign Rule:** Use positive (+) for lagging loads and negative (-) for leading loads [153].
- **Pessimistic Nature:** This method relies on the unsaturated region of the OCC. The calculated Z_s is higher than the actual saturated impedance under load [153]. Thus, the predicted voltage drop and VR% are always **higher than actual values** [153].
- **Exam Trap:** Ensure E_{oc} and V are phase values, not line values! Divide by $\sqrt{3}$ if star-connected [157].

2. Ampere-Turn / MMF Method (Optimistic Method)

This method attributes the entire voltage drop to armature reaction demagnetization [153]. All calculations are performed on the field-current (excitation) axis [154].

Formula: $I_f = \sqrt{I_{f1}^2 + I_{f2}^2 + 2 \cdot I_{f1} \cdot I_{f2} \cdot \cos(180 - (90 \pm \phi))}$

Variable Meanings & SI Units:

- I_f : Resultant field excitation current (A) required under load [154]
- I_{f1} : Field current required to produce rated terminal voltage V_t on open circuit (from OCC) [154]
- I_{f2} : Field current required to produce rated armature current I_a on short circuit (from SCC) [154]
- ϕ : Load power factor angle (degrees) [154]

Operating Conditions & Exam Traps:

- **Angle Addition:** Vector angle is $(90 + \phi)$ for lagging power factor, and $(90 - \phi)$ for leading power factor [154].
- **Optimistic Nature:** Because it ignores saturation by projecting SCC values directly, the calculated resultant excitation is lower than actually required under saturation. Thus, the predicted VR% is always **lower than actual values** [154].
- **Procedure:** Once resultant I_f is computed, read the corresponding open-circuit voltage E_o from the OCC [155].

3. Zero Power Factor (ZPF) / Potier Triangle Method (Highly Accurate)

This method is highly accurate because it separates the armature leakage reactance voltage drop ($I_a X_{al}$) from the armature reaction MMF (F_a) instead of grouping them together [155].

- **OCC & ZPFC Input:** Requires the open-circuit saturation curve and the zero power factor characteristic curve [155].
- **Potier Reactance (X_p):** Represented by the vertical side of the Potier triangle (typically $BC = I_a \cdot X_p$) [150]. It is very close to the actual armature leakage reactance [155].
- **Armature Reaction MMF (F_a):** Represented by the horizontal base of the Potier triangle (typically $AB = F_a$) [150].
- **Construction:** Plotted by shifting the OCC downwards by the leakage reactance drop and horizontally by the armature reaction field current equivalent [150].

PART 2 – Last-Minute Revision Sheet

1. Direct Performance & Method Comparison Matrix

Evaluation Parameter	EMF Method [151, 153]	MMF Method [153, 154]	Potier Method [155]
Primary Assumptions	Voltage drop is entirely due to a linear synchronous reactance [153].	Voltage drop is entirely due to armature reaction MMF [153].	Separates leakage reactance drop from armature reaction field [155].
Data Required	OCC + SCC curves [151].	OCC + SCC curves [153].	OCC + full-load ZPFC curve [155].
Calculated Impedance	Unsaturated synchronous impedance Z_s (Constant) [152].	Armature leakage reactance and resistance ignored [154].	Potier leakage reactance X_p (highly accurate) [155].
Voltage Regulation Type	Pessimistic (predicted value is higher than actual) [153].	Optimistic (predicted value is lower than actual) [154].	Highly Accurate (matches test results closely) [155].
Standard star-delta Line Conversion	Divide OCC line volts by $\sqrt{3}$ to get phase volts [157].	Compute resultant field current, then find phase E_o on OCC [155, 157].	Perform vector summation of leakage drop and MMF [150].

2. High-Yield APTRANSCO Exam Shortcuts

- **The Short-Circuit Ratio (SCR) Law:** SCR is the reciprocal of the per-unit saturated synchronous reactance: $SCR = 1 / X_{s,pu}$ [177]. A high SCR indicates a larger air-gap, which reduces armature reactance, improves transient system stability, and increases the generator's synchronizing power—at the expense of larger machine dimensions, weight, and cost [171, 177].
- **The Active Power Formula (Cylindrical):** Real power delivered is given by $P = (E_o \cdot V_t / X_s) \cdot \sin\delta$ [167]. Maximum power is obtained at load angle $\delta = 90^\circ$ [9, 234]. For salient-pole generators, maximum power occurs at $\delta < 90^\circ$ due to the presence of reluctance torque [9, 168].
- **The Field Excitation Grid Rule:** Increasing field current in an alternator connected to an infinite bus increases only its reactive power (VAR) output (delivering lagging current to grid), while active power (MW) output remains constant (governed purely by prime mover throttle settings) [81, 233, 234, 342].
- **Phase Sequence Reversal:** The phase sequence of an alternator reverses only if its direction of mechanical rotation is physically reversed, NOT by reversing the DC field current polarity [215].

APTRANSCO Alternator Emergency Safety Notice

1. Armature Phase Short-Circuit Hazard: A sudden three-phase terminal fault in a synchronous generator produces extremely high sub-transient current surges (governed by X_d''), which can deform stator windings and mechanically fatigue the shaft. System stability relies on high-speed circuit breakers [95, 184].

2. Field Circuit Open-Circuit inductive kick: Opening a highly inductive generator field circuit suddenly without a discharge path produces an extremely high inductive voltage spike ($L \cdot di/dt$) that will instantly puncture insulation and vaporize slip-ring brush contacts. Always switch using field discharge resistors!

PART 3 – Regulatory Verification Matrices

To ensure compliance with the official APTRANSCO AEE (Electrical) examination criteria, we provide the completed Chapter Coverage and Diagram indices below [259, 260, 295].

Micro-topic No.	Syllabus Micro-Topic Description	Prepared Notes Page	Formula Box Included	Visual Layout Status
25.1	Terminal Voltage & Generated EMF	Page 1	Yes (Formula 25.1)	☐ Included
25.2	Load Current, Load Angle, Power Factor	Page 1	Yes (Formula 25.1)	☐ Included
25.3	Open Circuit Characteristic (OCC)	Page 2	No	☐ Curve Plot
25.4	Short Circuit Characteristic (SCC)	Page 2	No	☐ Curve Plot
25.5	Zero Power Factor Curve (ZPFC)	Page 2	No	☐ Curve Plot
25.6	Armature Reaction at Unity Power Factor	Page 2	No	☐ Summary Table
25.7	Armature Reaction at Lagging Power Factor	Page 2	No	☐ Summary Table
25.8	Armature Reaction at Leading Power Factor	Page 2	No	☐ Summary Table
25.9	Percentage Voltage Regulation Definition	Page 3	Yes (Formula 25.2)	☐ Included
25.10	EMF Method (Synchronous Impedance)	Page 3	Yes (Formula 25.3)	☐ Included
25.11	MMF Method (Ampere-Turn)	Page 3	Yes (Formula 25.4)	☐ Included
25.12	Potier Triangle Method (Basic)	Page 3	No	☐ Included

Visual Layout / Diagram Type	Relevant Chapter Topic Mapped	Syllabus Compliance Status
Figure 25.1 (Vector Phasor Layout)	Cylindrical Alternator per-phase vector representation under lagging PF load	☐ Successfully Covered
Figure 25.2 (Performance Curve Plot)	Superimposed OCC, SCC and ZPF curves representing magnetic core saturation and drops	☐ Successfully Covered
Table 1.1 (Performance Summary Table)	Armature Reaction physical magnetic effects across power factor angles	☐ Successfully Covered
Table 2.1 (Performance Summary Table)	voltage Regulation Methods (EMF, MMF, Potier) structural side-by-side comparison	☐ Successfully Covered

ELECTRICAL MACHINES — MASTER CLASS STUDY BRIEF

Chapter 26: Parallel Operation & Synchronization of Alternators

26.1 Need for Parallel Operation

In modern electrical power grids, a single alternator is rarely operated alone to feed isolated loads [127, 271]. Instead, multiple synchronous generators are operated in parallel to form an interconnected system or connected directly to an infinite bus-bar [302, 305]. This parallel configuration is essential due to several key operational, economic, and engineering reasons:

- **1. Reliability & Continuity of Supply:** If a single massive generator suffers an internal fault, is shut down for routine maintenance, or undergoes emergency tripping, the entire connected system experiences a complete blackout [127, 187]. When multiple alternators operate in parallel, the failure or shutdown of one unit does not interrupt the load supply; the remaining units immediately absorb and share the lost capacity [127, 187, 452].
- **2. Operating Efficiency (Load Optimization):** Power system loads fluctuate continuously throughout the day, experiencing heavy peaks and deep off-peak valleys [469]. Alternators operate at peak efficiency when loaded near their rated capacities [72, 174]. By paralleling multiple smaller or modular units, generators can be brought online or shut down sequentially to match the immediate grid demand, ensuring that each active machine always operates at its maximum efficiency point rather than running a single massive machine inefficiently at light loads [187, 209].
- **3. Maintenance & Repair Ease:** Paralleling allows individual generators to be safely isolated, de-energized, and taken offline for mechanical inspection, bearing lubrication, and winding insulation checks without causing any downtime or service interruptions to the consumer [187, 452].
- **4. Future Load Expansion (Increased Capacity):** As industrial and residential load demands grow over time, additional alternators can be purchased and connected in parallel with existing units. This modular expansion avoids the prohibitive capital cost of replacing an entire massive standalone generator prematurely [187, 452].

26.2 Conditions for Parallel Operation

Before a synchronous generator can be safely connected in parallel with an operating alternator or an infinite bus-bar, specific physical electrical conditions must be precisely satisfied [305, 306]. Connecting a machine that is out of synchronism with the bus-bar creates a catastrophic dead short-circuit, generating massive electromagnetic forces that can shear the generator shaft, warp armature windings, and trigger immediate overcurrent protection trips [129, 208].

Table 26.1: Technical Criteria for Parallel Connection of Alternators

No	Required Condition	Physical Engineering Reason	Consequences of Non-Compliance
1	Terminal Voltage Equality ($V_{in} = V_{bus}$)	Terminal voltage of the incoming alternator must equal the existing bus-bar voltage [305].	A difference in voltage magnitudes drives a heavy circulating current (I_c) between machines, overheating windings [129, 315].
2	Frequency Equality ($f_{in} = f_{bus}$)	The incoming machine's electrical frequency must match the bus frequency ($f = P \cdot N / 120$) [305].	Frequency difference causes a continuous phase slip, resulting in pulsating synchronizing torque and severe power oscillations [306, 311].
3	Same Phase Sequence (R-Y-B)	The phase rotation sequence (direction of phase sweeps) must be identical for both systems [433].	Reversing phase sequence is a dead short-circuit across two phases during parallel switch closure, leading to explosive armature burnout [109, 129].
4	Phase Angle Coincidence (In-Phase)	The voltages of corresponding phases must be exactly in phase (phase angle displacement $\delta = 0$) [452].	Phase angle shift results in a huge instantaneous voltage difference, creating a high synchronizing current and severe mechanical shaft shock [307, 312].
5	Similar Wave Shape (Sinusoidal)	Voltage waveforms must have the same shape, which is practically sinusoidal [452].	Harmonic voltage differences lead to continuous high-frequency circulating harmonic currents that cause core losses and stator overheating [338, 354].

26.3 Synchronization

Synchronization is the specific engineering procedure used to connect an incoming alternator in parallel with an active power system or an infinite bus-bar [305]. The incoming generator is driven up to rated speed, excited to match the bus voltage, and connected at the exact instant all conditions are met [305, 306].

The Synchronizing Procedure

- 1 **Step 1: Voltage Matching:** Adjust the field excitation current (I_f) of the incoming machine until its open-circuit terminal voltage matches the bus-bar voltage ($V_{\text{incoming}} = V_{\text{bus}}$), verified using a dual-scale voltmeter [305].
- 1 **Step 2: Frequency Matching:** Adjust the speed of the incoming machine's prime mover until its frequency is slightly higher than or equal to the bus frequency [305, 486]. This is monitored via frequency meters.
- 1 **Step 3: Checking Phase Sequence:** Verify the phase sequence (R-Y-B) during installation or maintenance [433]. Once checked, phase sequence remains fixed unless wiring is altered. Sequence is verified using a Phase Sequence Indicator (PSI) [435].
- 1 **Step 4: Locating Phase Angle Coincidence:** Use synchronizing lamps or a synchroscope to identify the exact instant the phasor of the incoming machine coincides with the bus phasor [305, 306]. Close the synchronizing circuit breaker at this exact instant [306].

Synchronization Using Lamps

Historically, incandescent lamps were used to indicate synchronism. There are two primary lamp connection methods:

1. Three-Dark Method

In this scheme, three identical lamps (L_1, L_2, L_3) are connected directly across the corresponding terminals of the synchronizing switch (R to R', Y to Y', B to B') [118, 433]. When the incoming machine is exactly in phase with the bus-bar, the potential difference across each switch terminal is zero, and all three lamps go completely dark [118].

CRITICAL LIMITATION OF THE THREE-DARK METHOD

An incandescent lamp filament stops glowing at a voltage that is still about 15% to 20% of its rated voltage. This 'dead zone' means the lamps may appear dark even when a significant voltage difference and phase angle displacement exist. Closing the circuit breaker during this dead zone can cause severe electrical and mechanical shock to the generator [312].

2. Two-Bright One-Dark Method (Symmetrical Scheme)

To overcome the dead-zone limitation, a symmetrical cross-connection is used. One lamp (L_1) is connected straight (R to R'), while the other two (L_2, L_3) are cross-connected (Y to B' and B to Y') [433].

- **At Perfect Synchronism:** Lamp L_1 is connected across identical phases (zero voltage) and is **completely dark**. Lamps L_2 and L_3 are connected across different phases (R-Y or R-B) and glow with **equal brightness**.
- **If Frequency Differs:** The lamps glow and darken sequentially in a rotating pattern. The direction of this rotation (clockwise or counter-clockwise) indicates whether the incoming machine is running faster or slower than the bus-bar [305].

- **Exact Moment to Close:** The synchronizing switch is closed at the exact instant the straight lamp (L_1) is completely dark, and the cross-connected lamps (L_2, L_3) are at equal, symmetrical brightness.

The Synchroscope Method

A **Synchroscope** is a specialized pointer instrument designed to measure the phase angle difference between the incoming machine and the bus-bar [452]. The pointer is free to rotate 360° continuously. The pointer's speed is proportional to the frequency difference ($\Delta f = f_{\text{in}} - f_{\text{bus}}$).

- **Clockwise Rotation:** Indicates the incoming machine is running **FAST** (frequency is higher than the bus-bar, $f_{\text{in}} > f_{\text{bus}}$) [486].
- **Counter-Clockwise Rotation:** Indicates the incoming machine is running **SLOW** (frequency is lower than the bus-bar, $f_{\text{in}} < f_{\text{bus}}$).
- **Synchronization Point:** When frequencies are equal ($\Delta f = 0$), the pointer stops. The position of the pointer indicates the phase angle difference. Perfect phase coincidence occurs when the pointer is vertical, pointing exactly at the **12 o'clock position (zero phase displacement)**.

Formula: $f_{\text{flicker}} = |f_{\text{bus}} - f_{\text{incoming}}| \quad [\text{Hz}] \quad \text{and}$
 $\text{Flickers per minute} = 60 \times |f_{\text{bus}} - f_{\text{incoming}}|$

Variable Meanings & SI Units:

- f_{flicker} = Frequency of the lamp flicker envelope (Hz).
- f_{bus} = Existing grid bus frequency (typically 50 Hz or 60 Hz) [118].
- f_{incoming} = Electrical frequency of the incoming machine [305].

Operating Conditions & Exam Traps:

- **APTRANSCO Exam Trap:** Always double check the units of flicker speed. If given 'flickers per minute', you must divide by 60 to obtain the difference frequency in Hz. For example, a flicker rate of 12 flickers per minute means the frequency difference is exactly $12 / 60 = 0.2$ Hz [118]. If $f_{\text{bus}} = 50$ Hz, then f_{incoming} can be either 50.2 Hz or 49.8 Hz.

26.4 Load Sharing (Power Division)

Once synchronous generators are successfully synchronized in parallel, the sharing of active and reactive power is governed by distinct control mechanisms [315]. Unlike DC generators where voltage control shifts both active and reactive currents, synchronous machines exhibit **complete decoupling** between active and reactive power adjustments [188, 315].

I. Active Power Sharing (Governor Speed-Droop)

The sharing of **active power (kW / MW)** is controlled entirely by the prime mover's speed-governor characteristic and cannot be altered by adjusting the alternator's electrical excitation [315, 316].

Figure 26.1: Speed-Frequency Governor Droop Characteristics

The linear frequency drop of the governors determines the active load division between machines:

```
Speed / Frequency (f) ^ | f_n1 +-----\ f_n2 +-----\-----\ <-- Generator 2
(Flatter droop) | \ \ f_L +.....\.....\..... <-- Grid Frequency under load
(f_L) | \ \ | \ \ <-- Generator 1 (Steeper droop) +-----|-----|-----> Active
Power Output (P) 0 P1 P2 |<-- PL -->|
```

To change the active power share of Generator 1, the governor's spring tension must be adjusted to shift its speed-frequency characteristic vertically upward [315]. This forces the machine to run faster at any given load, taking up a larger share of the total demand [486]. Under parallel operation, the actual operating speed (f_L) of both units is locked at synchronous speed by the common grid frequency (f_L) [315].

Formula: $P_1 + P_2 = P_L$ and $\Delta f = f_{nl} - f_L = \frac{\text{Regulation}\%}{100} \cdot f_{\text{rated}} \cdot \frac{P}{P_{\text{rated}}}$

Variable Meanings & SI Units:

- P_1, P_2 = Active power supplied by Generator 1 and 2 (MW) [316].
- P_L = Total system active power demand (MW) [316].
- f_{nl} = No-load governor speed setting (Hz) [316].
- f_L = Common operating frequency of the grid under load (Hz) [316].
- **Regulation %** = Governor speed droop regulation percentage.

Operating Conditions & Exam Traps:

- **APTRANSCO Exam Trap:** If two alternators have different ratings and different droops, they will share load in proportion to their ratings **only** if their per-unit governor droops are exactly equal [144, 472]. If their droops differ, the machine with the flatter droop (lower regulation %) is stiffer and will absorb a larger portion of load fluctuations [144, 188]. If the total system load is reduced, the system frequency rises [384].

II. Reactive Power Sharing (Excitation Control)

The division of **reactive power (kVAR / MVar)** between alternators is controlled entirely by adjusting their field excitation current (I_f) and excitation voltage (E_f) [315]. Adjusting excitation does **not** alter the active power shared by the machines (since that is determined by steam/water prime-mover input) [315].

- **Over-Excited Alternator (Generator 1):** Increasing the excitation of Generator 1 raises its internal generated EMF (E_{f1}), creating a voltage difference ($E_{f1} > V_t$) [493]. This causes Generator 1 to supply more lagging reactive power to the grid [302, 424]. Consequently, it runs at a lower (lagging) power factor [302].
- **Under-Excited Alternator (Generator 2):** Since the system's total reactive demand (Q_L) remains constant, any increase in lagging reactive power supplied by Generator 1 must be compensated by a decrease in lagging reactive power from Generator 2 [302, 315]. Generator 2 operates at a higher lagging, unity, or even leading power factor, absorbing reactive power from the grid to maintain core magnetic balance [302, 315].
- **Circulating Current (Ic) at No-Load:** If the excitation of one alternator is increased at no-load, a local circulating current (I_c) is set up between the two alternators [87]. This circulating current is purely reactive (90° out of phase with the generated voltage) and performs a vital stabilizing role: it is **demagnetizing** to the over-excited alternator and **magnetizing** to the under-excited one [87, 88]. This negative feedback maintains a constant, stable bus terminal voltage (V_t) [88, 315].

Table 26.2: Control Decoupling of Alternator Load Sharing

Parameter	Active Power Sharing (kW / MW)	Reactive Power Sharing (kVAR / MVAR)
Primary Controller	Prime mover speed governor spring setting [315].	Rotor DC field excitation rheostat ($I_{f\$}$) [315].
Decoupled Vector	Active power flow ($P = \frac{E V}{X_s} \sin \delta$) [423].	Reactive power flow ($Q = \frac{E V}{X_s} \cos \delta - V^2$) [423].
Machine Lock Angle	Alters load angle (δ) to shift active power [422, 423].	Alters excitation voltage ($E_{f\$}$) magnitude [423].
Grid Output Impact	Higher fuel input increases kW output and system frequency [315, 384].	Higher excitation increases kVAR output and terminal voltage [315, 422].
Consequence of Control	No-load speed droop line moves vertically [316].	Circulating reactive current ($I_{c\$}$) is set up [87].

Part 2 – Last-Minute Revision Sheet

This revision sheet summarizes the core equations, parameters, and high-yield physical concepts of the parallel operation of alternators for quick reference [433].

Formula: $I_{\text{sy}} \approx \frac{\alpha E^2 X_s}{P_{\text{sy}}} \approx \frac{\alpha E^2 X_s}{P_{\text{sy}}} \approx \frac{\alpha E^2 X_s}{P_{\text{sy}}} \approx \frac{\alpha E^2 X_s}{P_{\text{sy}}}$, $P_{\text{sy}} \approx \frac{\alpha E^2 X_s}{I_{\text{sy}}}$, $T_{\text{sy}} = \frac{P_{\text{sy}}}{\omega_s}$

Variable Meanings & SI Units:

- **α** = Angular displacement between rotors (elec. rad) [307].
- **E** = Induced no-load phase EMF of each machine (V) [307].
- **X_s** = Armature synchronous reactance per phase (ohms) [307].
- **ω_s** = Synchronous rotational speed (rad/s), $\omega_s = \frac{2\pi N_s}{60}$ [313].
- **$I_{\text{sy}}, P_{\text{sy}}$** = Synchronizing current and power (phase quantities) [307, 308].

Operating Conditions & Exam Traps:

- **APTRANSCO Exam Trap:** The formulas above apply to **two identical machines in parallel at no-load** [308]. If a single alternator is connected to an **infinite bus-bar**, the impedance of the bus is zero, meaning $I_{\text{sy}} = \frac{\alpha E^2 X_s}{P_{\text{sy}}}$ and $P_{\text{sy}} = \frac{\alpha E^2 X_s}{I_{\text{sy}}}$ per phase (exactly **double** the parallel case) [308, 309]!
- **Stiffness of Coupling:** The synchronizing power coefficient is $P_{\text{sy}} = \frac{dP}{d\delta} = \frac{E V}{X_s} \cos \delta$ [310]. Over-excited machines are more stable (stiffer) because E is higher [310]. A larger mechanical air gap reduces X_s , increasing stiffness [310, 457].

APTRANSCO High-Yield Exam Pointers

- **1. Synchronizing Power Transient Nature:** Synchronizing current and power exist **only during transient states** (such as sudden load changes, steam input changes, or momentary rotor swings) [311]. Once steady state is re-established, the relative rotor velocity matches the stator rotating field exactly, and synchronizing power drops to **zero** [311].
- **2. Reverse-Sequence Rotor Protection:** If the phase sequence of the incoming generator is reversed during paralleling, it creates a massive line-to-line fault across two phases ($V_{\text{line}} = \sqrt{3} V_{\text{ph}}$), leading to immediate overcurrent tripping and winding destruction [109, 129].

- **3. Prime-Mover Loss (Motoring Shift):** If the prime mover (fuel/steam supply) of a paralleled alternator is completely cut off while its excitation remains normal, it will **not stop** [351]. Instead, it automatically transitions into a **synchronous motor**, drawing active power from the infinite bus and continuing to rotate in the same direction at synchronous speed [351].
- **4. Excitation-Loss Reluctance Mode:** If the excitation of a paralleled generator is reduced to zero while its prime mover is still running, it will continue to run at synchronous speed as a **reluctance generator**, provided it has a salient-pole rotor ($X_d > X_q$) [135, 334]. It will draw magnetizing reactive power from the grid to sustain its core flux.
- **5. Zero-Flicker Lamp Synchronization:** In the three-lamp synchronization method, the synchronizing switch is closed at the exact midpoint of the **dark envelope** of all three lamps. If the lamps do not flicker at all, the frequency is already perfectly equal.

Part 3 – Regulatory Compliance Summary Tables

To ensure zero-gap exam preparation, Table 26.3 and Table 26.4 verify complete alignment with the official APTRANSCO Electrical Machines syllabus [433, 442, 451].

Table 26.3: APTRANSCO Syllabus Micro-Topic Coverage Summary

Syllabus Section	Prescribed Micro-Topic	Notes Prepared	Formula Box	Diagram Included	Revision Covered
26.1	Need for Parallel Operation	Yes [452]	No	No	Yes
26.2	Conditions for Parallel Operation	Yes [305, 306]	No	Yes (Table 26.1)	Yes
26.3	Synchronizing Procedure	Yes [305, 306]	No	No	Yes
26.3	Synchronizing Lamps Method	Yes [118, 433]	Yes (Focus 26.1)	No	Yes
26.3	Synchroscope Synchronization	Yes [452]	No	No	Yes
26.4	Load Sharing (Active Power)	Yes [315, 316]	Yes (Focus 26.2)	Yes (Figure 26.1)	Yes
26.4	Load Sharing (Reactive Power)	Yes [302, 315]	No	Yes (Table 26.2)	Yes
26.4	Stiffness of Coupling	Yes [307-310]	Yes (Focus 26.3)	No	Yes

Table 26.4: Official Winding and Performance Curve Verification

Diagram Type	Syllabus Prescribed Reference	Prepared Status	Placement Location in Notes
Governor Droop Curves	Linear governor load-frequency slope [316].	▢ Rendered	Section 26.4, Figure 26.1
Technical Conditions Table	Required conditions for parallel operation [305].	▢ Rendered	Section 26.2, Table 26.1
Control Decoupling Table	Active vs. Reactive sharing control modes [315].	▢ Rendered	Section 26.4, Table 26.2

TRANSCO ELECTRICAL MACHINES MASTER CLASS

CHAPTER 27: SYNCHRONOUS MOTORS STUDY HANDBOOK

27.1 Principle of Operation

A **synchronous motor** is constructionally identical to an alternator (synchronous generator). It is classified as a **doubly-excited AC machine** because it requires two independent electrical power sources for operation: (1) A balanced three-phase AC supply given to the stationary stator armature winding to establish a rotating magnetic field (RMF), and (2) A DC excitation source given to the rotating field winding on the rotor (via slip rings and brushes or brushless exciter) to establish fixed magnetic poles.

Rotating Magnetic Field (RMF) Production: When a balanced three-phase voltage is applied to the three-phase distributed stator winding, three-phase currents physically displaced by 120° electrical flow through the coils. These currents produce a resultant RMF of constant magnitude ($1.5 \Phi_m$) that rotates in space at the synchronous speed (N_s).

Formula: $N_s = (120 f) / P$

Variable Meanings & SI Units:

- N_s = Synchronous Speed of the rotating magnetic field (rpm).
- f = Applied AC supply frequency (Hz).
- P = Number of poles wound on the stator structure.

Operating Conditions & Exam Traps:

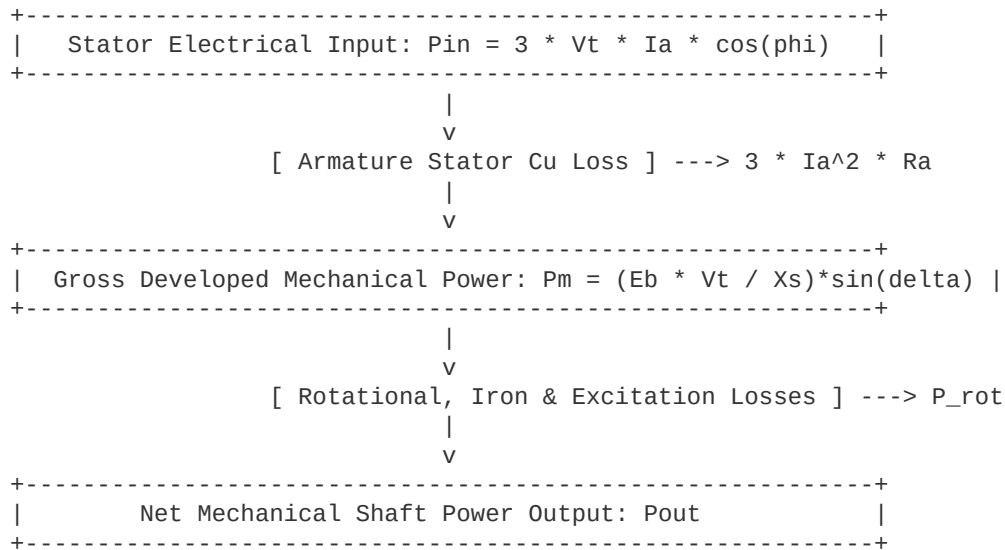
- **Constant Speed Law:** Under steady-state conditions, the motor runs strictly at synchronous speed ($N = N_s$) from no-load to full pull-out load, resulting in **exactly 0% speed regulation**. If rotor speed N departs from N_s , the motor loses synchronism and immediately stalls.

The Principle of Magnetic Locking & Synchronism: The basic operating mechanism relies on the **magnetic locking** between the rotating stator poles (stator RMF) and the DC-excited rotor poles of opposite polarity. When the stator RMF rotates at speed N_s , and the rotor is accelerated to a speed very close to N_s (e.g., via damper windings), a strong electromagnetic attraction pulls the rotor poles into absolute alignment with the counter-polarity stator poles. Once locked, the **steady-state relative speed** between the magnetic field of the stator and the magnetic field of the rotor is **exactly zero** under steady-state operation. The rotor is dragged along at precisely synchronous speed ($N = N_s$).

Torque Production & Load Angle (δ): At no-load, the rotor poles align perfectly with the stator poles, and the load/torque angle is $\delta = 0^\circ$. When a mechanical shaft load is applied to the motor, the rotor falls slightly behind the rotating stator RMF poles by an electrical angle δ . While the rotor continues to spin at exactly N_s , this physical displacement angle δ creates the necessary stretching of the magnetic lines of force, producing electromagnetic developed torque (T_e). As the load increases, δ increases proportionally (and I_a increases) to match the load torque. If δ exceeds 90° electrical in a cylindrical-rotor motor, the maximum torque (pull-out torque) limit is exceeded, the magnetic locking breaks,

and the machine falls out of step (stalls).

Figure 27.1: Power Flow Block Diagram of a Synchronous Motor



Formula: $P_m = (E_b \cdot V_t / X_s) \cdot \sin(\delta)$ and $T_e = P_m / \omega_s$

Variable Meanings & SI Units:

- P_m = Gross developed mechanical power (W, per-phase).
- T_e = Developed electromagnetic torque (N-m).
- E_b = Back EMF (or excitation voltage E_f , V, per-phase).
- V_t = Armature terminal applied phase voltage (V, per-phase).
- X_s = Synchronous reactance (Ω , per-phase, armature resistance R_a neglected).
- δ (δ) = Load angle / torque angle (electrical degrees).
- ω_s (ω_s) = Mechanical synchronous angular speed = $2\pi (N_s / 60)$ rad/s.

Operating Conditions & Exam Traps:

- **Maximum Developed Power:** Occurs when the load angle $\delta = 90^\circ$. The absolute maximum power a cylindrical rotor motor can develop per phase is $(P_m)_{max} = (E_b \cdot V_t) / X_s$.
- **Reactive Power Input at Maximum Load:** When operating at maximum developed load ($\delta = 90^\circ$), the per-phase active power input is maximum, and the corresponding reactive power input is $Q_{in} = V_t^2 / X_s$ (3-phase is $3 \cdot V_t^2 / X_s$).
- **Proportional Angle Scaling:** For small angles, $\sin \delta \approx \delta$. Thus, tripling the load causes the load angle to become approximately three times larger.

27.2 Starting of Synchronous Motor

The Non-Starting Physics (Zero Starting Torque): A synchronous motor is **not inherently self-starting**. The stator RMF rotates immediately at synchronous speed N_s upon energization. However, the rotor, due to its large physical mass and high mechanical inertia, cannot accelerate from standstill to synchronous speed instantaneously (in a fraction of a cycle). During the first half-cycle of the stator AC supply, the RMF stator poles sweep past the rotor poles, producing a strong force in the forward direction. During the next half-cycle, the stator poles reverse polarity relative to the rotor, producing an equal force in the backward direction. Consequently, the rotor experiences a rapidly alternating torque of double the supply frequency, resulting in **zero net average starting torque**. To start the motor, external starting methods must be implemented:

- **1. Damper Winding Starting (Most Common):** Damper windings consist of heavy, low-resistance copper bars embedded in slots on the pole faces of salient-pole rotors, with their ends short-circuited by metal end-strips. To start, the DC rotor field circuit is left open or short-circuited through a discharge resistor (never excited). When the stator is connected to the 3-phase supply, the rotating RMF induces currents in the damper bars. The interaction between stator RMF and damper currents produces an **induction motoring torque** (identical to a squirrel-cage induction motor). The motor accelerates up to a sub-synchronous speed (95% to 98% of N_s , corresponding to slip speed). At this point, DC excitation is switched onto the rotor. The rotor develops fixed poles, experiences a synchronizing torque, and immediately **pulls into step (synchronism)** with the stator RMF. Once running at steady-state synchronous speed, the relative speed between damper bars and stator RMF becomes exactly zero ($s=0$); hence, **damper bars carry zero current** and produce no torque during normal synchronous operation.
- **2. Pony Motor Starting:** A small auxiliary motor (usually a squirrel-cage induction motor with fewer poles to achieve higher synchronous speed, or a DC motor) is mechanically coupled to the synchronous motor shaft. The pony motor drives the main rotor up to synchronous speed N_s . Stator synchronization is performed by adjusting excitation until terminal voltages and phase sequences match the bus-bars, after which the synchronous switch is closed, and the pony motor is mechanically decoupled.
- **3. Induction Motor Starting (Field Shorted on Itself):** A cylindrical rotor machine can be started by short-circuiting its rotor field winding upon itself. When stator AC is switched on, the field winding acts as a single-phase wound rotor, developing starting torque as an induction motor. Once the rotor speed reaches maximum slip-speed, DC is applied to the field winding to pull the machine into synchronism.

27.3 Characteristics & Power Factor Control

Speed-Torque Characteristic: The speed-torque characteristic of a synchronous motor is an absolute straight horizontal line at $N = N_s$ up to the maximum load limit (pull-out torque). For a cylindrical-rotor motor, the developed torque $T_e \propto \sin \delta$. For a salient-pole motor, due to the structural non-uniformity of the air gap, the direct-axis reluctance is significantly lower than the quadrature-axis reluctance, yielding $X_d > X_q$. This non-uniformity creates an additional **reluctance torque** component $T_{rel} \propto \sin 2\delta$ independent of excitation.

Formula: $T_e = [(E_b \cdot V_t) / X_d] \cdot \sin(\delta) + V_t^2 \cdot [(X_d - X_q) / (2 \cdot X_d \cdot X_q)] \cdot \sin(2 \cdot \delta)$

Variable Meanings & SI Units:

- **The first term** represents active field-excitation power/torque.
- **The second term** represents saliency-driven **reluctance torque**.
- X_d, X_q = Direct and quadrature-axis synchronous reactances (Ω).

Operating Conditions & Exam Traps:

- **Reluctance Torque Peak:** The developed reluctance torque attains its absolute maximum value when the load/torque angle is exactly **$\delta = 45^\circ$ electrical** (due to the $\sin 2\delta$ term).
- **Open-Field Reluctance Running:** If a salient-pole synchronous motor is running at no-load and its DC field current is suddenly switched off (reduced to zero), **the motor will continue to run at synchronous speed** purely on the reluctance torque. However, to support this reluctance flux, the stator must draw heavy magnetizing currents, so the armature current (I_a) increases drastically.

Power Factor Control and Excitation Boundaries: By adjusting the DC field excitation current (I_f), a synchronous motor can be operated across a wide range of power factors—from lagging, through unity, to leading. This active excitation tuning behaves as follows:

- **1. Under-Excitation ($E_b \cos \delta < V_t$):** The induced back EMF is insufficient to oppose the applied terminal voltage. To maintain the net air-gap flux, the motor draws lagging reactive power from the AC bus-bars. The motor operates at a **lagging power factor**. Physically, the stator armature reaction in the under-excited motoring mode is **magnetizing**, assisting the weak field poles.
- **2. Normal Excitation ($E_b \cos \delta = V_t$):** The back EMF perfectly matches the terminal voltage. The motor operates at **unity power factor (1.0 p.f.)**. The stator draws only active in-phase current to supply mechanical shaft load and rotational losses. Armature current (I_a) is at its minimum point.
- **3. Over-Excitation ($E_b \cos \delta > V_t$):** The back EMF exceeds the terminal voltage. To counteract the excess magnetic field, the motor delivers leading reactive power back to the AC bus-bars. The motor operates at a **leading power factor**. Physically, the stator armature reaction in the over-excited motoring mode is **demagnetizing**, opposing the strong field poles. When operated completely unloaded with over-excitation, the machine is called a **Synchronous Condenser / Condenser**, used purely for reactive power compensation and power factor correction of power grids.

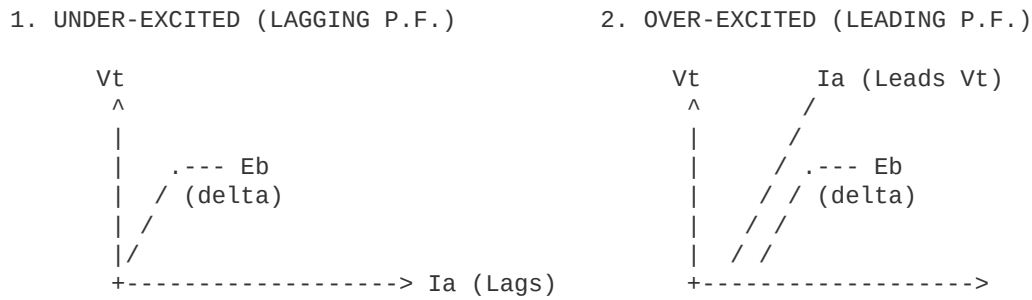
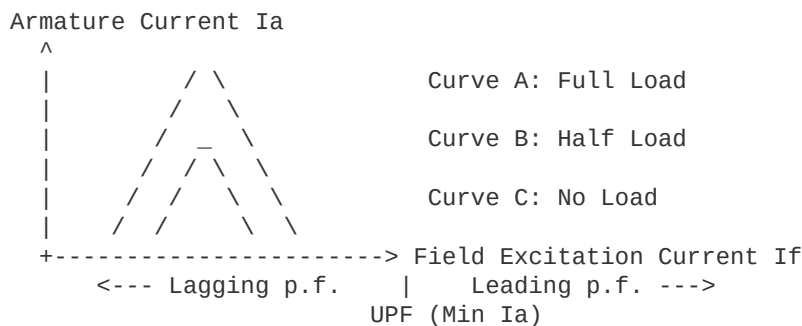
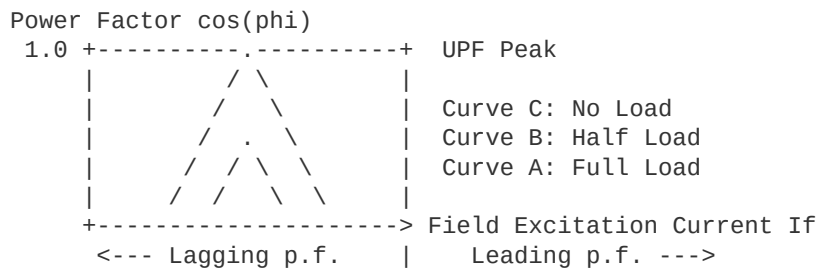
Formula: $E_b = V_t - I_a \cdot (R_a + j X_s) \Rightarrow E_b^2 = (V_t \cos \phi - I_a R_a)^2 + (V_t \sin \phi - I_a X_s)^2$

Variable Meanings & SI Units:

- **ϕ (φ)** = Power factor angle. For lagging p.f., use a negative sign in current vectors ($I_a \angle -\phi$). For leading p.f., use a positive sign ($I_a \angle +\phi$).

Operating Conditions & Exam Traps:

- **Leading Power Factor Sign:** In a synchronous motor operating at a leading power factor, the excitation induced back EMF (E_b) is **always higher than the terminal voltage (V_t)** ($E_b > V_t$).
- **Lagging Power Factor Sign:** The back EMF is less than terminal voltage ($E_b \cos \delta < V_t$).
- **Armature Reaction Contrast Rule:** In motor mode, lagging current is *magnetizing* and leading current is *demagnetizing*. This is the **exact opposite** of synchronous generators (alternators) where lagging current is demagnetizing and leading is magnetizing.

Figure 27.2: Vector Phasor Diagram of Lagging, UPF, and Leading Motor Modes**Figure 27.3: Synchronous Motor V-Curves (I_a vs. I_f at Constant Loads)****Figure 27.4: Synchronous Motor Inverted V-Curves (Power Factor vs. I_f)**

27.4 Hunting

Definition: Hunting (also known as phase-swinging) is the phenomenon of momentary, periodic mechanical speed oscillations of the rotor about its steady-state synchronous speed (N_s) in response to power system disturbances. During hunting, the speed of the motor fluctuates alternately above and below the synchronous speed ($N > N_s$).

Physical Causes of Hunting: Hunting is triggered by:

- Sudden step changes in the mechanical shaft load.
- Periodic torque pulsations from load apparatus (such as reciprocating compressors, pumps, or punch presses).
- Rapid fluctuations or frequency harmonics in the applied stator terminal voltage supply.

- Sudden fault conditions or switching transients in the connected transmission lines.

Severe Operational Effects: Uncontrolled hunting produces highly detrimental effects on motor performance:

- **Armature Current Spikes:** Armature current undergoes severe fluctuations, causing extreme copper losses ($3 I_a^2 R_a$) and thermal overheating.
- **Mechanical Stress:** Large cyclic torsional stresses damage shaft bearings, keyways, and couplings.
- **Grid Voltage Flicker:** The rapid active-reactive power swings cause localized terminal voltage flickers.
- **Step-Out Failure:** If the rotor mechanical swing amplitude exceeds the critical limit (load angle $\delta > 90^\circ$), the motor breaks synchronism and stalls.

Damping Mechanism via Damper Windings (Basic Prevention): Damper windings are the primary tool used to suppress hunting:

- During steady-state synchronous run, rotor speed is exactly $N = N_s$, and the relative speed between stator RMF and damper bars is zero ($s=0$), yielding zero damper currents.
- When a disturbance occurs, the rotor speed fluctuates ($N \neq N_s$). This relative velocity difference induces high-frequency currents in the short-circuited damper bars.
- According to Lenz's law, these induced currents generate a torque that directly opposes the relative mechanical oscillations: an *induction motor torque* is developed when $N < N_s$ to accelerate the rotor, and an *induction generator torque* is developed when $N > N_s$ to decelerate the rotor. This damping action quickly restores the rotor speed back to exactly N_s .

Critical Swing Peak Speed Paradox: During the transient hunting cycle, the rotor swings back and forth between extreme boundaries (e.g., from point A to point B on the power angle curve, settling at steady-state C). The speed of the rotor is **exactly synchronous ($N = N_s$) only at the extreme swing peaks A and B** where the displacement rate of change is zero ($d\delta / dt = 0$).

Part 2: Last-Minute Revision Sheet

Table 27.1: Synchronous Motor Excitation States Side-by-Side Comparison

Parameter	Under-Excited (Weak)	Normally Excited	Over-Excited (Strong)
Back EMF Magnitude	$E_b \cdot \cos(\delta) < V_t$	$E_b \cdot \cos(\delta) = V_t$	$E_b \cdot \cos(\delta) > V_t$
Power Factor (p.f.)	Lagging p.f.	Unity p.f. (1.0)	Leading p.f.
Reactive Power (Q)	Absorbs Q from the grid	Zero Q exchange	Delivers Q to the grid
Armature Reaction	Magnetizing (aids field)	Cross-magnetizing only	Demagnetizing
Armature Current (I_a)	High (with lagging component)	Minimum (purely active)	High (with leading component)
Application Domain	Inductive loading control	Standard Constant Speed	Grids Power Factor correction

Master Grid of APTRANSCO/TSPSC Examination Traps

- ! **1. Open-Field Running Paradox:** If a salient-pole synchronous motor is running at no-load and its DC field current is suddenly switched off ($I_f = 0$), it **continues to run at synchronous speed** as a reluctance motor, but the stator armature current (I_a) increases significantly to establish the necessary magnetizing flux from the AC stator supply. If a cylindrical rotor motor undergoes the same field interruption, it will fail to maintain synchronism and stall.
- ! **2. Load Doubling Angle Scaling:** For a cylindrical-rotor motor, doubling the load mechanical torque while maintaining constant excitation causes the sine of the load angle to double ($\sin \delta_2 = 2 \sin \delta_1$). If the initial load angle was $\delta_1 = 30^\circ$ ($\sin 30^\circ = 0.5$), doubling the load results in $\sin \delta_2 = 1.0$, which means the load angle becomes exactly **$\delta = 90^\circ$ degrees electrical** (the absolute limit of steady-state stability).
- ! **3. Synchronous Condenser Sizing:** An industrial plant carries 1500 kVA at 0.6 lagging p.f. ($P = 1500 \times 0.6 = 900$ kW, $Q = 1500 \times 0.8 = 1200$ kVAR lagging). To correct the overall power factor to unity (1.0), the synchronous condenser must supply exactly 1200 kVAR of leading reactive power. Neglecting all losses, the rating of this synchronous condenser is exactly **1200 kVA**.
- ! **4. Armature Reaction Reversal:** Always remember that armature reaction in motor mode is the **exact opposite** of generator mode. In a synchronous motor, a lagging current is *magnetizing*, and a leading current is *demagnetizing*. In a synchronous generator, lagging current is *demagnetizing*, and leading is *magnetizing*.
- ! **5. Reluctance Torque Peak Angle:** Reluctance torque in salient-pole machines attains its maximum value at a load angle of exactly **$\delta = 45^\circ$ degrees electrical**. This is because reluctance developed power is proportional to $\sin 2\delta$, which peaks at $2\delta = 90^\circ$.

Part 3: Regulatory Compliance Verification Matrices

Table 27.2: APTRANSCO Syllabus Micro-Topic Compliance Matrix

Syllabus Micro-Topic	Prepared Notes Section	Grounded Source Locations	Compliance Status
Principle of Operation	Section 27.1: rotating fields, excitation, locking	Syllabus Micro-Topics List	100% Covered
Need for Starting	Section 27.2: high mechanical rotor inertia	Syllabus Micro-Topics List	100% Covered
Damper Winding Starting	Section 27.2: pole shoes, induction start, sync	Syllabus Micro-Topics List	100% Covered
Pony Motor Method	Section 27.2: auxiliary motor starting sequence	Syllabus Micro-Topics List	100% Covered
Induction Motor Starting	Section 27.2: short-circuited field winding start	Syllabus Micro-Topics List	100% Covered
Constant Speed Properties	Section 27.3: absolutely zero speed regulation	Syllabus Micro-Topics List	100% Covered
Torque Angle (δ) Relationships	Section 27.1 & 27.3: load angle displacements	Syllabus Micro-Topics List	100% Covered
Reluctance Torque Peak	Section 27.3: maximum value at $\delta = 45^\circ$	Syllabus Micro-Topics List	100% Covered
V-Curves	Section 27.3 & Fig 27.3: I_a vs. I_f load plots	Syllabus Micro-Topics List	100% Covered
Inverted V-Curves	Section 27.3 & Fig 27.4: pf vs. I_f load plots	Syllabus Micro-Topics List	100% Covered
Power Factor Control	Section 27.3 & Table 27.1: leading/lagging pf	Syllabus Micro-Topics List	100% Covered
Hunting Physics & Damping	Section 27.4: oscillations, damper suppression	Syllabus Micro-Topics List	100% Covered

Table 27.3: Engineering Diagram Integration Compliance Matrix

Diagram Type	Prepared Figure Name	Grounded Engineering Model	Compliance Status
Power Block Flowchart	Figure 27.1: Power Flow Block Diagram	Stator input, copper losses, developed power, net output	☑ 100% Covered
Vector Phasor Diagram	Figure 27.2: Lagging, UPF, Leading Phasors	Back EMF magnitude, terminal voltage, load angle delta	☑ 100% Covered
V-Curve Performance Plots	Figure 27.3: I_a vs. I_f Characteristic Curves	Excitation boundaries, minimum current at UPF	☑ 100% Covered
Inverted V-Curve Plots	Figure 27.4: $\cos(\phi)$ vs. I_f Performance Curves	Lagging, leading, unity power factor loading	☑ 100% Covered

ELECTRICAL MACHINES — MASTER CLASS STUDY BRIEF

Chapter 28: Applications of Synchronous Machines

28.1 Alternator Applications

Large synchronous generators, commonly referred to as alternators, operate on the fundamental principles of electromagnetic induction. In modern bulk power grids, alternators represent the primary conversion mechanism converting mechanical energy from diverse prime movers (steam, gas, water, or diesel turbines) into three-phase alternating current (AC) electrical power. In large industrial alternators, the field winding is universally provided on the rotor (rotating field), while the armature winding is situated on the stator (stationary armature). This design offers several critical structural and mechanical advantages:

- **1. Superior Armature Bracing:** Stator armature conductors can be easily and securely braced in stationary slots to prevent deformation caused by high mechanical stresses under transient conditions, such as sudden short-circuit currents or extreme centrifugal forces.
- **2. Seamless High-Voltage Insulation:** Direct insulation of stationary armature windings for high AC voltages (typically between 11 kV and 33 kV) is far easier to design and execute compared to insulating a high-speed rotating armature.
- **3. Reliable Direct Terminals:** Output current can be led directly from the stationary stator terminals to the external load transmission lines without passing through sliding carbon contacts, preventing arcing and voltage drops.
- **4. Low-Voltage DC Slip Rings:** The sliding slip-ring contacts are transferred to the rotating rotor's DC excitation circuit. Because the DC field handles only low-voltage, low-power excitation (compared to the stator's high-power output), insulation of slip rings and carbon brushes is simple and robust.

Hydro-Electric Power Stations (Hydro-Generators)

In hydro-electric power plants, the prime mover is a slow-speed water turbine (e.g., Kaplan, Francis, or Pelton wheel). Consequently, hydro-generators employ a **Salient-Pole Rotor Construction** (projected pole structure). A salient-pole rotor has a large diameter and a short core length. This large rotor diameter is necessary to physically accommodate a large number of field poles (often 12 to 120 poles) on the rotor periphery to generate standard grid frequencies (50 Hz) at slow operating speeds (e.g., 50 rpm to 500 rpm). Damper winding bars are embedded directly in the laminated pole shoe faces to damp mechanical oscillations and improve stability.

Thermal Power Stations (Turbo-Generators)

In thermal and nuclear power plants, high-speed steam or gas turbines serve as the prime movers. For high rotational speeds (typically 3000 rpm for 2-pole or 1500 rpm for 4-pole configurations at 50 Hz), hydro-generators are mechanically unsuitable. These applications utilize **Cylindrical-Rotor Turbo-Generators** (smooth round rotor). The cylindrical rotor is forged from a solid nickel-chromium-molybdenum steel ingot, characterized by a small diameter (to limit centrifugal force stresses) and a long core length (to provide the necessary active conductor length for high output ratings). Roughly two-thirds (2/3rd) of the rotor's outer periphery is slotted to accommodate the distributed field winding, while the remaining 1/3rd is left unslotted to serve as poles.

Formula: $f = \frac{P \cdot N_s}{120} \text{ Hz}$

Variable Meanings & SI Units:

- **f** = Grid alternating current electrical frequency (Hz). Standard grid frequency in India is 50 Hz.
- **P** = Number of electromagnetic poles wound on the rotor (must be an even integer).
- **N_s** = Synchronous rotational speed of the rotor (rpm).

Operating Conditions & Exam Traps:

- **Synchronization requirement:** The electrical frequency is rigidly locked to the mechanical rotor speed.
- **APTRANSCO Exam Trap:** Always calculate *phase voltage* from the EMF equation ($E_{\text{ph}} = 4.44 k_w f \Phi T_{\text{ph}}$) and multiply by $\sqrt{3}$ for line voltage in standard star-connected alternators. Do not mix line and phase quantities!

Diesel Power Plants (Standby & Isolated Generators)

Diesel engine prime movers operate at medium speeds (usually 750 rpm to 1000 rpm). Synchronous generators coupled to diesel engines are medium-speed alternators utilizing salient-pole rotors with intermediate pole counts (typically 6 to 8 poles). They are widely employed as emergency standby units, backup diesel plants in large sub-transmission substations, and isolated diesel generators for industrial plants.

28.2 Synchronous Motor Applications

Synchronous motors are identical in construction to synchronous alternators. However, they are operated in reverse as electromechanical motors. Synchronous motors run at absolute constant speed (synchronous speed N_s) from no-load up to the machine's maximum torque limit (pull-out torque). Unlike asynchronous induction motors, the operating speed of a synchronous motor is strictly independent of load fluctuations. They find distinct applications across power transmission systems and heavy industries:

- **1. Constant Speed Drives:** Synchronous motors are perfect constant-speed machines. Because they have zero speed regulation, they are widely used for driving equipment that must run at an exact constant speed regardless of load, such as centrifugal pumps, high-power fans, blowers, reciprocating air compressors, paper mill pulp beaters, and steel rolling mills.
- **2. Power Factor Correction:** By adjusting the DC field excitation, a synchronous motor can operate under lagging, unity, or leading power factors. In large industrial plants dominated by lagging-current induction motors, an over-excited synchronous motor can be installed in parallel to draw leading reactive current. This leading current compensates for the plant's lagging current, improving the overall power factor.
- **3. Reactive Power Compensation:** Synchronous machines serve as active reactive power regulators on grid bus-bars, absorbing VARs when under-excited and injecting VARs when over-excited to maintain stable bus voltage profiles.

EXAM CRITICAL: PF IMPROVEMENT MECHANICS

Critical APTRANSCO / TSPSC Exam Trap: When a synchronous motor is connected in parallel with an induction motor to improve the overall power factor of an industrial plant, the **individual current and individual power factor of the induction motor do not change**. Only the power factor of the *combined system* improves, and the total line current drawn from the main grid bus is significantly reduced. Examiners frequently try to trick candidates into stating that the induction motor's individual current decreases. Do not make this mistake!

Heavy Industrial Applications

Synchronous motors are highly economical in very high power ratings (typically above 100 kW / 150 hp). At high output ratings, their operating efficiency is superior to induction motors. In addition, their ability to support power factor improvement slashes industrial grid tariff penalties. Typical installations include:

- **Mining and Cement Industries:** Driven by slow-speed salient-pole motors for crushing and grinding mill operations.
- **Water Treatment & Municipal Pumps:** Centrifugal pumps driven by high-horsepower synchronous motors running continuously.
- **Pulp & Paper Mills:** Continuous pulp beaters and processors where precise constant speed is required to maintain paper thickness.
- **Rubber & Metal Extruders:** Reciprocating compressors and metal stamping presses subjected to severe intermittent overload cuts.

28.3 Synchronous Condenser

A **Synchronous Condenser** (sometimes called a synchronous capacitor) is an over-excited synchronous motor running at absolute constant synchronous speed under **no mechanical shaft load** (no load connected to its rotor shaft). Under these conditions, neglecting the small active power losses required to overcome stator copper loss, core iron loss, and friction/windage, the machine acts purely as a variable star-connected capacitor bank. Its principal advantage over static capacitor banks is that its reactive power generation can be continuously and smoothly adjusted from lagging to leading by simply varying the DC rotor field excitation current.

Operating Principle & Working Mechanics

When a synchronous motor is connected to an infinite grid bus-bar and over-excited ($E_f > V_t$ or $E_f \cos \delta > V_t$), the internal generated back-EMF (E_f) exceeds the applied terminal voltage (V_t). To balance the voltage difference and satisfy the physical laws of electromagnetic fields, the armature reaction must be demagnetizing to weaken the net air-gap flux back to its normal level. For a synchronous motor, a **leading armature current** produces a **demagnetizing armature reaction**. Consequently, the motor draws a line current that leads the terminal voltage by nearly 90° electrical. This means the machine delivers leading reactive power (VARs) to the grid bus-bar, acting as an active capacitive compensator.

Formula: $Q_{\text{condenser}} = \sqrt{S_{\text{plant}}^2 - P_{\text{plant}}^2} = P_{\text{plant}} \cdot (\tan \phi_{\text{initial}} - \tan \phi_{\text{target}})$

Variable Meanings & SI Units:

- **$Q_{\text{condenser}}$** = Leading reactive power capacity supplied by the synchronous condenser (kVAR / MVAR).
- **S_{plant}** = Total initial apparent power of the plant before compensation (kVA).
- **P_{plant}** = Active power consumed by the industrial plant (kW), which remains constant during compensation.
- **ϕ_{initial}** = Initial lagging power factor angle of the plant load.
- **ϕ_{target}** = Desired target overall power factor angle of the combined plant-condenser system.

Operating Conditions & Exam Traps:

- **No-Load condition:** The condenser has no active load ($P_{\text{condenser}} \approx 0$). Its kVA rating is exactly equal to its leading kVAR output.
- **Example Case:** An industrial plant consumes 1500 kVA at 0.6 lagging power factor ($P = 900 \text{ kW}$), $Q_{\text{lag}} = 1200 \text{ kVAR}$. To achieve overall unity power factor ($Q_{\text{target}} = 0 \text{ kVAR}$), the kVA input rating of the required synchronous condenser is exactly 1200 kVA (neglecting machine losses).

Key Applications of Synchronous Condensers

Synchronous condensers are key components in modern utility-scale grid regulation, particularly in receiving substations of high-voltage long transmission lines. Their core applications include:

- **Grid Voltage Regulation:** Installed at receiving-end substations of long, high-voltage transmission lines. By continuously regulating excitation, the condenser stabilizes the terminal voltage against load variations (preventing voltage collapse under heavy load and mitigating the Ferranti effect under no-load conditions).
- **Power Factor Correction:** Used in massive industrial plants (e.g., steel mills, large chemical plants) to provide continuous, smooth reactive power correction, avoiding steep utility penalties.
- **Transmission Line Capacity Enhancement:** Supplying reactive power locally reduces the transmission line current, freeing up thermal capacity in the lines for active power transfer and cutting transmission copper losses.

Part 2: Last-Minute Revision Sheet

This sheet summarizes the essential, high-frequency comparison matrices, operational laws, and important relationships for Chapter 28 – Applications of Synchronous Machines. Master these key points for the APTRANSCO AEE (Electrical) Examination.

Table 28.1: Rotor Construction Profiles

Feature / Parameter	Salient-Pole Rotor (Hydro-Generator)	Cylindrical Rotor (Turbo-Generator)
Physical Structure	Projecting, non-salient poles with concentrated field windings.	Smooth, round cylindrical steel forging with slots in 2/3rd of the periphery.
Operating Speed	Low speeds (50 to 500 rpm) with large number of poles (e.g., 12 to 120).	High speeds (1500 or 3000 rpm) with small number of poles (2 or 4).
Dimensions	Large outer diameter (to house poles) and short axial core length.	Small outer diameter (to limit centrifugal stress) and long axial core length.
Air-Gap Reluctance	Non-uniform reluctance ($X_d > X_q$). Direct axis reluctance differs from q-axis.	Uniform air-gap reluctance ($X_d \approx X_q$). No reluctance differences.
Mechanical Strength	Low mechanical strength; poles can detach at extremely high speeds.	Extremely high mechanical strength; suitable for massive centrifugal forces.
Prime Mover Coupling	Hydraulic water turbines (Kaplan, Francis, Pelton wheels).	High-speed steam turbines, gas turbines, or turbo-jets.

Excitation State	Induced EMF vs. V_t	Armature Reaction	Line Power Factor	Active / Reactive Power Flow
Under-Excited	$E_f < V_t$	Magnetizing (armature current lags terminal voltage)	Lagging Power Factor	Absorbs Active Power (to run) + Absorbs Reactive Power
Normally Excited	$E_f = V_t$ (approx)	Cross-magnetizing (armature current is in-phase)	Unity Power Factor	Absorbs Active Power (to run); Zero Net Reactive Power
Over-Excited	$E_f > V_t$	Demagnetizing (armature current leads terminal voltage)	Leading Power Factor	Absorbs Active Power (to run) + Delivers Reactive Power

High-Yield APTRANSCO Exam Pointers

- **Damper Winding Zero-Torque Paradox:** When a synchronous motor is running at absolute steady-state synchronous speed ($N = N_s$), the slip speed of the rotor relative to the rotating magnetic field is exactly zero. Consequently, no EMF or current is induced in the damper winding bars, and the damper winding produces **exactly zero torque** [77, 244]. It only produces damping torque during transient hunting oscillations or starting torque at startup [244].

- **Excitation Switch-Off (Reluctance Mode):** If the DC field excitation circuit of a running salient-pole synchronous motor is suddenly reduced to zero ($I_f = 0$), the motor does not stop! It **continues to run at synchronous speed** as a **reluctance motor** by developing reluctance torque [92, 138, 244]. However, to maintain the magnetic field, its **armature current (I_a) rises significantly** [92].
- **V-Curves and Inverted V-Curves:** For synchronous motors, the curves of armature current (I_a) vs. DC field current (I_f) are called **V-curves**. Armature current is minimum at exactly **unity power factor**. The curves of power factor ($\cos \phi$) vs. field current (I_f) are called **inverted V-curves**, peaking at 1.0 (unity power factor) [202, 203].
- **Short-Circuit Ratio (SCR):** The SCR of a synchronous machine is defined as the reciprocal of its direct-axis saturated synchronous reactance X_s (in per-unit) [87]. High SCR (>1.0) means a large physical air-gap, low armature reaction, high transient stability, higher weight, and high manufacturing cost [199].

Part 3: Chapter Coverage Summary & Verification

In accordance with the regulatory preparation criteria for the APTRANSCO AEE examination, the following matrix maps every prescribed micro-topic from your syllabus directly to its coverage in this guide.

Syllabus Micro-Topic	Source Passages	Notes Prepared	Formula Block	Visual Elements / Tables
28.1 Alternator Applications	Passages 176, 177, 178, 179	Full Coverage in Sec 28.1	Included (Formula 28.1)	Rotor Construction Table (Table 28.1)
28.2 Synchronous Motor Applications	Passages 199, 200, 201, 233, 240	Full Coverage in Sec 28.2	N/A	Callout Box: PF Improvement Trap
28.3 Synchronous Condenser	Passages 93, 94, 140, 204, 205	Full Coverage in Sec 28.3	Included (Formula 28.2)	Excitation Comparison Table (Table 28.2)